🤿 **Challenger Deep**

Ghost Ships and the Rubber Duck Armada

Derelict vessels, ocean drift science, and what the sea gives back

Organization SuperResearcher
Date 2026-10-30

A ship without a crew is not one thing but three: a legend when nobody can check the story, a report when somebody dates and places a sighting, and a case file when an institution takes custody of the facts. This report follows vessels and objects that outlived their owners' intentions: a Hudson's Bay Company steamer abandoned in Arctic ice in 1931 and reported drifting for three decades; a Japanese fishing vessel carried uncrewed across the Pacific by a tsunami; sneakers and bath toys that became scientific instruments when they washed ashore; and the lost nets, evasive tankers and contested wrecks that still drift, fish and hide with nobody aboard. The connecting thread is observation. Drift became measurable once lost cargo started arriving with dates, positions and serial numbers: beachcombers' finds let oceanographers test drift models, instrumented buoys turned anecdote into an archive, and the 2011 Japan tsunami debris field — visible from orbit for a month, then lost to view — showed both the reach and the limits of that science, with few of thousands of reported objects ever confirmed. The same machinery explains where plastic gathers: more than five trillion pieces at the surface, a Pacific accumulation zone growing faster than early estimates suggested, a microplastic fraction still being revised upward. Where observation ends, institutions take over: derelict vessels accumulate because authority to seize and remove them is fragmented, and the model programme built for the western United States shows what a complete system — registration, prevention, turn-in schemes, removal, dedicated funding — requires. The final chapters turn to disappearance as strategy and as fate. Lost fishing gear keeps catching for years. Satellites now expose fleets that never transmit their positions, from illicit squid boats to tankers moving sanctioned oil under convenience flags. Sunken warships pose the legal form of the same question — whether decades of official silence amount to abandonment — and courts generally hold that governments surrender little by doing nothing. In the Mediterranean, forensic reconstruction of rescue incidents documents how blame migrated onto the rescuers as the crossing grew deadlier. The report closes with the sea's own violence: one rogue wave, measured in a North Sea storm, and the models still learning to say what the water will do next.

The Ghost Ship Phenomenon

Famous derelicts and the stories attached to them

A derelict becomes famous when a story attaches to it — and the stories that survive are usually the ones that left paperwork: a museum accession ledger, a coast guard photograph, a removal bill. The cases gathered in this report range from a Hudson’s Bay Company steamer that drifted in Arctic ice for decades, to a Japanese fishing vessel that crossed the Pacific uncrewed in about a year, to an aging crab boat in Oregon whose fame is a cautionary line in a policy report. What they share is not mystery for its own sake. It is the gap between a vessel’s working life and its long, uncontrolled afterlife — and the people who kept encountering, boarding, salvaging, photographing, and eventually paying for that afterlife.

The archetype here is SS Baychimo, the Hudson’s Bay Company cargo steamer abandoned in the Beaufort Sea icepack and described by the researchers who reconstructed her history as “the legendary abandoned vessel that floated in the Beaufort Sea icepack for over 30 years,” said to make “occasional appearances for over 30 years in the Alaskan Chukchi and Beaufort Sea region” [1]. That phrasing matters: the 30-year career as a ghost ship is reportage preserved in the literature the researchers cite, not a continuous documented track. Her last working season already foreshadowed the drift. Sea ice in the summer of 1931 was difficult, and travelling east Baychimo “only rounded Point Barrow on 22 August, the latest she had ever entered the Beaufort Sea on her annual run to Coronation Gulf” [1]. She reached Fort Hearne on 5 September, spent four days offloading supplies and taking on cargo — including eight crates of filmmaker Richard Finnie’s personal gear and ethnological specimens — and then turned for home [1].

The homeward leg never completed. Outward-bound, Baychimo passed Point Barrow travelling southwest on 18 September after difficult navigation through thick ice in the Beaufort Sea, but by 25 September she “was locked in ice in Peard Bay near Franklin Point” [1]. By mid-October it was clear the vessel was trapped; most passengers and

crew were evacuated by air to Nome, while a small contingent built a shore camp to wait through the winter and keep watch on the ship [1]. After a late November storm, the ship had vanished, and the shore crew assumed she had sunk [1]. She had not. In early December the abandoned ship was spotted by Iñupiat hunters and trappers near Skull Cliff, south of Barrow, who salvaged several bales of furs from her hold and notified the shore party; by February 1932 she had disappeared again, and the remaining shore party was evacuated by air to Kotzebue [1]. In subsequent years, "the vessel, locked and drifting with the pack ice, was spotted and occasionally boarded," and some cargo was retrieved, although she rarely appeared close to shore [1].

One documented boarding shows how a ghost ship's story gets carried ashore in pieces. On 11 August 1933, Baychimo was visited by the crew of MS Trader while anchored off the village of Wainwright — owner Ira Rank, the brothers Peter and Kari Palsson as captain and mate, and British traveller Isobel Wylie Hutchison [1]. The crew tied up to Baychimo and tried to break her free from the ice to tow her somewhere she could be salvaged, but she was too large for the smaller vessel to free from the ice pack [1]. They nevertheless recovered several items from her [1]. Hutchison, who was in the Arctic ostensibly collecting Alaskan botanical specimens for the Royal Herbarium at Kew, recorded acquiring Copper Inuit parkas and artifacts "from a sack of curios" aboard Baychimo, objects later donated to the Cambridge University Museum of Ethnology [1]. Finnie himself later confirmed the connection, writing that he recognised the costumes in a photograph in her book: "they were part of the ethnological collection I had entrusted to the Baychimo" [1].

That paper trail is why Baychimo can be told as history rather than only as legend. A small collection of Canadian Arctic ethnographic artifacts entered the University of Alaska Museum in 1934, given accession number Acc. 514 and placed in boxes labelled "From Canadian Arctic taken from the 'Beychimo'" [1]. The assemblage then sat in the museum's archaeological collections for 81 years "with no recognition of its historical importance or its relationship to the abandoned Hudson's Bay Company cargo steamer, SS Baychimo" [1]. It was rediscovered only in October 2015, when researchers combing the collections for a circumpolar archaeology class opened a drawer labelled "Acc. 514 Canadian Arctic" and found specimens typical of early 20th-century Copper Inuit and Netsilik groups [1]. The collection itself is small — only 14 artifacts in total, characteristic of specimens collected from the Copper and Netsilik Inuit in the Northwest Territories and Nunavut [1] — but it anchors the famous ship to physical, catalogued objects, from a musk-ox horn blubber pounder to a copper ulu with a caribou antler handle [1].

Eighty years after Baychimo, a very different ghost ship made the same point in a single photograph. The Ryou-Un Maru was a 50-metre "ghost ship" discovered in Canadian waters about 200 km off the coast of Haida Gwaii on 20 March 2012, photographed by the Canadian Coast Guard [2]. She was one of the first very large pieces of Japan tsunami debris to reach the west coast of North America in the first half of 2012, alongside a shipping container carrying a motorcycle found on Haida Gwaii on 18 April 2012 and the large concrete dock that washed up on Agate Beach, Oregon, on 5 June 2012 [2]. These objects shared significant windage, which "is believed that this has played an important role in their early arrival at the coast" [2]. Unlike Baychimo, the Ryou-Un Maru arrived inside a monitoring era — and the monitoring still mostly failed to see her coming: there had been no monitoring of even large pieces of debris, so there was no advance warning of these arrivals, and the Oregon dock was first reported floating offshore only the day before it beached [2].

The tsunami context also shows how thin the line is between a famous derelict and ordinary debris. NOAA received 1,385 reports of tsunami debris sightings as of 18 October 2012 — 620 shoreline sightings and 765 at-sea sightings — yet only 14 objects had by then been definitively confirmed to originate from the Tohoku tsunami [2]. One of the first confirmed reports came from the Russian sailing vessel STS Pallada, which on 22 September 2011, about 800 km northwest of Midway Atoll, encountered a small fishing boat registered in Fukushima, Japan, and found radiation levels below detection limits when it tested the boat [2]. The Government of Japan's own catalogue of at-sea sightings recorded some 97 sightings from April through November 2011, most of them larger objects such as small vessels and containers, with many observed vessels reported capsized or partially submerged [2]. A derelict, in other words, is also a data problem: most floating objects are never identified at all.

Not every famous derelict drifts romantically. The policy literature preserves the F/V Western, a 69.9 ft long, 78 gross ton, wood-hulled former crabbing vessel from 1934, as a case study in how derelicts are made on purpose-adjacent neglect [3]. Denied moorage at the Port of Coos Bay, Oregon, in 2014 because of her condition, the boat anchored unauthorised in Coos Bay, was battered by storms and beached at least twice, and finally sank in 2015 near a busy navigation channel in sensitive fish and invertebrate habitat [3]. The arithmetic is the story: while floating, she would have cost $30,000 to remove; once sunk, removal cost $95,000 [3]. The same report generalises the pattern — vessels become abandoned or derelict through aging and weathering, neglect, technological change, own-

ers unable to keep up with maintenance costs, maintenance costs exceeding the vessel's commercial value, and damage from an incident exceeding the vessel's value [3].

Even a famous ship's cargo can out-drift the ship. When the Hansa Carrier, en route from Korea, was hit by a storm on 27 May 1990 at 48 degrees north, 161 degrees west, 21 containers were swept off deck, five of them holding 80,000 Nike shoes [4]. The shoes washing up on Oregon beaches — and the beachcomber swap meets held to match lefts with rights — became the founding anecdote of modern drift science, because each Nike tongue carried a serial number distinguishing the shipwrecked shoes from ordinary losses [4]. Oceanographer Curt Ebbesmeyer used those numbers to deduce that only four of the five containers had likely opened, spilling 61,280 shoes into the sea [4]. The Hansa Carrier herself is barely remembered; her spilled cargo is famous.

That is the pattern this report follows. A famous derelict is rarely famous as a hull. Baychimo is remembered through a filmmaker's crates, a museum drawer, and a traveller's memoir [1] [1]; the Ryou-Un Maru through a coast guard photograph and a tsunami debris count in which she was one of only 14 confirmed objects among 1,385 reports [2]; the F/V Western through a removal-cost comparison in a blue-ribbon policy report [3]. The sections that follow take each thread in turn — the Arctic ship and her scattered collection, the drift science that spilled cargo made possible, the tsunami debris fleet, and the legal and practical machinery for dealing with vessels nobody claims — because a ghost ship is never just a ship. It is the trail of encounters, objects, and costs left behind when a vessel goes on without its people.

Exhibit 1. Documented cases anchoring this report (compiled from the cited sources)

Vessel / event	Year	What happened	Evidence grade
SS Baychimo	**1931**	**Hudson's Bay Company steamer abandoned in the Beaufort Sea icepack; sighted and boarded intermittently for over 30 years [1]**	**Documented — museum reconstruction**
Hansa Carrier (Nike spill)	**1990**	**Storm at 48°N, 161°W swept 21 containers overboard, five holding 80,000 Nike shoes [4]**	**Documented — shipping/ court record**
Friendly Floatees spill	**1992**	**~29,000 bath toys lost on 10 January 1992 at 44.7°N, 178.1°E; landfalls tracked by beachcombers [4]**	**Documented — landfall reports + drift model**
Ryou-Un Maru	**2011–2012**	**Japanese fishing vessel torn from its moorings by the Tōhoku tsunami; photographed off Haida Gwaii in March 2012 and sunk by the U.S. Coast Guard off Alaska [2]**	**Documented — coast guard record**
Point Estero	**2017**	**Grounded near Cayucos State Beach, California; oil removed but hull left — removal estimated at $70,000, owner uninsured, no program funding; still in place as of 2019 [3]**	**Documented — state program record**



Exhibit. Fig. 4. Above: The SS Baychimo trapped in ice in Peard Bay off of Point Franklin, northern Alaska. Below: Aircraft landing in the winter of 1931 near Wainwright, Alaska, to rescue passengers and crewmembers of Baychimo (in the background). (Unpublished photogr. Source: S0010.

SS Baychimo: the Arctic ghost ship that kept being sighted

Baychimo's fame rests on a paradox of evidence. She is, in the words of the museum researchers who reconstructed her afterlife, "the infamous 'ghost ship,' an abandoned icebound vessel said to make occasional appearances for over 30 years in the Alaskan Chukchi and Beaufort Sea region" — and the operative word in that sentence is said [1]. The sighting history that made her famous is not a ship's log or a coast guard file. It is a reconstruction, assembled decades later from secondary accounts — principally Dalton (2006) and Norton and Graves Gaylord (2004), the two sources cited again and again for every movement of the drifting ship [1] [1] [1]. Understanding Baychimo means understanding what kind of record a sighting history is: a chain of encounters, each reported by whoever happened to be on that coast, in that season, looking at the ice.

The mechanism behind the appearances was simple and is the one part of the story documented at the time. The vessel was "locked and drifting with the pack ice," so her positions were the ice's positions: she surfaced in the record when the pack carried her near a settlement, a hunting party, or another vessel, and vanished from it when the ice took her away again [1]. The researchers add the logistical corollary that explains the gaps: "she rarely appeared close to shore and transportation was difficult," so even a known sighting did not mean a reachable ship [1]. A ghost ship, in the Arctic case, is what an ice-drifting hull looks like to a sparse human network of observers — intermittent by geography, not by anything uncanny.

Who those observers were is itself part of the record. The people who kept encountering Baychimo were the working population of the western Arctic coast: Iñupiat hunters and trappers (whose December 1931 sighting was noted in the previous section), the shore party left to watch her, and the crews of the small trading vessels that worked the coast. The best-documented of those working vessels is MS Trader, described in the museum paper as "a small wooden trading vessel owned by Ira Rank and the U.S. Mercantile Company in Nome" [1]. Trader was no expedition ship; she was the coast's ordinary transport, and her crew's 1933 encounter with Baychimo entered the record precisely because it was routine traffic meeting a drifting derelict. One of her crew, Peter Palsson, is identified by the researchers as "one of the last people known to have boarded Baychimo" [1] — a quietly important superlative, because it marks where the documented sighting history effectively ends: not with a sinking, but with the last boarding anyone wrote down.

The fragility of that record is visible in its smallest details. When the museum researchers went looking for the origin of their Baychimo artifacts, they found the donor's name written in a handwritten accessions ledger as "Peter Balssan," with no address or other details, and the ship's name rendered "Beychimo" on the collection boxes [1]. An extensive literature and archival search for an archaeological site named "Beychimo," or for any donor named "Balssan," was unfruitful; only recurring references associating collector Otto William Geist with a Peter Palsson in 1930s northern Alaska broke the deadlock [1]. The researchers conclude it is "highly likely that both Baychimo and Peter Palsson's names were misspelled in the accessions ledger, written at least 16 years after the donation of the materials to Geist" [1]. A famous ghost ship, it turns out, can be misspelled out of her own archive: for decades the museum held physical pieces of her cargo under a name that matched nothing in the literature.

The researchers' reconstruction of how the sightings connect to the surviving objects is correspondingly explicit about its own inferential steps. They lay out the chain they believe the evidence supports — materials acquired by Finnie in 1930–1931 from a Canadian Arctic Inuit group or groups, procured by Palsson in 1933, and donated to the museum through Geist in 1934 [1] — and elsewhere mark its weakest link with an open caveat: the museum artifacts were "very likely taken directly from Baychimo, but we cannot exclude the possibility that they were also acquired from locals at Wainwright who had already salvaged several items from the vessel" [1]. That distinction matters for the sighting history too. A derelict that is repeatedly boarded and salvaged by different parties does not produce one story but several, and the cargo disperses along each of them; some of Baychimo's material went to Cambridge with a traveller (as the previous section described), and some stayed on the coast to be traded, handed over, and eventually accessioned under a misspelled name [1] [1].

Even the ship's visual record predates her ghost career and comes from her working life. The photograph the museum paper reproduces as its Figure 1 — "The Hudson's Bay Company steamer, the Baychimo," from the McCord Museum in Montreal — was taken by Captain George E. Mack sometime between 1910 and 1926, years before her abandonment [1]. The images of her trapped in ice in Peard Bay, and of the aircraft landing near Wainwright in the winter of 1931 to rescue her passengers and crew, survive as unpublished photographs from 1936 in the Alaska and Polar Regions Collections at the University of Alaska Fairbanks [1]. In other words, the famous Arctic ghost ship is il-

lustrated almost entirely by photographs of the ship she was before she became a ghost, and of the single season in which she was lost.

What, then, did the sightings actually establish? On the evidence in these sources: that an abandoned, ice-locked steamer persisted in the Beaufort and Chukchi pack ice after 1931, was seen and boarded more than once, shed cargo into several hands, and was still being boarded in August 1933 by a named crew whose members can be traced into museum records [1] [1] [1]. The further claim — appearances "for over 30 years" — is the literature's summary of a much longer, thinner trail, and should be read as reportage layered over that documented core, not as a surveyed track [1]. That layered structure is not a defect unique to Baychimo; it is the normal epistemology of ghost ships. The next sections follow the two hardest pieces of evidence she left behind: the ethnographic collection that was loaded onto her, and the one boarding that can be reconstructed almost hour by hour.

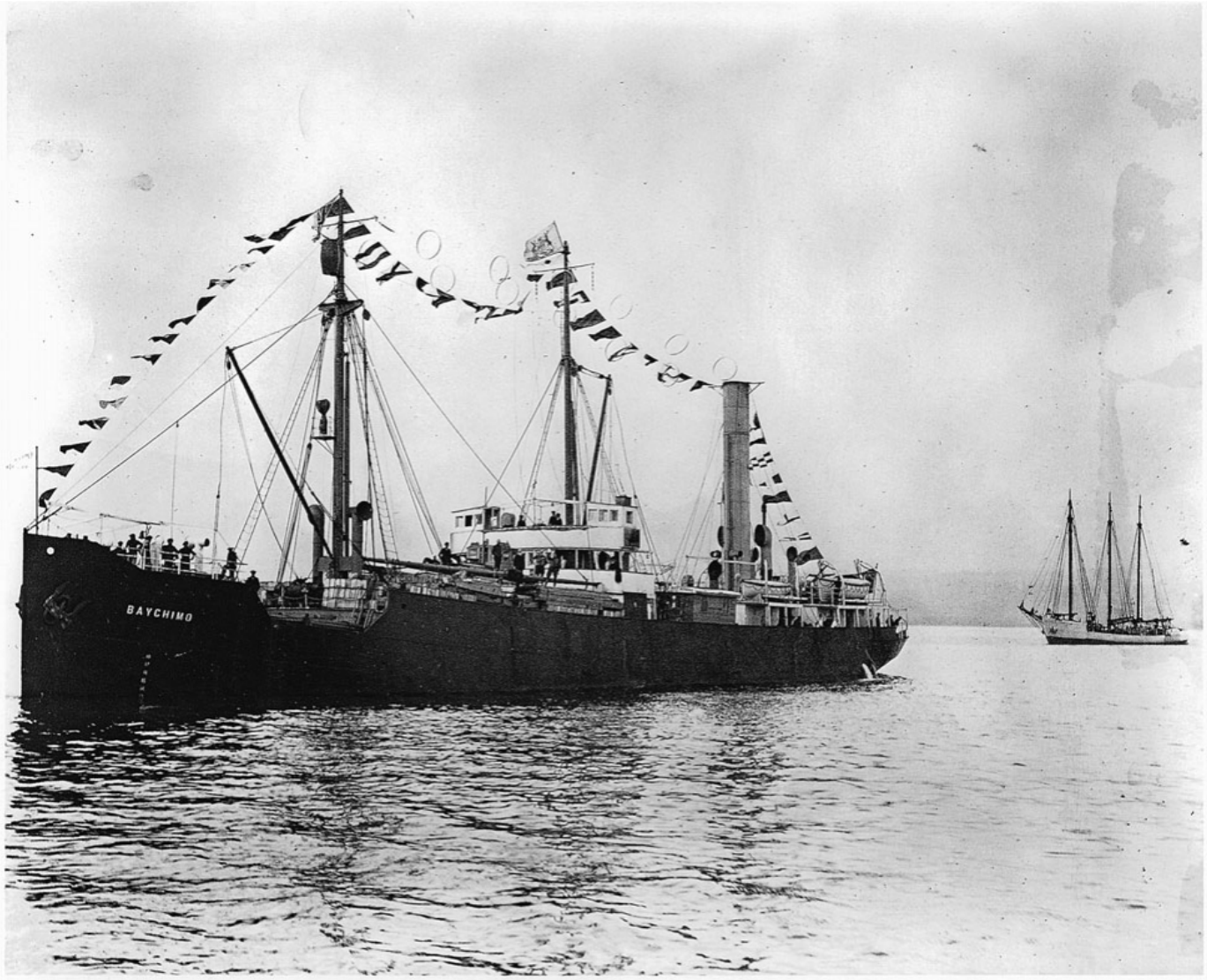


Exhibit. Fig. 1. The Hudson's Bay Company steamer, the Baychimo; photo taken by Captain George E. Mack between 1910–1926. Source: S0010.

Richard Finnie's ethnographic collections and SS Baychimo

The objects that make Baychimo historically recoverable exist because a government sent a filmmaker north to collect a culture it believed was about to disappear. In the summer of 1930, the Canadian Department of the Interior commissioned filmmaker and photographer Richard Finnie to spend a year in Canada's western Arctic as a Special Investigator [1]. His main objective, the museum researchers summarise, was to document as much as possible of the "'pristine' culture and ways of life of the Copper Eskimos that had yet to be transformed by Euro-American contact" (citing Geller 1996) [1]. That salvage premise — collect now, before contact changes everything — is the founding assumption of the collection, and the irony of its fate is that the collection itself was then scattered by exactly the maritime world it was meant to outlast.

Finnie's itinerary put him aboard Baychimo at the start of the work. He travelled north down the Mackenzie River to the Arctic Ocean, arriving at Herschel Island in late July 1930, and after a short stay embarked on the Hudson's Bay Company supply ship Baychimo, which was delivering provisions to trading posts in Coronation Gulf and as far east as Cambridge Bay [1]. He left the ship in late August at Cambridge Bay and spent the next year exploring and travelling to various western Arctic destinations, including a pioneering flight over the north magnetic pole [1]. During that year he acquired the large collection of ethnological artifacts and specimens from around Coronation Gulf that this report has been tracking; the intended destination was specific — "the ethnographic department at the Canadian National Museum in Ottawa" [1]. The epilogue of the museum paper confirms the same destination and date: "Richard Sterling Finnie collected the materials in 1930 for the Canadian National Museum in Ottawa" [1]. The collection never arrived there. Finnie himself had flown back to Ottawa earlier in the summer of 1931, while his eight crates travelled on without him — the loading of those crates onto Baychimo at Fort Hearne was described in Section 1 and is the hinge on which everything after turns [1].

Finnie the collector deserves a precise description, because the collection's later significance depends on who he was. He went on to make several photographic and film documentaries about life in northern Canada, including the documentary film on the Copper Inuit made during the 1930–1931 trip, Among the Igloo Dwellers (Geller 1996) [1]. The museum paper characterises his work as unique among his contemporaries: his depictions of the early industrial development of the north and of the intermingling of Inuit and Euro-American cultures made him "a pioneer in early film documentary in northern Canada" [1]. He later told the Baychimo story himself in print — his 1940 book Lure of the North is the source the researchers cite for the crates' intended shipment [1] [1]. Scholarship has since treated him as a subject in his own right: Geller's 1996 Film History article examines "Richard Finnie's views of natives and development in Canada's 'last frontier'" [1].

The researchers are notably restrained about the collection's ethnographic value, and the restraint is informative. The period of Finnie's trip, the epilogue notes, "was a period of major cultural and social transitions in Arctic communities, with increasing industrial resource exploitation and at the tail end of the heyday of northern scientific expeditions," and several researchers and explorers had made extensive collections from the Copper and Netsilik Inuit before him [1]. The verdict follows directly: "In and of themselves, the objects in the UAMN Baychimo collection do not contain great revelations about Copper and Netsilik Inuit groups, as many of the types of artifacts are represented in the earlier collections of Birket-Smith, Burwash, Jenness, Noice, and Stefansson" [1]. Finnie was, in collecting terms, a late arrival working ground that Stefansson (1908–1912), Jenness (1914–1916), Noice (1919–1921), Birket-Smith (1921–1924), and Burwash (1925–1926) had already covered — the comparative collections against which the museum artifacts were identified [1].

What the collection has instead of revelations is a chain. "The historical significance of the UAMN Baychimo collection," the epilogue states, "lies in the chain of linked relationships between Richard Finnie, legendary Baychimo, Peter Palsson and the crew of Trader, Isobel Wylie Hutchison, and finally Otto Geist and the United States Department of the Interior-Alaska College Archaeological Expedition" [1]. That is a remarkable reframing: a collection commissioned as a record of Copper Inuit culture derives its modern significance from a network of Euro-American and maritime relationships — collector, ship, trading crew, traveller, and a second collector — through whose hands it passed. The paper's own title announces the same conclusion, framing the work as "rediscovering lost relationships" between Canadian Inuit and Iñupiat communities through materials recovered from the ghost ship [1].

The chain also had a fork in it, and the researchers close it explicitly. "We are aware of only one other institution that has Finnie's ethnographic materials from Baychimo: the collection made by Hutchison and donated to the Cambridge University Museum of Ethnology (now, the Museum of Archaeology and Anthropology)" [1]. So the surviving Finnie material from the ship exists in exactly two places — Fairbanks and Cambridge — each half arriving by a different person's salvage from the same derelict. The rest of what was in the crates, so far as these sources record, is unaccounted for.

As for the ship that carried the collection: "The fate of Baychimo is unclear, but it most likely eventually succumbed to the ravages of the sea ice and the Arctic Ocean" [1]. No wreck, no confirmed sinking, no final position — the researchers' careful "most likely" is the whole ending. The collection outlived its ship and its collector's intentions alike: gathered in 1930 for Ottawa, it spent eight decades in obscurity in Fairbanks, and its rediscovery, the epilogue concludes, "brings these 80 year-old ethnographic materials out of obscurity, and places them in a wealth of historical context with a revived contemporary meaning" [1]. The next section reconstructs the encounter that set that detour in motion — the day a small trading vessel tied up to the ghost ship in the ice off Wainwright.



Exhibit. Richard Finnie poses with Inuit actors from his film Among the Igloo Dwellers at Tikeraq, Coronation Gulf, Nunavut, in April 1931. Source: S0010.

MS Trader visits Baychimo

The boarding itself — Trader's crew tying up to the icebound steamer off Wainwright on 11 August 1933, failing to break her free, and recovering several items — was told in Section 1. What has not been told is everything around that day: what else was aboard the derelict, who else had been taking things from her, and the oddly specific paper trail by which the Trader men's salvage reached a museum. The paper's own abstract widens the frame first: when Baychimo was abandoned, "large amounts of scientific and navigational instruments and gear and personal items were left aboard, among them an ethnographic collection gathered in 1930 from Inuit groups in the Canadian Arctic by Richard Sterling Finnie" [1]. Finnie's crates, in other words, were a fraction of a whole ship's abandoned contents, and the ship "was boarded several times over the next three years with items being salvaged by locals from nearby Wainwright and Barrow" [1].

One of those local salvages reached Finnie himself, by post, and his account of it is the most human document in the Baychimo record:

> One day I received a note from Charles Brower, veteran whaler and trader of Point Barrow, enclosing an old letter I recognized as having been in one of those cases on the derelict [Baychimo]. One of the local Eskimos who had been plundering the ship had handed it to him, and now he sent it along with the suggestion that I might like to have it as a souvenir. [1]

Finnie quotes the episode in his 1940 book (p. 47), and the museum researchers use "plundering" without embarrassment — salvage from an unguarded derelict was ordinary coastal practice, and it is why the researchers cannot be certain the museum artifacts came straight off the ship: they were "almost certainly recovered from Baychimo at the same time, either by Ira Rank or the Palsson brothers," but acquisition from Wainwright locals who had already salvaged items from the vessel cannot be excluded [1]. Hutchison's published account of the Trader visit, in her 1937 memoir North to the Rime-ringed Sun (pp. 110–115, 240), is the researchers' cited source for what the Trader party took and saw [1] [1].

The Trader connection did not end at the ice edge, because Trader and her people were woven into the coast's scientific traffic. Otto Geist — the collector who would receive the Baychimo specimens — had been sponsored since 1926 by President Charles Bunnell of the Alaska Agricultural College and School of Mines, and ran a sequence of major expeditions: the Bunnell-Geist Bering Sea Expedition (1927–1929), the Alaska Bering Sea Expedition (1931–

1933), and the US Department of the Interior–Alaska College Archaeological Expedition (1934–1935), including one of the largest excavations in Alaska to that date, at the Kukulik site near Savoonga on St. Lawrence Island [1]. From 1933 to 1937 Geist was in frequent contact with Rank and Palsson: his crews regularly travelled to Gambell and Savoonga on Trader and bought supplies from Rank's U.S. Mercantile Company [1]. The Trader men were, in effect, the expedition's local logistics.

That intimacy is what makes the handover reconstructable — and its remaining uncertainty instructive. The Trader party left Hutchison at Barrow on 9 September 1933 after the boarding [1]. Geist, meanwhile, left Savoonga on the US Coast Guard cutter Northland on 23 September 1933 and was waylaid in Nome for a month by weather; the researchers are "uncertain if Trader reached Nome by the time of Geist's departure in October 1933," but conclude they "have no reason to believe that the Baychimo collection was given to Geist during this time, and not in 1934 as stated in the UAMN ledger" [1]. The 1934 season supplies the candidate moment. On 11 July 1934 Geist asked Nome Harbor Lighterage, U.S. Mercantile, and Lomen Commercial Companies for a price to move eight people and baggage, including an umiak skin boat, from Nome to Savoonga [1]. On 17 July the party split for lack of space: Geist and three crew flew by Northern Air Transport floatplane to a lake near Gambell, while the other four members and the cargo went on a "small trading vessel" [1]. The expedition's accounting report shows a U.S. Mercantile Company boat — "which we assume is Trader" — carried that party and baggage from Nome to Savoonga on 17–18 July 1934 for $200 [1]. The researchers' conclusion is stated as a surmise, and should be quoted as one: "We surmise that on 17 July 1934, during Trader's voyage to Savoonga, that Palsson gave one of Geist's crew the ethnographic materials from Baychimo. Geist subsequently deposited the artifacts at the nascent University of Alaska Museum in Fairbanks" [1]. No other records show Trader transporting Geist or his crew in 1934 beyond that 17 July run, and Geist left Savoonga on Northland on 30 September 1934 [1].

So the most famous boarding of the Arctic's most famous ghost ship resolves, in the archive, into a $200 transport line in an expedition accounting report and a surmised handover on a routine trading voyage. That is not a deflation of the story; it is the story's evidentiary character. The abstract of the museum paper — published in Polar Record 52 (265): 464–473 in 2016, received January 2016 and first published online 7 March 2016 — summarises the whole arc in one sentence: in 1934 Palsson "gave several ethnological specimens to members of the United States Department of the Interior-Alaska College Archaeological Expedition," and that year the collection was accessioned to the nascent museum, where its relationships "remained obscure" for over 80 years [1]. The next section opens the drawer where those specimens waited.

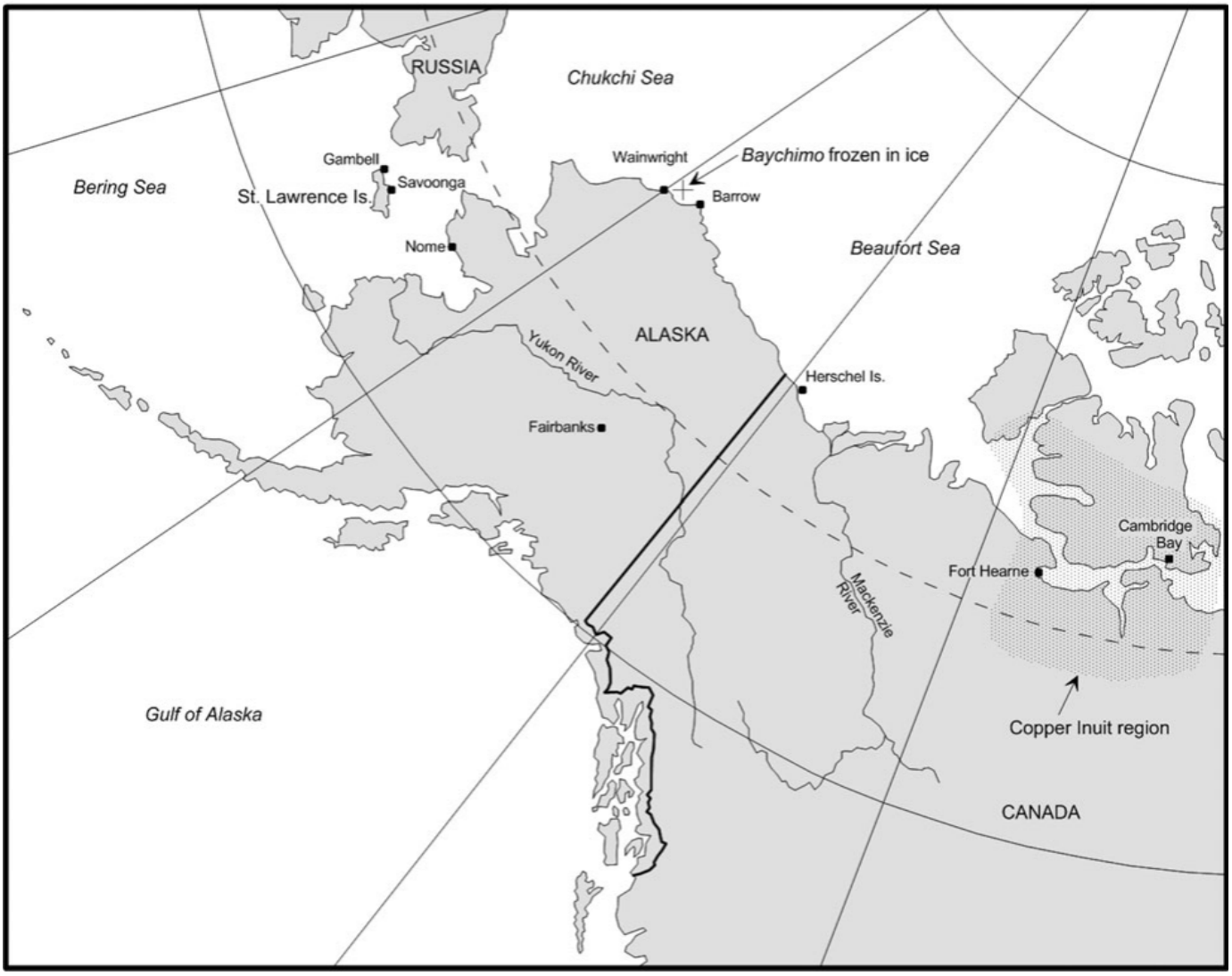


Exhibit. Fig. 3. Map of the western North American Arctic, showing locations mentioned in the text. Source: S0010.

Discovery of the Baychimo materials at the University of Alaska Museum

The drawer opening in October 2015 was described in Section 1. What made the discovery a discovery — rather than a routine inventory sighting — is the institutional history that had buried the collection, and the identification work that followed. Both are in the museum paper, and both are more specific than "forgotten for 81 years."

The burial was administrative. The artifacts were added to the museum in 1934 by Otto William Geist, described by the researchers without varnish as "the notorious collector and naturalist" [1]. In the museum's early days, "archaeological and ethnological specimens were grouped into a single anthropological collection, and differentiation between the two was not made until the early 1960s when the Archaeology and the Ethnology and History Departments became distinct entities within the museum" [1]. When the departments split, "the ethnographic artifacts under Acc. 514 were placed into the archaeological collections after the establishment of the two departments" [1]. An ethnographic collection from a named ship thus spent its museum life filed as archaeology, in the wrong department, under a misspelled ship's name — three layers of misfiling, each sufficient on its own to hide it from anyone searching by ship, by donor, or by discipline.

The rediscovery therefore required identification, not just location. The researchers describe the collection object by object, matching each of the 14 artifacts against the comparative Canadian Arctic collections of Stefansson, Birket-Smith, Jenness, Noice, and Burwash [1]. The work shows in the details they were able to certify. Two fish rakes (UA514-5250 and -5251) carry copper shims stabilising their hooks and, in one, an iron rivet repairing a cracked head; a complete rake Jenness collected at Bernard Harbour in 1914–1916, lashed to a wooden handle, matches their heads, and Damas records the type's use: put under water and pulled up with a jerk to catch several Arctic char on the spikes at each pass [1]. A copper straight-edged "man's knife" (UA514-5247) measures 22.5 cm, its slightly curved blade tang inset into a caribou antler handle [1]. Scissors (UA514-5248) run 17 cm, with antler handles and iron sheet-metal blades attached by iron rivets; their prototypes, the researchers note, probably arrived as early

as the mid-19th century through the Hudson's Bay Company and whalers, and they were used for cutting hair and sewing [1]. A skin scraper (UA514-5249), 30 cm long, pairs a cold-hammered copper blade with a caribou antler handle whose rear point forms a "fork" preventing the hand from slipping during use [1].

Two objects resist tidy identification, and the researchers say so — a useful marker of the paper's evidentiary discipline. The bone pins (UA514-5253), six objects originally logged in the 1930s–1950s ledger as "Wound pin… Bone… 6 lashed together" (in fact only two are lashed together), are mostly consistent with wound pins used to plug wounds in seals to stop bleeding so the animal could be dragged across ice (Kemp 1984) — yet individual pieces match published parallels described variously as meat skewers, wound pegs, and wound needles, with lengths from 12 to 14.5 cm for the flat, oar-shaped forms and 14 cm for the lashed nail-form pair [1]. And one piece (UA514-5252) defeats functional assignment altogether: a 6.5 cm rectangular hide patch with 12 strands of seed beads ending in caribou incisors, 18 cm in total height, whose "actual function of this piece is unclear" — possibly a brow band, hairpiece, part of a belt, or an amulet [1]. The leister side-prong (UA514-5254) is more tractable: 39.5 cm long, with alternating barbs — 10 on one side, 11 on the other — and a proximal knob to constrain its lashing to the spear shaft [1].

Read as a group, the identifications are what converted a mislabelled drawer into the Baychimo collection. Every object is an early 20th-century Copper Inuit or Netsilik type with published parallels collected between 1908 and 1926 — consistent with Finnie's 1930 Coronation Gulf collecting and inconsistent with an Alaskan archaeological site, which is what the archaeology filing had implied [1] [1]. The discovery, in short, was an act of comparative scholarship performed on the museum's own holdings: the ship had been in the building for 81 years, and it took a photograph session for a circumpolar archaeology class, a drawer label, and a shelf of expedition-era ethnographies to see her. The acknowledgments record the rediscovery's collaborative character: Fawn Carter at UAMN photographed the collection, Dave Norton discussed Arctic exploration and Baychimo with the authors, and Angela Linn provided insight on the museum's history [1].

The paper closes the loop on what the discovery means. The materials, it concludes, provide "a remarkable story of historical congruence during the early 20th century Arctic explorations," and their addition to the record brings 80-year-old ethnographic materials "out of obscurity" with "a revived contemporary meaning" [1]. With the Baychimo arc complete — ship, collector, boarding, museum — the report now steps back to ask what kind of thing a "ghost ship" is, and how much of the category is legend, how much reportage, and how much documented case.



Exhibit. Fig. 7. Left: MS Trader arriving at St. Lawrence Island, 1939. Right: Pete Palsson, Captain of the Trader, St. Lawrence Island, 1939. (O.C. and Ruth Connelly photograph album, Archives and Special Collections, Consortium Library, University of Alaska Anchorage. Source: S0010.

Ghost ships as a category: legend, reportage, and documented cases

“Ghost ship” is not a legal or scientific term. It is a storyteller’s label laid over at least three different kinds of evidence, and the first five sections of this report have now exhibited all three. Keeping them separate is not pedantry: each tier supports different claims, justifies different spending, and fails in different ways.

The first tier is **legend**: a vessel’s afterlife as transmitted by retelling, where the duration and drama grow because no continuous record constrains them. Baychimo’s “over 30 years” of appearances is this report’s standing example — a figure the literature repeats and no log substantiates. Legend is not worthless; it is what makes a derelict culturally visible at all. But it cannot be audited, and it systematically prefers the mysterious version: a ship that “kept being sighted” over a ship that mostly sat locked in ice, rarely near shore, being picked over by working salvors.

The second tier is **reportage**: sightings, journalism, memoirs, and catalogues produced near the events by identifiable observers. Hutchison’s memoir and Finnie’s 1940 book sit here, as does the Government of Japan’s catalogue of 97 at-sea tsunami debris sightings. Reportage is checkable in principle — observers can be named, dates compared — but it is opportunistic: it records what witnesses happened to encounter. The tsunami review makes the structural point about its own reportage tier, noting that the distribution of reported sightings likely reflects “the frequency of human passage rather than the actual distribution of the tsunami debris” [2]. Reportage draws the map of where people look, not where derelicts are.

The third tier is the **documented case**: a vessel that enters an administrative record — an inventory, a removal file, a cost estimate, a court or agency decision. This tier has its own governing vocabulary, and it is not “ghost ship” but “abandoned and derelict vessel” (ADV). The Blue-Ribbon Program report defines the pivot of the whole documented tier in one sentence: “Authority refers to the legal ability of a governing agency to declare a vessel ‘abandoned’ and thus remove and dispose of it” [3]. A documented ghost ship is one a government has formally declared abandoned — a legal act, with due-process consequences — and the report stresses that “a legal process for an aquatic land custodian to accomplish seizure, impoundment, removal or custody of an ADV through due process is currently lacking across most west coast jurisdictions” [3].

The documented tier, where it exists, replaces atmosphere with arithmetic. The Pacific States/British Columbia Oil Spill Task Force identified ADVs as “a common threat and a critical, emerging issue in 2017,” formed an ADV Workgroup across Alaska, California, Hawaii, Oregon, and Washington in 2018, and published a White Paper in March 2018 documenting scope and scale [3]. Its conclusions redraw the category completely. “The problem of ADVs includes both commercial and recreational vessels”; “the majority of ADVs are recreational, yet commercial vessels are typically larger and on a per vessel basis, can cost several orders of magnitude more than recreational vessels to remove”; most states face “an increasing backlog of existing or ‘legacy’ ADVs” alongside a steady stream of new ones [3]. And the institutional map is bleak: “In the US, there is no comprehensive federal program”; state programs “vary widely,” with only Washington “comprehensive” and most underfunded; in Canada the federal Abandoned and Wrecked Vessel Act is “comprehensive yet underfunded” [3]. Five US federal agencies hold fragments of ADV authority — NOAA, the US Army Corps of Engineers, the US Coast Guard, the EPA, and FEMA — and “federal authorities and responsibilities frequently end once contamination is” addressed [3].

A single documented case shows how the tiers collapse into one another. In 2017 the Point Estero ran aground near Cayucos State Beach, California; the US Coast Guard removed the oil and departed; removal was estimated at $70,000; “the uninsured owner walked away”; the vessel was ineligible for the state’s recreational ADV program and no commercial program existed; both the county and the State Lands Commission had authority to remove her but no funding — “as of 2019, the vessel remains” [3]. No legend will attach to the Point Estero. She is a pure documented case, and the documentation’s conclusion is that nothing happened.

The category work also exposes why documented derelicts matter beyond their stories. ADVs “break up, sink, or block navigation channels,” carry “harmful quantities of oil, lubricant, and other toxic substances,” and settle on bottoms where they scour or crush “sensitive habitats like eelgrass beds and kelp meadows” [3]. Their persistent organic pollutants “are fat-soluble and eventually accumulate in animal fat, becoming concentrated in top predators like orca whales and otters” [3]. None of that is visible from the legend tier, where a derelict is a silhouette in fog; all of it is routine in the documented tier, where a derelict is a leaking chemical inventory with a hull around it.

Finally, the tiers feed each other, and the policy literature knows it. The Blue-Ribbon report’s model outreach plan instructs agencies to build websites offering “case studies and interesting stories” and to practice “leveraging ADV stories, especially on topics such as removals, turn-in events, and problem ADVs” [3]. Legend and reportage are, in

the documented tier's own strategy, fundraising instruments — the fame of a few derelicts is meant to pay for the removal of the anonymous many. The category even extends below the waterline and down in scale: the international fishing-gear literature treats derelict gear and derelict vessels together, with a 2004 APEC seminar defining derelict gear as a critical problem because of "ghost fishing," habitat loss, and navigation safety, and the 2005 Bali Plan of Action supporting efforts "to address derelict fishing gear and derelict vessels" in the same breath [5]. A net that keeps fishing without a fisher is a ghost ship in miniature — crewless, uncontrolled, and documented mainly when someone with a form encounters it.

The rest of this report works mostly in the documented and reportage tiers: drift models that turn sightings into physics — an object with equal surface area above and below the waterline drifts at 2.5% of the 10-metre wind speed, in the wind's direction rather than the current's [2] — debris catalogues, removal programs, and gear-retrieval surveys. But the legend tier never disappears. It is waiting at the start of the next chapter too, in its most durable form: an object in a bottle, cast on the sea by an unknown hand.

Reading the Ocean's Surface: Drift Science

Message in a bottle: studying objects cast up on our shores

The oldest instrument in oceanography is a sealed container thrown into the sea by someone who will never see it again. Its modern practitioner, as profiled in Smithsonian in July 2001, is Curtis Ebbesmeyer of Seattle — an oceanographer who, in the article's summary, traces ocean currents "by studying objects cast up on our shores" [4]. Ebbesmeyer calls himself a "filter feeder on floating objects," and his method is a deliberate refusal of the field's instrumentation: where most oceanographers track currents with satellites and high-tech buoys that cost thousands of dollars apiece and are "deployed in limited numbers," he studies "movements of random junk" reported by "his own vast, ragtag worldwide army of beachcombers" [4] [4]. His one-word advice to budding oceanographers is "plastics," followed by a catalogue of what the sea actually carries: roll-on antiperspirant balls, computer monitors, lightbulbs, armadas of toys, toilet seats, bales of rubber, surfboards, coconuts, aircraft, "the occasional human body, and a surprising number of genuine messages in bottles" [4].

The method has an epistemology, and Ebbesmeyer states it plainly: "The literature of things that float from here to there is so scattered it makes no sense until you compress it. Then it begins to take on a glow, like radium" [4]. A single beached object is an anecdote; thousands of beached objects, logged with positions and dates by a network, become a current meter. The network's raw material is the beachcombing culture of the Pacific Northwest, where converging currents and winds make the region "one of the world's most productive beachcombing regions" — its serious combers prize handblown Japanese glass fishing floats worth at least $3,000 for the rarest, monitor 24-hour weather radio for two or three days of southerly winds to push debris in from the gyre, and turn out at 3 A.M. tides with lanterns and headlights [4]. One of Ebbesmeyer's unpaid "research assistants," Forks plumber John Anderson, had backpacked 30,000 drift-net floats off beaches into a pile by his garage, and later planted a crane superstructure in his yard and covered much of its 40-foot height with foam fishing floats — "my monument to beachcombing" [4] [4].

Ebbesmeyer came to this from orthodox practice. He started in the mid-1960s as a Mobil Oil roustabout, then became an oceanographer after the University of Washington, figuring out how to tow icebergs — "the largest floating objects in the world" — away from drilling platforms over the Grand Banks; in 1974 he joined Evans-Hamilton, a consulting firm specialising in measuring ocean currents, with projects including sewage outfall placement and oil-slick tracking [4]. His applied work already ran on floatables: on the Exxon Valdez spill he learned that oranges are good experimental surrogates for spilled oil because thousands dumped in seawater "distribute themselves exactly like spilled oil" [4]. He and colleagues had also produced a key climate paper by the same compressive method, first pointing out that a single sudden 1976 Pacific climate/circulation swing had affected dozens of seemingly unconnected phenomena — water-oxygen content, Arctic sea-ice extent, Andean glaciers, and the life cycles of corals, crabs, oysters, and salmon — assembled, as the profile notes, "like so much beachcombed drift" from existing studies [4].

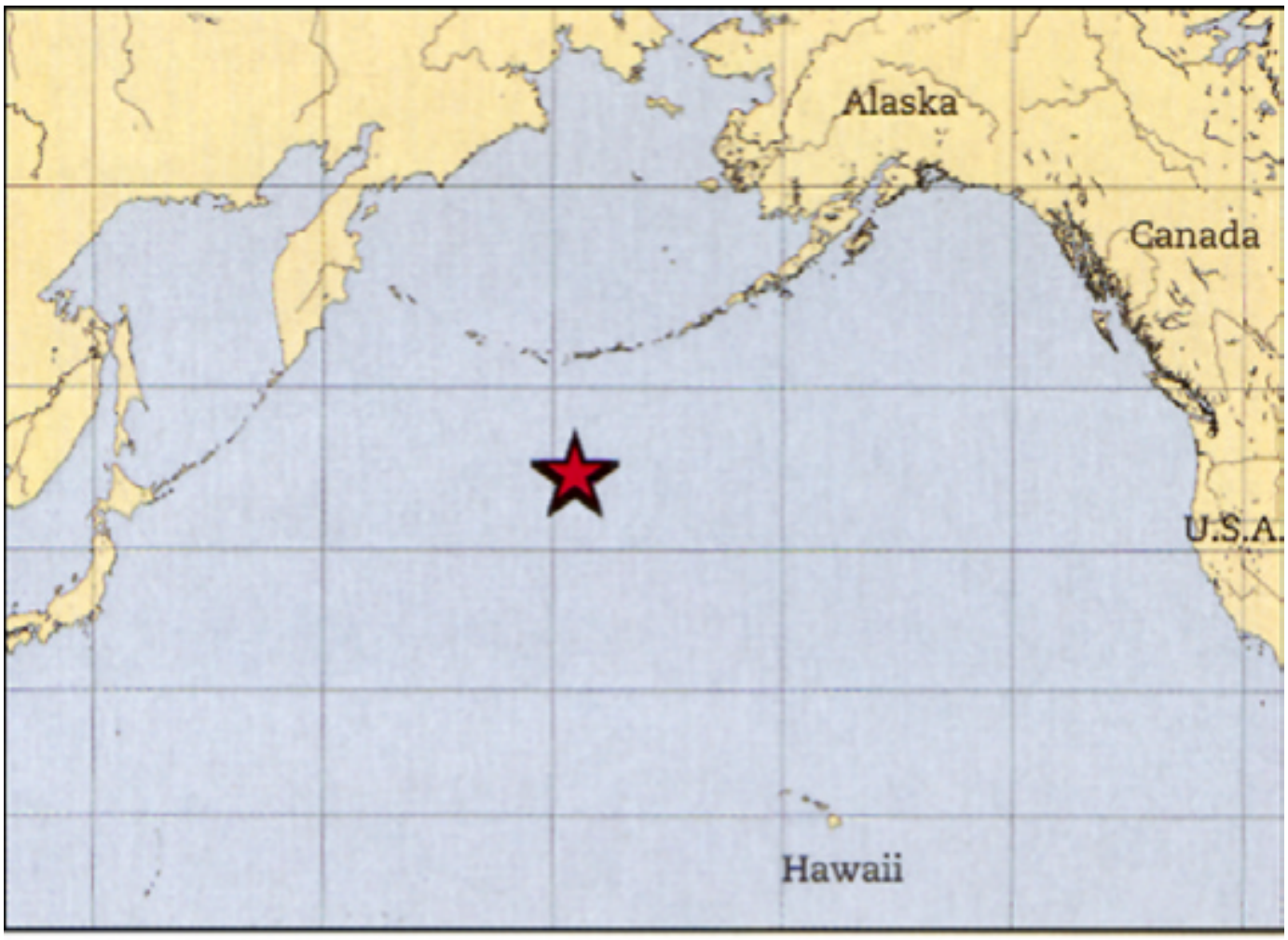
The ocean these objects traverse is structured, and the profile lays out the structure the whole drift literature depends on. Each ocean hosts one or more huge gyres shaped by prevailing winds, Earth's rotation, and bordering landmasses; in the Pacific, the Kuroshio Current arcs away from Japan, becomes the North Pacific Drift, turns south off Washington and Oregon as the California Current, and returns past Hawaii toward the Philippines, China, and Ja-

pan — a circuit of roughly 14,000 miles that an object may take six years to complete [4]. Unless it is sidetracked: storms drive high-windage floaters off track, the counterclockwise Alaska Current can snag objects toward Siberia or through the Aleutians and Bering Strait into the Arctic, and objects can "hang out for years" in the calm eyes of the gyres [4]. The Arctic has its own conveyor: trees falling from Siberian and Canadian riverbanks enter sea ice and ride the Transarctic Drift clockwise, in oceanographer Eddy Carmack's telling, for as long as 30 years before being "spit out near Greenland" [4]. The same surface layer is also, in the words of Ebbesmeyer's collaborator Jim Ingraham, chaotically variable: two toys dumped at the same moment in the same spot may end with one in Hawaii and the other frozen in an Arctic ice floe [4].

Against that background, the literal message in a bottle becomes a data point with a story attached — exactly this report's recurring structure, at the scale of a single object. In December 1991 a friend sent Ebbesmeyer a bottle found on southern Vancouver Island; inside were dampened sheets with Chinese characters — a printed appeal for the release of the dissident Wei Jingsheng, by then imprisoned 12 years [4]. It took Ebbesmeyer and six colleagues a year of phone calls, helped by "a friendly American ex-intelligence man," to decipher the waterlogged message and establish that Taiwanese frogmen had been aiming propaganda bottles toward the mainland for years; the evidence suggested this one had likely been released in summer 1980 near the island of Quemoy, when coastal currents were favourable — and had obviously gone off course [4]. Ingraham ran the OSCURS current model to plot its probable route, through an island chain below Japan and then across the Pacific, and they published the reconstruction in the geophysical journal Eos, ending with the hope that "the magic of bottle notes" would help free Wei [4]. Wei was freed in 1997 — "though no bottle connection is known" — and, passing through Seattle, met Ebbesmeyer, who gave him a copy of the paper [4]. The caveat is the method in miniature: a bottle can prove a route, not a rescue.

The profile's author supplies a second, smaller bottle, and its anticlimax is equally instructive. Beachcombing at Leadbetter Point in Washington's Willapa National Wildlife Refuge, he logged 302 items — beer cans, a 55-gallon drum, shotgun shells, a fluorescent light tube, countless nurdles — and found "a real live message in a bottle": a translucent plastic vitamin bottle holding a photocopied memorandum dated 20 October 1999 about lifeboats aboard the USS Camden, an informal survival Q&A ("With water, how long could a man survive without food? A: Three weeks or longer") [4]. The Camden was a combat-support ship based near Seattle whose recent missions were routine coastal exercises; everyone aboard was safe, and the bottle "probably did not drift far" [4]. Most cast-up objects are like this: short journeys, mundane contents, no sender in distress. The science is in aggregating them anyway — the Nike spill of the previous sections yielded 1,600 reliably documented wash-ups, a 2.6 percent recovery rate the researchers call "very respectable," and gave Ingraham's previously theoretical OSCURS model its first real-life test [4].

That is the transition this chapter now makes. The bottle era of drift science — one object, one beach, one story — becomes, in the next section, an armada: 29,000 identical bathtub toys released at a known point on a known day, the largest controlled drift experiment never designed as one.


Alaska
Canada
U.S.A.
Hawaii

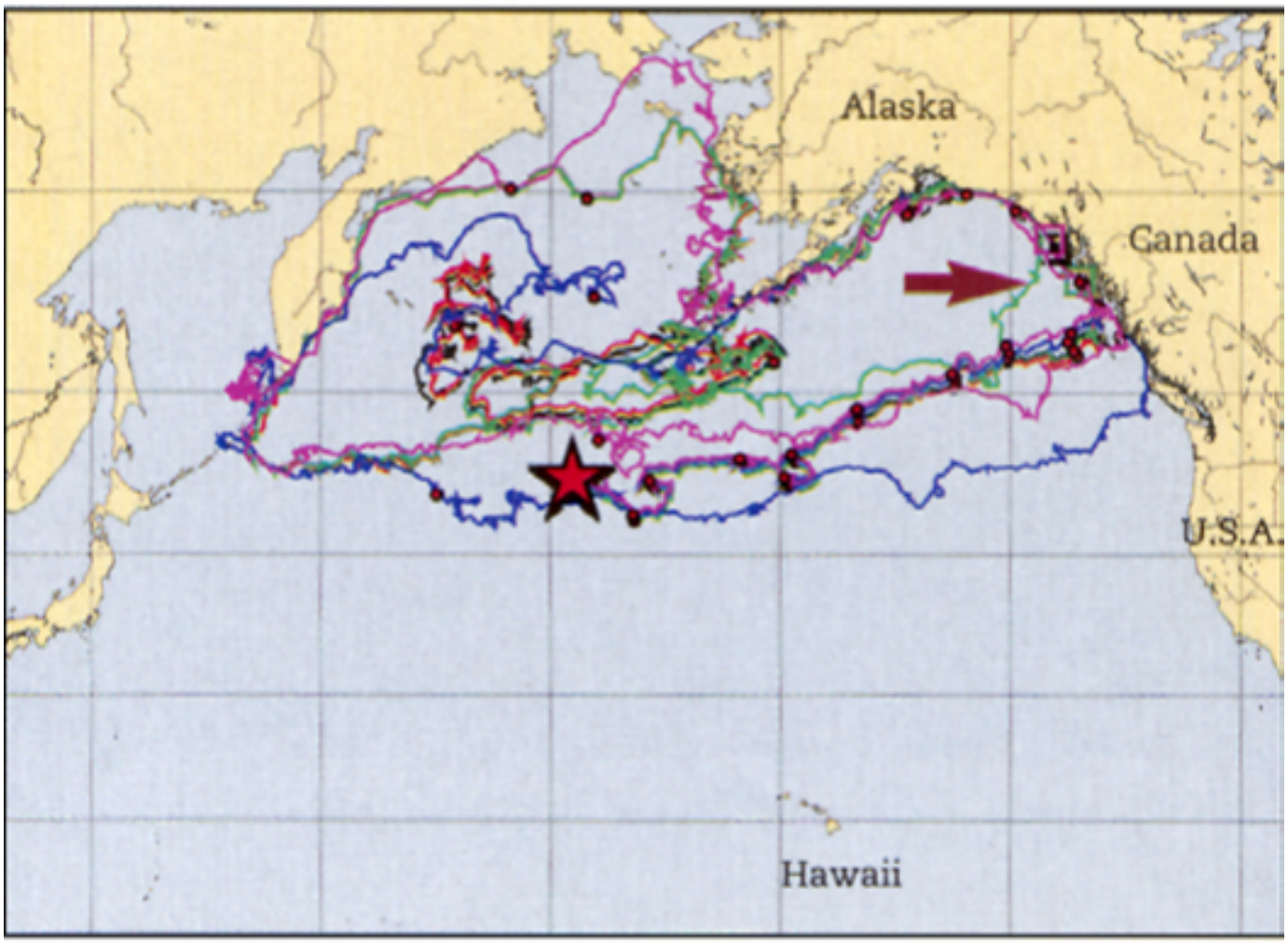

Alaska
Canada
U.S.A.
Hawaii

Exhibit. BATHTUB TOYS' DRIFT JIM INGRAHAM/NOAA/NATIONAL MARINE FISHERIES SERVICE (3). Source: S0001.

Bathtub toys' drift: the Friendly Floatees as accidental oceanographers

The Nike spill had proved the method; the bathtub toys made it a science with a control group of 29,000. The first news reached Ebbesmeyer as a 1993 article from the Sitka Sentinel in Alaska, reporting that "scores of tangerine-size yellow ducks, green frogs, blue turtles and red beavers had beached in Alaska coves" — another big container spill, "bathtub toys, no less" [4]. Where the shoes had been footwear with serial numbers, the toys were better instruments still: four distinct, brightly coloured, near-indestructible shapes, released together, that no one could mistake for anything else on a beach.

Ebbesmeyer confirmed some 400 sightings along 530 miles of Alaska shoreline, and this time he ran the experiments the Nike windfall had not allowed [4]. He obtained samples from the toys' US distributor, Kiddie Products: each four-toy set came in a plastic shell with cardboard backing, but soaking in seawater dissolved the packaging, "meaning the toys had probably been released to go their separate ways within 24 hours" of the spill [4]. Some salvaged toys appeared gnawed by sea otters, so he drilled otter-tooth-size holes in his samples to judge what portion of the originals might have sunk: none did — "Very high quality bathtub toys... Buoyant even when half-waterlogged" [4]. He put them in his freezer to test whether Arctic cold would crack them: "Nope" [4]. The Floatees, in other words, were nearly perfect drifters — individually released, effectively unsinkable, freeze-proof, and unmissable.

The spill's exact origin, the dataset's anchor point, arrived by telephone. A minor shipping executive called to tell Ebbesmeyer "his ship had come in," directing him to a Tacoma dock, where he spent four hours with the captain, who opened the ship's log to the spill: "29,000 toys dumped on January 10, 1992, at 44.7 degrees north, 178.1 degrees east" [4]. A known number, a known day, a known point in the North Pacific: with that, Ingraham's OSCURS model "got huge amounts of new data" [4]. The profile's figure caption summarises the output: after 400 toys were found, Ingraham "used a computer model to track the objects' route and extrapolate where the remainder would go," with the charts projecting the toys' travels through 1995 and beyond [4].

The validation loop ran both ways. For the Nikes, the model's predictions from spill and beaching coordinates had matched observations — OSCURS "matched nicely" — which "allowed Ingraham to refine OSCURS, and draw computer-animated Nike paths, spreading like ganglia in the ocean and balling up in coastal currents" [4]. Then, using historical weather data, they simulated identical spills for 27 May of every year from 1946 to 1991: "The Nikes went a different way each time — a tribute to the ocean's variability," prompting Ingraham's remark that "the surface layer is a strange place" [4]. That is the deepest scientific result in the profile: not a map of the route, but a demonstration that no single route exists — each year's ocean is a different experiment, and only ensembles of floatables can describe it.

The toys' strangest publication venue followed. Kiddie Products printed the duckies' calculated, twisty 15,000-mile route on the back of every new bathtub-toy package — "By far our best publication," Ebbesmeyer says [4]. The spill had already made the pair "minor media stars" before the Nike study even appeared in Eos ("what reporter could resist science based on oceangoing sneakers?"), and hundreds of people phoned Ebbesmeyer about other odd floatables they had found; failing to find even one other scientist pursuing such work, he decided he was the man for the job, cut his paid workweek by 40 percent, and waited for his "fondest hope: another big container spill" [4]. The toys institutionalised the practice: he founded the nonprofit Beachcombers' and Oceanographers' International Association, with the Beachcombers' Alert! newsletter (500 paid subscribers, from as far as West Africa and New Zealand), which has documented spills "of everything from onions to hockey gloves" [4]. His unpaid research assistants included Barry Tweed, a retired contractor who built two seaside houses from salvaged surf logs, and Vern Krause, a school bus driver who found a mysterious stainless-steel sphere that the military confiscated after Ebbesmeyer sought help identifying it — deemed safe and returned the next week [4].

The armada's final lesson in the profile is darker, and it points toward this report's plastic chapters. In summer 1999 McLeod shipped for three weeks on an Algalita Marine Research Foundation vessel into one eye of the North Pacific Subtropical Gyre — "dubbed the Garbage Patch, because it collects so much junk" — to help census marine trash [4]. About as far from land as one can get, they found piles of rope, fishing gear, liquid containers, a volleyball, even a refrigerator; but "the main haul was nurdles," with fine-net drags indicating 129,000 per square mile, six times the mass of the collected zooplankton, and a mahimahi caught for dinner was full of them [4]. Ebbesmeyer's response was a diagnosis: "How do we measure the sea's wildness? Like a doctor sizing up a patient, I'd look her in the face. She's breaking out in a nasty rash" [4].

The Friendly Floatees thus worked as accidental oceanographers in both directions. Downstream, their beachings calibrated a current model and demonstrated its year-to-year variability. Upstream, they revealed the observer network — captains opening logs, executives making phone calls, distributors printing drift routes on packaging, hundreds of beachcombers reporting finds — that turns industrial accidents into oceanographic data. The next sections take up the instrumented version of the same idea: the global fleet of purpose-built drifters whose tracks now define what is known about where floating things, ships included, actually go.

The observational global drifter dataset

The bathtub toys were accidental drifters: superb ones, but finite, unrepeatable, and silent between beachings. The instrumented successor is the Global Drifter Program, whose archive is the observational foundation for the debris science in the rest of this report. In the key methodological paper for this chapter — van Sebille, England, and Froyland's study of garbage-patch origin and evolution — the dataset is introduced in one dense paragraph: "Within the observational Global Drifter Program (Niiler 2001, Lumpkin 2003, Lumpkin et al 2012), buoys that get advected with the near-surface flow have been released since the early 1980s; these can be used to study where and over what timescales marine debris accumulates in the global ocean" [6]. Three decades of continuous releases convert the one-off spill experiment into a standing observatory.

The archive's resolution and coverage are what make it usable for debris questions. The drifters "have a battery life of up to 5 yr and the post-processed data yields geo-locations of the buoys every 6 h" [6]. Coverage is "reasonably comprehensive, with more than 85% of the ocean surface having over 100 location fixes per 1° × 1° grid box" (van Sebille et al 2011, Maximenko et al 2012) [6]. Compare the observational basis of the earlier chapters: for the tsunami debris field, comprehensive observations were "lacking" and the distribution "unobservable"; for the toys, roughly 400 beach recoveries. The drifter archive is the first dataset in this report in which the surface ocean is sampled densely, repeatedly, and by design rather than by accident and eyesight.

It is not, however, a single instrument, and the paper is careful about its heterogeneity. "All buoys used here are deployed with a drogue at 15 m, but many of them lose their drogue during their lifetime, so that 48% and 52% of the data is from buoys with and without drogues, respectively" [6]. The two populations measure different oceans: "non-drogued buoys are more sensitive to direct wind forcing and wind drift, whereas drogued buoy trajectories track the ocean flow at 15 m depth, and so are more representative of the upper ocean geostrophic flow and Ekman transport" (Poulain et al 2009, Grodsky et al 2011) [6]. A drogue is a sea anchor; losing it turns a current meter into something closer to a piece of floating debris. The saving fact, established by Maximenko et al (2012), is that "non-drogued buoys largely aggregate in the same region as drogued buoys" — their destinations agree even where their dynamics differ [6].

The study's decision to combine both populations is argued on explicitly debris-centred grounds, and it is the most important methodological idea in this section. First, pragmatically, combining "allows us to work with a much larger data set, which greatly improves the accuracy of our method" [6]. Second, and more interestingly, the mixture is realistic: "the marine debris in the real ocean is also a combination of plastics in the neuston (the layer where the wind has a very large effect) and near-neutrally buoyant plastics in the mixed layer" (Kukulka et al 2012, Morét-Ferguson et al 2010, Cole et al 2011) [6]. A drogued drifter models the second population; an undrogued one, ridden by the wind, models the first — the high-windage fraction that, in the tsunami case, crossed the Pacific fastest. Third, the vertical story does not stop at two layers: "biofouling on the smallest plastics increases their density and results in some being submerged (Andrady 2011) or even sink to the ocean floor (Bergmann and Klages 2012)" [6]. Hence the conclusion: "the combination of drogued and non-drogued buoys most closely resembles the variety of vertical positioning of surface layer marine debris" [6].

Two limitations should be carried forward with the dataset. The buoys are short-lived relative to the questions asked of them: even at up to five years per buoy, individual trajectories cannot resolve connectivity between ocean regions over the interannual-to-centennial timescales on which debris accumulates — the problem the transit-matrix method of the next section exists to solve [6] [6]. And the drifters, like bottles and toys before them, describe the ocean that floating things experience: near-surface flow, wind exposure in the top metres, and convergence zones. What sinks, beaches, or is eaten leaves this dataset as surely as it left the beachcomber record.

With those caveats, the dataset powers the study's headline application: using the observed trajectories "in a particle-trajectory tracer approach that includes the seasonal cycle to study the fate of marine debris in the open ocean from coastal regions around the world on interannual to centennial timescales" [6]. The observational era of

drift science thus runs in a clean line — bottle, shoe, toy, buoy — each generation trading narrative charm for sampling density. What the buoys collectively reveal about where the surface ocean takes things is the subject of the next two sections.

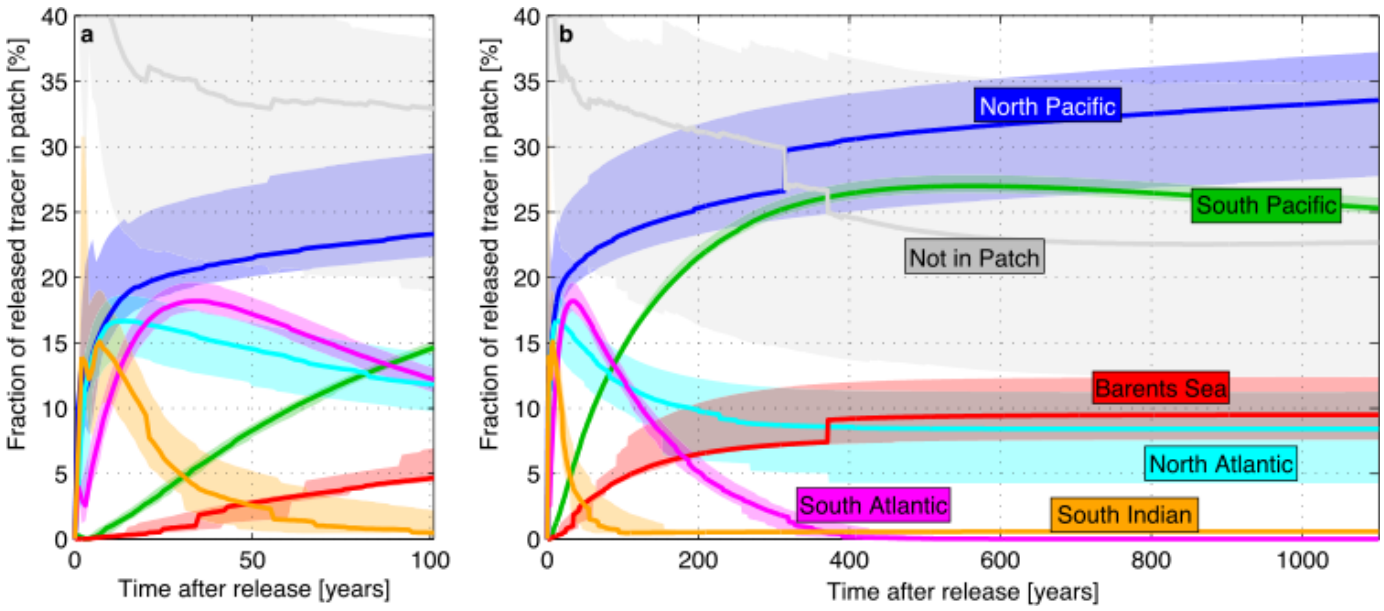


Exhibit. Figure 2. The evolution of the amount of tracer within the six garbage patches. Source: S0005.

The transit matrix approach

The drifter archive has a structural defect for debris science: its instruments die young. "Observed buoy trajectories are too short lived to resolve the connectivity between ocean regions on the timescales we are interested in here" [6]. A buoy that reports for at most five years cannot testify about a century of accumulation, and debris questions — how fast coastal material reaches a patch, how leaky the patch is, where its contents end up — are century questions. Van Sebille, England, and Froyland's answer is to stop following buoys and start following their statistics: "we use the buoy trajectories as input for a tracer matrix, a statistical method of analysing observational data that has been shown to adequately capture the observed dynamics of the ocean circulation on both regional and global scales" (Froyland et al 2007, Dellnitz et al 2009, van Sebille et al 2011, Maximenko et al 2012) [6].

The construction is exact. "All available buoy trajectories are gridded onto a 1° × 1° grid and then used to create a set of transit matrices **P**_b. These transit matrices give, for each grid box, the probability that particles move to any other grid box in 60 days" [6]. The 60-day step is a compromise with the archive's geography: "Using periods shorter than 60 days yields transit matrices with less than 10 crossings in some regions of the oceans where drifter coverage is low, whereas using periods longer than 90 days means that the seasonal cycle is not adequately resolved" [6]. Seasonality is then built into the matrix set rather than averaged away: "each transit matrix uses data from only a two-month period. For instance, **P**_1 is the transit matrix from all buoy trajectories with 60 day segments starting in January or February, **P**_2 is the same but then for March and April, and so on through to **P**_6 for November and December" [6]. Iterating the matrix equation — the tracer field advanced by the seasonally appropriate matrix every 60 days, cycling through the six — "yields... the advection of tracer concentrations as a function of time," and "with relatively little computational burden, this tracer transit matrix can then be integrated forward in time for many centuries, allowing for the investigation of global tracer evolution on millennial timescales" [6]. The method converts three decades of short buoy lives into a renewable statistical ocean: no buoy survives the simulation, but the transition probabilities distilled from all of them do.

Two design choices define what the method can and cannot say. The first is conservation. "The matrices **P**_b are row normalized, which means that the amount of tracer is conserved. In these experiments, therefore, we do not consider the beaching of marine debris (Lumpkin et al 2012) or the abyssal sinking of particles due to heavy biofouling; focusing instead on the evolution of that part of the marine debris inventory that remains in the open ocean for decades to centuries" [6]. The second is coastal scope: "we focus on the open ocean pathways of marine debris, which means we do not explicitly account for near-shore flows through which the marine debris has to migrate from the land to the open ocean" [6]. The transit matrix is thus deliberately a model of the debris that stays at sea — the

derelicts and plastics of the open ocean — not of the much larger fraction that strands, sinks, or never leaves the coast.

The release experiment bolted onto the matrices is equally explicit about its proxies. “Ideally, the tracer would be initialized using actual observed marine debris fluxes into the ocean, but such data are not available on a global scale. Human population is arguably the most relevant and best-constrained proxy for marine debris fluxes (Marxsen 2001, Halpern et al 2008)” [6]. Tracer in each ocean-adjacent grid box is therefore “scaled to the local population density within 200 km from the coast,” using a gridded population product, on the reasoning that additional socio-economic indices “do not necessarily relate to coastal ocean debris input, as waste disposal efforts vary widely from region to region” [6]. Where the drifter archive itself is too thin to build a matrix — “particularly in the Indonesian Archipelago and in the North Sea” — no tracer is released at all; and “no tracer is released within the Mediterranean Sea since no observational drifters cross the Gibraltar Strait, and hence this flow cannot be resolved by the tracer transport calculations” [6]. The model’s blind spots are inherited directly from the observations: where no buoy has travelled, the matrix has no probabilities, and the honest response is exclusion rather than invention.

Measurement of the output is by a single statistic. “A tracer amplification factor (TAF) is used to identify patches… $TAF_(x,y) = c(x,y)N/C$,” so that a grid box with TAF 20 holds “twenty times more tracer… than if all tracer is uniformly distributed over the global ocean,” and “patches are defined as connected areas within the different ocean basins wherever TAF > 2.0” [6]. The release protocol is a tracer pulse, not a continuous emission: “Tracer is released as six pulses over the course of 1 yr… this provides a clearer picture of how the tracer patches accumulate in time and then disperse,” with multi-year releases recoverable afterwards because “the tracer concentrations resulting from multi-year releases can be constructed using a linear superposition of concentrations resulting from all earlier pulses” [6].

Set against its predecessor, the advance is precise. Maximenko et al (2012) “incorporated an idealized initial state with surface debris uniformly spread over the global ocean” and “assumed the advection of tracer to be constant in time and thereby ignored the seasonal cycle of the surface ocean circulation” [6]. The transit-matrix study repairs both: debris enters “at the coastal margins where industrial activity and waste disposal is most highly concentrated,” the circulation keeps its seasons, and the payoff is the ability “to study the fate and mixing of marine debris in a more realistic scenario than previously achieved… [and] examine, for the first time, how each of the garbage patches are connected, how inter-ocean exchange mixes debris from different regions, how ‘leaky’ these patches are and how they will evolve over century timescales” [6]. What the matrices found when run — six patches, their leakiness, and the North Pacific’s role as the system’s main attractor — is taken up in the sections on basins, patches, and budgets that follow.

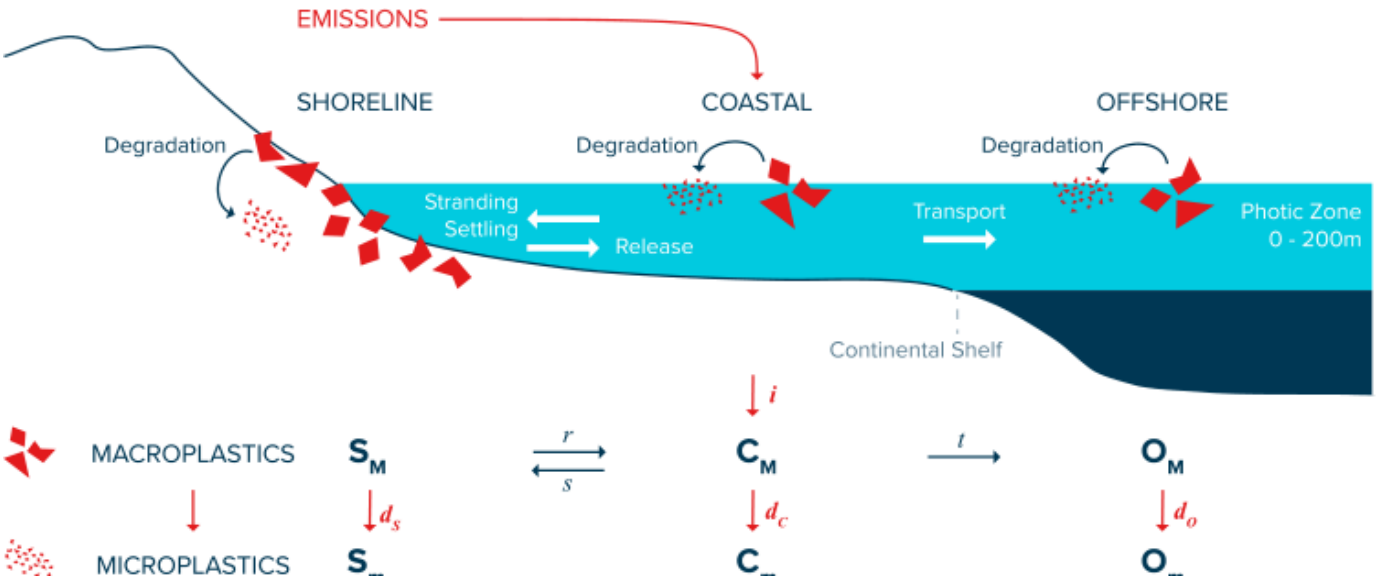


Exhibit. Figure 1. Predicting quantities of positively buoyant macroplastic debris in the global ocean. Source: S0006.

Ocean surface-layer basins and currents

Everything that floats in this report — derelict hulls, tsunami debris, shoes, toys, drifters, plastic — is sorted by the same machinery: the wind-driven circulation of the ocean’s surface layer. This section assembles what the corpus’s

sources agree that machinery looks like, because the patch inventories, budgets, and models of the following sections all presuppose it.

The organising fact is convergence. “Since the surface currents preferentially converge and subduct in certain locations, the debris in the near-surface layer tends to collect in several relatively confined regions. These regions of accumulation are commonly known as the great ocean garbage patches” [6]. The mechanism is named in the same source: “To first order, the formation of the patches is governed by the well-established dynamics of Ekman pumping in the subtropical gyres (Maximenko et al 2012), where wind-driven convergence of the surface flow leads to accumulation of surface water in the centre of the gyres. As the debris is generally less dense than seawater, it floats and accumulates in the regions of strongest convergence near the surface” [6]. A basin, for a floating object, is therefore not a bowl of water but a circulation cell: material enters at the margins, spirals inward under Ekman transport, and resides where the surface flow converges and downwells — while the material itself, being buoyant, stays at the top of a water column that is slowly sinking beneath it.

The geography that results is fivefold, with a sixth outlier. Observations assembled by Cózar and colleagues — 3,070 samples, with plastic present in 88% of open-ocean surface samples — produced a distribution that “agreed with those predicted from ocean surface circulation models, confirming the accumulation of plastic debris in the convergence zone of each of the five large subtropical gyres” [7]. The five accumulation zones carried comparable plastic concentrations per surface area, but “the North Pacific Ocean contributed importantly to the global plastic load (between 33 and 35%), mainly owing to the size of this gyre” [7]. The transit-matrix study of the previous sections finds the same five subtropical patches and adds “one previously unreported patch in the Barents Sea” [6] — a high-latitude accumulation with no subtropical gyre to explain it, examined in detail later in this report.

Within and between the basins, the currents have names and the sources use them. The western arm of the North Pacific Gyre, the Kuroshio Current, carried “exceptionally high” measured surface plastic concentrations, “including the highest reported for nonaccumulation regions” — plausibly because “the high human population on the eastern coast of the Asian continent, the most densely populated coast in the world, with one-third of the global coastal population,” feeds it [7]. The Great Pacific Garbage Patch itself sits in “the eastern part of the North Pacific Subtropical Gyre,” has been described as “a gyre within a gyre,” and its concentrations are “mostly attributed to a connection to substantial ocean plastic sources in Asia through the Kuroshio Extension (KE) current system as well as intensified fishing activity in the Pacific Ocean” [8]. The material properties set the terms of transport: “Around 60% of the plastic produced is less dense than seawater,” and buoyant plastic “can be transported by surface currents and winds, recaptured by coastlines, degraded into smaller pieces by the action of sun, temperature variations, waves and marine life, or lose buoyancy and sink” [8].

The hemispheric comparison is where the simple basin picture first cracks. Eriksen and colleagues found plastics of all sizes “in all ocean regions, converging in accumulation zones in the subtropical gyres, including southern hemisphere gyres where coastal population density is much lower than in the northern hemisphere,” and — the anomaly — “the total amounts of plastics determined for the southern hemisphere oceans are within the same range as for the northern hemisphere oceans… which is unexpected given that inputs are substantially higher in the northern than in the southern hemisphere” [9]. Their candidate explanations define the open questions of basin dynamics: either “plastic pollution is moved more easily between oceanic gyres and between hemispheres than previously assumed,” redistributing items across basins; or there are unaccounted southern sources, “such as currents from the Bay of Bengal that cross the equator south of Indonesia”; or surface losses are disproportionately larger in the north, with stranding on local shores “more important in the northern than in the southern hemisphere” [9]. The transit-matrix results lean toward the first explanation — on centennial timescales “the different basins are much better connected than previously thought and… inter-ocean exchanges play a large role in the spreading of marine debris” [6] — but the sources leave all three mechanisms in play.

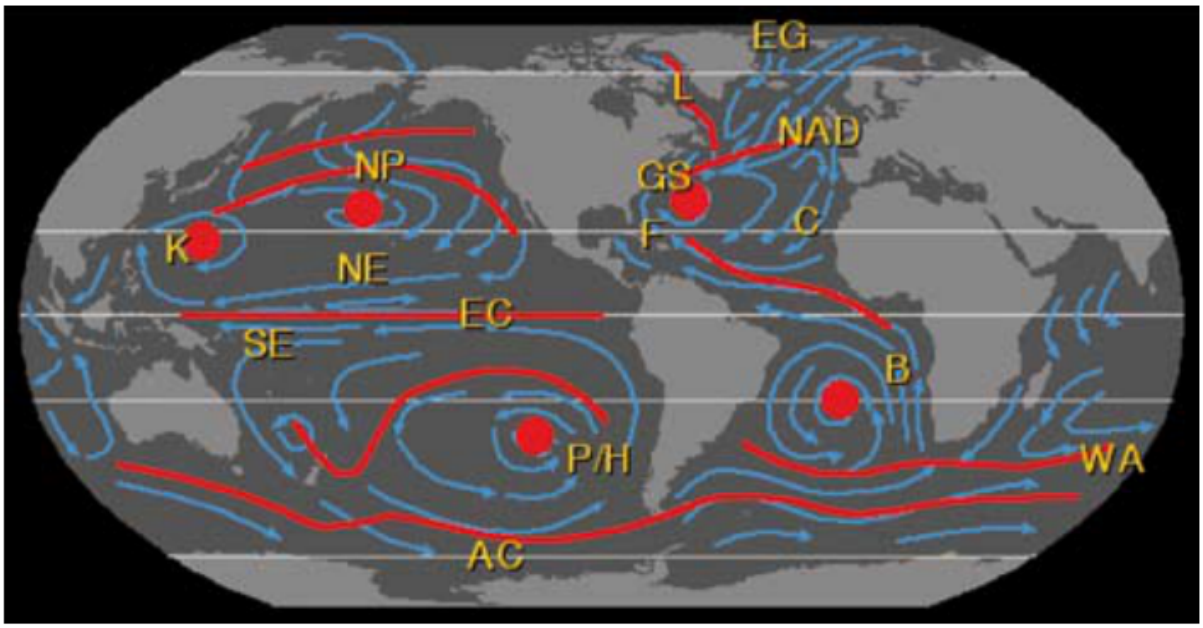
The surface layer is also not a closed budget, and the basin picture must include its exits. Eriksen et al list the removal processes operating on floating plastic: “UV degradation, biodegradation, ingestion by organisms, decreased buoyancy due to fouling organisms, entrainment in settling detritus, and beaching” [9]. Their scale check is sobering: against 288 million tons of plastic produced worldwide in 2012, “our estimate of the global weight of plastic pollution on the sea surface, from all size classes combined, is only 0.1% of the world annual production” [9]. Where the other 99.9% resides — coasts, seafloor, biota, fragments below net size — is the budget problem taken up in Section 16. Even the modelling conventions encode rival pictures of the surface layer: the Eriksen study’s particle model takes surface-layer velocities from the HYCOM system (surface layer only, of 32 vertical layers) and “assumes that debris particles are mostly submerged in the water and extra forcing on potentially emerged parts of the debris is neg-

lected" [9] — a submerged-debris ocean, quite different from the windage-driven ocean that delivered the Ryou-Un Maru and the high-windage tsunami debris to North American shores.

What the sources jointly establish is a working map: five subtropical convergence basins fed by named boundary currents, cross-basin and cross-equator exchanges strong enough to erase a hemispheric input gradient, a possible Arctic accumulation in the Barents Sea, and a surface layer that continuously leaks its contents to shores, depths, and food webs. Linear Ekman theory, as van Sebille et al caution, "does not predict the timescales of patch formation from debris entering at the coastal margins, nor how eddy mixing or other non-linear processes can counteract the accumulation of debris and provide a flux out of the patch locations" [6]. The map is settled; the fluxes across it are not. The next sections quantify what is actually out there.

FIGURE 5

Examples of ocean convergence zones



The red dots indicate where marine litter may accumulate

Exhibit. Map of ocean convergence zones, with red dots marking the regions where marine litter is expected to accumulate. Source: S0017.

Where Debris Gathers: Garbage Patches and the Plastic Ocean

Plastic pollution in the world's oceans: more than 5 trillion plastic pieces

The number that defines public understanding of ocean plastic comes from a single paper's title: Eriksen and colleagues' "Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea," published in PLoS ONE 9(12), e111913, on 10 December 2014 [9]. The author list itself maps the field's institutions — Marcus Eriksen of the Five Gyres Institute, Laurent Lebreton, Martin Thiel, Charles J. Moore of the Algalita Marine Research and Education foundation (whose vessel carried McLeod into the gyre in Section 8), Francois Galgani, Peter G. Ryan, and Julia Reisser among them [9]. The underlying data were published openly: "These data are available at figshare.com... Plastic Marine Pollution Global Dataset" [9].

The study's claim to priority is scope across sizes. "This is the first study that compares all sizes of floating plastic in the world's oceans from the largest items to small microplastics" [9]. The supporting information shows the sampling architecture: "24 expeditions from 2007–13 contributed data collected at 1571 field locations, with count and weight data in four plastic size classes from six regions: North Pacific (NP), North Atlantic (NA), South Pacific (SP), South Atlantic (SA), Indian Ocean (IO), Mediterranean Sea (MED)" [9]. The four size classes run from small microplastics to macroplastics: 0.33–1.00 mm, 1.01–4.75 mm, 4.76–200 mm, and >200 mm, with weight density meas-

ured at 1,333 stations from net tows and survey transects [9]. For the largest class, visual survey transects recorded "4,291 macroplastic items (>200 mm) in nine categories... in the North Pacific, South Pacific, South Atlantic, Indian Ocean, and Mediterranean Sea," and mean item weights were derived from beached macroplastic collected in coastal surveys in Chile, western South Africa, the US east coast, and the Hawaiian Islands, then applied to the items observed drifting at sea [9]. Measured regional means were checked against the study's particle model — HYCOM surface currents coupled to the Pol3DD tracker described in Section 11 — with "generally a good correspondence between the measured and modeled means for each region" [9] [9].

The findings, beyond the headline total, are three. First, ubiquity with structure: "Plastics of all sizes were found in all ocean regions, converging in accumulation zones in the subtropical gyres, including southern hemisphere gyres where coastal population density is much lower than in the northern hemisphere" [9]. Second, the hemispheric anomaly examined in Section 11: southern totals rival northern totals despite much higher northern inputs, implying easier inter-gyre and inter-hemisphere transport than previously assumed, or unaccounted southern sources, or disproportionate northern losses [9]. Third — and most consequential for the budgets that follow — the size structure does not add up. "The comparison of size classes and weight relationships suggests that during fragmentation plastics are lost from the sea surface" [9]. If large items simply broke into ever more small ones, small-particle counts should swell accordingly; instead the surface ocean holds far less microplastic than fragmentation arithmetic predicts. Cross-checks sharpen the point: for the North Pacific, an 11-year dataset estimated 21,290 metric tons of floating microplastic against this study's 12,100 metric tons for the same region, and the global microplastic study it compares with put the total floating microplastic load at 7,000–35,000 metric tons against Eriksen's 35,500 — while that same study "found a 100-fold discrepancy between expected microplastic weight and abundance and their observations, indicating a tremendous loss of microplastics" [9]. The authors' conclusion: "the similarities between our results and those of this study... support our hypothesis that the ultimate fate of buoyant microplastics is not at the ocean surface" [9].

The removal mechanisms are enumerated, and they split into export and transformation. Export: ingestion by organisms — which can package microplastics into fecal pellets, "thus enhancing sinking" — decreased buoyancy from fouling organisms, entrainment in settling detritus, and beaching [9]. Transformation: UV degradation and biodegradation, the latter strengthening as particles shrink because "at decreasing particle size the surface area:volume relationship is increased dramatically and oxidation levels are higher, enhancing their biodegradation potential" [9]. And a measurement exit: fragmentation of already brittle microplastics may proceed so fast that particles pass below the sampling nets' 0.33 mm mesh and become unavailable to the very surveys that count them [9]. Even the vertical dimension is partly an artefact of weather: the study applied "a correction for vertical distribution to all samples related to wind-driven turbulence," while noting that "downwelling at convergence zones may also influence the vertical distribution of slightly buoyant particles such as microplastics" — the gyres that concentrate plastic at the surface also push some of it down [9].

The scale framing closes the paper's discussion and frames this whole chapter: with 288 million tons of plastic produced worldwide in 2012, the entire floating inventory — more than 5 trillion pieces, over 250,000 tons — amounts to "only 0.1% of the world annual production" sitting on the sea surface [9]. Five trillion pieces is simultaneously an enormous count and a rounding error in the production ledger. That tension — a surface ocean at once saturated and missing most of the plastic ever released into it — is the quantitative puzzle the next sections take apart: what the open-ocean surveys actually measured, what the best-studied patch contains, and where the missing mass must be.

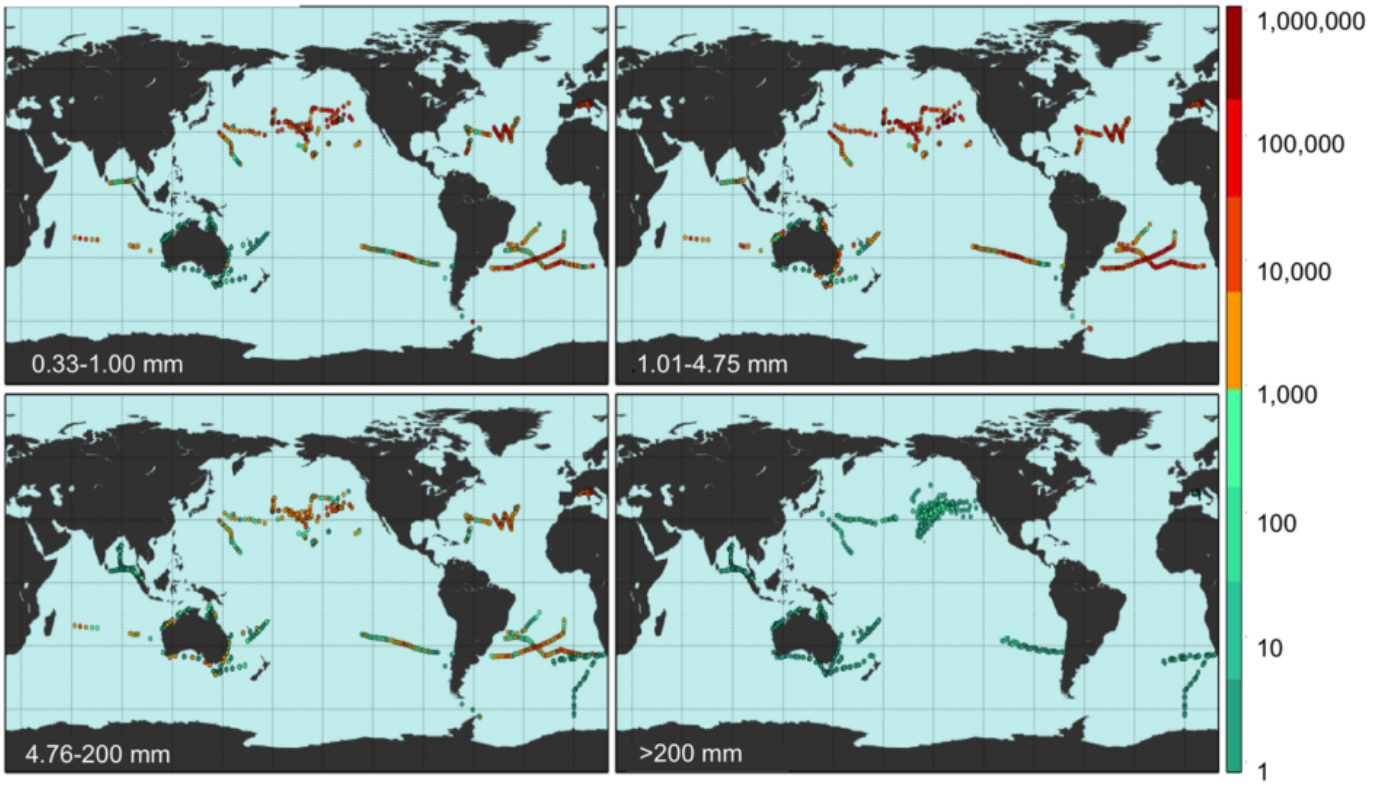


Exhibit. Figure 1. Field locations where count density was measured. Source: S0002.

Plastic debris in the open ocean

Cózar and colleagues' "Plastic debris in the open ocean" (PNAS 111: 10239–10244, 2014) is the austere counterpart to Eriksen's headline inventory: where Eriksen counts everything afloat, Cózar asks why the open ocean holds so little. The dataset is a global compilation — 3,070 samples, with plastic in 88% of open-ocean surface samples and concentrations spanning four orders of magnitude — organised by 15 major zones classified as nonaccumulation, outer accumulation, and inner accumulation areas in each basin [7] [7]. The method corrects for the ocean's own interference: concentrations were converted between volume and area using tow depths by net type, samples reported without wind correction were re-screened against CCMP satellite winds, discarding collections made above 5 $m \cdot s^{-1}$ wind "above which the effects of wind stress can be significant," and measurements were averaged over 2° grid cells to avoid overweighting heavily sampled areas [7]. The wind-corrected dataset comprises 442 grid cells and 1,127 net tows; the uncorrected one, 851 cells and 3,070 tows [7].

The resulting load estimate is strikingly small. "Using the high and low ranges of spatial concentrations measured within 15 major convergence/divergence zones in the global ocean, we estimate the amount of plastic in the open-ocean surface between 7,000 and 35,000 tons" [7]. The paper's own summary rounds the same result: "The current plastic load in surface waters of the open ocean was estimated in the order of tens of thousands of tons (10,000–40,000)" [7]. Against Eriksen's >250,000 tons for all floating plastic, the difference is definitional as much as empirical — Cózar counts the open-ocean surface layer only, excluding coasts, the Mediterranean-scale semi-enclosed seas, and large items poorly sampled by nets — but the paper insists the shortfall is real: "even our high estimate of plastic load, based on the 90th percentile of the regional concentrations, is considerably lower than expected, by orders of magnitude" [7]. The input side makes the gap vivid: the US National Academy of Sciences estimated the plastic flux to the oceans at 45,000 tons per year already in the 1970s, when that was 0.1% of global production; production has since quintupled to 265 million tons per year in 2010, around 50% of produced plastic is buoyant, and an estimated 60–64% of the terrestrial floating load is exported from coastal to open-ocean waters [7].

The paper's central evidence is not the total but the size spectrum — a fingerprint of removal. Fragmentation is a fractal process: cracking patterns in photodegraded plastic appear "at multiple scales, from centimeters to few microns," so progressive fragmentation should yield ever more fragments toward small sizes, with the abundance–size distribution following a power law of exponent 3 in steady state [7]. The Malaspina circumnavigation samples (4,184 plastic items in calm-condition tows used for the size analysis, from 7,359 items overall) fit that expectation almost exactly above 5 mm — "a power exponent of 2.93 ± 0.08, similar to the expected value, but only for size classes larger than 5 mm" [7] [7]. Below 5 mm the observed distribution falls away from the fragmentation model: abundance

peaks around 2 mm, with “a pronounced gap below 1 mm,” and the assessed losses concentrate “around sizes of 2.2 mm” [7]. A sampling artefact is explicitly excluded: “the size distribution of nonplastic particles in the same samples followed the characteristic power distribution, with increasing abundances toward smaller sizes” [7]. The plastic is not being missed; it is being removed, size-selectively, at the millimetre scale — and the lack of any observed increase in surface concentrations over time, despite rising inputs, means “the removal rate of the presumed sink would also have needed to increase” in step [7].

Four candidate sinks are weighed. Shore deposition is discounted for the missing sizes: “A selective washing ashore of the millimeter-sized fragments trapped in central areas of the open ocean is unlikely” [7]. Nanofragmentation remains open: the sub-1 mm gap “could indicate a fast breaking down of the plastic fragments from millimeter scale to micrometer scale,” possibly a two-phase fragmentation in which photodegraded millimetre fragments break down acceleratedly — with scanning electron micrographs suggesting oceanic bacterial populations contribute to degrading particle surfaces [7]. Biofouling offers a reversible exit: small fragments with high surface-to-volume ratio can be ballasted by epiphytic growth until they reach seawater density and drift or sink, hovering where their density matches the water column — but “field experiments have shown that biofouled plastic debris undergoes a rapid defouling when submerged, causing the plastic to return to the surface” [7]. Large objects are a separate loss term the fragmentation model omits: they “are commonly observed on the seafloor,” and their cavities — “bags, bottles” — facilitate the ballasting that sinks them [7].

Ingestion receives the most quantitative support, because the missing sizes are exactly prey sizes. “The size interval accumulating most of plastic losses corresponds to that of zooplankton (mainly copepods and euphausiids)”; reported plastic incidence in stomachs runs 1–29% in epipelagic zooplanktivorous fish and 9–35% in small mesopelagic fish, and “the most frequent plastic size ingested by fish in all these studies was between 0.5 and 5 mm, matching the predominant size of plastic debris where global losses occur in our assessment” [7]. Mesopelagic fish — the ocean’s most abundant zooplanktivores, at densities near one individual per square metre even in the oligotrophic gyres, migrating nightly from 200–1,000 m to feed at the surface — carry a standing load of plastic “on the same order of magnitude as our estimates of free plastic on the surface,” with turnover times from a year upward; their viscous feces form spheroids “sinking at high velocities (around 1,000 $m \cdot d^{-1}$),” a direct conveyor from surface to bottom [7] [7].

The paper ends in disciplined uncertainty. “Large loads of plastic fragments with sizes from microns to some millimeters are unaccounted for in the surface loads. The pathway and ultimate fate of the missing plastic are as yet unknown. We cannot rule out either of the proposed sink processes or the operation of sink processes yet to be identified” [7]. Nanoplastic abundance “has still not been quantified in the ocean,” deep-ocean microplastic measurements are scarce, yet available observations “point to a significant abundance of microplastic particles in deep sediments,” implying vertical transport by biofouling or ingestion [7]. And the authors are explicit about where their own estimate is weakest: it “could be greatly improved through joining sampling efforts particularly in semiclosed seas (e.g., Mediterranean) and the southern hemisphere, where existing data are scarce” [7]. The open ocean, on this accounting, holds tens of thousands of tons — and functions less as a reservoir than as a waystation, passing plastic downward, inward, and into food webs at rates the surface counts alone cannot reveal.

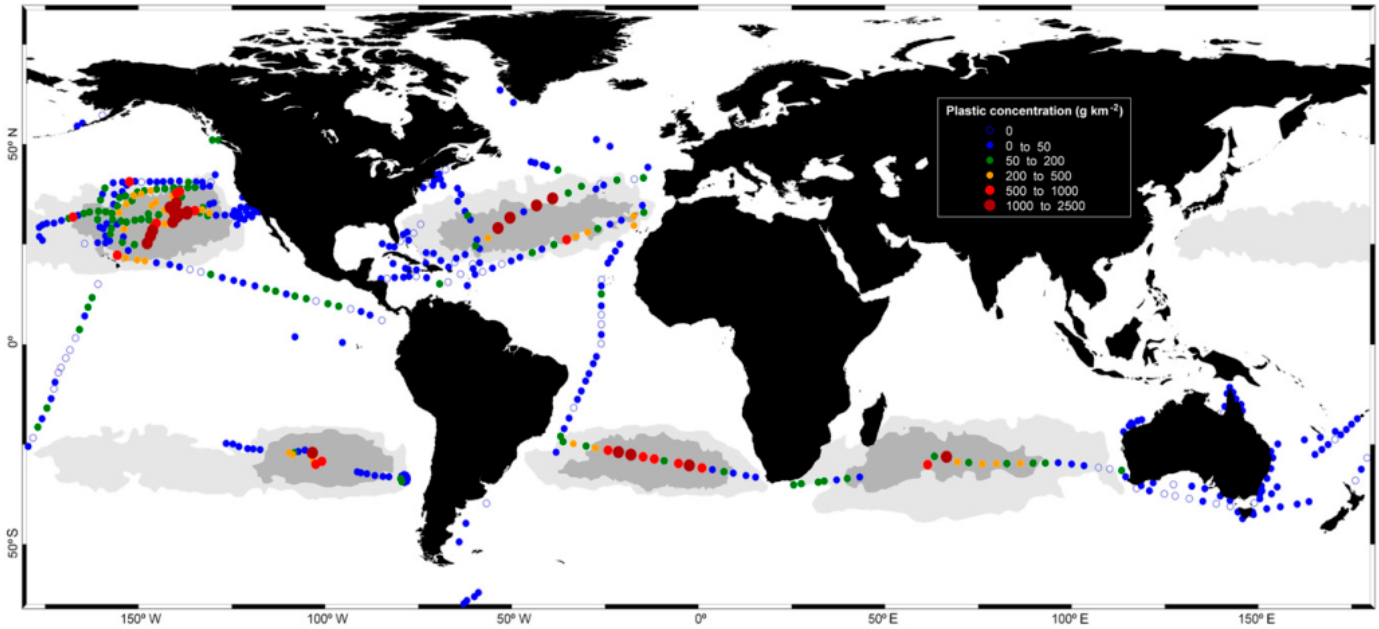


Exhibit. Fig. 1. Concentrations of plastic debris in surface waters of the global ocean. Source: S0003.

Evidence that the Great Pacific Garbage Patch is rapidly accumulating plastic

Lebreton and colleagues' 2018 Scientific Reports study is the most complete single-patch census in the corpus, and it revises the Great Pacific Garbage Patch upward by an order of magnitude — mostly, the authors argue, because previous surveys could not see the big pieces. The study (Lebreton, Slat, Ferrari et al, Scientific Reports 8:4666, 2018) defines its object precisely: an accumulation zone "formed in subtropical waters between California and Hawaii," surveyed by multi-vessel trawls and aircraft and modelled against a calibrated particle simulation [8]. The field effort combined 350 single Manta-net tow locations by 17 participating vessels (July–September 2015), 76 paired Manta and Mega net locations by RV Ocean Starr (July–August 2015), and 31 aerial mosaic segments from a C-130 Hercules (October 2016) [8].

The headline result: "at least 79 (45–129) thousand tonnes of ocean plastic are floating inside an area of 1.6 million km²; a figure four to sixteen times higher than previously reported," with the difference explained "through the use of more robust methods to quantify larger debris" [8]. The trawl haul itself was almost pure plastic: "more than 99.9% of the 1,136,145 pieces and 668 kg of floating debris collected by our trawls" [8]. The 1.6 million km² boundary — where modelled concentrations run from tens to hundreds of kg per km² — "comprises ~87% of the ocean plastic material present in our model domain (120°W–160°W, 20°N–45°N)" [8]. Inside it, the estimated 1.8 (1.1–3.6) trillion pieces divide by size into 1.7 trillion microplastics (6.4k tonnes), 56 billion mesoplastics (10k tonnes), 821 million macroplastics (20k tonnes), and 3.2 million megaplastics (42k tonnes) [8].

The mass distribution inverts the count distribution, and that inversion is the study's signature finding. "More than three quarters of the GPGP plastic mass was contained in the upper size classes (>5 cm)," with macroplastics contributing 25% and megaplastics 53% [8]. Microplastics were "8% of the total mass but 94% of the estimated 1.8 trillion pieces" [8]. By type, hard plastics, sheets and films (type H) made up 47% of mass and nets, ropes and lines (type N) 52%, with megaplastic mass overwhelmingly type N [8]. The single largest carrier: "at least 46% [of the GPGP mass] was comprised of fishing nets" in the abstract's accounting, and for megaplastics specifically, "86% of their 42 k tonnes contribution was carried by fishing nets" [8] [8]. Mean measured mass concentrations rank the same way — megaplastics 46.3 kg km^{-2}, macroplastics 16.8, mesoplastics 3.9, microplastics 2.5 — while numerical abundance ranks oppositely, microplastics averaging 678,000 pieces km^{-2} against 3.5 for megaplastics [8]. Polymer analysis by FTIR found polyethylene and polypropylene "by far the most common polymer types," and most particles were unidentifiable fragments; identifiable objects included containers, bottles, lids, caps, packaging straps, eel trap cones, oyster spacers, ropes, and fishing nets [8].

The debris carries dates and addresses. Of 50 items with a readable production date, 1 was from 1977, 7 from the 1980s, 17 from the 1990s, 24 from the 2000s, and 1 from 2010 — the patch is an archive of decades, not a snapshot of recent litter [8]. "386 objects with recognizable words or sentences written in 9 different languages" split with roughly one third Japanese (115 objects) and another third Chinese (113), and a production country could be identified on 41 objects across 12 countries [8]. The comparison with earlier estimates is stated as a methods correction

with an accumulation signal inside it: the ~79k-tonne estimate is “nearly sixteen times higher than a previous study (~4.8 k tonnes) that used net trawl data only and four times higher than another assessment (~21 k tonnes) that combined net trawl data with vessel-based visual surveys,” a gap the authors attribute mainly to better large-debris quantification — while allowing that “differences between estimates could also be attributed to increasing levels of ocean plastic pollution in the area, and particularly plastic inputs from the 2011 Tohoku tsunami” [8]. On the tsunami’s contribution they put numbers: an estimated 4.5 million tonnes of debris washed to sea, ~70% sinking rapidly per the Japanese government, leaving ~1.4 million tonnes transportable at the surface; Japan was the main production country (34%) among labelled collected objects, and the dispersal model suggests the tsunami contributed “around 10–20% of the debris input into the GPGP post 2011” [8].

The “rapidly accumulating” claim of the title rests on the time series: microplastic mass concentrations from surface net tows across decades, fitted within the GPGP by an exponential (R^2 = 0.92) rising faster than the fit for surrounding waters (R^2 = 0.78) [8]. The abstract’s conclusion is the same: “ocean plastic pollution within the GPGP is increasing exponentially and at a faster rate than in surrounding waters” [8].

Three interpretive findings matter for the rest of this report. First, persistence is selective: “only certain types of debris have the capacity to persist and accumulate at the surface of the GPGP” — rigid PE/PP and bundled nets dominate, while plastic films, ~37% of PE/PP waste generation, “were rarely found,” plausibly because low volume-to-surface items sink via biofouling or fragment below surface layers before arriving [8] [8]. Second, the source balance is upside down relative to models: “at least half of the collected GPGP plastics was composed of objects from marine based sources,” against model predictions that land-based plastic would dominate offshore — a discrepancy the authors assign to beaching preferentially removing coastal plastic, to marine gear’s engineered durability, and possibly to misestimated source shares (their model assumed fishing, aquaculture, and shipping at 28.1% of global inputs) [8]. Third, the patch is a low-windage trap: the null-windage model scenario best matched measured concentration gradients (R^2 = 0.52, declining to 0.39–0.21 as windage was added), ghostnets “generally have a draft of several metres, and therefore are unlikely to be influenced by wind transport,” and high-windage debris is spread widely, likelier to beach or exit the patch — consistent with foam’s near-absence in the GPGP against its abundance in Alaska beach clean-ups, and with high-buoyancy Tohoku debris stranding rather than accumulating [8] [8].

Finally, the missing-plastic problem survives even here. “Considering currently accepted plastic inputs from land- and marine-based sources, our global model predicted millions of tonnes of ocean plastic to be within the GPGP region, while we only found tens of thousands of tonnes” — a two-orders-of-magnitude gap implying removal from the surface and/or fragmentation below the quantified 0.5 mm floor, on top of the ~40% of plastic demand that is not buoyant at all and “should sink to the floor soon after release” [8]. And the authors insist their own number is conservative: applying the outer 0.1 kg km^{-2} contour, “our model predicted over 100k tonnes of ocean plastics,” before accounting for high-windage debris passing through uncounted, vessel-wake and tow-angle sampling biases, and aerial detection misses [8]. The best-measured patch on Earth is thus simultaneously 79,000 tonnes, growing exponentially — and, against inputs, still almost empty.

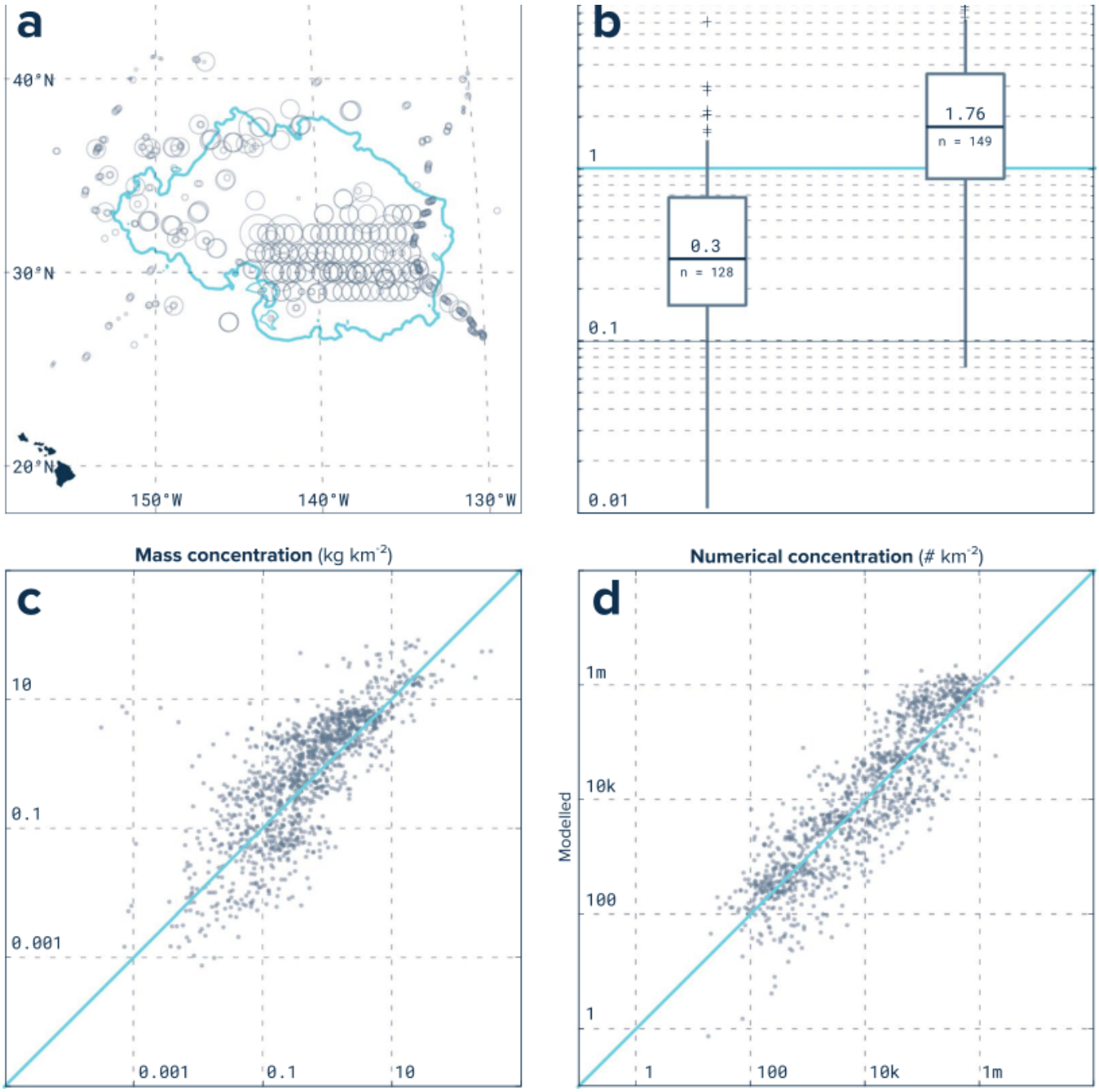


Exhibit. Figure 2. Numerical model calibration. Source: S0004.

Origin, dynamics and evolution of ocean garbage patches from observed drifters

Sections 9 and 10 described the instrument and the method; this section reports what van Sebille, England, and Froyland's transit-matrix experiment actually found when coastal tracer was released into the observed surface circulation and integrated for 1,100 years. The headline is that garbage patches are not permanent features with fixed contents. They are standing patterns in a flow — each with its own life history of growth, leakage, and decay — and on long enough timescales the system has a single main attractor.

The evolution unfolds in dated frames. "After 1 yr, the fast-flowing western boundary currents and equatorial jets of the oceans have advected much of the tracer into the open ocean" [6]. After 10 years, "well-defined regions where TAF > 2.0... can be seen in the five subtropical gyres, although the patch in the South Pacific remains relatively small at this stage, largely due to its remoteness from the most densely populated coastlines" [6]. By 50 years the patches have diverged in character: "The South Indian patch has decreased in size, while the South Atlantic patch has grown slightly. The South Pacific patch is still relatively small in spatial extent but has increased in maximum TAF values. Finally, there is a newly formed patch in the Barents Sea" [6]. The Barents accumulation has no subtropical gyre beneath it: "As there is no Ekman-driven convergence in the Barents Sea, accumulation of tracer there might be partly related to slow surface convergence due to deep-water formation, and possibly also to surface buoys be-

coming grounded in the (seasonal) sea-ice," and the authors note that observations of Arctic debris accumulation "are not yet available" [6]. The discussion adds the likely driver: the Barents patch "is most likely due to the enhanced release of tracer in highly populated areas around the North Atlantic Ocean" [6].

The long integration separates the patches into survivors and transients. The Barents patch "can still be observed after 100 yr, while the South Indian patch is considerably smaller than 50 yr earlier. After 1000 yr there are no longer any elevated levels of tracer in the South Atlantic and South Indian patches, while the North Pacific patch is continuing to expand" [6]. Quantitatively, "all patches except for the North Pacific patch reach a maximum size within the first 500 yr, after which they decay over a range of timescales," and the decay is fast for two of them: "the South Atlantic and South Indian patches completely disperse within 400 and 100 yr, respectively," while "the North Atlantic and Barents Sea patches reach an equilibrium size after approximately 300 yr" and "the South Pacific patch slowly decays after reaching a maximum size at 500 yr" [6]. Not everything is ever in a patch: "The amount of tracer not within any of the six main patches stabilizes at approximately 22% of total tracer released," most of that residue — 13% of all tracer — sitting "on the fringes of the patches, where the TAF < 2.0," with small local accumulations such as the Bay of Biscay [6].

The dynamics are a balance, not a trap. "The time evolution of the garbage patches is governed by the balance between accumulation due to Ekman convergence and dispersion due to other processes," with mesoscale eddies as the principal disperser: regions of highest sea-surface-height variability "coincide with the edges of the patches where relatively low spatial gradients in TAF appear" [6]. Eddy geography explains patch definition — "the South Pacific patch is farthest away from regions of high eddy activity and also the best-defined patch of all" [6]. The abstract quantifies the surprise: "With the exception of the North Pacific, all patches are much more dispersive than expected from linear ocean circulation theory, suggesting that on centennial timescales the different basins are much better connected than previously thought and that inter-ocean exchanges play a large role in the spreading of marine debris" [6].

Leakage was measured directly by reseeding each patch with its own tracer and running two years forward. "For the two Pacific patches, tracer that leaks away remains in the respective basins. For the other four patches, in contrast, there is clearly a preferred direction of leaking. For example, tracer from the North Atlantic patch gets advected northeastward towards the Barents Sea, while tracer from the Barents Sea patch gets advected southwestward towards the North Atlantic patch" [6] — a two-way exchange between the Atlantic and Arctic accumulations. In the southern hemisphere, "the two subtropical gyres in these basins span the so-called 'super-gyre' (Speich et al 2007), and our results provide direct observational evidence that this super-gyre also manifests in ocean surface flow" [6]. The discussion generalises the pathways: "Debris from the South Atlantic and South Indian patches ultimately migrates to the South Pacific patch, from where it slowly crosses the Equator to the North Pacific," and the southern patches' rapid dispersal is plausibly tied to "the proximity of the Southern Ocean, which facilitates rapid inter-ocean exchange via the Antarctic Circumpolar Current, vigorous eddy activity and mixing (Sallée et al 2008), as well as the super-gyre spanning the three basins" [6]. Even the leakage is partly an illusion of residence: the convergence zones are "rather leaky, with tracer being expelled from the patches within a matter of years, although for most of the patches this leakage is then rapidly re-accumulated" [6].

The endpoint of the system is the North Pacific. "The North Pacific patch is ultimately the largest 'attractor' of surface tracer of all, and our study suggests that a significant fraction of the marine debris reaching the open ocean outside of the North Atlantic will eventually end up in that patch" [6]; the abstract states the multi-millennial version — "a significant amount of the debris released outside of the North Atlantic will eventually end up in the North Pacific patch, the main attractor of global marine debris" [6]. The North Pacific patch of Section 14 — 79,000 measured tonnes and rising — is thus not merely the largest accumulation but the surface ocean's long-run destination, which reframes its exponential growth as both a local input story and a global convergence story.

Two cross-checks close the study. Against Maximenko et al (2012), who used the same drifter observations with a uniform start and no seasons, "incorporation of a seasonal cycle and a more representative source function do not seem to strongly affect the main features of the garbage patches, except for an additional sixth patch in the Barents Sea" [6] — the patch geography is robust to method; its dynamics are what the new method adds. And the policy corollary is drawn explicitly: "on centennial timescales the marine debris problem becomes a global problem, as the leakiness of the patches leads to most ocean-rim nations contributing at least some proportion of debris to most of the patches," so that managing the patches "require[s] global coordination and international activity" [6] [6]. Independent evidence from the GPGP census points the same way at shorter timescales: the patch's position shifts with climate indices — Lebreton et al found 2002 and 2004 analogous to their survey conditions in ENSO and PDO terms,

and calibrating to those years improved their model fit, "suggesting that climatic events such as ENSO or PDO influence the average position of the GPGP" [8]. A garbage patch, in short, is a climate-modulated, leaky, centuries-long sorting process — not a place where the ocean's debris simply stops.

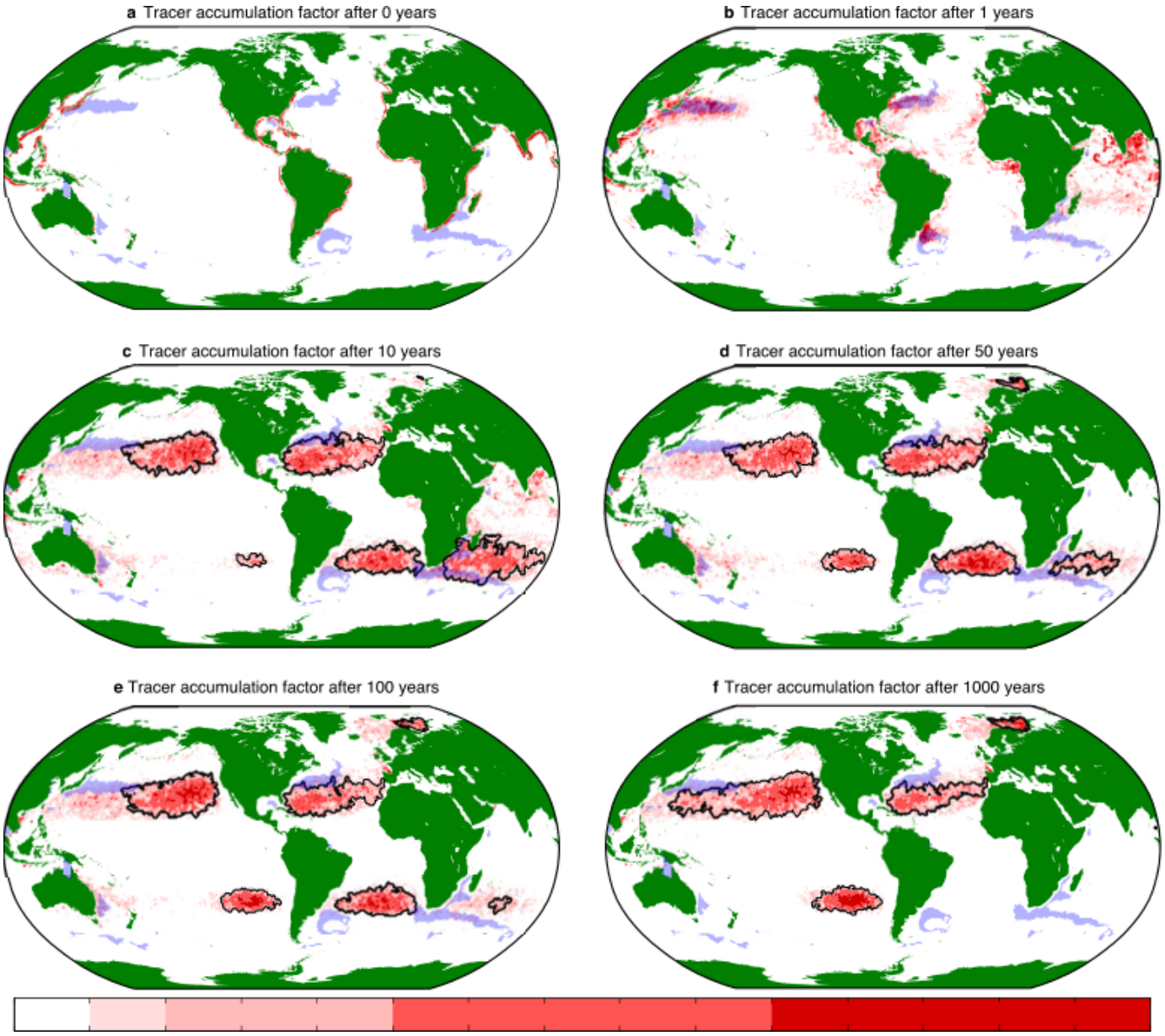


Exhibit. Figure 1. The locations and sources of the six garbage patches. Source: S0005.

A global mass budget for positively buoyant macroplastic debris

The missing-plastic problem has two rival solutions, and Lebreton, Egger, and Slat's 2019 mass-budget study (Scientific Reports 9:12922) exists to choose between them. The first solution, from Koelmans et al (2017), holds that the surface ocean is a fast transit zone: modelling emission, degradation, and settling for macro- and microplastics, it concluded that "99.8% of the plastic mass that has entered the marine environment since 1950 has degraded into micro- and nanoplastics and has subsequently settled below the surface layer," and predicted that under zero emissions "almost all plastic would be removed from the ocean surface layer within three years" [10]. The budget paper objects on observational grounds: "Dispersal models and observations suggest that debris released from terrestrial sources and accumulating in offshore oceanic gyres requires on average a minimum of several years to reach such remote oceanic areas. Thus, if positively buoyant plastic were persisting less than three years on the surface layer, accumulation in offshore subtropical gyres would not occur" [10]. Add the dated debris of Section 14 — "a significant number of decades-old objects, dating as far back as the 1970s" in the North Pacific gyre — and the fact that "the degradation of plastic initiated by solar UV radiation is severely retarded when floating in seawater," and the fast-sink solution looks untenable for macroplastics [10].

The alternative is a box model built to make the shoreline the main character. The model takes synthetic polymer production from 1950 to 2015, keeps the positively buoyant fraction — 65.5% of total production — and ages it through the market-sector lifetime distributions of Geyer et al (packaging, transportation, construction, electrical/electronic, consumer products, industrial machinery, other) to compute each year's discarded mass [10]. The marine environment is three domains — shoreline (S), coastal surface waters (C, defined by bathymetry shallower than the ~200 m photic depth), and offshore surface waters (O) — each with a macroplastic (>0.5 cm) and a secondary-microplastic compartment, with mass conserved across all six [10]. Five parameters run the system: the input fraction i, stranding/settling probability s, shoreline release probability r, offshore transport probability t, and the degradation rate d into microplastics, applied per compartment [10]. The input fraction is anchored to Jambeck et al's 2010 estimate of 4.8–12.7 million metric tons reaching the ocean from 274 million metric tons of plastic waste generated, giving i = 1.7–4.6% [10].

The parameters are not free: two independent constraints discipline them. The first is the age distribution of real gyre debris. Using the 2015 GPGP expedition's dated macroplastics — 664 kg landed from 83,144 collected pieces >0.5 cm, of which 427 had recognisable inscriptions yielding 11 languages and 50 production dates — the authors show that "only the degradation rate d significantly impacts the relative age distribution of positively buoyant macroplastics in offshore surface waters," because degradation is the only process that selectively removes old mass [10]. The best fit is a slow rate: "minimum value is found for d = 3%," with Pearson p-values of 0.009 at 3% against 0.0133, 0.0361, and 0.0310 at 10%, 50%, and 90% [10]. A rapid degradation sink of >90% per year "cannot reproduce observations of plastic age distribution at sea" [10]. The under-representation of 2010–2015 objects in the gyre sample is itself informative — explained "by the minimum time required for plastic debris to travel to oceanic gyres (~5 years)" [10]. The second constraint is Lagrangian: tracking 2,510,918 particles from coastal releases, the study finds that with no wind influence, "96% of the particles investigated have spent at least two consecutive days near the shoreline" after one year — their operational proxy for stranding — rising to 98% at a 2% windage coefficient [10]. The convergent parameter set is s = 97%, t = 33%, r = 1%: nearly everything strands, a third of what floats escapes offshore per cycle, and almost nothing that strands is released in a given year [10].

The budget that emerges relocates the missing plastic from the deep sea to the coast. "Plastic accumulated in offshore surface waters is highly persistent. Accumulated quantities are less than initially expected because of the capacity for the global landmass to trap and filter marine litter inducing a delay — likely on the order of decades — for fragmented buoyant plastic to reach offshore accumulation zones" [10]. Most of the mass that has entered the ocean since the 1950s "has not disappeared from the ocean surface by degradation but is stranded or settled on its way to offshore waters, possibly slowly circulating between coastal environments with repeated episodes of beaching, fouling, defouling and resurfacing" [10]. The age structure follows: "Most of the modeled macroplastic mass floating in coastal waters is composed of relatively new objects while older objects are better represented in the open ocean" [10]. The authors name the process a natural sorting: "the shoreline to capture the major part of floating material and where only a small fraction eventually escapes and accumulates in offshore waters," with the convergence condition $s \gg r$ implying "that the landmass likely is storing a major fraction of the missing plastic debris," including debris "buried under sediments… stored for unknown duration" and liable to eventual transport to deep water by sedimentary gravity flows [10].

The model also reaches one generation down the size cascade. Its predicted stock of degraded (secondary microplastic) material for 2014 is 0.28–0.75 million metric tons, against observed global surface microplastic estimates of 0.093–0.236 million metric tons for the same year — suggesting "at least two-third of these microplastics have disappeared from the ocean surface layer likely by settling, ingestion, aggregation, stranding or degradation into even smaller particles," with the relative contributions "largely unknown" [10]. And because degradation is slow and transit slower, "the microplastic contamination resulting from the formation of secondary particles in the marine environment is mostly representative of the degradation of objects from the 1990s and earlier" [10] — today's microplastic weather was made by last century's macroplastic. The authors project future buoyant macroplastic and degraded stocks under three emission scenarios — growth at the 2005–2015 average rate, constant emissions from 2020, and emissions stopped from 2020 — with the coastal reservoir guaranteeing that offshore stocks respond to any of them only on long lags [10].

The paper's humility is structural: five whole-ocean average parameters, assumed constant across years, polymers, object types, and places, with degradation rates that "will also likely differ between polymer composition and dimension of objects" and input rates tied proportionally to consumption since the 1950s [10] [10]. Its recommendation is correspondingly empirical — "the systematic collection of origin and [age data]" on debris, because age distributions are what discriminate between rival budgets [10]. For this report's through-line, the budget reframes the derelict prob-

lem once more: the ocean's missing plastic, like its missing ships, is mostly not sunk and gone. It is beached, buried, and waiting in the coastal margin — the same margin where abandoned vessels accumulate, uncounted, in the ADV inventories of Section 6.

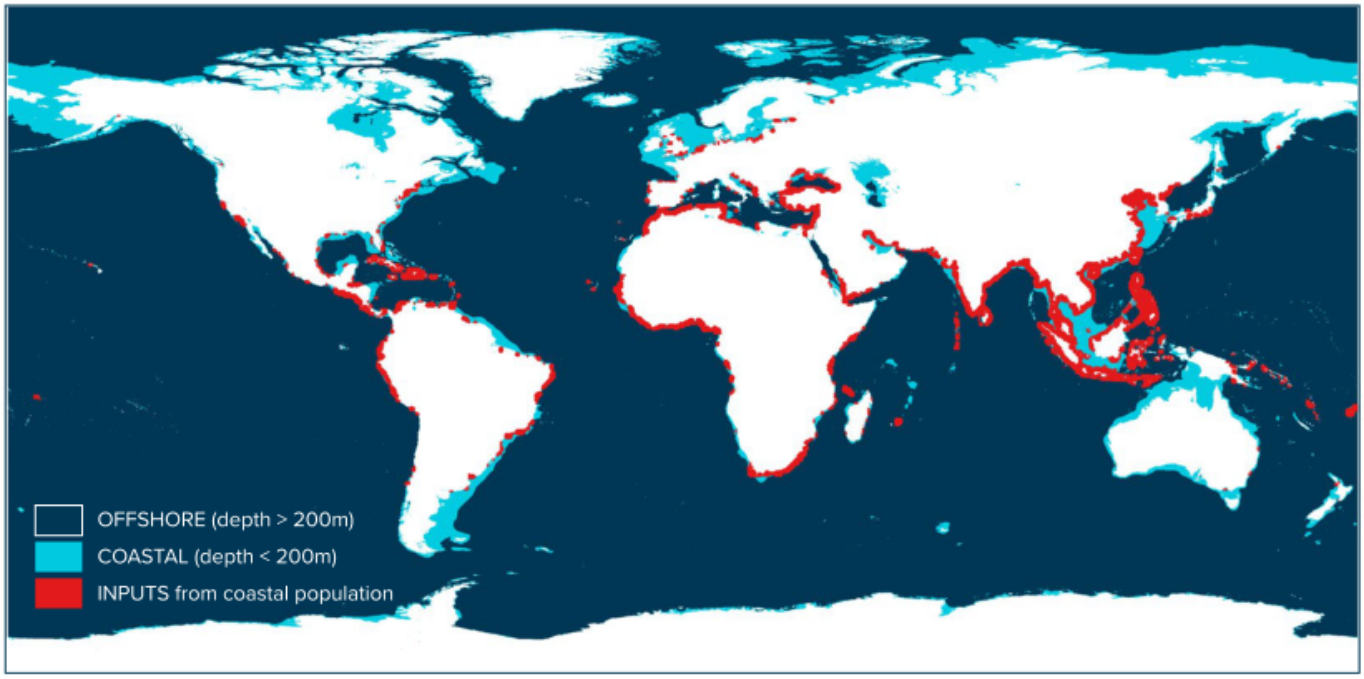


Exhibit. Source locations for the Lagrangian dispersal model and the global ocean surface model domains used to track positively buoyant macroplastic debris. Source: S0006.

The growing plastic smog: microplastics across the world's oceans

The inventories in the preceding sections were snapshots: a census of pieces and tonnes in a given year, in a given place. The "plastic smog" study — Eriksen and colleagues, published in PLOS ONE on 8 March 2023 under the title "170 trillion plastic particles are afloat in the world's oceans" — asks a different question: not how much plastic is in the ocean surface layer (OSL) at one moment, but how that amount has changed across four decades [11]. The distinction matters because earlier attempts to fix the global OSL mass produced only a wide bracket — previous estimates ranged from 93,000 to 578,000 tonnes — and because spatial and temporal data gaps, plus variability in station selection, sampling, and analysis, make any single snapshot hard to interpret and a trend harder still to establish [11].

Building a forty-year dataset from other people's trawls

The trend rests on a compiled dataset of 11,777 sampling stations drawn from published literature and unpublished sources, including previously unpublished data from 5 Gyres Institute expeditions [11]. The underlying samples were collected with three different sea-surface devices — manta trawls, AVANI trawls, and rectangular neuston nets — with net openings between 0.5 m and 1 m [11]. Because mesh size determines what counts as a particle, the authors filtered the data to samples with a lower mesh limit between 53 µm and 505 µm; in practice the dataset is dominated by one gear class, with 94.7% of samples (n = 11,154) collected at 320–350 µm mesh, against 2.5% at 200 µm (n = 303), 2.5% at 500–505 µm (n = 293), and just 0.2% at 53 µm (n = 27) [11]. Laboratory practice was laborious and manual: microplastics were separated from natural debris under a dissecting microscope after sieving, counted individually, and then weighed together within each size category; counts were expressed per square kilometre of trawled surface, and mass was estimated with a single literature conversion of 1.36×10^{-2} g per particle, from Morét-Ferguson et al. [11]. The modelled results for each year are averages of daily model results, and the lower and upper estimates reported are two times the standard error of the model [11].

The 2019 headline: a smog counted in trillions

For 2019, the model estimates that 82–358 trillion plastic particles — mean 171 trillion, primarily microplastics — were afloat in the world's oceans, weighing 1.1–4.9 million tonnes with a mean of 2.3 million tonnes [11]. The asymmetry between those two numbers is the essence of the "smog": an almost uncountable population of particles whose combined modelled mass is modest next to the macroplastic stocks discussed earlier in this report, because the mass figure is built by multiplying a particle count by a single small per-particle conversion [11]. Methodologically, the estimate also differs from its predecessors: rather than distributing modelled concentrations back onto ocean-model grids, the authors used an oceanographic model to estimate concentrations by location, extracted a time trend, and multiplied the mean concentration prediction by the size of the ocean — a choice made partly because extrapolating into sparsely sampled basins would have misrepresented the model's certainty [11].

The shape of the trend: flat, then a cliff

The forty-year curve has three distinct regimes. From 1979 to 1990 there are few samples, the confidence intervals are correspondingly wide, and no clear trend can be detected [11]. Sampling density increased through the last decade of the twentieth century, yielding reliable trends between 1990 and 2015; within that window, the period through about 2004–2005 shows substantial variability that could be read as stagnation or even a decreasing trend [11]. From 2005 onward — the paper also states the break as 2006 onward — the fluctuation ends and is replaced by a consistent, rapid, dramatic increase in plastic abundance [11]. After accounting for wind, site selection, and undersampling biases, the authors conclude that the increase since the turn of the century is significant [11].

Earlier, narrower records point the same general direction, though only for northern oceans ringed by industrialised countries: archived Continuous Plankton Recorder samples show microfibres increasing since the 1960s and macroplastic entanglement increasing since the late 1950s; Day and Shaw reported rising North Pacific microplastic between 1976 and 1985; and Wilcox et al. found the western North Atlantic increasing from 1986 to 2015 at a rate paralleling global cumulative plastic production — while other studies, the authors note, have found no evidence of a rise over time at all [11].

Why the curve bent: law first, then volume

The paper's most provocative suggestion is that the flat decades may not be an accident of sampling. A cluster of binding international interventions preceded the millennium: the 1972 Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter (the London Convention), the 1982 United Nations Convention on the Law of the Sea, and — most directly — MARPOL Annex V in 1988, which created legally binding agreements among 154 countries to end the discharge of plastics from naval, fishing, and shipping fleets [11]. Alongside the treaties came a voluntary industry measure: in 1991 the Plastic Industry Trade Association launched "Operation Clean Sweep," aiming at zero loss of plastic pellets, powders, and flakes from factories, after which decreasing pellet ingestion in biota was observed [11]. The authors are careful to frame this as possibility rather than proof — the stagnating trend "may have benefitted" from these measures [11].

Whatever held the line before 2005 stopped holding it. The authors attribute the rapid increase from 2005 onward to several compounding drivers: exponential growth of plastic production as it relates to inputs; changes in terrestrial waste generation and management; and the continued degradation and fragmentation of older macroplastics adrift at sea or stranded on shorelines and in rivers, which feeds the microplastic count from the existing stock [11]. The globalisation of raw plastic materials at the turn of the century accelerated imports, exports, and domestic production of plastic products and packaging, while recycling remained historically low even in countries with highly developed waste-management systems; the international trade in low-value waste plastics then left much of that material as mismanaged waste in receiving countries [11]. The turn of the century also brought increased plastic emissions from large fishing fleets and artisanal fisheries [11]. The synthesis is stark: increased international economic activity, together with the fragmentation and resuspension of degraded macroplastics, has overwhelmed both the natural mechanisms that export plastic out of the OSL and whatever positive effect the early binding interventions had [11].

How much to trust it

The authors stress the corroboration first: the post-2005 increase parallels trends estimated independently for global beaches, drawn from completely separate observations, which strongly suggests that ocean plastic pollution over the past fifteen years has reached unprecedented levels [11]. The caveats are equally explicit. The vast majority of samples come from the North Pacific and North Atlantic, so the results are biased toward those basins; the South Atlantic, South Pacific, Mediterranean, and Indian oceans remain data-poor [11]. The coefficient of variation across the data spans 0.447 to 10.558 with a mean of 2.602, underscoring how difficult accurate time-trend modelling remains [11]. Particle counts are sensitive to the lower size limit a net can retain, and the count-to-mass conversion is admittedly simple, with machine-learning methods for better conversions still in development [11]. The smog, in other words, is real and rising on the evidence available — but it has been measured most carefully in the two northern basins where the instruments have always been, and its smallest particles are still, by construction, the ones the nets cannot see [11] [11].

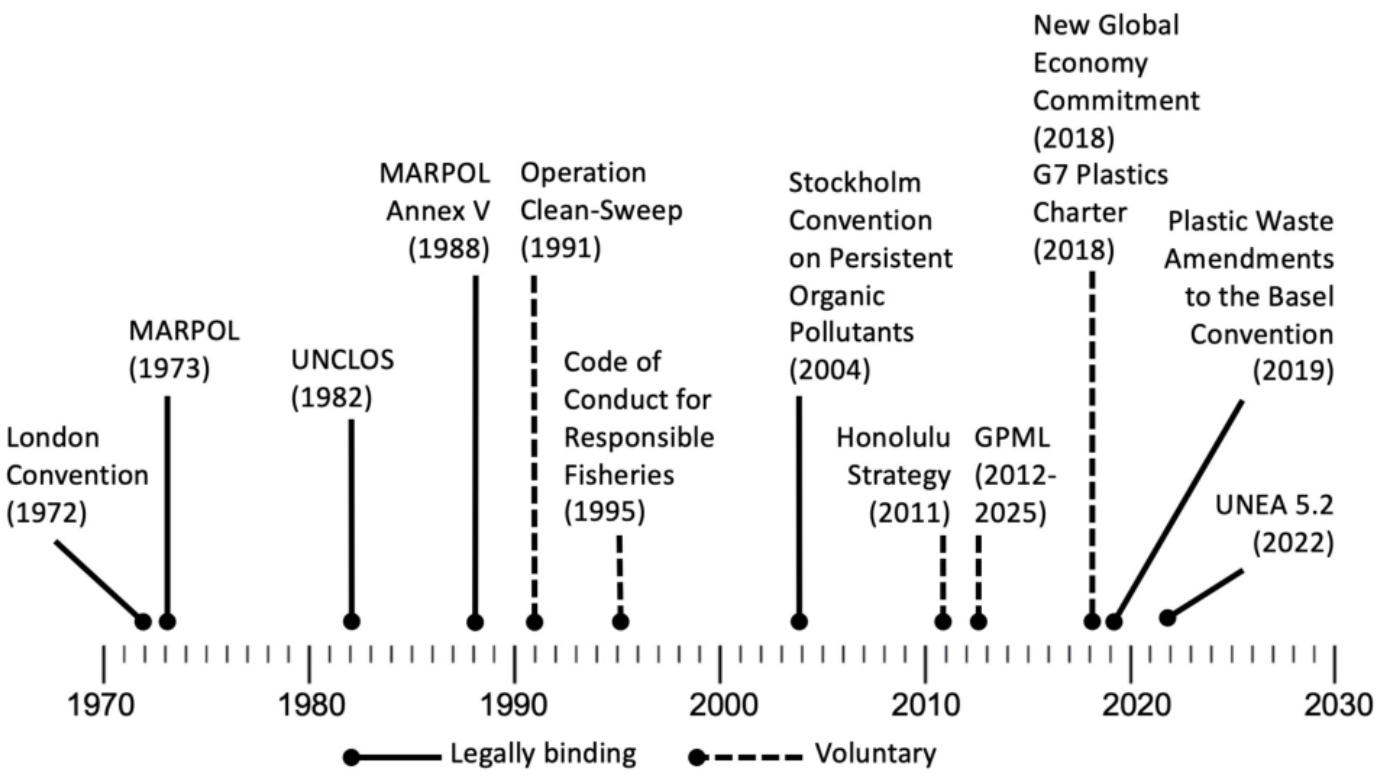


Exhibit. Fig 4. International policy interventions and maritime law. Source: S0007.

Breaking the Plastic Wave: scenarios and system interventions to 2040

The science chapters of this report end in an uncomfortable place: the surface-layer particle count has been rising rapidly since 2005, and the best available budget says the missing mass is parked along the world's shorelines, still fragmenting. Breaking the Plastic Wave — a global assessment built on a stochastic model of the entire plastic system — asks what it would take to change that trajectory by 2040, and its companion policy analysis in the plastic-smog literature frames the stakes plainly: without substantial, widespread policy changes, the rate at which plastics enter aquatic environments will increase approximately 2.6-fold from 2016 to 2040 [11]. Existing international policy, that analysis notes, is fragmented, business-oriented, short on specificity and measurable targets, and — the 2019 Plastic Waste Amendments to the Basel Convention excepted — non-binding, voluntary, and weak in monitoring and enforcement; it is "not stemming the tide" [11].

Two versions of 2040

The report models a Business-as-Usual (BAU) future against a System Change Scenario (SCS) in which eight interventions are applied concurrently, ambitiously, and starting immediately [12]. The headline contrast is leakage: under BAU, 29 million metric tons of plastic leak into the ocean in 2040 — 26 million tons macroplastic and 3 million tons microplastic — against 5 million metric tons under System Change (4 macro, 1 micro), an ~80% reduction in ocean

leakage, with leakage to land also reduced by more than 80% [12] [12]. Virgin plastic production tells the same story from the upstream end: 400 million metric tons in 2040 under BAU, against 181 million under System Change [12]. Put against the baseline, that is a fall of 11% relative to 2016 — significant because today 95% of plastic demand is fulfilled by virgin plastic, and the calculation counts only plastic in municipal solid waste [12]. Employment rises rather than falls: 11 million jobs under BAU against 12 million under System Change [12]. The report's summary of the endpoint is that growing global demand for plastic utility in 2040 could be met with roughly the same amount of plastic in the system as today, decoupling plastic growth from economic growth [12].

The wedges: every tonne gets one fate

The model's accounting device is the "wedge": any plastic that enters the system has a single fate, and the wedges partition the whole flow, macroplastic and microplastic alike [12]. Under System Change in 2040, the Reduce wedge — plastic utility fulfilled without physical plastic, through elimination (130 million tons, 30% of the flow) plus consumer reuse and new delivery models — is the largest single share [12]. Substitute follows at 71 million tons (17%), utility met by paper, coated paper, and compostables; Recycle accounts for 84 million tons (20%), split between mechanical recycling (closed- and open-loop) and chemical recycling back to plastic; Dispose — chemical conversion to fuel, landfill, and incineration — takes 101 million tons (23%); and the Mismanaged wedge, everything that escapes into open burning, terrestrial pollution, and ocean pollution, shrinks to 44 million tons (10%) [12]. The economics are asymmetric: Reduce levers are often the most economical to implement, while plastic substitutes are typically more expensive [12]. The cost figures are techno-economic — capital and operating costs across the whole value chain, subsidies and taxes excluded — and they deliberately omit the externalities of mismanaged waste, which the authors did not price [12].

Eight interventions, applied together or not at all

The scenario rests on eight named interventions: (1) reduce growth in plastic consumption; (2) substitute plastics with suitable alternative materials; (3) design products and packaging for recycling; (4) expand waste collection rates in the Global South; (5) increase mechanical recycling capacity globally; (6) scale up global capacity for chemical conversion; (7) build safe waste disposal facilities; and (8) reduce plastic waste exports [12]. Responsibility is distributed across the chain — consumer-goods brands and retailers own the first three, local governments collection, waste-management companies recycling and (with the petrochemical industry) chemical conversion, and national governments disposal capacity and export reduction [12]. The concurrency condition is not decorative: the modelled 80% reduction depends on all interventions being applied ambitiously and at once, and the interventions are load-bearing for each other — collection precedes recycling, landfilling, and incineration, and design for recycling is what makes mechanical recycling economically viable and scalable [12] [12].

Where money is limited, the report offers a priority order. Reduction — elimination, expanded consumer reuse, new delivery models — ranks first on every axis: it is the most attractive solution environmentally, economically, and socially, delivers the biggest pollution reduction, often represents a net saving, and offers the highest greenhouse-gas mitigation [12]. Mechanical recycling ranks next, ahead of chemical conversion and substitutes on cost, climate, technology readiness, and regulatory grounds; each metric ton of mechanically recycled feedstock offsets 48% in GHG emissions, and plastic should be designed to be mechanically recycled wherever possible [12]. Substitution is a case-by-case judgement — substitutes are typically more expensive, their carbon impact can be better or worse by material and geography, and designing for reuse beats swapping one single-use material for another [12]. Chemical conversion to plastic earns a conditional place: it returns feedstock to the petrochemical process as virgin-like plastic and could create an economic sink for low-value plastic, but it is unproven at scale, costlier and more energy- and emissions-intensive than mechanical recycling, and its expansion should be contingent on decarbonised energy [12]. Controlled disposal — landfill, incineration, plastic-to-fuel — is the last resort: not circular, resource-hungry, and expensive once collection costs and externalities are honestly counted [12].

Co-benefits, and the people already doing the work

The scenario's side effects are substantial. The eight interventions together cut cumulative plastic-related greenhouse-gas emissions 14% below BAU over 2021–2040, and annual emissions 25% by 2040 — driven by reduced virgin production and conversion, which together account for 80% of life-cycle plastic emissions, and by reduced open burning [12]. Open burning itself, a major source of carcinogens and toxins, falls from 133 million metric tons per

year in 2040 under BAU to 23 million [12]. Governments save US$70 billion in present value between 2021 and 2040, and 700,000 net new formal jobs are created in middle- and low-income countries by 2040 [12]. The report also insists on recognising the workforce the current system already runs on: 11 million waste pickers are currently responsible for 60% of global plastic recycling, largely unrecognised and underpaid, and integrating them into formal systems is presented as both a social obligation and a practical necessity [12].

What even success leaves behind

System Change is a ceiling, not a cure. Even under the scenario, 5 million metric tons of plastic still flow into the ocean in 2040, and a cumulative 248 million metric tons will have entered the ocean between 2016 and 2040 — which is why the authors urge stakeholders to push beyond the maximum the model assumes [12]. The conclusion is blunt about the hierarchy of effort: "Breaking the Plastic Wave" is about fighting plastic pollution, not plastic itself, but recycling and waste-management scale-up — the cornerstone of a circular economy — will not be enough at current production levels, let alone expected growth; on a downstream-only path, plastic's 2040 emissions of 1.6 Gt-CO_2e would nearly double against 2016 and claim some 15% of the allowable emissions budget under the IPCC's RCP 2.6 pathway [12]. Reduction and appropriate substitution are essential, governments must realign the incentives that currently advantage virgin feedstock over recycled material, and industry must remove avoidable single-use and hard-to-recycle plastic from the market [12]. The governance track moved while the report circulated: early in 2022, at the UN Environment Assembly 5.2 in Nairobi, all member states adopted a resolution committing to a legally binding global agreement covering the full life cycle of plastic — production, design, and disposal — by 2024 [11]. The smog study's verdict on what such an agreement needs is the note this section ends on: standardised monitoring frameworks to track global trends, and binding, enforceable agreements that restrict emissions of plastic pollution in the first place — because cleaning up plastic already in the environment has, in its words, limited merit [11].

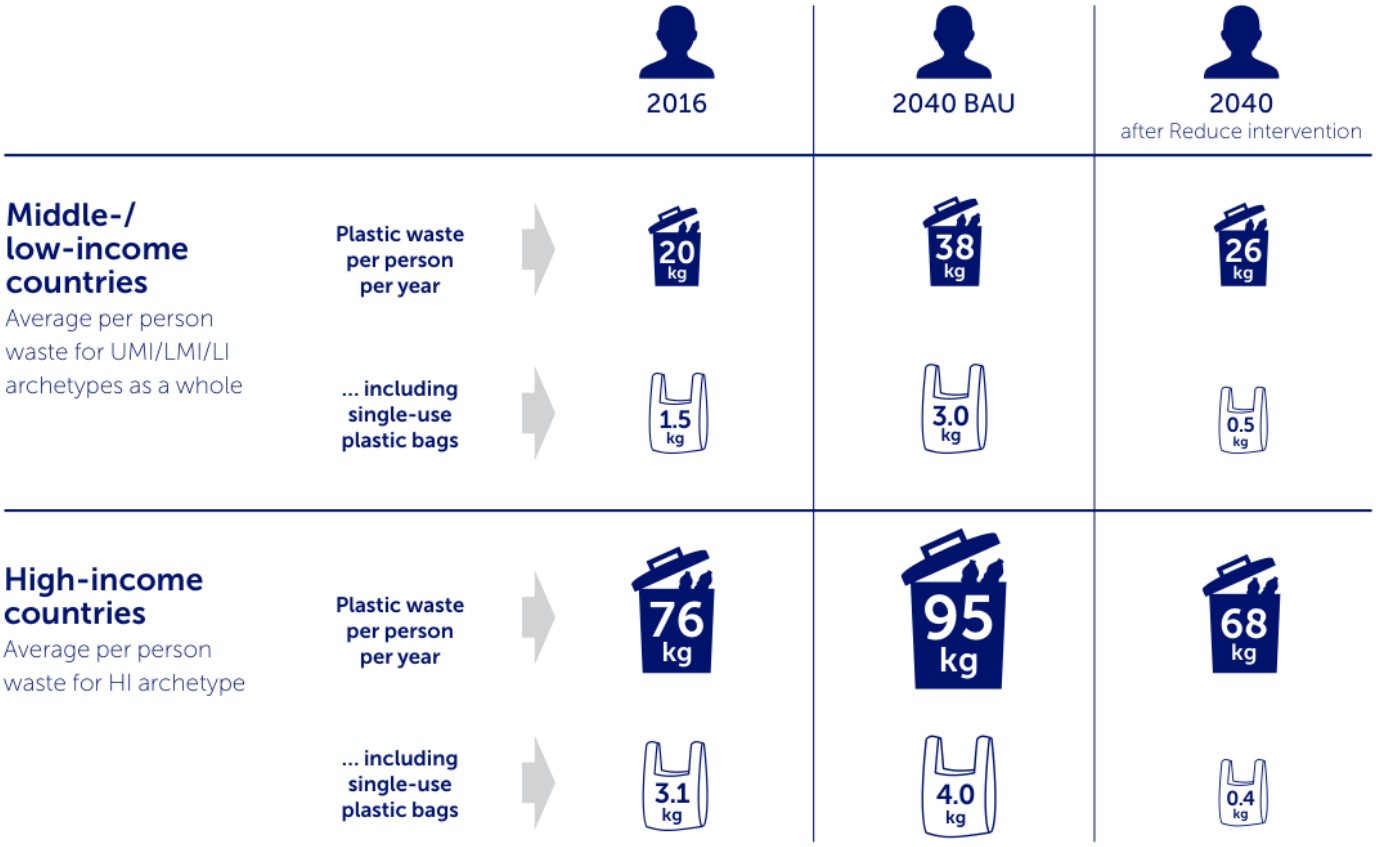


Exhibit. Change in total plastic waste generation and per-capita plastic bag consumption under the Business-as-Usual Scenario and after Reduce interventions are applied. Source: S0008.

The 2011 Tsunami Debris Field

A severe marine debris event: the Japan tsunami debris report

Most marine debris arrives the way the plastic of the previous chapters does — gradually, anonymously, and in pieces too small to name. The Japan tsunami marine debris (JTMD) event was the opposite: a single, dated catastrophe that converted a coastline's built environment into a debris field in a matter of hours, and then required the United States to organise a federal response to flotsam. NOAA declared it a Severe Marine Debris Event, and its Severe Marine Debris Event Report documents what federal agencies did between March 2011 and May 2013 [13]. The legal machinery is specific: under the Marine Debris Act (33 U.S.C. 1951 et seq., as amended by Title VI of Public Law 112-213), NOAA must develop, with the Interagency Committee, plans for timely response to events the Administrator determines to be severe marine debris events, including assessment of impacts on human health, navigation safety, natural resources, tourism, and even livestock and aquaculture [13].

The event

On 11 March 2011 a magnitude 9.0 earthquake — centred under the ocean about 70 km off Japan's east coast — generated a tsunami with waves up to 130 feet that devastated over 200 miles of the Tōhoku coast [13] [2]. As the water receded, it washed much of what stood in the inundation zone back into the ocean; the resulting debris field was captured in satellite imagery and aerial photographs of the waters around Japan immediately afterwards [13]. The monitoring problem was apparent from the start: the debris area was so large, and high-resolution satellite coverage so narrow, that a full survey was neither possible nor practical [13]. Analysts worked through imagery from ENVISAT, LANDSAT, SPOT, and ASTER, supplied via the U.S. Geological Survey, the International Disaster Charter, European Space Agency archives, and a joint NASA/JAXA site, mostly at 15–30 m resolution; the National Geospatial-Intelligence Agency later provided 1–5 m imagery for targeted areas [13] [13]. Even that had limits: the debris cloud was visible as patches in satellite imagery for only about a month, and by 14 April 2011 it had dispersed past visibility; when NOAA examined high-resolution imagery of a 50 km by 60 km patch of ocean where models placed the debris in December 2011, it detected nothing — the debris was either smaller than the 1–2 m resolution or not where the forecast put it [2]. The Canadian science review drew the honest conclusion: the overall distribution of the tsunami debris in the ocean was unobservable and effectively unknown [2].

What the models could say was structural rather than precise. Debris items sink, disperse, and break up en route; objects of different buoyancy and wind exposure leave the same starting point at different speeds and in different directions, so simulations showed the debris spreading across a vast area of the North Pacific, thousands of miles from land [13]. Updated modelling that folded in two years of wind and current data suggested debris could reach the Pacific Northwest coast during winter 2011–2012 — a hypothesis the report says was supported by the small boats, pallets, and large docks that came ashore in Oregon and Washington — while the bulk of the debris was likely still dispersed north of the Hawaiian Islands [13].

The response machine

NOAA's Marine Debris Program led the federal effort, and its first problem was coordination. NOAA coordinated with the Governments of Japan and Canada, briefed Congress, and ran a bi-weekly call open to federal, state, and local agencies whose participant list exceeded 200 representatives, spanning affected coastline from Midway Atoll and Hawaii to British Columbia and King Salmon, Alaska [13]. Sightings from the public and from vessels were funnelled to a single address, DisasterDebris@noaa.gov, entered into a tracking database, and loaded into the Environmental Response Management Application (ERMA) to map at-sea and shoreline reports; anything hazardous or obstructing navigation was passed immediately to the Coast Guard [13]. In December 2011 NOAA also formed a Subject Matter Expert group of modellers — the leads of the University of Hawaii's SCUD model, NOAA's OSCURS, and NOAA's GNOME — and later work integrated U.S. Navy HyCOM blended winds into GNOME, run as a hindcast on real weather and current data and refreshed every two weeks [13]. GNOME's advantage was behavioural range: it carried particles with a spread of windage values drawn from the Coast Guard's windage/leeway library, representing everything from low-windage objects to the high-windage items that travel fastest [13]. The report is careful with the epistemics — a hindcast shows where debris may be now, not where it will be, and "a model is not reality" [13].

Around NOAA sat the rest of the state. The Department of State handled bilateral and multilateral coordination, holding high-level meetings with Japan's Ministry of Environment and receiving Japanese government data on debris

types, quantities, and modelled trajectories [13]. State also coordinated the human edge of the event: items of personal significance found in the United States could be referred through Japanese consulates, which would try to identify the owner and determine what should be done with the item [13]. Removal authority, by contrast, was a patchwork the report lays out statute by statute: the Army Corps of Engineers leads on obstructions in federally maintained navigable channels under Section 19 of the Rivers and Harbors Act (33 U.S.C. § 414); the Coast Guard, as Federal On Scene Coordinator for the coastal zone, leads removal of oil and hazardous-substance threats under the National Contingency Plan; EPA takes emergency removals in the inland zone under CERCLA § 104 and the Clean Water Act; and debris that is neither a pollution threat nor a navigation hazard falls to state, local, or tribal jurisdictions [13]. The Coast Guard also issued Broadcast Notices to Mariners for debris hazards and worked with the Corps on channel obstructions [13]. The underlying navigation risk, as the Canadian review framed it, was concentrated in large objects floating mostly below the waterline — partially submerged vessels, containers, docks — largely invisible to radar and crews [2].

Refuges, parks, and the species problem

The receiving coastline was substantially public land. The Fish and Wildlife Service's refuge system includes 180 refuges routinely affected by marine debris, and at least 32 National Wildlife Refuges in Alaska, Washington, and Oregon were expected to receive tsunami debris over the following years; the FWS drafted tsunami debris response plans for visitor, volunteer, and responder safety and habitat protection [13]. Tsunami-attributed debris was observed at Alaska Maritime and Kodiak refuges in Alaska, at Dungeness and Willapa in Washington, and at Midway Atoll — whose staff, given four hours' notice, executed their tsunami plan without injury, though the wave killed tens of thousands of nesting seabirds and left birds and sea turtles entangled in debris for staff to rescue [13]. The Alaska Maritime refuge alone encompasses more than 2,400 islands across thousands of miles of hazardous Aleutian water [13]. The National Park Service counted 85 ocean and Great Lakes units with over 11,000 miles of shoreline, 31 of them potentially affected, and issued a Tsunami Debris Response Policies and Procedures handbook together with the Response Protocols for Biofouled Debris and Invasive Species [13].

Biofouling was the event's most distinctive scientific risk. Debris crossing the Pacific arrives alive, carrying Japanese coastal organisms, and the FWS Aquatic Invasive Species program worked on mitigation plans against the full catalogue of invasion consequences — competition and predation on native species, hybridization, pathogens and parasites, food-web disruption, habitat engineering, and fouling of aquaculture and energy structures [13]. The statutory frame was the Nonindigenous Aquatic Nuisance Prevention and Control Act of 1990, amended by the National Invasive Species Act of 1996, which established the Aquatic Nuisance Species Task Force co-chaired by the FWS Director and the NOAA Administrator [13]. The risk stopped being abstract in December 2012, when a dock over 65 feet long and 20 feet wide, weighing more than 185 tons, washed onto Mosquito Beach in Olympic National Park — a designated wilderness shore overlapping the Olympic Coast National Marine Sanctuary — similar to a dock the Government of Japan had identified as tsunami debris [13]. A team from the park, the Washington Department of Fish and Wildlife, the NSF-funded JTMD Invasive Species Team (Williams College, Oregon State University, and the Los Angeles County Museum), and NOAA reached the site to sample its non-native species, and tracking beacons were attached to the dock [13].

What the public was told

NOAA's outreach brochure compressed the response into four verbs — grab it, toss it, be careful, report it — with sightings sent to the same DisasterDebris address [13]. Confirmed tsunami items by then included vessels, buoys, sports balls, and a floating pier, and the expected arrivals included fishing nets, lumber, plastics, household items, foam, and possibly chemical or oil drums [13]. Two reassurances were repeated everywhere: radiation was highly unlikely to be a hazard — debris tested in West Coast states and Hawaii, including items known to be from the tsunami, showed no contamination above normal levels — and not every object on a Pacific beach was tsunami debris, since ordinary debris washes up constantly and only unique identifying information distinguishes the two [13] [2]. That identification problem ran through the whole event: without a name or registration number, tsunami debris is effectively indistinguishable from the everyday debris of Hawaii, Alaska, and the West Coast [13]. The report's own framing is the section's verdict: the tsunami debris, severe as it was, "adds to an already existing problem" — and the response it justified was built to outlast the event that triggered it [13].

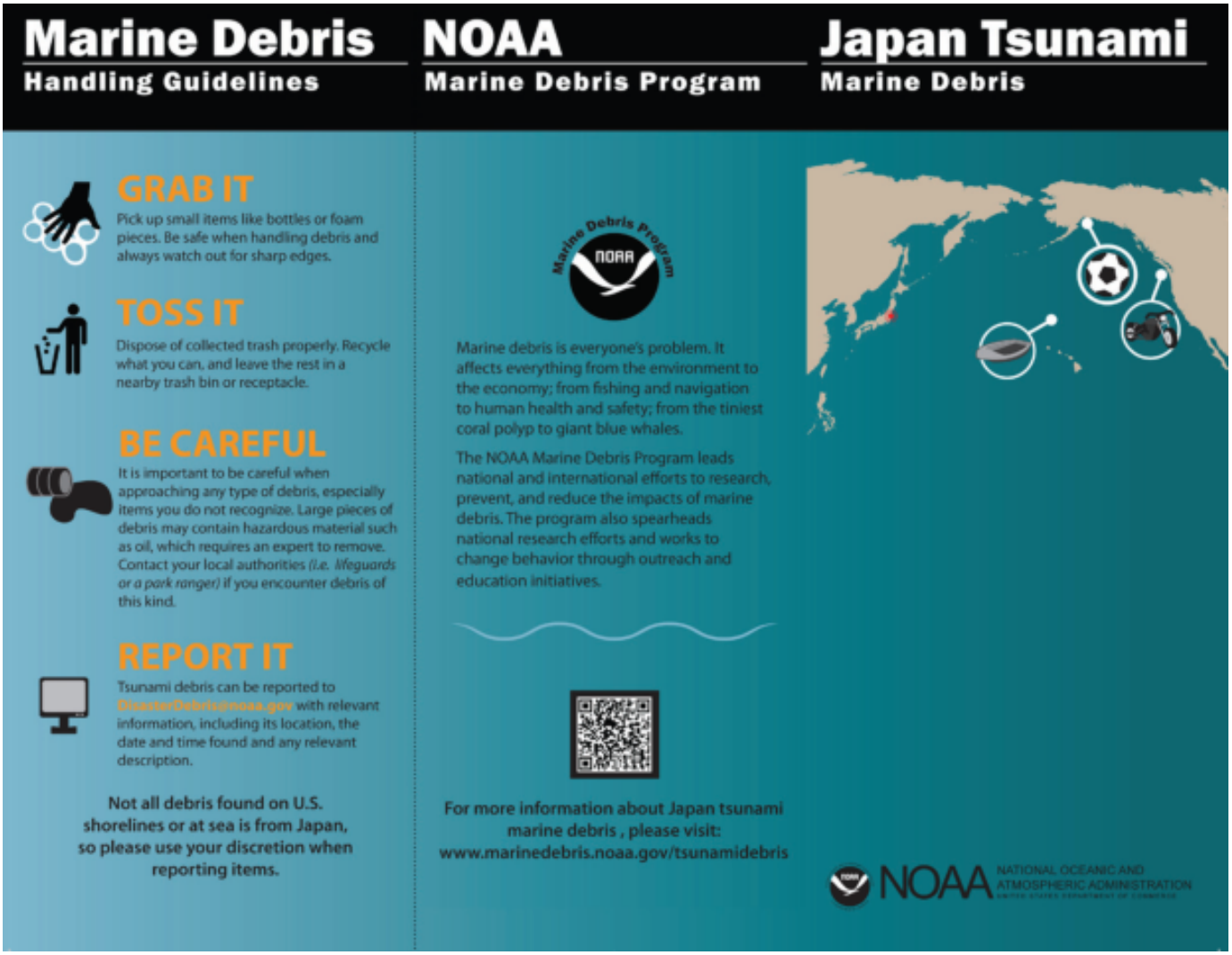


Exhibit. NOAA communications and public outreach on Japan tsunami marine debris, which drew sustained media and public interest over the two years covered by the report. Source: S0011.

Transport of Japan tsunami marine debris to the coast of British Columbia

The American response described in the previous section was built to receive debris; the Canadian question was narrower and harder: when would the debris arrive in Canadian waters, where would it land, and how much of it would there be? The answer took the form of a Fisheries and Oceans Canada Science Response — an updated review (March 2013) that superseded the original advisory document because the evidence had changed: debris had been observed at sea, significant pieces had reached the North American coast, and new model simulations had replaced the earlier zero-windage, climatology-driven runs with simulations using the actual winds over the Pacific from March 2011 through August 2012 and, crucially, varying the windage of the debris [2].

What was actually afloat

The starting numbers had already collapsed once. Initial estimates put the mass of debris swept into the ocean at 20–25 million tonnes; the Government of Japan's updated estimate was that about 70% of it sank on Japan's continental shelf, leaving roughly 1.5 million tonnes afloat as of March 2012 [2] [2]. Composition was not known in detail, but the preponderance was thought to be construction material from the houses swept to sea from coastal towns — which makes lumber a likely major constituent of the field — and individual pieces ranged from small fragments to objects weighing hundreds of tonnes [2]. Only the most buoyant and durable objects would survive the crossing at all [2].

The conveyor: currents, bifurcation, and seasons

The transport route is a relay of currents. Debris entered the ocean along the northern coast of Honshu, where the Oyashio and Kuroshio Currents merge into the Kuroshio Extension, a swift eastward flow that carried material away

from Japan toward the central North Pacific [2]. There the flow broadens and slows into the North Pacific Current (the West Wind Drift), which ferries debris across the basin toward North America [2]. Approaching the coast, the current bifurcates — drifter studies put the split at about 50°N, with year-to-year variability — into the northeastward Alaska Current and the southeastward California Current; debris entering Canadian waters comes overwhelmingly via the northern branch [2]. Currents alone do not beach anything: they bring debris close to shore, and deposition is finished by retreating tides, onshore winds, and waves [2]. The seasons gate the process — summer winds along the B.C. coast are generally northerly, driving upwelling and offshore transport that protects the coast, while winter's southerly winds produce downwelling and onshore transport that favour deposition [2].

Meanwhile the field was dissolving. The ocean's eddy field spreads any compact cluster: with lateral eddy diffusivities of 5,000–10,000 m^2/s reported for the Kuroshio Extension, the dispersion scale reaches roughly 400 km after six months — a circle of some 500,000 km^2, an area the review compares to Alberta or California — and within that expanse the debris is not uniform but patchy, gathered in clumps and elongated filaments [2].

Windage splits the field into fleets

The review's central physical point is that "the debris" was never one thing moving at one speed. An object's velocity is the sum of the surface water velocity and a wind-driven component set by how much of it stands above the waterline; floats, drums, buoys, containers, docks, and vessels — including partially submerged ones — have significant windage [2]. Such objects were probably a small fraction of the total load, but a disproportionately important one, because the large, potentially hazardous objects are exactly the high-windage ones [2]. New simulations — from NOAA, from the Government of Japan, and from Maximenko and Hafner at the University of Hawaii, using observed winds and altimeter-derived time-varying currents — ran the field as separate windage classes from 0 to 5% [2].

The classes behaved like different fleets. Simulated debris at very high windage (4–5%) reached or neared the North American coast during winter 2011–2012, nine months to a year after release, and the westerlies pushed it further north than the zero-windage debris; by spring 2012 it had either washed ashore or lay near the coasts of Alaska, British Columbia, Washington, and Oregon — consistent with the high-windage objects that actually arrived in early 2012, the ghost ship found off Haida Gwaii among them, its windage estimated at about 3.5% [2]. Debris at zero to moderate windage (≤2%) — likely the bulk of the field — was still well offshore in April 2012 and further south, a widely dispersed cloud whose greatest density sat northwest of Hawaii around 40°N [2]. The leading edge of the ~2% class approached the coast in spring–summer 2012, but summer winds veered it south and away from British Columbia; by September 2012 much of it lay spread between California and Hawaii, and the trade winds were set to carry it west through winter 2012–13, toward Hawaii and ultimately back toward Asia [2].

The forecast for British Columbia

Extrapolating the classes forward gave the review its conclusions. Near-zero-windage debris — the lumber-heavy majority — would begin arriving on the North American coast in the first half of 2013; most of it would then move south with the California Current, some washing ashore in Washington, Oregon, and possibly California before heading toward Hawaii, while a smaller portion entered the Alaska Current — and it is from that northern load that low-windage debris would wash ashore in British Columbia and Alaska, potentially for a number of years [2]. A Washington State review (Miller and Brennan 2012) similarly spread landfall over a few years [2]. British Columbia, meanwhile, was also forecast to receive considerable amounts of the earlier, high-windage fraction, because that fleet's track ran further north [2]. The great bulk of the low-windage debris, however, would never land at all: models showed most of it remaining in the ocean for many years and collecting in the North Pacific Garbage Patch — and debris caught there was unlikely ever to reach the B.C. coast [2] [2].

At the shoreline scale, the review refused precision. The models could not identify specific B.C. beaches; instead, existing deposition patterns would govern — collector beaches that receive debris today, particularly on the west coast and around Haida Gwaii, would receive the tsunami debris too [2]. Little debris was expected inside the Strait of Georgia, protected by the seaward-flowing surface currents at the mouth of Juan de Fuca Strait and the barrier of the Gulf and San Juan Islands [2]. Most of what arrived would be small pieces rather than large objects, added to the debris that floats into Canadian waters every day; and since most pieces carry no identifying marks, the only measurable signature of the tsunami's arrival might be an increase in the weight of debris washing ashore relative to the long-term average [2]. Even the sighting record carried a warning label: the geographic distribution of reported sightings likely maps the frequency of human passage more faithfully than the distribution of the debris itself [2].

Uncertainty as the finding

The review's most quoted quality is its candour about what could not be said. The models had not been adequately tested, and their outputs — arrival rates, trajectories, landfall quantities — depended sensitively on the windage of the debris, which was poorly known: nobody could say what fraction of the field had 3–4% windage, what fraction 2%, and any "nowcast" of the debris distribution would inherit whatever distribution was guessed [2] [2]. With no satellite tracking since April 2011, reliable data on the location, pathways, and drift rates of the debris were simply lacking, and the field was presumed highly diffuse across a vast expanse of ocean [2]. For British Columbia, then, the science could name the currents, the seasons, the beaches, and the physics — and could bound the timing by windage class — but it could not honestly promise a rate or a quantity. The debris would arrive the way the review said the models showed: sorted by the wind, spread over years, and mostly indistinguishable from what was already washing ashore [2] [2].

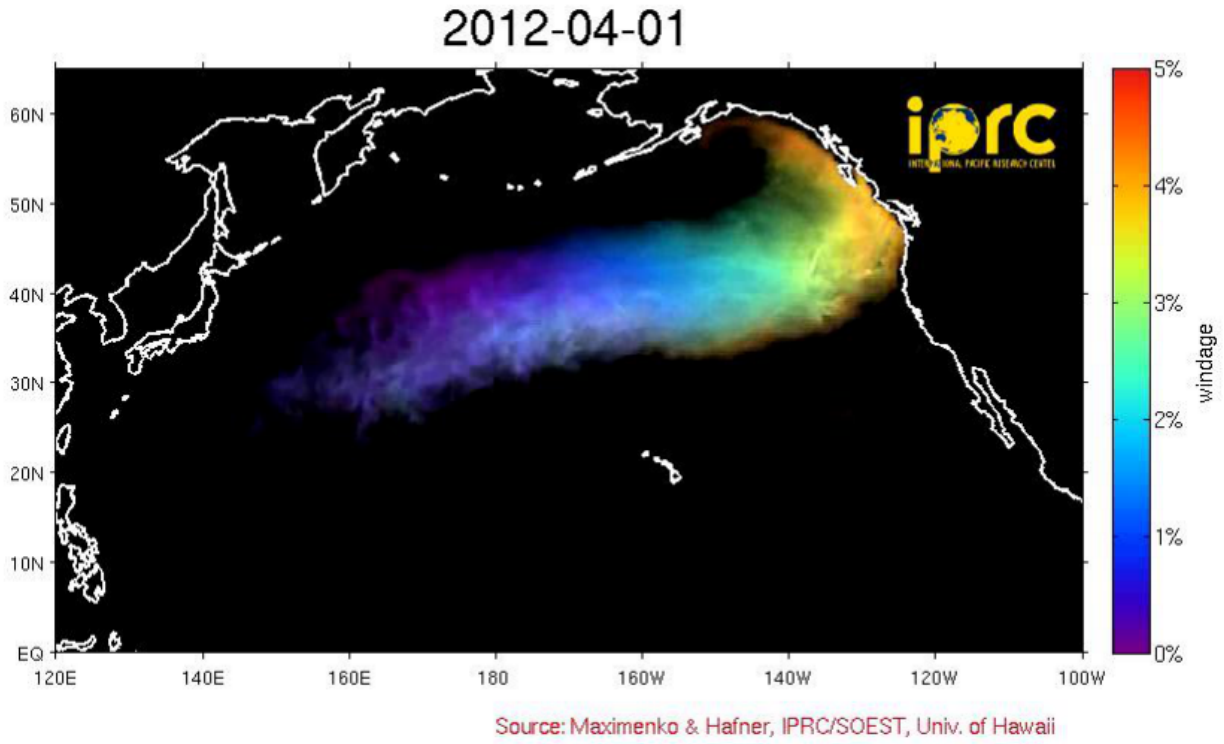


Exhibit. Figure 5. Results from model simulations of debris with varying (0-5%) windage. Source: S0012.

Model simulations of debris distribution

Every distribution map in this report after April 2011 is, in a strict sense, a drawing of something nobody could see. Once the tsunami debris cloud dispersed beyond satellite resolution, the Canadian review states the position plainly: with direct observations unavailable, "it is necessary to rely on models to estimate the present distribution of the debris and to forecast its distribution in the future," while the extent to which models can do this accurately "is an open question" [2]. The simulations built for the tsunami debris — and, in parallel, for the garbage patches of the earlier chapters — are therefore best read as instruments with known biases, not as photographs. This section examines how those instruments were constructed, what they agreed on, and exactly where they could not be trusted.

Two families of simulation

The tsunami work split into two families. The first is the climatological model: a statistical construction built from the trajectories of a large number of surface drifters deployed over a 30-year period, representing the motion of objects that travel with the top layer of the water column (Maximenko et al. 2012) [2]. Because drifters ride essentially awash, the model applies to debris with effectively zero windage — objects floating mostly below the waterline — and not to high-windage objects exposed directly to the wind [2]. Its virtue is reach: climatology can be integrated years into the future, which makes it the tool for long-term fate [2]. Long-term forecasts were produced both at the University of

Hawaii (IPRC) and by NOAA, and although the two models were differently formulated, their results were generally consistent with each other [2]. The second family works from the opposite end: simulations driven by the actual winds observed over the Pacific since March 2011, resolving debris into windage classes so that the distribution of each class can be estimated separately — the approach whose class-by-class behaviour Section 20 traced [2]. The two families answer different questions: the climatological model says where the field is going over years; the observed-wind model says how the sorting by windage explains what has been seen so far.

What the climatological runs showed

Run forward from the Japanese coast, the drifter-statistics simulation disperses the debris over a large area during its Pacific crossing [2]. Its leading edge reaches the North American continent about 2 to 2.5 years after deposition, putting the start of low-windage arrivals in the first half of 2013 [2]. Arrival is not a pulse but a ramp: after the first appearance, the rate of coastal arrival increases gradually, and debris continues to be carried close to the coast for a number of years [2]. For British Columbia specifically, the simulation places peak debris proximity to the coast at about March 2014 — three years after release — with debris remaining present along the B.C. coast for at least five years, albeit at reduced levels [2]. The long tail dominates the budget: the simulation shows most of the low-windage debris — over 90% — still in the ocean after five years, with a large fraction accumulated in the Subtropical Convergence Zone northeast of Hawaii, the North Pacific Garbage Patch, where convergence of wind-driven surface currents traps floating debris for many years [2]. The tsunami's litter, in other words, was forecast to become a minor, slow addition to the same accumulation system the drifter science of Sections 9–15 described — a connection the August 2013 congressional update of the federal debris report also drew in presenting the event to legislators [13].

The nowcast problem

If forecasting is hard, saying where the debris is on a given day proved harder. A "nowcast" of the distribution requires partitioning the debris field among windage classes — assuming how much of the field sits at 3–4%, how much at 2%, and so on — and that partition is unknown and can only be guessed, while the nowcast's output depends sensitively on the guess [2]. The observed-wind simulations are explicit about their own status: they render debris as relative concentration — pieces per unit area — and are "illustrative of the behaviour of objects with different windages," not simulations of the actual distribution in the ocean [2]. The result is a peculiar epistemic position that runs through the whole tsunami literature: the models could reproduce the pattern of arrivals convincingly while remaining unable to state the field's location, total afloat, or arrival rate at any particular shoreline.

The same machinery, pointed at the patches

The garbage-patch modelling of Section 14 used a sibling instrument, and its construction shows what calibration looks like when observations exist. There, Lagrangian particles were advected by sea-surface currents, wave-induced Stokes drift, and winds, released continuously from 1993 to 2012 from land-based sources (coastal population hotspots and major rivers) and sea-based sources (fishing, aquaculture, shipping), with forcing scenarios built up stepwise — currents alone, then added wind exposure at windage coefficients of 0%, 0.1%, 0.5%, 1%, 2%, and 3% — using HYCOM currents, NCEP/NCAR winds, and WAVEWATCH III Stokes drift [8]. Calibration against trawl measurements produced a defined object: the GPGP boundary was drawn at the 1 kg km^{-2} modelled microplastic concentration contour, enclosing 1.6 million km^2, and validated against measurement — stations inside the contour showed a median measured concentration of 1.8 kg km^{-2} against 0.3 kg km^{-2} outside [8]. Notably, the best model fit came from the currents-only scenario (R^2 = 0.52 across 277 cells), with fit declining as windage forcing was added (R^2 = 0.39 down to 0.21) [8]. Confidence intervals were formulated to carry both sampling uncertainty (vertical-mixing corrections, aerial-mosaic mass estimation) and modelling uncertainty into the final loads [8].

The two literatures then cross-checked each other. The GPGP team reported that the windage dependence of particle trajectories in their model was "in good agreement with sightings and modelling of debris originated from the 2011 Tohoku tsunami": the first identified Japanese items to reach North American shores, after 10 to 12 months, were high-windage objects — buoys, boats, and floating docks — with debris also reaching the Hawaiian Islands at 18 months [8]. A plastic-pollution model calibrated on trawls in 2015–2016 thus reproduced, from the forcing side, the arrival ordering that the tsunami simulations had predicted from the release side. That convergence — two differently built models, one event — is the strongest validation either could offer, given that the tsunami field itself could never be surveyed.

What the simulations could not do

The limits are structural, and both literatures state them. A debris simulation is only as good as its initial conditions: the source distribution along the coast, the mass released, and above all the partition of the field among behaviours — windage classes for the tsunami debris, size and type classes for patch plastic. Those initial conditions are precisely what cannot be observed once the debris is at sea [2] [2]. The models' agreement with each other and with the sparse sighting record justifies their qualitative conclusions — dispersal, sorting by windage, accumulation in the convergence zone, a multi-year arrival tail — and the Canadian review relied on exactly those. But every quantitative promise (a rate, a quantity, a date for a specific beach) inherited the uncertainty of the guesses underneath, which is why the review that used these simulations most heavily also refused, in the end, to convert them into numbers for British Columbia [2].

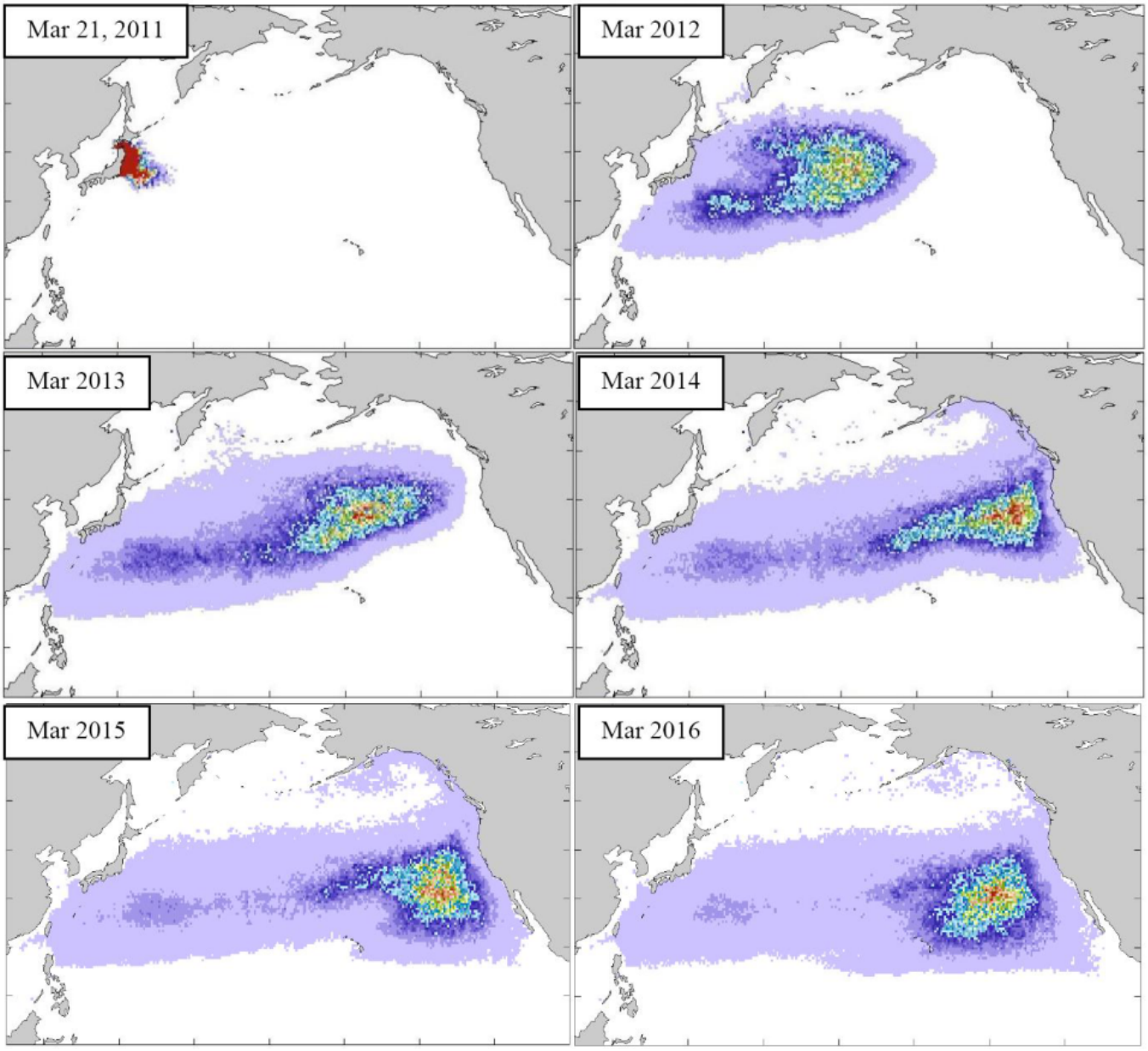


Exhibit. Modelled distribution of tsunami debris at yearly intervals, from the climatological drift model of the North Pacific. Source: S0012.

Tsunami debris arrivals and sightings by state

The sightings system described in Section 19 generated volume; confirmation generated the record. Against the thousands of reports logged by October 2012 — of which only a small fraction could ever be definitively confirmed — the federal report's appendix preserves the harder currency: a numbered catalogue of identified tsunami items and where each one was found [2] [13]. Read by jurisdiction rather than by date, the catalogue of 26 entries is the ar-

rivals map of the event's first two years — and its composition is itself a finding. Half the entries are vessels. Thirteen of the 26 are derelict vessels: found in the open North Pacific, off Alaska, and ashore in British Columbia, Washington, Oregon, Hawaii, and on Midway Atoll [13]. The rest is a miscellany of the identifiable — the objects with names, registrations, or unmistakable provenance: fishing buoys and a soccer ball and volleyball in Alaska; a steel tank and a derelict vessel in British Columbia; a blue plastic bin in both Hawaii and Washington; a soccer ball in Washington; a large yellow buoy in Hawaii; a plastic sign on Midway Atoll; and the two floating docks, in Oregon and Washington [13]. Everyday debris — the lumber, foam, and fragments that made up most of the field — is absent from the list for the reason Section 19 established: without identifying marks, it could not be confirmed, so the confirmed record systematically over-represents the large, the durable, and the labelled [13].

Alaska: the northern shore and the foam problem

Alaska's confirmed entries cluster at the top of the catalogue — the Ryou-Un Maru, listed as a derelict squid vessel found offshore Alaska, two fishing buoys, and the two sports balls [13]. The state's own debris documentation, maintained by the Marine Conservation Alliance Foundation's monitoring programme alongside public sightings recorded with NOAA's Marine Debris Program, showed a great deal of debris on Alaska's beaches and in its waters, and its composition was distinctive: a large amount of it was Styrofoam, some as buoys and the rest as insulation from buildings — a material the Alaska plan singles out as particularly bad, because it breaks apart on beaches and is ingested by animals or worked into the environment [14]. The timing also ran ahead of the current-only projections used in early planning, which had put first landfall in spring 2013: debris identified as possibly tsunami-generated made landfall on the west coast in December 2011, eight months after the tsunami, with further sightings following in Washington and British Columbia — an earliness the plan attributes to high-windage debris riding wind as well as the roughly six-mile-a-day surface current [14]. Alaska's most famous arrival ended as a disposal operation: the Ryou-Un Maru, a 164-foot shrimp vessel, was sunk by the U.S. Coast Guard in April 2012 off Alaska in 6,000 feet of water, and the Alaska plan recommends that similar floating vessels be treated the same way where environmental harm can be minimised — rather than letting them wreck on shore and impose a costly removal [14]. An independent line of evidence corroborates Alaska's foam-heavy arrivals from the other side of the basin: the GPGP census found negligible foam inside the patch but a high abundance of foam removed from Alaska during beach clean-ups, supporting the conclusion that high-windage debris does not accumulate in the gyre but is spread widely and beached instead [8].

Washington: the organised state

Washington's catalogue entries — three derelict vessels, a blue plastic bin, a soccer ball, and a floating dock — understate its role, because Washington is the state whose response the federal report documents in most detail [13] [13]. The state worked with federal agencies and Indian Tribes to develop a state response contingency plan for JTMD, with NOAA supporting a workshop to solicit input on the plan's main elements and helping to draft it [13]. The Washington JTMD team, which includes NOAA, carried the plan outward: 17 presentations at community public forums, spanning tribal meetings, workshops and conferences, community events, and agency meetings [13]. The effort had a measurable price: as of February 2013, Washington had spent over $100,000 on JTMD expenses, including outreach, a dedicated response phone line, cleanup and removal operations, and biofouling and aquatic invasive species assessment and remediation [13]. NOAA led shoreline monitoring in the state at 11 sites, and state and federal teams collaborated on removals — including the Misawa dock that came ashore within the Olympic Coast National Marine Sanctuary and Olympic National Park — while Washington also worked with experts to identify and monitor invasive species attached to debris, with NOAA sharing information and sightings of interest [13].

Oregon, Hawaii, Midway, and British Columbia

Oregon's three catalogue entries are two derelict vessels and its floating dock — the Agate Beach dock of June 2012, already told in Section 1 — and the state's experience otherwise ran through the refuge and park systems described in Section 19 [13]. Hawaii's five entries are the most varied of any state: a blue plastic bin, a large yellow buoy, and three derelict vessels, one of them found broken into pieces [13]. Midway Atoll, the event's most remote American shore, recorded a derelict vessel and a plastic sign [13]. British Columbia's confirmed list — a derelict vessel and a steel tank alongside the container found on Haida Gwaii in April 2012 — reflects the Canadian review's expectation, set out in Section 20, that the province's west coast and Haida Gwaii collector beaches would take the northern, high-windage share of the field [13] [2]. Reports, as distinct from confirmations, came from every jurisdiction the debris

touched — British Columbia, Alaska, Washington, Oregon, California, and Hawaii — a geography of reporting that the Canadian review warns maps human passage and population as much as debris [2].

What the by-state record shows

Laid side by side, the state records trace the physics of the previous two sections onto a map. The early, confirmed arrivals are concentrated in the north — Alaska and British Columbia first, in the winter and spring of 2011–12 — because the high-windage fleet that carried the identifiable objects ran the northerly track [14] [2]. What each state actually received differed by exposure and by material: Alaska's beaches accumulated the foam that the gyre rejected [8] [14]; Washington's shore, fronting the sanctuary and the national park, produced the event's most organised state apparatus and its most expensive single removals [13]. And beneath the confirmed catalogue sits the unconfirmed mass: the sightings from six jurisdictions, the everyday debris indistinguishable from tsunami debris, and the low-windage bulk whose arrival, the models said, would not peak until 2013–2014 and would register — if it registered at all — only as extra weight on the collector beaches [2]. The arrivals record is therefore best read as two documents at once: a precise list of 26 identified objects, and, behind it, the outline of everything that could not be identified at all [13].



Exhibit. Japan tsunami marine debris arrivals and sightings reported by state, from the NOAA severe marine debris event report on the 2011 Tohoku tsunami. Source: S0011.

Ghost ships among the tsunami debris

Of everything the tsunami put to sea, the vessels are what the record remembers. Half of the federal report's 26 confirmed items were derelict vessels — found in the open North Pacific, off Alaska, and ashore in British Columbia, Washington, Oregon, Hawaii, and on Midway Atoll [13]. That dominance is partly an artefact of identifiability, since a hull carries a name and a registration where a plank of lumber carries nothing [13]. But it is also physics. A vessel floats high, presents a large sail area to the wind, and so travels with the high-windage fleet — the objects the Canadian review identified as the field's fast, northerly vanguard, and simultaneously its most dangerous component, since large, potentially hazardous objects are precisely the ones with significant windage [2]. The ghost ships of the tsunami debris were therefore the first arrivals, the most visible arrivals, and the arrivals that most concentrated official minds.

The archetype

One vessel stands for the class. The Ryou-Un Maru — the 50-metre Japanese fishing vessel photographed by the Canadian Coast Guard off Haida Gwaii in March 2012, and sunk by the U.S. Coast Guard off Alaska the following month — is treated elsewhere in this report (Sections 1, 20, and 22) and is not re-told here [2] [14]. What belongs to this section is the doctrine her case produced. Alaska's cleanup plan prints her photograph under the heading "Ghost Ships" and draws the operational conclusion: similar floating vessels should be disposed of at sea in the same way,

where environmental harm can be minimised and disposal conducted safely, because the alternative is to let the vessel shipwreck on shore — causing environmental damage first and a costly removal afterward [14]. A drifting, unmanned hull is, in this framing, a wreck on a delayed fuse: the only choice responders actually get is whether it comes to rest in 6,000 feet of water or on a beach [14].

Why the vessels were the problem case

Three properties made derelict vessels the debris category that response plans had to be written around. The first is hazard to navigation: a hull floating mostly below the waterline is largely invisible to ships' radar and crews, and the Canadian review ranked such partially submerged vessels — with containers and docks — as the highest navigational risk in the field [2]. The second is what a vessel carries: Alaska's plan records reports of large objects in the water and containers with quantities of fuel on shore, and assigns the Coast Guard responsibility for safety of navigation and for oil pollution in exactly this context [14]. The third is where vessels come ashore. Alaska's chapter of the federal report notes debris densities in some areas as much as 1,000% over pre-tsunami baseline data points, confirmed by a state-funded aerial survey of the Gulf of Alaska that showed high densities of light debris — and stresses that Alaska's rugged shoreline makes detection, assessment, and removal of what arrives there unusually challenging, with volunteer engagement often limited compared to other states [13]. A 50-metre hull on a roadless Aleutian beach is not a cleanup; it is a salvage expedition.

The institutional response in Alaska took that shape. Ongoing cleanups were run by organisations including the Gulf of Alaska Keeper, the Marine Conservation Alliance Foundation, the Center for Alaskan Coastal Studies, and the Island Trails Network, supplemented by consistent shoreline monitoring from NOAA's Auke Bay Labs; NOAA also contributed $50,000 toward removal of tsunami debris in Prince William Sound [13]. Beyond the standing effort, the state built a prioritisation scheme: debris-density data from the 2012 aerial surveys, combined with impact input from land and resource managers, would geographically prioritise cleanup and mitigation actions for 2013 and then expand for long-term use — a triage system for a coastline too long to clean everywhere [13].

Ghosts with paperwork

It is worth pausing on the word "ghost." The derelicts of Sections 1–6 earned the label through absence — crews gone, owners silent, whereabouts a matter of sightings and legend. The tsunami's vessels invert the formula. They were photographed from the air and catalogued at sea — the Government of Japan's sighting record is full of small vessels, many reported capsized or partially submerged — they were tracked through a sightings database, matched against registrations, and in the Ryou-Un Maru's case assessed, engaged, and deliberately sunk by a coast guard acting on a written doctrine [2] [14]. They are ghost ships with complete paperwork: unmanned and unsteered for an ocean crossing, yet never anonymous. That is what makes them the hinge between this report's two halves. The Baychimo tradition asked what story attaches to a vessel no one can find; the tsunami fleet asked a colder question — what a state does with a vessel everyone can identify, no one owns up to, and no one can afford to let ashore [13] [14].

The unconfirmed majority hangs over the catalogue, as it does everywhere in the tsunami record. Thirteen confirmed vessels imply a larger uncounted company: the sightings that were never resolved, the hulls that capsized and sank en route, the vessels absorbed into the everyday derelict traffic of the North Pacific, where an unmarked wreck is just debris [2]. The models of Section 21 give those missing ships a statistical afterlife — the low-windage fraction settling, over years, into the convergence zone — but as individual objects they exited the record the way the Baychimo entered it: seen occasionally, reported imperfectly, and otherwise left to the ocean [2].



Exhibit. The Japanese fishing vessel Ryou-Un Maru, a 164-foot shrimp boat carried across the Pacific by the 2011 tsunami, sunk by the US Coast Guard off Alaska in April 2012. Source: S0013.

High-windage debris and large-item strike teams

The Alaska cleanup plan — prepared by the Marine Conservation Alliance Foundation (MCAF) in 2012 — is organised around the same physical sorting that governed the forecasts. Its table of contents reads like the drift models translated into operations: a first phase for high-windage debris (survey, then cleanup), a second phase, "2013 and Beyond," for current-driven debris, subdivided into small and mid-sized objects and large-item strike teams, and closing sections on ghost ships, invasive species, hazardous materials, and long-term monitoring [14]. The plan's premise is that the two fleets identified by the modellers arrive on different timetables and demand different kinds of labour.

Windage, in the plan's working definition, is the sum effect of wind on an object in the water: a high windage factor comes from a large sail area and low drag — a piece of Styrofoam, or a buoy with no lines attached — while a low factor comes from low sail area and high drag, such as a piece of wood [14]. The high-windage fleet was the early one, and the plan expected its arrivals to decrease late in 2012; with the majority of the wind coming from the south, most of that debris was expected to come to rest on Gulf of Alaska beaches [14]. The current-borne fleet would follow from the spring of 2013 onward, on the current's schedule rather than the wind's — and unlike high-windage debris, it might stay with the current, pass Alaska's shores by, circulate until it landed at a later date, or follow currents into the Bering Sea [14].

The manpower problem: small and mid-sized debris

For the current-driven mass of small and mid-sized objects, the plan is blunt about the resource required: manpower. The expected landfall items are the contents of a built environment broken small — refrigerators, commercial fishing nets, even bowling balls — scattered along the remote and often rugged beaches of the Gulf of Alaska [14]. Prioritisation would come from the 2012 aerial survey of the Gulf of Alaska combined with historic knowledge of catchment areas, identifying the coastline segments most affected [14]. The executing institution had a track record: within MCAF alone, 83 projects had cleaned over two million pounds of debris from Alaska's coast since 2003 [14]. Its experience was that successful cleanups hinge on local partners — groups expert in their own region's land, water,

protected species, weather, and safety precautions — ranging from small fishing communities in Southeast Alaska to community groups in Prince William Sound and Kodiak, and professional salvers [14].

Method followed terrain. Cleanup methods were primarily low-impact human power, as documented in a presentation at the 5th International Marine Debris Conference in March 2011; ATVs and trucks were used where possible, but most rocky-beach sites were limited to human power and hand tools — methods chosen in part because they are biologically and ecologically gentle [14]. The working year was short: generally May to mid-September, constrained by wind, sea, sea-ice conditions, and in some instances marine mammal migrations; cleanups for current-driven debris were to begin in 2013 and continue for a number of years [14]. Project selection was designed in two tracks: an impact-zone solicitation, which lets local knowledge prioritise areas of importance and keeps work near communities — more economical, and focused on places that matter to the people who depend on ocean resources — and a pre-identified-area solicitation, in which priority coastline segments are chosen from the aerial survey and past cleanup knowledge by debris concentration, environmental harm, coastline sensitivity (critical habitat, marine mammal rookeries, bird breeding grounds, proximity to endangered species), and accessibility, then divided into segments and put out for bid, with proposers required to bid on entire segments [14]. The remote segments, the plan concedes, would be less economical — but would carry the most environmental benefit [14].

Strike teams: the items people cannot carry

The large items were a different problem class entirely. When the tsunami hit Japan, entire towns washed out to sea — houses, cars, trailers, and boats among them [14]. The plan names the specimens already in view: a white transport container, of the kind attached to the back of a truck, which washed ashore at Naikoon Provincial Park in Canada, and two black tanks spotted in the Gulf of Alaska — photographed 20 miles off Hinchinbrook Island on 5 June 2012, seagulls catching a lift on them [14]. Items of that dimension and weight could not be removed by cleanup crews working off the road system — and in Alaska, the road system reaches only a small fraction of the state. Instead, salvers, helicopters, and landing craft would be needed [14].

The strike-team concept is therefore an exercise in deferred, routed logistics. Rather than chasing individual items as they landed, the plan proposed to chart where the large items came to rest and then conduct one large-scale removal effort in 2015 or 2016 — the most economically feasible method [14]. The work plan had three steps: survey to catalogue GPS coordinates; prioritise and map a collection route; then contract with salvers, landing craft, helicopters, and other needed resources [14]. The survey step could run on voluntary reporting, but was best accomplished by repeating the 2012 aerial survey in 2015/16, which would fix the large items' coordinates and provide a basis of comparison against the earlier survey [14]. The collection route would be prioritised by the environmental harm posed by each item, the sensitivity of the coastline where it lay (critical habitat, rookeries, breeding grounds, proximity to endangered species), its accessibility, and its distance from other large debris — with cars, which contain plastics and fuels and threaten marine and terrestrial species, ranked above scrap metals or wood unless contamination concerns dictated otherwise [14]. The degree of removal required would not be known until the current-driven debris had made landfall — "if it is one car or a fleet of 2,000, the removal method will be the same" was the plan's position as the strategy extended into the current-driven phase [14].

The plan's deeper logic is worth stating plainly, because it closes the tsunami chapters of this report. Every prior section treated windage as a property that predicted arrival — which fleet, which coast, which winter. The Alaska plan treats windage as a property that predicts work: what can be picked up by hand in a short subarctic season, what must wait years for a helicopter and a landing craft, and what — in the ghost-ship case of Section 23 — should never be allowed to land at all. The debris field's physics became, in the end, a labour schedule [14] [14].



Exhibit. Large-item strike teams handled the biggest tsunami debris washing ashore in Alaska, including houses, cars, trailers, boats and transport containers carried out to sea from Japanese towns. Source: S0013.

A plan for the clean-up of tsunami debris off Alaska

The document at the centre of this section is not a government plan. A Plan for the Clean Up of Tsunami-Related Marine Debris off Alaska (2012) was written by the Marine Conservation Alliance Foundation (MCAF) — the organisation that had been conducting marine debris cleanups throughout Alaska since 2003, sponsoring 83 projects that removed more than 2.1 million pounds of debris from the Alaska coastline between 2003 and 2011, alongside forums, conferences, and a public-awareness programme for fishermen, students, and the general public [14]. The foundation presented the document as "thoughts and recommendations for a strategy to systematically and effectively address the tsunami debris problem," and it was explicit about the vacancy it was filling: the United States had developed an opportunistic user reporting programme and internally discussed options, but no public plan had been announced [14]. Section 24 detailed the plan's operational phases; this section treats the plan whole — its problem statement, the Alaskan stakes, its specialist tracks, and the long-term programme meant to outlast the debris itself.

The problem as the plan states it

The plan's arithmetic is the event's arithmetic, stated for a state audience. The tsunami created more than 20 million tons of debris, of which five million tons was estimated to have been washed into the ocean; the Japanese Ministry of the Environment estimated that 70% of that amount sank, leaving 1.5 million tons — over 3 billion pounds — adrift [14]. The debris fields were dense enough to track by satellite until 14 April 2011, after which agencies turned to modelling to estimate when and where debris might go [14]. What the monitoring then documented in Alaska was a shoreline accumulation problem with a distinctive composition — a large amount of Styrofoam, as buoys and as building insulation, a material that breaks apart on beaches, is ingested by animals, and works itself into the environment [14].

The stakes were jurisdictional as much as ecological. Both the State of Alaska and the federal government held responsibilities toward the animals inhabiting the beaches, including federally listed threatened and endangered marine mammals and seabirds that the debris put at greater risk — the plan's context includes the Steller sea lion and the short-tailed albatross — and much of the affected land was owned and managed by state and federal governments themselves [14]. The federal land dimension is larger in Alaska than anywhere else in the country: the U.S. Forest Service holds jurisdiction over the two largest National Forests in the United States, both Alaskan, adjacent to 16,700 miles of salt-water shoreline that tsunami debris could impact — the 5.4-million-acre Chugach National Forest in Southcentral Alaska, the nation's second-largest, including Prince William Sound, and the Tongass National Forest, at nearly 17 million acres (80% of Southeast Alaska) the nation's largest [13]. The majority of those lands are off the road system, accessible only by boat, aircraft, or foot [13]. The Forest Service's debris role ran through its ordinary machinery: annual monitoring within ranger patrols documenting debris-impacted areas in places such as the Nellie Juan–College Fiord Wilderness Study Area and the Sitka Ranger District and Admiralty Island National Monument; a special use authorization potentially required before surveys or cleanups on National Forest lands, with permits weighing wilderness designations, heritage sites, group size, and equipment; coordination with the Alaska Department of Natural Resources, under an existing memorandum of understanding, for cleanup at or below mean high tide; and an Alaska Region representative engaged in the Biofouling Framework [13].

The programme around the cleanups

Around the phased removal operations of Section 24, the plan wraps three specialist tracks. For invasive species, an Alaska workgroup made up of the Smithsonian and federal and state agencies tracked the risk, with samples and sightings to be reported using a standard instrument — the Standard Protocol for Collection and Preservation of Non-Native Species from Marine Debris Form [14]. For hazardous materials, the plan identifies a gap and proposes the fix: the State of Alaska and the federal government need to designate one agency that will remove hazardous materials found on the shore, with designated cleanup partners reporting hazardous-material sightings for later removal and agency representatives joining ongoing cleanup teams to be on hand for removal and disposal [14]. For radiation — the concern that caught public interest — the plan reports the monitoring result directly: MCAF monitors in a spring 2012 study confirmed no indications of radioactivity on any of the beaches or objects monitored [14]. The removal machinery itself was to run on contracts: requests for proposals distributed to clean designed routes, executed by professional salvers and landing-craft operators experienced at removing large items from Alaska's shores — the plan cites an 80-foot steel commercial vessel off St. Paul Island, and supersacking debris for helicopter transport to a waiting landing craft, as at Gore Point [14].

The long term: baselines and five questions

The plan's final movement is its most distinctive. Long-term monitoring sites were established along the coast to serve as baseline data against which the change from incoming tsunami debris could be tracked [14]. With continued data collection, the plan argues, five essential questions about the tsunami debris in Alaska could be answered: (1) in the five years after the tsunami, how much Japanese tsunami debris, by weight, made landfall in Alaska; (2) what was the timeline for wind-driven and current-driven debris to make landfall; (3) what is the composition and concentration of debris per year; (4) which cleanup tactics were most effective for each wave of debris; and (5) what safety and health concerns or other challenges resulted [14]. A report should then be issued summarising the conclusions — "and hopefully provide a guide that can be used by other regions affected by debris from tsunamis in the future" [14]. The cleanup, in other words, was designed as an experiment whose results would be published: Alaska's shoreline as the instrumented test case for the next event, wherever it occurs.

The seam the plan sits in

The plan's existence is explained by a legal fact stated in the federal report. The government's response authority under CERCLA may be invoked for a release or threatened release of a hazardous substance, or of a pollutant or contaminant presenting imminent and substantial endangerment — and to the extent debris includes neither, that authority is limited; debris determined to be non-radioactive solid waste, the report concludes, may most appropriately be addressed through state solid waste management authorities [13]. Most tsunami debris is exactly that: non-radioactive solid waste, arriving on public land, in a state whose coastline defeats ordinary logistics. The MCAF plan is what fills the space between a federal reporting programme and a state waste authority — a private foundation's operating strategy, built on a decade of cleanup work, proposing routes, contracts, monitoring baselines, and a five-

year research programme for debris that no single statute had assigned to anyone [14] [13]. The next part of this report turns to the older, slower version of the same problem — vessels and debris that are nobody's legal responsibility — where the gaps the tsunami exposed turn out to be permanent features of the law [13].



Exhibit. State contingency planning and other proactive measures taken to prepare for Japan tsunami debris arriving on US shores, from the NOAA severe marine debris event report. Source: S0011.

Abandoned and Derelict Vessels: The Modern Ghost Fleet

Abandoned and derelict vessels defined

The tsunami chapters dealt with vessels that became derelicts in an afternoon, in full view of governments. The standing problem of the western states is the opposite: vessels that become derelicts one at a time, over years, in marinas and anchorages — and the first obstacle is not removal but definition. In the framework of the Blue-Ribbon Program report prepared for the Pacific States / British Columbia Oil Spill Task Force, everything turns on a legal act: "authority" means the legal ability of a governing agency to declare a vessel "abandoned" and thus remove and dispose of it [3]. A derelict vessel is therefore not simply a boat in bad condition. It is a boat whose condition, history, and ownership have crossed a threshold that a statute recognises — and the thresholds, as this section shows, are drawn differently almost everywhere. (The Blue-Ribbon report collects its own formal definitions of "abandoned" and "derelict" in its Appendix A; the working picture below is drawn from how the term operates across the jurisdictions it surveyed [3].)

Why the category matters: the harms

The definition exists because of what an unmanaged vessel does. Through deliberate action or negligence, abandoned and derelict vessels (ADVs) break up, sink, or block navigation channels [3]. They often contain harmful quantities of oil, lubricant, and other toxic substances, both in the materials used to construct the vessel and in cargo aboard; those chemicals can injure or kill marine mammals, waterfowl, and other aquatic life, and contaminate aquatic lands, shorelines, and water bodies [3]. Many of the toxics do not dissolve in water and persist for lengthy periods: persistent organic pollutants are fat-soluble and accumulate in animal fat, concentrating in top predators such as orca whales and otters [3]. A vessel that settles on the bottom keeps working damage, scouring or crushing sensitive habitats like eelgrass beds and kelp meadows [3]. The report's summary is categorical: ADVs are a growing global problem that harm aquatic health and the health of humans and wildlife, threaten commerce and navigation safety, and deplete resources communities depend on [3]. The causes of dereliction itself — ageing, neglect, maintenance costs outrunning value — were catalogued in Section 1 and are not repeated here.

Who defined the problem

The definitional work on the West Coast was done by a specific institution. The Pacific States / British Columbia Oil Spill Task Force identified ADVs as a common threat and a critical, emerging issue in 2017, and in 2018 formed the ADV Workgroup — ADV experts and programme leads from the five Task Force jurisdictions: Alaska, California, Hawaii, Oregon, and Washington [3]. The Workgroup's first task was to document the scope and scale of the problem in each state and to identify successful efforts elsewhere in the United States and Canada; in March 2018 it published its findings as The Current State of Abandoned and Derelict Vessels on the West Coast – White Paper [3]. Its second task produced the Blue-Ribbon report itself: a comprehensive "model" state programme, built from 33 recommendations, organised around five elements that any comprehensive programme must contain — Authority; Prevention; Public Outreach and Education; Removal and Deconstruction; and Funding [3] [3]. Because Canada addresses ADVs primarily at the federal rather than provincial level, the Blue-Ribbon recommendations address only the western U.S. states [3].

The White Paper's map of the category

The White Paper's conclusions amount to a survey of how the category is — and is not — governed. The problem includes both commercial and recreational vessels; the majority of ADVs are recreational, yet commercial vessels are typically larger and can cost several orders of magnitude more per vessel to remove [3]. Alongside a steady stream of newly abandoned vessels, most states face an increasing backlog of existing "legacy" ADVs [3]. In general, government policies have not been created to address the problem — the report instances the significant discrepancies between how abandoned cars and abandoned vessels are addressed [3]. In the United States there is no comprehensive federal programme; the few involved federal agencies (the Coast Guard and the Army Corps of Engineers among them) have limited roles [3]. State programmes vary widely: only Washington can be considered comprehensive, and most state programmes have insufficient funding [3]. Canada's federal Abandoned and Wrecked Vessel Act is comprehensive yet underfunded, and takes precedence over provincial programmes [3]. And no jurisdiction has a comprehensive outreach and education programme associated with ADVs [3].

Where the definitions fragment

The gaps in authority double as a map of how the definitions fail. Based on a review of many U.S. states, the report identifies five gaps in state and local authority [3]. First, a narrow focus on recreational vessels only: some agencies' statutes or programme designs exclude commercial vessels — fewer in number, but more expensive to deal with and with a larger single-point environmental impact [3]. Second, a narrow focus on reasons for removal: some agencies may remove only in emergencies or threats to human safety — criteria most ADVs will eventually meet, but which leave no flexibility for nuisance vessels that have not yet become emergencies [3]. Third, a limited geographic focus: removal authority confined to public aquatic lands leaves private aquatic landowners to rely on their state's lost-property or trespass laws, with little to no help from state or federal agencies [3]. Fourth, limited authority for local jurisdictions: where removal approvals are made at state level against state priorities, localities are effectively removed from setting priorities in their own jurisdictions — a local agency needs state approval to remove a vessel on state-owned aquatic lands even when it offers its own funding [3]. Fifth, limits on private property owners: owners who find ADVs on their property usually cannot access state funds for removal and — NOAA Marine Debris Program grants aside — lack the legal authority to remove the vessels themselves, having to appeal to the state to declare the property lost; Alaska is the exception, authorising private property owners to declare vessels abandoned or derelict and thus subject to removal [3].

Underneath the five gaps sits the threshold problem. Most Task Force jurisdictions already have sufficient authority to declare vessels abandoned or derelict — but a legal process by which an aquatic land custodian can accomplish seizure, impoundment, removal, or custody of an ADV through due process is lacking across most west coast jurisdictions [3]. The custodian's authority must be broad enough for a wide and partly unforeseeable variety of situations, yet specific enough to prevent the taking of property without due process — exacting in procedure, flexible in application [3]. At the federal level, five agencies hold ADV authorities — NOAA, the Army Corps of Engineers, the Coast Guard, EPA, and FEMA — each governing specific situations, which limits their effectiveness; as the Point Estero case (Section 6) showed, federal responsibility frequently ends once contamination is removed [3].

The threshold in practice: at-risk versus derelict

One state's programme design shows the threshold as a line with programmes built on both sides of it. Florida's Vessel Turn-In Program accepts only "at-risk" vessels — those whose titled owners have received a citation or written warning under section 327.4107, Florida Statutes — and states the boundary explicitly: "Derelict vessels are not eligible for this program" [15]. An applicant's vessel must be floating on state waters, in at-risk condition, with clear title and no liens, and must not be in a derelict condition or under a derelict-vessel investigation at any point up to removal [15]. The programme's premise is that "at-risk" and "derelict" are successive legal states of the same hull, and that intervening on the near side of the line is cheaper and cleaner than acting on the far side — a premise Section 30 examines in full [15].

A definition, then, is not a description but a gate. Whether a decaying vessel can be seized, by whom, from which waters, at whose expense, and with what process owed to its owner — every operational question in the sections that follow is downstream of how "abandoned" and "derelict" are defined and declared. The next section takes up the gate's weakest point: the missing legal process itself [3] [3].



Exhibit. The F/V Western, an abandoned former crabbing vessel in Coos Bay, Oregon, photographed for the Blue-Ribbon Program report on abandoned and derelict vessels. Source: S0014.

Gaps in authority: no legal process for seizing and removing derelicts

Section 26 ended on a paradox: most west coast jurisdictions can declare a vessel abandoned or derelict, yet lack a legal process by which a custodian can seize, impound, remove, or take custody of it through due process. This section opens that paradox up. The gap is not a single missing statute but a pattern repeated at every level of government: authority over derelict vessels exists in fragments — fenced by geography, by purpose, by vessel type, and by funding — and the fragments do not join up into a conveyor that carries a hull from declaration to disposal. The Blue-Ribbon report's authority chapter is the autopsy, and its recommendations are the proposed repair.

The federal patchwork, agency by agency

Five federal agencies hold ADV authorities, and the report's summary table shows why none of them amounts to a removal service [3]. NOAA's primary geographic focus for addressing ADVs is within National Marine Sanctuaries; beyond them, it supports other marine environments through Marine Debris Program grants for private and public lands — but since ADVs are only one kind of marine debris, the grant programme "cannot be considered a true ADV-removal program" [3]. The Army Corps of Engineers addresses ADVs only in federally recognised navigation channels, and only where the vessel impacts the maintenance or navigation of the channel; it is authorised to remove the vessel but has no funding to do so [3]. The Coast Guard (on the coast) and EPA (inland) act on the pollution threat: where a vessel presents an environmental or public-health threat they remove the pollution — petroleum removal is paid from the Oil Spill Liability Trust Fund — but the vessel itself would have to be removed from the agency's own budget, and neither agency has dedicated ADV removal funding [3]. FEMA's ADV authorities arise from the Stafford Disaster Relief and Emergency Assistance Act, and it can fund removals only under declared emergencies [3]. Read together, the table describes a system in which a derelict can be defuelled by one agency, ignored by another for want of a channel, and left in place by all of them — legally, at every step. (The dereliction causes the report lists extend Section 1's catalogue in one respect worth noting: alongside ageing, neglect, and costs outrunning value, it counts federal and state fishery-reduction and disaster-relief programmes that render a vessel's original purpose obsolete [3].)

The state-level procedural gap

Below the federal layer, the missing process is the core defect. The report insists that the procedure by which state agencies gain the legal right to remove, deconstruct, sell, or use a vessel should be spelled out in state statute "in straightforward language comprehensible to any agency staff member" — in most places, it is not [3]. A second defect follows from scarcity: state agencies prioritise removals because resources are limited, but that prioritisation

"can have the unintended consequence of limiting the ability of a local entity (city, county or private property owner) to act" [3]. A vessel too low on the state's list is not merely delayed; in many jurisdictions it is frozen, because nobody else holds the procedural key.

The repair: strict in process, unrestricted in capability

The report's first authority recommendation is architectural: ensure the aquatic land custodian has authority to remove a hazard, nuisance, or threat while protecting the owner's rights and due process — "the best structure for an ADV removal program would be strict in process and unrestricted in capability," with legal authority free of gaps in jurisdiction, clear in process, and able to adapt to changing circumstances [3]. Programmes should not constrain agencies' ability to enact and enforce ordinances or to take any action authorised by federal or state law [3]. Local authorities should have a voice in removal prioritisation and, when appropriate, authority to seize, impound, remove, or gain custody of recreational or commercial vessels; and authority should be "unburdened with undue governmental bureaucracy," so that a city or a private owner can initiate a removal on local need rather than competing with other localities for state or federal approval [3].

Washington State supplies the working model — and the residual flaw. Its Derelict Vessel Removal Program prioritises by threats to human safety and the environment, as limited resources require; but Washington's statutes allow any authorised public entity to remove vessels within its jurisdiction, so local entities such as ports can remove a vessel that is a low priority for the state [3]. Yet Washington also illustrates the private-owner limit from Section 26: counties typically hold ADV removal authority on private property, but if the county refuses to exercise it, the property owner cannot use the state programme to remove the vessel [3]. The conveyor exists, but one of its gates can still be shut by inaction.

Condition, not origin: the 41-foot counterexample

The report's fifth authority recommendation attacks the recreational/commercial fence directly: the agency with removal authority should be able to remove any vessel, with decisions based on the vessel's current condition and situation — not its as-built intended use or ownership status [3]. The case study is pointed. Washington DNR's Derelict Vessel Removal Program posted notice on, and took into custody, a former U.S. Coast Guard 41-foot utility vessel that was legally owned by the U.S. Bureau of Indian Affairs; after taking custody, the programme sold the vessel and deposited the funds into the Derelict Vessel Removal Account [3]. The reasoning is stated flatly: the ex-Coast Guard vessel posed the same threat to human safety and the environment as a similarly sized recreational vessel; the DVRP had jurisdiction over State Owned Aquatic Lands, the vessel met the definition of abandoned, and state statutes therefore gave it authority to remove and sell, provided legal notice requirements were followed — the vessel's as-built intended use "was not a consideration" [3]. The underlying population makes the principle necessary: most commercial vessels are eventually sold for private recreational purposes, or drop off the registration rolls once maintenance and seaworthiness become cost-prohibitive — yet most states still regard them as commercial [3]. "All vessels deteriorate and eventually pose the same risk to human safety and the environment, so their current condition and situation should be the only removal priority criterion" [3].

Disposal, private owners, and enforcement

Three further recommendations close the procedural chain. On disposal, an agency should be able to sell, deconstruct, recycle, or use a removed vessel in whatever way provides the best public benefit — but statutes should not encourage or prioritise sale regardless of condition, precisely to discourage the same vessel being abandoned repeatedly under multiple owners; proceeds typically reimburse the removing agency, with any remainder going into a fund for future removals, and deconstruction should be as environmentally friendly as possible with costs in mind [3]. On private property, owners should be empowered to declare vessels abandoned or derelict and subject to removal, because ADV damage does not stay on the parcel: contaminants migrate by ocean currents and sediment movement and through the food chain, so early removal by an empowered owner prevents larger environmental impacts [3]. And on enforcement, ticketing authority for vessel registration and other aquatic laws should be extended to all appropriate state agency personnel [3].

The picture that emerges is of a problem misdiagnosed as scarcity of power. Power exists — five federal agencies, fifty state codes, counties, ports. What is missing is the choreography: a single, due-process-clean sequence that

runs from declaration through seizure, custody, and disposal, exercisable at the level of government closest to the vessel. Section 28 turns to the document that assembles that sequence into a model programme [3] [3].

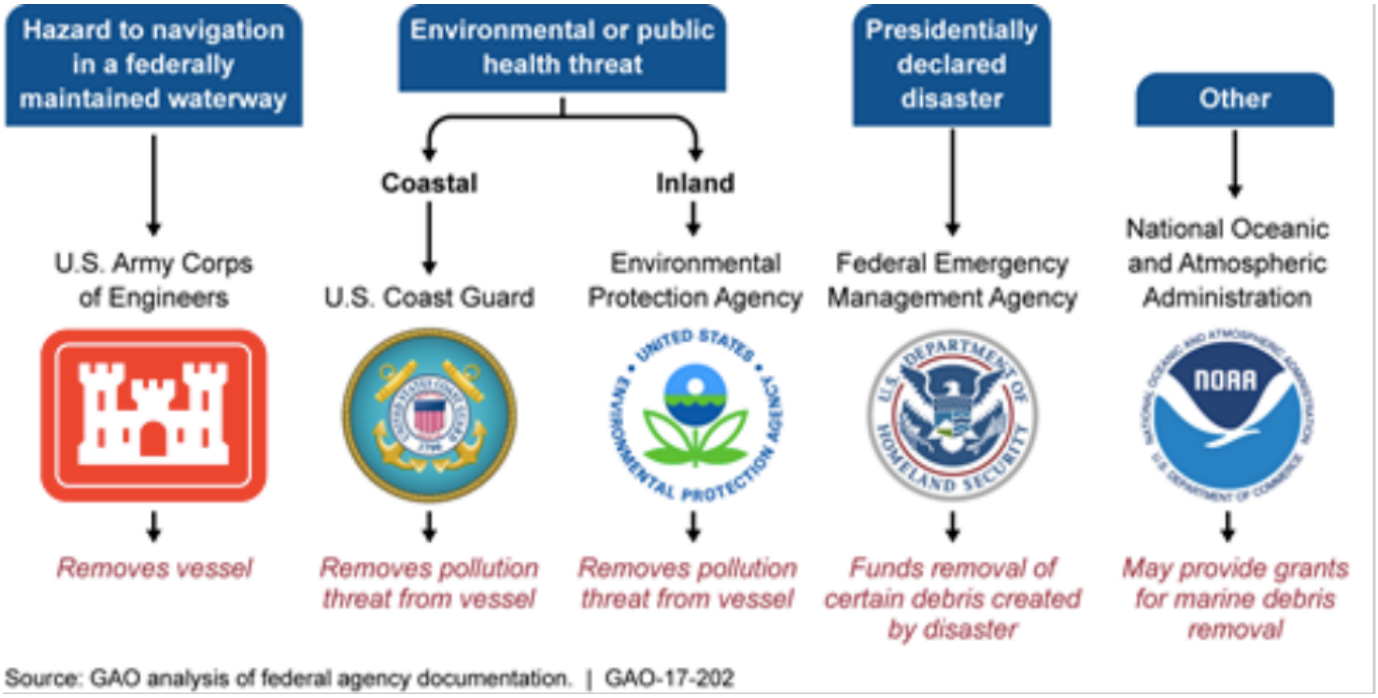


Exhibit. Figure 1. U.S. Federal Agency ADV Authorities State and Local Authorities. Source: S0014.

The Blue-Ribbon Program for the western states (AK, CA, HI, OR, WA)

Sections 26 and 27 worked through the Blue-Ribbon report's diagnosis — what an abandoned or derelict vessel is, and why the legal machinery for dealing with one is full of holes. This section steps back to the document itself: how the programme was assembled, what its architecture is, and the definitions on which the whole edifice rests — definitions the report's evidence has so far pointed to without quoting.

Genesis: from scope survey to model programme

The Workgroup's initial task, when the Task Force formed it in 2018 from ADV experts and programme leads across the five jurisdictions, was empirical: "to document the scope and scale of the problem of ADVs across each jurisdiction" [3]. The March 2018 White Paper summarised that survey, and its conclusions — no comprehensive policies in general, no comprehensive U.S. federal programme, state programmes varying widely with only Washington comprehensive and none sufficiently funded, Canada's federal act comprehensive but underfunded — set the Workgroup's next task: "to develop a comprehensive, blue-ribbon or 'model' state/provincial-level program to address ADVs" [3]. The resulting paper "consists primarily of recommendations that may be implemented by states to address their own ADV issues," and the Workgroup's claim for it is strong: adoption of each recommended element "would result in a comprehensive ADV program that would successfully remove legacy ADVs and prevent and remove future ADVs" — though implementation "will vary by state due to differences in existing programs and extent of the problem" [3]. One scope limit is underlined in the document itself: because ADVs are addressed primarily at the federal level in Canada, the blue-ribbon paper includes only recommendations for western U.S. states [3].

Architecture: five elements, thirty-three recommendations

The programme is organised around the White Paper's five necessary elements of a comprehensive state programme — authority; prevention; public outreach and education; removal and deconstruction; and funding [3]. For each element, the paper follows the same three-part structure: a high-level overview of the topic; a summary of the gaps and issues, drawn primarily from the White Paper's findings; and a list of recommendations for states [3]. A total of 33 recommendations are included in the report, with a summary in Appendix B [3]. The architecture extends beyond the states in one direction: "to address the numerous gaps identified in the 'authorities' section, this paper also

includes a list of recommendations for the Task Force's federal partners" [3] — Section IX, recognising that the federal patchwork dissected in Section 27 cannot be repaired by state statute alone [3].

The definitions, at last

Section 26 noted that the report's formal definitions live in its Appendix A; the appendix text is worth setting out, because the programme's whole logic turns on a distinction it draws in two sentences. **Abandoned** is "the legal condition of the vessel, in terms of ownership"; **derelict** is "the physical condition of the vessel, in terms of seaworthiness" [3]. One word describes a legal state, the other a material one — which is why a vessel can be either without being both, and why removal law has to bridge the two. The supporting definitions cash the distinction out: **seaworthy** means the vessel and its equipment are "physically fit and in full working order, able to encounter and withstand the ordinary perils of the sea during its contemplated use, and suitable for its intended purpose" [3]. The stage on which custodians act is **aquatic lands** — "all tidelands, shorelands, harbor areas, and the beds of navigable waters, including lands owned by the state and lands owned by other public or private entities" — and the **aquatic land custodian** is the owner or lessee of that land, "either fresh or salt water" [3]. Vessel categories follow the Coast Guard: a **commercial vessel** is any vessel "engaged in commercial trade or that carries passengers for hire," while a **recreational vessel** is non-commercial watercraft used or capable of being used as transportation on the water, other than a seaplane — and not inner tubes, air mattresses, or the small rafts and flotation toys used by swimmers [3]. Two financial-legal definitions complete the working vocabulary: **financial responsibility** is the demonstrated ability of a responsible party to pay spill costs and damages up to a specified amount, evidenced by insurance, P&I club documents, surety bonds, guarantees, letters of credit, or qualification for self-insurance; and **secondary liability** is the responsibility that arises when the party directly liable fails to perform a duty [3]. Even the fuel gets a definition — **Bunker C**, the residual oil left after the lighter products are distilled out of crude [3].

One tension inside the appendix deserves note. The report defines **commercial/recreational** as "the original intent of the vessel usage as built (e.g. an old tug converted to a houseboat would be a commercial vessel)" [3] — classification by as-built intent. Yet Section 27's fifth authority recommendation insists removal decisions should ignore exactly that, resting on current condition instead [3]. The definitions classify; the recommendations prioritise. The programme lives with both: what a vessel is follows its birth, but what is done about it follows its state.

The weakest element, and the road after publication

Of the five elements, public outreach and education is the one where the White Paper found the least to build on. Its key findings: none of the ADV programmes within the Task Force jurisdictions has a comprehensive public outreach and education component; jurisdictions currently rely on websites, social media, printed materials, and speaking engagements; most programmes do not conduct the target-audience research needed for effective campaigns; most rely heavily on partnerships with local municipalities and organisations for public engagement (the Alaska Marine Safety Education Association is the example given); and NOAA's Marine Debris Program is "a go-to resource for funding ADV outreach and education efforts" as well as for strategy [3]. Stand-alone efforts exist in support of particular programme aspects, such as vessel turn-in programmes, but nothing integrated [3]. (The prevention element, treated in Section 29, reaches further into daily practice than its name suggests — far enough to cover the lease terms marinas operate under and even the states' own habit of auctioning off their surplus vessels.)

The report closes with an implementation agenda rather than a peroration. Once the programme is finalised, the Task Force will engage its partners: meeting individual federal agencies to discuss the federal recommendations and next steps; hosting a round-table of ADV experts to map a path forward; developing a cohesive outreach and education programme that each jurisdiction can tailor — possibly including a suite of "'Spills 911'-type outreach materials" for recreational vessel owners, distributed through existing networks such as Dockwalkers, plus materials for state legislators and others in key positions; and reporting on implementation and results [3]. A model programme, in other words, that knows it is only a model: five elements, thirty-three recommendations, and a definitions appendix that quietly carries the conceptual load [3] [3].



Exhibit. January 14, 2020 This blue-ribbon program is the product of the Pacific States/British Columbia Oil Spill Task Force ADV (ADV) Work Group. Source: S0014.

Prevention through registration, fees and enforcement

Every preceding section has been about paying for the past — the legacy backlog, the removal, the deconstruction. Prevention is the element that stops the account from refilling. The White Paper identified three prevention gaps on the west coast: a lack of vessel registration requirements for both recreational and commercial vessels; a lack of insurance requirements, especially for wreck removal; and a lack of vessel turn-in programmes for vessels in serious disrepair [3]. The Blue-Ribbon programme's prevention chapter answers each in turn — registration first, because registration is what turns an anonymous hull into a known liability with an owner, a fee stream, and a paper trail.

Registration: the oil-and-tyres model

The first prevention recommendation is to establish a vessel registration system, including fees and adequate enforcement, for both recreational and commercial vessels [3]. The report's template is mundane and deliberate: an effective system "would be similar to that currently used for oil and tire disposal" — vessels, like motor vehicles, could be required to register and renew annually, with fees collected from both sectors every year [3]. Washington has the most established programme for registration revenue collection, though the report faults its commercial criteria: Washington does not capture large container ships that make one port call per year, focusing instead on commercial vessels that spend more time in the state — criteria the report says should be expanded [3]. In Washington law the machinery sits in identifiable statutes: boat registration under Chapter 88.02 RCW, derelict vessels under RCW 79.100, and the watercraft excise tax under RCW 82.49 [16]. And registration does double duty: beyond identifying owners, it "provides an opportunity to collect a fee to fund an ADV program" [3].

The fee numbers, from the report's funding chapter, show how small the unit charges are and how they add up. Washington collects about $800,000 annually for ADV purposes from vessel registration — $3 per year from each of roughly 250,000 recreational vessels, and $1 per linear foot per year from about 1,700 commercial vessels [3]. California takes a different route to the same end, collecting ADV funds from a marine fuel surcharge — "apportioning payment based on vessel use, not vessel ownership" — which raises about $2 million per year and helps fund California's vessel turn-in programme; a per-gallon fee on marine diesel would extend the logic into a "pay to play" system in which larger, more active boats pay more than smaller or less-used ones [3]. Other states fall back on annual appropriations of general funds or on clean-water programme funds [3]. The report's funding recommendations add the plumbing: a quick, flexible mechanism to move state money to local agencies — upfront where needed, never requiring local agencies to front removal costs and seek reimbursement later — and authority for agencies to recover removal costs from financially viable responsible parties, "the current, and potentially past, vessel owners" [3].

The database: knowing which hulls are sinking

Registration names the fleet; the second recommendation watches it. States should establish a comprehensive database to track — and potentially prioritise — ADVs, and the report calls identifying vessels of concern and maintaining that database "one of the most important aspects of prevention" [3] [3]. A robust tracking system records location and condition assessment and supports prioritisation for disposal decisions; ranking vessels for state-funded removal stretches limited resources, and vessels should be prioritised by risk, impact, and ease of removal — to the point that removing several lower-risk vessels for the cost of one mid- or higher-risk vessel "is not unreasonable," since it achieves the largest reduction in overall threat [3]. But the prioritisation warning from the authority chapter applies with full force: ranking can block local authorities when a vessel does not appear on the state-wide priority list, so local entities must be involved in the ranking process [3]. Reporting should be seamless from the field — a free mobile app or a hotline, with a consistent set of questions for the initial reporter — while never encouraging the public to board or enter a vessel, and with the hosting agency alert to the fact that a vessel may look abandoned to a reporter while its owner disagrees: "tracking ADVs is a related but separate subject from vessel ownership and seizure" [3]. Marinas and moorage owners feed the system, collecting and maintaining vessel and owner information for annual submittal to the state titling and registration authority; a reporting form can ride along with the annual rent and insurance certificate submitted to the leasing agency [3]. The economic engine underneath is a multiplier the report states plainly: removal costs are often three times higher for sunken vessels than for floating ones, so addressing vessels before they sink cuts removal cost, pollution, liability, and navigation risk together — and a maintained database "will assist agencies in preparing/planning for potential cleanups," a coordinated effort among local, state, and federal agencies, marinas, and ports [3].

Insurance: wreck removal above the value of the vessel

The third recommendation closes the gap the White Paper flagged most sharply: require wreck removal insurance above the value of the vessel, for both recreational and commercial vessels [3]. The logic is the dereliction arithmetic of Section 1 — when removal costs more than the boat is worth, an uninsured owner walks away and the public pays; insurance priced to the removal liability, not the hull value, keeps that cost with the owner. Working practice exists at the leasing level: Washington DNR's Northwest Region verifies protection-and-indemnity coverage with pointed questions — is the policy current, does it provide wreck removal, has the agency seen incidents where similar policies failed to pay — requires insurance certificates naming the state as additionally insured, and accepts performance bonds in lieu of insurance from lessees, drawable if the lessee fails to remove a derelict or abandoned vessel within the specified timeframe [3]. Alongside insurance sit the lease terms themselves: the report's ninth and tenth prevention recommendations would shorten lease periods on state-owned lands and waters — Oregon's 15-year leases make compliance checks difficult, and the state has had to mandate site visits at least once every three years; Washington DNR has moved to shorter terms, with five years the Workgroup's suggested figure — and would restrict state agencies from auctioning or surplussing their own dilapidated vessels into the problem, as California's OSPR was once required to do, with Washington's RCW 47.01.470 restrictions as the counter-model [3].

Enforcement, and who ultimately pays

Registration without enforcement is a mailing list. The authority chapter already recommended extending ticketing authority to state agency personnel; the funding chapter pays for it — a mechanism to fund or compensate local law enforcement for the time and equipment needed to enforce vessel registration and aquatic laws and to issue civil penalties, in recognition that local enforcement capacity is limited, plus state grants to local law enforcement agencies for ADV-cleanup time and expenses [3] [3]. The deepest recommendation reaches past owners altogether: states should develop programmes requiring "the companies that built, and make profit, from commercial vessels" to finance some or all of the vessels' future disposal — because the common national story is of commercial vessels built by profitable, financially capable companies and sold on to progressively smaller, less capable owners, until formally commercial vessels sit with individuals or small businesses "ill-suited to deal with the liability and cost of deconstruction and disposal"; such a programme fits best at the federal level, with reciprocal agreements among adjacent states as the fallback [3]. The case study is the Black Kite: in 2014 and 2015 the Coast Guard spent $90,000 responding to the Black Kite and Black Hawk, two 122-foot tugs found derelict and drifting in San Francisco Bay and then sinking at the dock; the tugs reverted to the City of Richmond, which had the U.S. Marshals auction them, and the Black Kite sold for $1 to a local artist — within months, California and the Coast Guard were responding to an oil-spill threat from the vessel again [3]. A one-dollar tug is the registration system's counterfactual: no fee, no insurance, no

database entry — and the public paying, twice, for a hull that changed hands for less than the cost of its own paperwork [3] [3].

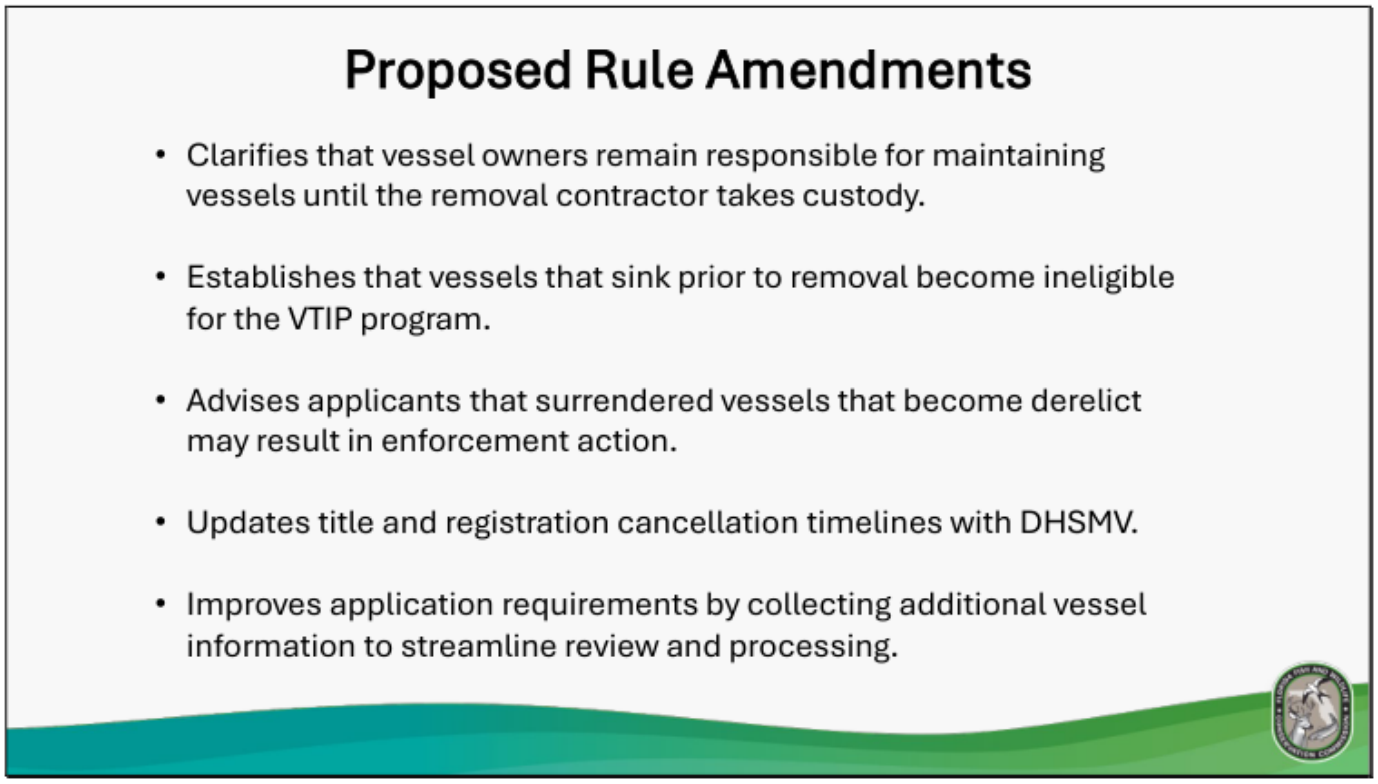


Exhibit. Florida Vessel Turn-In Program results showing continued demand for the program and successful completion of approved vessel removals. Source: S0016.

Vessel turn-in programs: Florida's VTIP and Washington's DVRP

A turn-in programme is prevention with a cash register: the state pays a small, certain sum now to remove a vessel that would otherwise cost a large, uncertain sum later. The Blue-Ribbon report's eighth prevention recommendation is simply "establish a vessel turn-in program," and its summary table names two working examples — Washington State DNR and California State Parks' Division of Boating and Waterways [3]. The funding chapter supplies the rationale: ongoing funding needs "could be minimized with effective vessel turn-in programs and other preventative measures," against a cost gradient that runs from as little as $100 to remove a small floating recreational boat that fits on a trailer, through an average of about $3,000 for a recreational vessel, to well over $1 million for a large sunken commercial vessel [3]. California's inventory of the Sacramento–San Joaquin Delta prices the alternative: $30 million to remove 52 known commercial ADVs, with new commercial abandonments arriving at one to two per year — roughly $4 million needed annually, indefinitely [3]. Turn-in programmes exist to intercept vessels at the cheap end of that gradient. Florida's VTIP shows how the interception is engineered; Washington's DVRP shows the custody machinery it plugs into. (The evidence is asymmetric, and worth flagging: the Florida programme is documented here in operational detail, while Washington's turn-in role appears through its exemplar listing and its surrounding statutes.)

Florida: the gate is the programme

The Florida Fish and Wildlife Conservation Commission created the Vessel Turn-In Program under section 327.4107(6), Florida Statutes, as one component of a broader Derelict Vessel Prevention Program enabled by the same legislation; VTIP began in July 2022 [15]. Its premise is stated as a bargain: an at-risk vessel "may be voluntarily turned in for removal and destruction without further penalty" [15]. The programme prevents vessels from becoming derelict by removing them while they are merely at risk, producing cost savings for taxpayers and fewer derelict vessels on Florida waters; owners who take the offer avoid additional fines for future at-risk violations, the expense of bringing the vessel into compliance, criminal charges if the vessel becomes derelict, and thousands of dollars in removal-cost reimbursement should the state have to remove it as a derelict or public nuisance [15]. Participation is

voluntary and free to the owner — FWC pays the removal, destruction, and disposal fees to the contractor from derelict vessel prevention funds, on completion and submission of all documents [15].

Eligibility is the design. From Section 26's threshold — a vessel must be at risk, not derelict — the guidelines build a full gate: the participant must hold clear title in their own name, verified by government-issued photo ID; the vessel must be floating on state waters; it must be in an at-risk condition, meaning the owner has received a citation or written warning from law enforcement at least once for a violation of section 327.4107; it must have no lienholders on file with the Department of Highway Safety and Motor Vehicles; and it must not be in a derelict condition nor the subject of an ongoing derelict vessel investigation — "Derelict vessels are not eligible for this program" [15]. All vessels must remain eligible until they are removed and destroyed, and where a vessel has multiple owners, all must consent and sign the waiver and release that protects the State and the Commission [15].

The process runs in a deliberate sequence. The owner submits an application with photographs of the vessel, copies of the title and government ID, a waiver of property interest, and copies of any at-risk citations or warnings; programme staff review and prequalify the removal, then send instructions for junking the vessel's title; once staff receive the completed application and the junked title, they hire a contractor to remove, destroy, and dispose of the vessel — with staff assisting owners through the application and the title-junking throughout [15]. The junked title is the programme's point of no return: ownership is extinguished on paper before the hull is touched, so no removed vessel can drift back into the fleet under a new owner.

The programme's own numbers, reported to the Commission in August 2026, show steady demand: since VTIP began, 501 applications have been received and 295 vessels approved for removal; 261 vessels have been removed to date; and in the 2026 contract year alone, 90 applications had been received, with 29 under review and removals ongoing [15].

The edge case: sinking between approval and removal

VTIP's failure mode is temporal. The Commission's proposed amendments to the programme rule (Florida Administrative Code Rule 68D-15.003) exist because of a specific experience: "vessels have sunk after being approved for the program but before removal could occur, creating additional costs and operational challenges" [15]. The amendments clarify that owners remain responsible for maintaining their vessels until the removal contractor takes custody; establish that a vessel that sinks prior to removal becomes ineligible — a vessel that becomes derelict or sinks before removal "is no longer eligible for the program, even if the applicant has been approved for participation"; warn applicants that a surrendered vessel that becomes derelict may result in enforcement action, including derelict-vessel charges; update title and registration cancellation timelines with DHSMV; and collect additional vessel information (type, colour) to streamline review [15]. Staff recommended the Commission approve the amendments for advertising and adoption, to be filed under section 120.54(3) without further public hearing unless one is requested [15]. The lesson is built into the gate itself: a turn-in programme is a race against buoyancy, and its rules have to price the interval between yes and gone.

Washington: turn-in inside a custody programme

Washington's Derelict Vessel Removal Program — the report's exemplar for broad authority, for any-vessel removal, and, with California's DBW, for turn-in [3] [3] — approaches the same interception from the custody side: its notice-and-custody machinery, seen in Section 27 taking a former Coast Guard vessel into state hands, is what a surrendered vessel enters voluntarily rather than by enforcement. The prevention lattice around it reinforces the handoff: the summary table pairs turn-in (Recommendation 8) with a disposal bond required for commercial vessels at initial construction and registration (Recommendation 6), and with secondary-liability laws plus a required seaworthiness survey for larger and older vessels before sale (Recommendation 7 — Washington statute, RCW 79.100.040) [3]. The registration base beneath it is large and motorised-only: Washington requires registration of most motorised boats through the Department of Licensing (government boats and some under-16-foot motorised boats are exempt; most human-powered boats need none), and owners registered an average of 230,000 boats per year from fiscal 2018 through 2023 [16].

The two programmes are the same idea at different depths of the pipeline. Florida built a narrow, brightly lit gate at the last cheap moment — floating, titled, cited, not yet derelict — and pays contractors to walk vessels through it before gravity and water do their more expensive work [15] [15]. Washington built the receiving institution: custody, ac-

counts, and statutes into which a turned-in vessel can be absorbed [3] [16]. Both answer the Delta arithmetic the same way — a vessel surrendered is a vessel that never has to be salvaged [3].

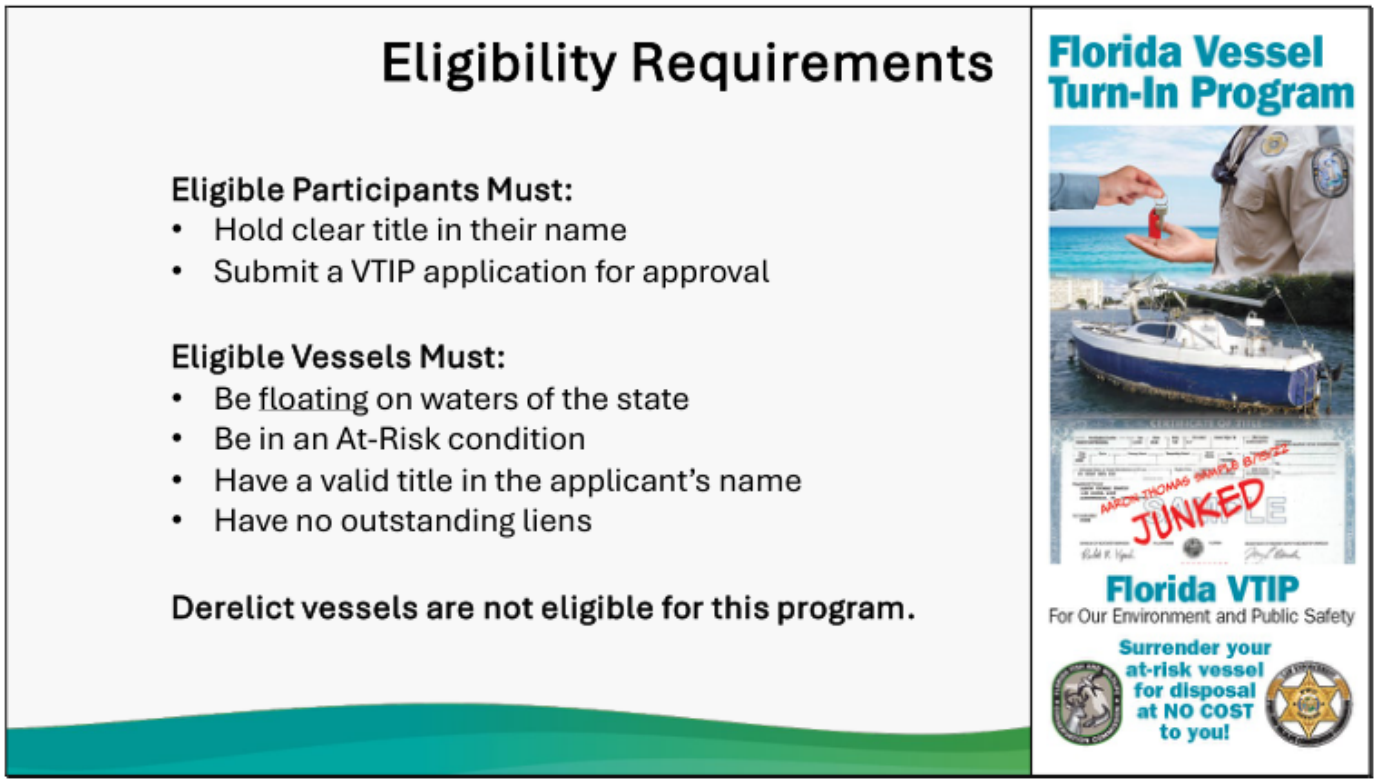


Exhibit. Eligibility requirements for the Florida Vessel Turn-In Program, under which owners with clear title can surrender an unwanted vessel and junk its title with staff assistance. Source: S0016.

Removal, deconstruction and vessel recycling

Authority declares, prevention intercepts — but at the end of every ADV story there is a physical act: a hull has to be lifted, towed, cut apart, and disposed of. Section VI of the Blue-Ribbon report, Removal and Deconstruction, is where the programme stops being law and becomes logistics. Its seven recommendations are unglamorous — contractor lists, liability cover, permitted yards, storage sites, landfill negotiations — and they are the difference between a removal programme on paper and one that can actually take a vessel out of the water [3] [3].

The contractor problem

The first recommendation is procurement reform: publish and maintain a list of qualified — licensed, bonded, insured — vessel removal and recycling contractors, and establish a statewide price agreement; Washington already has such a list, which could serve as a model, and the summary table credits Washington and California both [3] [3]. The problem it solves is procedural decay. State agencies may want to contract directly, but procurement rule limits often impede them, forcing a formal Request for Proposal process "which is labor-intensive and often yields no bids and meanwhile worsening condition for the vessel of concern" [3]. Flat-rate contracts through each state's head contracting agency, with pre-approved and vetted contractors, would save time and attract a larger pool of candidates; central vetting expands both the contractor list and the range of projects contractors may support; and a streamlined direct approach "would especially help in emergency situations or with smaller vessel removal" [3]. California's emergency contracting procedures are offered as a second model other west coast jurisdictions could copy [3].

Two further recommendations de-risk the work itself. States should ensure that responding agencies involved in cleanup operations have adequate liability protection [3]. And they should encourage development of temporary permitted facilities for vessel deconstruction, including large commercial vessels — because long-term permitted facilities prevent delays in procuring approval for shipbreaking at non-permitted or ad hoc locations, and provide better environmental protection through hazardous materials management [3].

Passing the contractor: the federal hand-off

The fourth recommendation is the operational answer to a gap Section 27 documented — the federal role that ends when the pollution does. States should coordinate with the Coast Guard to establish a practice of "'passing' contractors from USCG employment during the pollution control phase of a response, to state or local control for the vessel salvage and deconstruction phase" [3]. The economics are mobilisation costs: the equipment already deployed with federal funding during pollution control stays on site instead of demobilising and being re-hired, reducing what state and local agencies pay for the deconstruction phase [3]. The practice has already been done for small vessels in some instances; the recommendation is to expand it [3]. Where Section 27's table showed the Coast Guard defuelling a vessel and departing, this is the choreography that keeps the crane on the dock.

Batching, stockpiling, and the landfill constraint

Removal is cheapest in bulk. The fifth recommendation: plan targeted local ADV removal events to leverage the cost of a single mobilisation of contractors, gathering abandoned vessels from a small defined area for bulk deconstruction and disposal [3]. The sixth establishes the buffer between gathering and destruction — safe and secure shoreside vessel storage, with identified places for vessels to be stockpiled for bulk deconstruction and disposal [3]. Stockpiling is also a freight strategy: it lets shipments of oil waste to landfill be timed with local shipping fluctuations "to take advantage of empty return barges or trucks, which are often available at lower rates" [3].

The disposal end of the chain is constrained by law and geography. With the restriction of EPA's Ocean Dumping Permit requirements, ADVs are most often deconstructed and disposed of in landfills — but the nearest landfill is often too small to accept the volume of waste from vessel deconstruction, even for relatively small vessels [3]. The constraint bites hardest in Alaska, where communities are not always connected by roadways, making shipment of solid waste extremely expensive; the report suggests local governments negotiate a reduced rate at the community landfill — especially where it is government owned and operated — for deconstruction and disposal of vessels before they sink [3]. Security matters too: stockpiled vessels, especially stored whole, must not become magnets for dumping of trash or hazardous materials, or shelter for individuals [3]. And stockpiling has a hard precondition: to reduce pollution risk, vessels should only be stockpiled once hazardous materials, especially liquids, have been removed — older vessels and larger commercial vessels are much more likely to carry lead and PCB-laden paint and asbestos, many landfills are not permitted to accept RCRA or hazardous waste, and shipping hazmat to permitted landfills can substantially increase disposal costs [3].

Recycling: the waste stream as a resource

The seventh recommendation looks past the landfill. States should establish a vessel recycling waste stream pilot project: vessel recycling "shows promise both as an environmentally responsible method of disposal for ADVs, as well as an economic resource for rural communities" [3]. The precedents are industrial. Some states have successfully used wood and fiberglass waste as fuel for concrete and paper mill plants; other endeavours use wood and fiberglass waste as fiber material in new composite items — barriers, light poles, and manhole covers [3]. Rhode Island is exploring the feasibility of using fiberglass from vessels to heat kilns for fired cement [3]. Washington DNR is requesting an appropriation of about $150,000 from the state legislature to conduct a pilot project to create a vessel recycling waste stream — the project would find possible waste streams and test their viability with actual vessels, and if successful "could serve as a model elsewhere" [3]. The summary table pairs the two: Rhode Island and WA DNR [3]. States should support pilot programmes and research into other creative disposal options to reduce the waste stream the ADV problem generates [3].

Section 27's disposal recommendation — sell, deconstruct, recycle, or use a removed vessel for the best public benefit — reads, in this chapter, as an industrial process rather than a legal permission: a vetted contractor pool, a permitted yard, a federal hand-off that keeps equipment on site, a stockpile timed to empty barges, a landfill that can take the waste, and, at the frontier, a kiln that burns the fiberglass [3] [3]. The ghost ship's last voyage is a supply chain.



Exhibit. A former US Coast Guard 41-foot utility vessel, legally owned by the US Bureau of Indian Affairs, taken into custody by the Washington Derelict Vessel Removal Program. Source: S0014.

Washington State recreational boating programs and derelict vessel funding

Every model needs an audited example. The Blue-Ribbon report calls Washington the only comprehensive ADV programme among the western states; in 2025, Washington's Joint Legislative Audit and Review Committee (JLARC) published the fiscal anatomy of the recreational boating system that programme lives inside — an update study directed by the Legislature in the 2023 operating budget (ESSB 5187), approved for distribution on September 17, 2025, and notable for what auditors rarely produce: the Legislative Auditor makes no recommendations [16]. The numbers describe a state that funds boating the way the Blue-Ribbon funding element prescribes — from boaters themselves, through dedicated accounts, with the derelict vessel programme holding a statutory share.

The money in and out

Recreational boating generated $108 million in state revenue in the 2021-23 biennium — $307 million across the three most recent biennia — and taxes and registration fees contributed 75% of boating-related revenue, the remainder coming from sources including state park user fees, boater safety certification fees, shellfish license endorsements, aquatic invasive species prevention permits, and grants; the state also pays for boating activities from accounts unrelated to boating, such as state bonds [16] [16]. Against that revenue, state agencies spent $86 million on recreational boating activities in 2021-23, across five categories: infrastructure and access, $52.7 million (61%); environmental protection, $20.9 million (24%); boater safety, $6.0 million (7%); marine law enforcement, $5.9 million (7%); and registration administration, $0.3 million (0.4%) [16] [16]. Six agencies share the work — Parks, the Recreation and Conservation Office, DNR, the Department of Fish and Wildlife, Ecology, and the Department of Licensing — coordinating through an Agency Boating Committee formed in 2009, with roles largely unchanged since JLARC's 2010 study; 87% of expenditures were drawn from seven accounts, including the Derelict Vessel Removal Program account [16] [16]. Washington is also unusual in pushing money downward: of the $86 million spent, $19.2 million was transferred or awarded as grants to local governments, making Washington one of few states to offer state funding

to local governments for recreational boating — with a match required, in money, labour, materials, land, or other contributions [16] [16].

The DVRP's transformation: a quarter of the excise tax

The derelict vessel programme sits inside environmental protection, and its recent history is a funding story. DNR spent $10.6 million administering the Derelict Vessel Removal Program in the 2021-23 biennium [16]. In 2022, the Legislature increased the programme's funding by allocating 25% of the watercraft excise tax to it — and programme expenditures increased 367% after the increase [16]. The output followed the money: during the 2021-23 biennium, DNR removed 226 boats from Washington's waters, and owners turned in 91 more voluntarily — the turn-in channel of Section 30, running at audited scale [16]. Set beside the registration-fee stream the Blue-Ribbon report documents (about $800,000 a year from the $3 recreational and $1-per-foot commercial fees), the excise-tax allocation shows the difference between a fee that symbolises cost-sharing and a tax share that funds a programme: the same state's removal capacity multiplied when the Legislature gave the DVRP a quarter of a broad-based boating tax rather than a flat per-vessel charge [3] [16].

The rest of environmental protection shows the programme's company. Parks spent $3.6 million administering the Clean Vessel Program, distributing $2.4 million in grants to install or maintain marine sewage disposal facilities — nine ports and four private marinas received grants in the biennium [16]. The Department of Fish and Wildlife spent $5.2 million: $4.7 million on its Aquatic Invasive Species programme (mandatory boat inspection stations, free decontamination sites including a mobile unit) and $554,000 on derelict fishing gear removal — a programme that collects crab pots left out on closed days of the summer crab fishery; in 2023, officers recovered 1,283 crab pots in Puget Sound, releasing 2,104 crabs trapped inside, and DFW also contracts with the Northwest Straits Foundation to retrieve derelict shellfish pots and lines when seasons are closed [16]. Ecology spent $1.5 million across invasive-plant and algae grant programmes and spill-prevention outreach [16]. The derelict-gear line is worth pausing on: it is the same abandonment logic as the DVRP's, applied to fishing gear rather than vessels — and it is where this report turns next, to ALDFG.

Enforcement, safety, and the comparative picture

Marine law enforcement's $5.9 million splits between two agencies: Parks spent $1.7 million, funded by a U.S. Coast Guard federal grant, coordinating and certifying training for marine law enforcement officers and distributing federal funds to sheriff's offices and police departments with approved boater safety programmes (in 2021-23 it also funded new boats for five entities); DFW, the state's primary marine law enforcement agency, spent $4.2 million on on-water patrols, boat-ramp enforcement and education, and aquatic invasive species inspection stations [16]. Boater safety programmes took $2.8 million from two state agencies plus $3.3 million transferred to local governments [16]. Registration itself is cheap to run — $0.3 million of DOL administration for a fleet averaging 230,000 registered motorised boats a year [16] [16].

Comparatively, Washington is typical in its rules and atypical in its generosity. Its vessel registration, boater education, and life jacket laws resemble those of other states; like Washington, 27 of the 44 surveyed administrators operate from a natural resource agency rather than a law enforcement agency; and Washington is one of only four states that specifically use data in safety programmes to evaluate and address equity — Parks targets its boater safety outreach using health disparity data, free-and-reduced-price-lunch proportions, and drowning incident reports, though no Washington agency specifically considers racial equity in recreational boating [16] [16]. The audit's silence is itself the finding: the Legislative Auditor made no recommendations, and none of the six agencies or the Office of Financial Management submitted comments [16]. Washington's system — registration base, excise-tax share, dedicated accounts, audited removals and turn-ins — is what the Blue-Ribbon funding element looks like when a state actually builds it [16] [16].

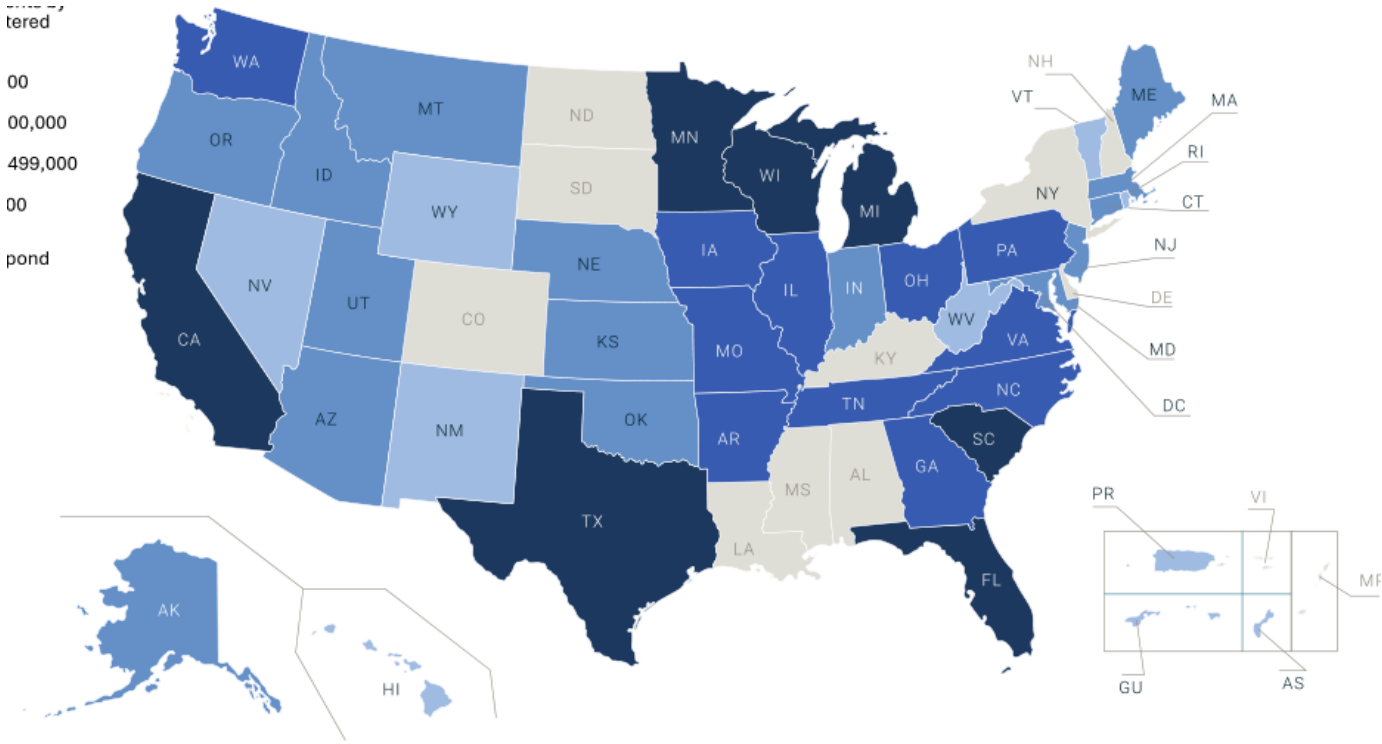


Exhibit. Forty states, the District of Columbia, and three territories responded to the JLARC survey of recreational boating programs. Source: S0015.

Gear That Keeps Fishing: Abandoned, Lost and Discarded Fishing Gear

Magnitude and composition of ALDFG

The report now changes scale. A derelict vessel is a single large object with an owner, a title, and a legal process; abandoned, lost, or otherwise discarded fishing gear — ALDFG, "ghost gear" — is millions of small objects with none of those things. A ghost net, in the FAO/UNEP review's footnote definition, is "a net that continues fishing after all control of the fishing gear is lost by a fisher" [5]. The Macfadyen, Huntington and Cappell review for FAO and UNEP is the part's foundation document, and its second chapter is an exercise in measuring something designed to disappear: there is no census of gear in use, direct reporting of loss by operators is rare, and most of what is known comes from fisher interviews, retrieval programmes, and beach surveys [5]. The result is a magnitude told in fragments — percentages of fleets, kilometres of net, tonnes recovered — which this section assembles.

The global frame

Breaking the Plastic Wave puts the global leakage of ALDFG at an estimated 640,000 to 1,150,000 metric tons per year, expected to increase as fishing effort and aquaculture grow, and ranks ghost gear among the most damaging sources of ocean plastic precisely because its entanglement risk is a design feature: gear built to trap and kill keeps doing so, and it is lost where fishing — and therefore wildlife density — is concentrated [12]. Loss propensity varies sharply by gear type: an estimated 29 per cent of lines are lost each year, 8.6 per cent of all traps and pots, and 5.7 per cent of nets; among nets, gillnets carry the highest risk of loss, bottom trawls are low risk, and purse seines and midwater trawls lowest [12]. But the denominators are missing — no comparable data exist on how much of each gear type is produced or used annually, so landed catch serves as an imperfect proxy, and on that basis gillnets and bottom trawls are the two highest-priority gear types for research and prevention [12]. The regional record shows what those percentages mean on the water.

Gillnets: the Baltic, the Atlantic, the Mediterranean

In the Baltic, FANTARED 2's study of Swedish gill-netters in 1998 found regular loss concentrated among demersal fishers (turbot and cod) using bottom-set gillnets in open sea: about 1,500 nets a year, 155–165 km of netting, 3.6–3.8 nets per active vessel — yet less than 0.1 percent of nets deployed [5]. Fishers recovered only about 10 percent of

losses themselves; gear conflicts were the main cause of loss; and eventual ghost nets surfaced in trawl hauls in two forms — longer nets near the conflict area, and small remnants scattered over a wider, less defined area [5].

Norwegian shelf fisheries lose mostly offshore nets (spawning saithe), again under 0.1 percent of the sector's nets, with high fisher recovery rates of 80–100 percent — yet between 1983 and 1997 Norwegian retrieval programmes still recovered 6,759 gillnets set for Greenland halibut, evidence that even in a highly regulated fishery with mandatory loss reporting and controls on net length and soak time, retrieval surveys remain necessary [5]. Around the United Kingdom, tangle netting lost most: 263 nets per year from 18 vessels, about a third recovered; the hake métier lost 62 nets per year from 12 vessels, half recovered; the wreck métier seldom lost whole gear but reported 884 snagging incidences from 26 vessels, shedding portions of net sheets [5]. In Brittany, wreck gillnetting showed the largest proportional loss — just under 3 percent of nets — while the flatfish and monkfish métier lost the most length, just under 5 km of net per vessel per year; on Portugal's Algarve coast, interviewed fishers reported mean losses of 3.2, 6.0, and 7.4 net sheets per boat per year in the local, coastal, and hake categories, the hake rate higher for its distance from shore and depth [5]. Most French gillnet fisheries lost under 0.5 percent of nets annually (the seabass fishery was an outlier at 2.11 percent), and in Spain's Cantabrian fleet — some 645 vessels — average loss ran to 13.3 nets per vessel per year, with the worst métiers at 27.9 [5].

The deepwater Northeast Atlantic is the review's darkest region. Building on FANTARED, the DeepNet project found vessels in the deepwater and upper-slope net fisheries often physically unable to carry their nets back to port — the net stores are used to hold fish — so only headline and footropes come ashore while net sheets are bagged, burnt, or dumped at sea, with evidence of illegal dumping north and northwest of the UK and Ireland [5]. Anecdotal evidence from one shark vessel suggests some 600 sheets of 50 m netting (30 km) are routinely discarded on a typical 45-day trip; scaled by regional effort, a crude estimate gives 1,254 km of sheet netting lost per year, and the deep-slope fisheries were estimated to lose about 15 nets (750 m) per vessel per day [5].

The Mediterranean, despite extensive small-scale gillnet, trammel net, and trap fisheries and the very large small-scale fleets of Greece and Italy, has attracted limited attention; only the French hake gillnet fishery has a total-loss estimate — the 65-vessel offshore fleet loses about 0.2 percent of its nets annually (36 to 73 nets), the 32-vessel coastal fleet some 9 to 17 nets a year [5].

Traps, pots, and the Pacific

The trap fisheries supply the wildest variance in the review. In Queensland's blue swimmer crab fishery, surveyed crabbers reported losing an average of about 35 pots per year (range 0 to 400), implying over 6,000 pots lost annually across the fishery — though researchers estimated fewer than half remain in the environment, the rest trawled up or stolen [5]. For the Bristol Bay king crab fishery, three studies of the same fishery produced annual loss estimates of 31,600 pots (Kruse and Kimker, 1993), 20,000 (Paul et al, 1994), and 7,000 (Stevens, 1996) — a fourfold spread that is itself a finding about the state of measurement [5]. British Columbia's Dungeness crab trap loss rate was estimated at 11 percent per year [5].

In the Northwest Pacific, the densities are measured rather than modelled: a Korean survey located an estimated 18.9 kg of marine litter per hectare in fishing grounds, 83 percent of it fishing nets and related materials such as ropes; a six-month survey of the Incheon coast found 194,000 m³ of debris weighing 97,000 tonnes, mainly of fisheries origin; and nationally coordinated Korean cleanups recovered 10,285 tonnes of fishing-related debris from coastal areas over 2000–2006 [5]. Up to 1,000 tonnes of ALDFG are recovered from the Sea of Japan annually, mostly bottom gillnets and pots, apparently mainly of non-Japanese origin [5]. The U.S. National Marine Fisheries Service estimated driftnet loss at 0.06 percent of nets per set — small per set, but compounding to 12 miles of net lost per night of the season and 639 miles per year in the North Pacific alone [5]. Hawaii's cleanup surveys of 1998–2002 found debris dominated by trawl and seine nets (83.6 percent), with monofilament and multifilament gillnets at 5.2 and 3.2 percent; over 600 metric tonnes of ALDFG have been removed from the Hawaiian archipelago by NOAA and its partners [5]. In Australia's Gulf of Carpentaria, a circulating gyre traps ALDFG in a repetitive cycle of fishing, washing ashore, and washing back out with storms and spring tides — nets arriving on the eastern shore with the November–March monsoon and on the western shore with the May–September trade winds [5].

The Caribbean exposes the floor of knowledge: a 2007 NOAA/State Department workshop in Key West reached no agreement on whether ALDFG is a significant regional issue, whether its causes are storms, missing disposal facilities, or outside sources — one participant held that most ALDFG was from outside the region — though fish traps and gillnets were agreed to hold the greatest ghost-fishing potential [5]. In the Northwest Atlantic, the first docu-

mented work is Way's 1977 Atlantic Canada study, retrieving 148 and 167 net fragments in 48.3 and 53.5 hours of grappling trawls [5].

Composition, then, is consistent even where magnitude is not: gillnets and traps dominate every regional account — the two gear families designed to fish unattended. The estimates disagree by factors of four because they are built from interviews, retrievals, and strandings rather than from any registry of what was deployed. Why the gear leaves the vessel — conflict, weather, economics, disposal — is the next section's question [5] [12].

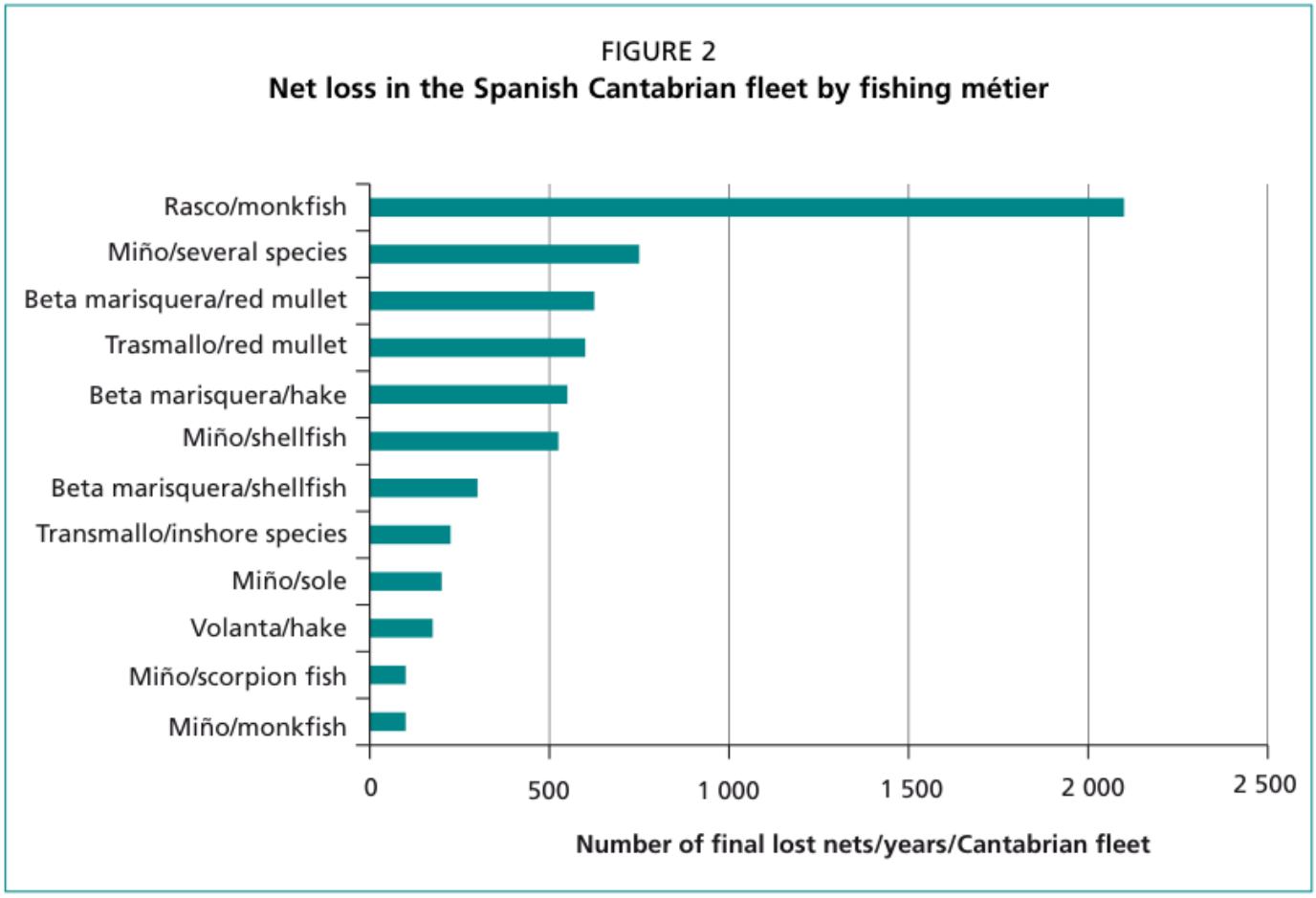


Exhibit. Net loss in the Spanish Cantabrian fleet broken down by fishing metier, including rasco for monkfish, mino and beta marisquera for red mullet and shellfish, and trasmallo. Source: S0017.

Reasons gear is abandoned, lost or otherwise discarded

The three verbs in ALDFG are three different behaviours, and the FAO/UNEP review's fourth chapter — with the Poseidon (2008) typology it reproduces — insists they not be merged. Gear is abandoned when a fisher deliberately does not retrieve it; discarded when disposal at sea is chosen over disposal onshore; lost when it goes accidentally. Each has its own causes, and the causes matter because the cures are cause-specific — though the review concedes the literature on causes is limited, "a potentially significant omission," and that in any one fishery several causes usually operate together [5] [5].

Lost, I: gear conflict

The most commonly reported accidental cause is conflict between gears — typically mobile, towed gear passing through ground where static gear is set, though anchored gillnets are also lost to merchant shipping [5]. In UK tangle net fishing, FANTARED 2 reported the most significant losses as whole-fleet or partial-fleet losses from gear conflicts: a partial loss runs from one net to several, a whole-fleet loss averages 30 nets, in an operation hauling an average of 12 km of netting per day [5]. The mechanics are exposure: vessels patrol their nets at night but cannot while hauling, leaving nets vulnerable to towing vessels, and whether the loss is whole or partial depends on the towing vessel's approach — trawl, scallop dredge, or beam [5]. Dahns and end ropes are particularly vulnerable to shipping in intense areas such as the English Channel and can be cut, leaving an entire fleet with no surface indication of po-

sition; intermediate buoy lines are the mitigation [5]. Conflicts between static gears — netters, liners, and potters competing for ground — are generally much less serious, because the gear is not usually moved any distance and temporarily lost gear is easier to find [5].

Conflict is not static. In the Baltic, losses of static nets to trawling have fallen in recent years through improved communication between skippers in the two sectors; in the Channel and Western Approaches, hake net fishers report greater losses because developments in trawl ground gear let trawlers tow areas previously inaccessible to them — modern sonar, 3-D mapping, and differential GPS allow trawlers to fish within 25 m of wrecks [5]. Brown et al (2005) found gear conflict a main cause of ALDFG in the Baltic cod net fishery and in many Greek fisheries (between mobile and static gear, and between part-time or recreational and professional fishers), and Hareide et al (2005) identified it as important in the deepwater Northeast Atlantic [5]. Where it is managed, it is managed socially: in the English/French Western Channel, communication between skippers and producer organisations keeps conflicts comparatively rare, underpinned by a formal gentlemen's agreement between the French and English associations allocating "blocks" to static or mobile gear, swapped every six weeks — an arrangement that "functions well and reduces gear loss considerably" [5]. For the most part, the review concludes, conflict ALDFG is unintentional [5].

Lost, II: weather, ground, and the price of holding on

Operational factors and the natural environment are "a very significant cause" — sometimes unintentional, sometimes intentional but unavoidable [5]. Poor weather, currents, seabed conditions, strong winds, and swell can overwhelm a vessel's ability to deploy, work, and retrieve gear; a common pattern for permanent loss is rough bottom combined with strong currents, snagging nets on the bottom — in the Western Channel net fisheries, Brown et al found losses mainly from weather and bottom snagging and very little from conflict [5]. Loss comes as fragments, or as whole fleets cut adrift for safety in very bad weather, or gear snagged on an obstruction the vessel cannot free [5]. Gear quality compounds it: old gear breaks and tears, and is not replaced — in the Gulf of Mexico wire-trap blue crab fishery, deterioration of buoys, lines, and knots, negligent assembly and maintenance, and plastic jugs used as floats are all cited (Perry et al, 2003), and the review stresses old gear is a developed-country problem too, wherever fishing is financially marginal and upgrading is unaffordable [5].

The limiting case is the Radiant. On 10 April 2002, fishing some 45 miles northwest of the Isle of Lewis with about 1,735 m of warp out in roughly 730 m of water, her port trawl door snagged on an underwater obstruction; the vessel effectively anchored herself to the seabed, power to the winches was lost, the loaded port warp dragged her into a heavy list, the engine room flooded, and she capsized while trying to free the gear — one crew member was lost, five rescued (MAIB Report 2/2003) [5]. Gear loss, the review notes, "may take the form of losses of complete vessels": the decision to cut away or hold on is made under exactly the conditions — snagged gear, bad weather, a straining vessel — where holding on can be fatal [5].

Aquaculture sits exposed in the same weather: its gear often cannot be removed from the sea ahead of a storm. Hasan Hanawi, a seaweed farmer in Bira, South Sulawesi, lays 20 anchored longlines of about 60 m and loses around 10 percent of his equipment to storm damage each year — washed onto land, rarely salvaged, the attached seaweed lost with it [5]. And disaster itself manufactures hazards: Ahmad Saiful, skipper of a 20 GT purse seiner out of Lampulo, Banda Aceh, lost two purse seine nets in two years — each worth about US$19,000 — damaged on tsunami wreckage in areas familiar to him; an ADB-funded programme has since sonar-mapped the debris field with some 30 local vessels and equipped them with GPS [5]. With climate change predicted to make weather more frequent and extreme, the review expects weather-driven loss to grow, and forecasting and forewarning to matter more [5] [5].

Discarded: the economics of letting go

Deliberate disposal has an arithmetic. Retrieval may be technically complicated or time-consuming with variable, uncertain results — pieces of netting, badly tangled bundles — and then discarding can be the rational act: quicker to discard entangled gear than to interfere with hauling and lose fishing time; pointless to retrieve gear whose value is negligible; not worth retrieving what costs more to recover than it is worth [5]. Floating fish aggregating devices may likewise be deliberately abandoned [5]. Section 33's deepwater case is the structural version: when a vessel's net stores must hold fish, the net sheets cannot come home at all [5]. What happens onshore decides much of the rest. In France, unwanted gear goes to waste collection centres for sorting and recycling, back to manufacturers, or out in municipal "big bags"; in the UK, harbour skips absorb disposal into harbour dues, but industrial-waste charges push

some nets into normal waste, community tips, or fly-tipping on land — and in neither country, on fisher interviews, do fishers appear to discard unwanted nets at sea (Brown et al, 2005) [5]. Disposal infrastructure and its pricing, in other words, can move the discard decision off the water entirely.

Abandoned: gear left because retrieval is the risk

The Poseidon typology reserves abandoned for deliberate non-retrieval driven by illegality: IUU fishing, enforcement pressure, and illegal gear — the fisher leaves the gear because coming back for it means meeting the authorities; the matched measures are enforcement and improved port state measures [5]. Against all three categories stands a counterweight the review is careful to record: gear is capital, fishers invest considerably in it and typically do not want to lose it, spending significant time searching for what is lost [5].

Why causes dictate cures

The typology maps each type to its remedies: abandoned gear to port state measures and enforcement; discarded gear to mandatory gear return, retrieval programmes, and gear marking; lost gear to zoning schemes, loss reporting, and avoiding setting gear when extreme weather is identified or predicted [5]. Measures run from avoidance (preventative, cause-specific, most effective) to cure (retrieval, applicable across causes but costlier), and the consistent conclusion of FANTARED, DeepNet, and subsequent workshops is that "prevention is better than cure" — in cost-effectiveness studies as in environmental terms — though retrieval can still be cost-effective against the alternative of gear left ghost fishing in situ [5]. Measures also fail in isolation: gear marking only bites with a monitoring, control, and surveillance regime behind it; waste facilities only work with incentives to use them [5]. And a residue is irreducible: weather loss from fixed and unattended gear may be almost impossible to eliminate, aquaculture is growing, severe weather is increasing — "some degree of ALDFG is therefore inevitable" [5]. What the abandoned gear then does, unattended, is the subject of the next section [5].

FIGURE 18
Types of ALDFG, their causes and measures to address

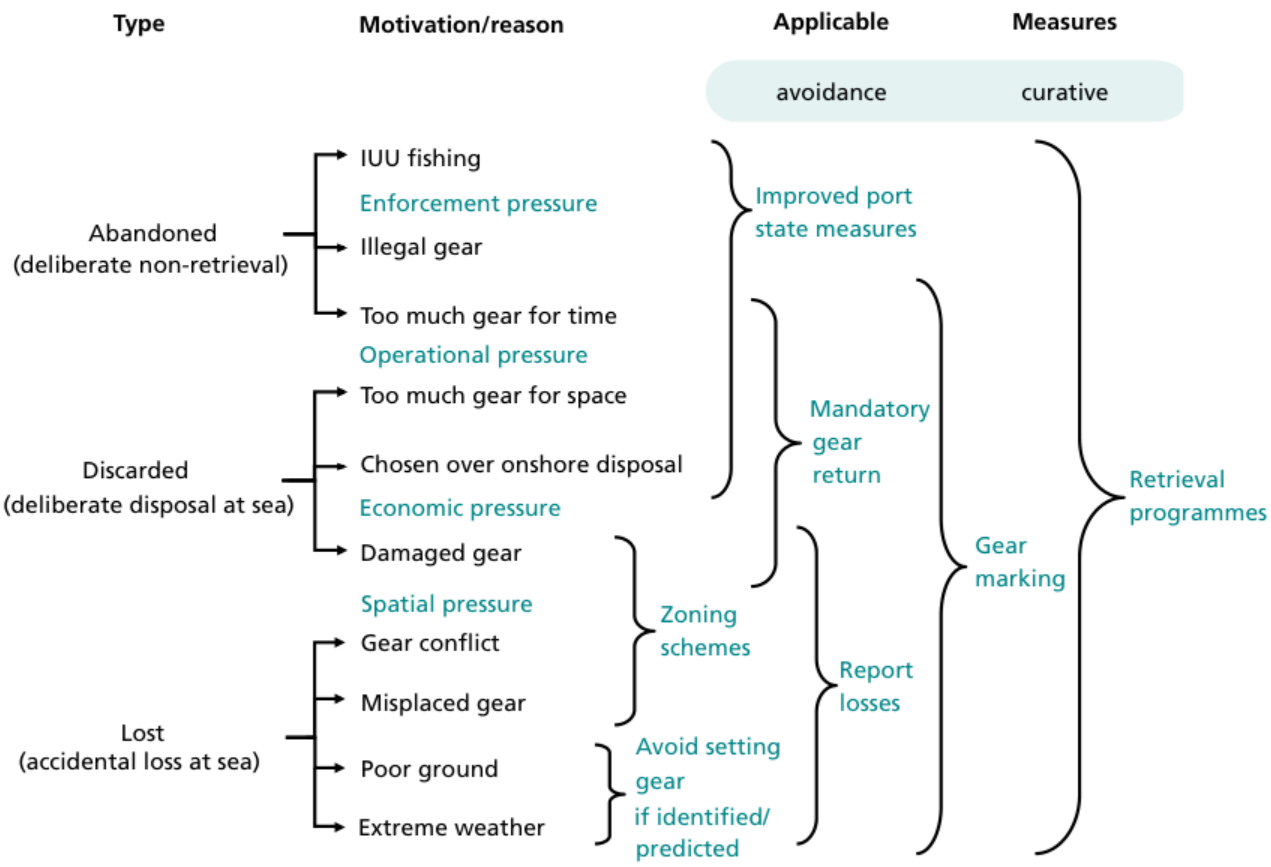


Exhibit. Types of abandoned, lost or otherwise discarded fishing gear, their causes, and the avoidance and curative measures that address them, including enforcement responses to IUU fishing. Source: S0017.

Ghost catches: continued catching of target and non-target species

Ghost fishing is the impact that makes ALDFG more than litter. The FAO/UNEP review calls the ability of lost gear to keep catching "one of the most significant impacts of ALDFG," and stresses how conditional it is: on the gear type, on whether it was abandoned as a set gear maximised for fishing or discarded where it is less likely to fish, and on the local environment — currents, depth, location [5]. A ghost catch is a catch with no fisher: no landing, no record, no quota, no market — just mortality, running unattended for as long as the gear holds together.

Efficiency is a trajectory

The review's central analytic point is that ghost fishing efficiency is not a property of the gear but of its biography. How a net or pot changes "during its progression from initial loss of control to its eventual demise" — and its state and position when that progression starts — determines what it catches [5]. Abandoned gear may be set for maximum fishing efficiency and, if well anchored, slow to collapse, giving high ghost catches; discarded nets may collapse immediately, with low initial efficiency, and may be discarded where they have little potential to ghost fish [5]. Gear that snags on rock, coral, or wrecks is held in a fixed fishing position, preserving its efficiency [5]. And the trajectory can reverse: once a net has shed its burden of captured fish and marine growth, it "has the potential to regain its shape and start fishing again" [5].

As control is lost, the gear's selectivity for its original target mutates through four mechanisms: the mesh distorts; marine growth changes the gear's transparency and detectability (itself a function of depth, water transparency, and productivity); the gear is translocated to different environments; and accumulated catches act as bait, drawing other species into entanglement or entrapment [5]. The result is that ALDFG "typically increasingly catches other fish and shellfish species that may or may not have a commercial value" [5]. Fish dying in nets attract scavengers, which are then caught themselves — cyclical catching, in which the gear baits itself [5]. The review's calibration is honest: overall ghost fishing catches are probably very low compared with controlled fishing (Brown et al, 2005) — but they vary by gear type and conditions, they are unselective in ways controlled fishing is not, and nobody is counting them [5].

Pots: the self-baiting experiment set

Pots and traps ghost fish through their bait. A lost pot is usually baited when set; over time the bait, or the lost catch, attracts scavengers — some commercially important — which become entrapped and die, forming new bait for the next arrivals [5]. Entrapped animals die of starvation, cannibalism, infection, disease, or prolonged exposure to poor water quality such as low dissolved oxygen (Van Engel, 1982; Guillory, 1993); the effects of derelict blue crab traps on other species, including terrapins and commercially important finfish, are documented (Smolowitz, 1978; Guillory, 1993; Guillory and Prejean, 1998) [5]. But the experiments refuse to generalise. Catching efficiency proved "as variable as pot loss rates," dependent on gear design, species behaviour, and seasonality — entry, escapement, and mortality are dynamic processes, and pots, unlike nets, depend on bait: once it is eaten or degraded, catch rates decline sharply [5]. The FANTARED project found Norwegian crab traps and Portuguese octopus traps showed no significant degradation over the study (wire fish traps excepted), and in the Portuguese octopus and fish traps there were almost no catches three months after deployment — though fish escape poorly, octopus and king crab escape readily, and post-escape mortality after days or weeks of retention is possible, little studied, and flagged as warranting further work [5].

Numbers exist where experiments were run. In Chesapeake Bay, simulated "self-baiting" blue crab traps caught slightly more than double the unbaited controls — mean catch rates of 0.785 against 0.385 crabs per trap per day (Havens et al, 2006) [5]. Off Wales, Bullimore et al (2001) set 12 pots to simulate ghost fishing: the original bait was consumed within 28 days, yet the pots continued fishing — mainly spider crab and brown crab — with catches declining to a minimum between nine and ten months, then rising again, possibly with water temperature; non-target Ballan wrasse appeared, especially late in the 398-day dive-survey series, as crustacean numbers fell [5]. Against that, Godøy et al (2003) deliberately "lost" pots for periods from five days to a year: in a string of four rectangular collapsible pots, all 92 tagged crabs had left after four months while 61 new crabs entered, very few dead crabs were found, and the authors concluded lost pots do not substantially contribute to crab mortality in those fisheries

[5]. In Oman, lost wire fish traps showed an estimated ghost fishing mortality of 1.34 kg per trap per day, decreasing over time, modelled at 67.27 kg per trap over three months and 78.36 kg over six (Al-Masroori et al, 2004) [5]. The honest summary is the review's own: efficiency is fishery-specific, design-specific, season-specific — measurable locally, unknown globally.

The wider impact ledger

Continued catching heads a longer list. ALDFG interacts with threatened and endangered species — seabirds, turtles, seals, cetaceans — through entanglement or ingestion, with entanglement generally considered far more likely to cause mortality, and persistent synthetic materials the durable vector [5]. Physical impacts on the benthos are gear-specific: gillnets alone do little, but nets dragged by currents or during retrieval can harm sponges and corals, deep-water nets need weights over 100 kg with localised effects, and traps bear hardest on seagrass beds and emergent coral and sponge habitat rather than sand and mud [5]. At the general level, UNEP's Global Programme of Action estimates as much as 70 percent of marine litter input sinks to the seabed, where accumulations can smother benthic communities [5]. Modern plastics can persist up to 600 years in the marine environment, and their degradation yields the microscopic fragments Thompson et al (2004) found particularly abundant in subtidal sediments — a potential source of toxic chemicals in the food web whose impact is not known [5]. The socio-economic ledger — navigational hazard to marine users above all — is real but, the review notes, the literature is scarce, costs resist rating or comparison, and the costs of compliance, rescue, and research "do not seem to have been attempted" [5].

That uncounted quality is the section's conclusion. Controlled fishing, however wasteful, is at least enumerated; ghost fishing is mortality outside every accounting system the report has examined — which is why the measures that follow divide into finding the gear (Section 36), and designing it to stop (Section 39) [5] [5].

Locating lost gear and retrieval programmes

Retrieval is the curative end of the ALDFG toolbox — the measure of last resort that Sections 34 and 35 have been building toward. It faces a problem no other cleanup does: the target was designed to be inconspicuous, it sits on the seabed in the dark, and in most fisheries nobody recorded where it went down. The FAO/UNEP review's measures chapter therefore splits the work in two — first locating the gear, then recovering it — and is candid that the first fishers on the scene are the owners themselves: fishers "will make every possible attempt to locate and recover their own gear as it has a significant economic cost in most fisheries," weighing that cost against the time and fuel a search consumes [5]. Programmes begin where self-recovery ends.

Finding what was designed to disappear

Surveys span a cost spectrum from the volunteer to the research vessel. Land-based beach surveys are the low-cost end, and the most appropriate form where the key impact is onshore — entanglement or littering on, for example, turtle nesting beaches; the Ocean Conservancy's International Coastal Cleanup provides the widely adopted guidelines for beach survey and subsequent clean-up [5]. In the United States between 1986 and 2002, that cleanup removed 89 million pounds of debris from more than 130,000 miles of shoreline, and from 1995 more than 108,000 divers collected a further 2.2 million pounds from over 3,900 miles of underwater habitat [5].

Sea-based surveys hunt gear that may still be ghost fishing or damaging habitat. Where no accurate location information exists, modelling techniques, local knowledge, and anecdotal information are used to identify potential hotspots and target the survey: the towed-diver surveys of the Northwestern Hawaiian Islands were aimed by identifying High Entanglement Risk Zones (HERZ) — places where oceanographic conditions collate marine debris in the same waters as high densities of sensitive species, in that instance monk seal nursery areas (Donohue et al, 2001) [5]. Side scan sonar, a seabed-mapping technology that has become more accurate and more affordable, suits relatively large or readily distinguishable items such as pots and traps — on its images a sport trap reads as a square, a commercial trap as a circle [5]. NOAA's Gulf of Mexico Marine Debris Project uses sonar from survey vessels to retrieve large debris and is also deploying an autonomous survey vessel (maximum depth 100 m, used mainly in shallow water under 50 m) to detect and map submerged wrecks and other navigation hazards — its deployment strictly controlled so the survey vehicle does not itself become one [5]. Skipper interviews and interpretation of vessel monitoring system (VMS) plots round out the information sources [5].

Recovery programmes

Recovery itself is usually mechanical and unglamorous: curative programmes typically work with a creeper or grapnel to snag nets, as in the net-fishery retrieval programmes undertaken in Sweden and Poland and the routine Norwegian programme (whose survey results anchor Section 37), which led to Norwegian, English, and Irish collaborative projects in the Northeast Atlantic deepwater gillnet fishery [5]. In the United States, many location and retrieval programmes run with NOAA Marine Debris Program support. The California Lost Fishing Gear Recovery Project, launched in July 2005 by the SeaDoc Society at the UC Davis Wildlife Health Center, invites any ocean user to report lost gear — by online form or free phone line — and hires experienced, certified SCUBA divers to remove it from near-shore waters in a safe and environmentally sensitive manner; since May 2006 it has retrieved nearly 11 tonnes of gear [5] [5]. Programmes target either gear types that remote sensing can find, or known hotspots where gear accumulates or habitat is particularly sensitive, such as marine mammal or bird colonies [5].

The Gulf of Mexico trap programme is the review's model of organised cure. The Gulf States Marine Fisheries Commission produced guidelines for developing derelict trap removal programmes (2003) whose elements are transferable to other fisheries and areas; coordinated through the Commission, Gulf states arrange annual voluntary short-term closures and targeted clean-up operations in trap fisheries with the fishers' own assistance — between 2004 and 2007, over 183 boats retrieved nearly 16,000 traps from 1,405,708 acres of U.S. coastal waters, and Louisiana's wildlife agency reported the closures and cleanups "very successful" in retrieved traps, volunteer participation, and acceptance by all user groups [5]. In Australia, the Carpentaria Ghost Net Programme took AU$2 million (US$1.9 million) of government funding to Gulf of Carpentaria coastal communities, which formed a network to clean beaches and run a coordinated information-recording process building a picture of ghost net quantities, impacts, and likely origins across northern Australian waters [5]. And in the Sea of Japan, recovery is continuous rather than episodic: national fisheries patrol vessels bring any ALDFG they identify to shore, as do fishing vessels chartered by fisheries organisations and local government under central government subsidy (Inoue and Yoshioka, 2004) [5].

The honest arithmetic of cure

How much of the problem can retrieval actually remove? The review's effectiveness survey gives the two best-documented fractions, and they are sobering. After DeepNet's crude estimate of 1,254 km of gear lost per year in the Northeast Atlantic deepwater fishery, a follow-up retrieval programme led by Irish authorities recovered approximately 35 to 40 km — around 3 percent of estimated annual losses [5]. At Port Susan in the United States in 2006, a side scan sonar survey of about 95 percent of the known coastal fishing grounds identified 403 items; 73 percent could be investigated by divers, and 174 items — 43 percent of those identified — were removed [5]. Effectiveness, the review notes, has mostly been assessed by expert judgement because baselines showing the scale of the problem rarely exist; in their absence, acceptability to stakeholders and enforceability become the working proxies, since fishers who find a measure unacceptably costly or restrictive will not comply with it [5]. Expert workshops for the Baltic and the Western Channel agreed on which measures are relevant but diverged on what is acceptable or enforceable in each fishery — there is no one-size-fits-all — while converging on one rejection: acoustic detection systems, biodegradable nets, and alternative gears were considered unacceptable by fishers in both (a verdict Section 39 will complicate) [5].

Retrieval also runs into law. A U.S. National Research Council report (2008) noted that recovery of derelict fishing gear in the United States "may be inhibited by prohibitions against tampering with abandoned gear, the application of cabotage laws and burdensome certification requirements for vessels that transport DFG, and fishery regulations that prohibit vessels from carrying gear that is not a gear type permitted under their license endorsement" [5] — the same fragmented-authority problem that governed derelict vessels, in miniature: the gear is legally someone's property even while it fishes on without them. The conclusion the measures chapter draws is the one Section 34 anticipated — retrieval is necessary, cost-effective against doing nothing, and structurally incapable of keeping pace; it buys time for the preventive measures that follow [5] [5].

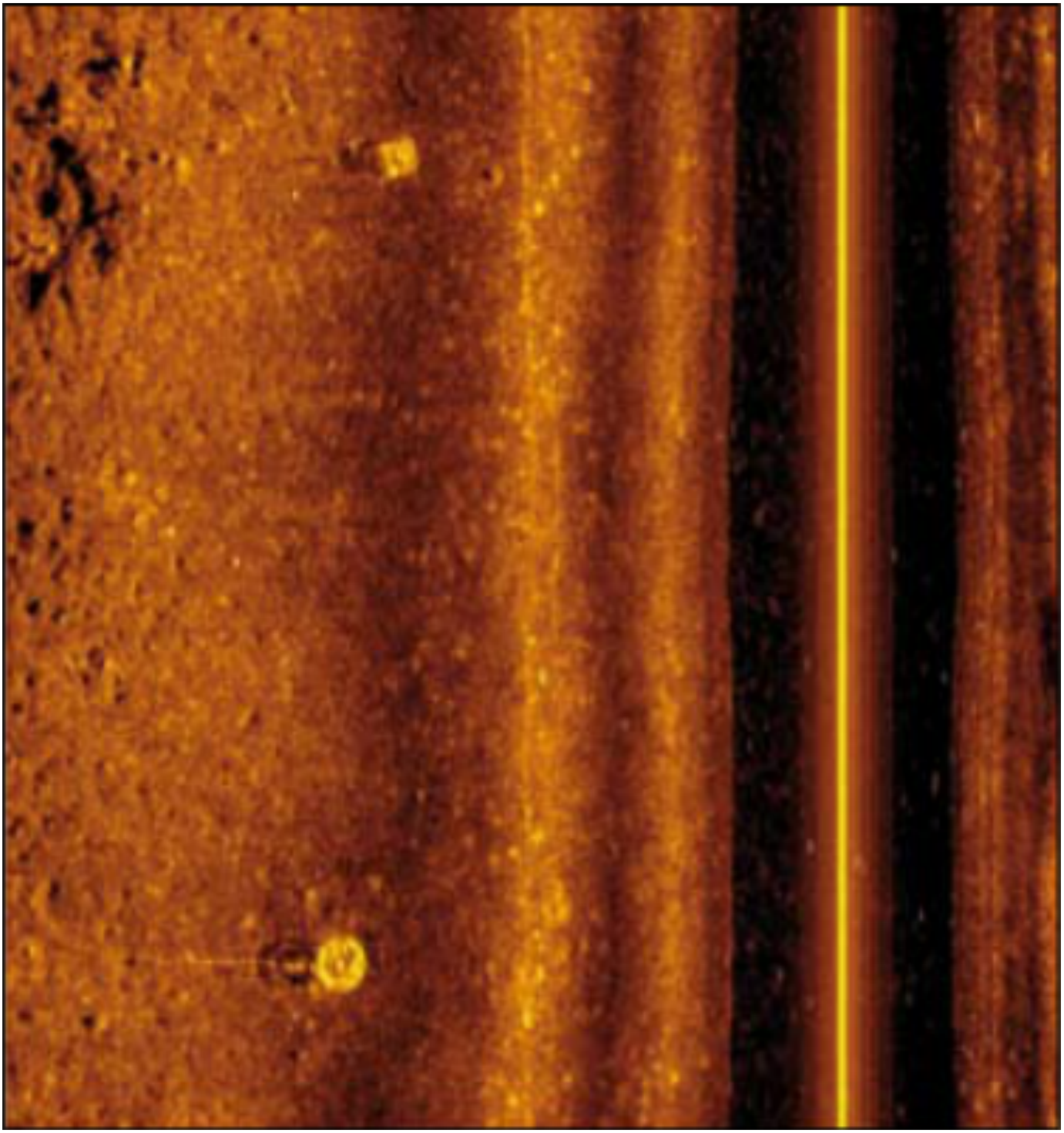


Exhibit. Side-scan sonar image of abandoned, lost or discarded traps on the seabed, from the Innerspace Exploration Team in the United States. Source: S0017.

The Norwegian retrieval survey and the Baltic retrieval programme

If Section 36 surveyed retrieval in general, this section looks at its two longest-running northern European institutions — and at what the FAO/UNEP review actually documents about them, which is revealing in itself: less their day-to-day operations than their cost anatomy. The review's Appendix D exists because feasibility, for retrieval, is an economic question — it tabulates the cost of the Norwegian gear retrieval survey (Table 11), estimated costs for a deepwater pilot retrieval survey (Table 12), and the process and costs of the Baltic retrieval programme conducted by Sweden (Table 13) [5]. The two programmes are the proof that retrieval can be routinised — annual, budgeted, run with the industry rather than against it — and their numbers show why routinised cure still cannot outpace the disease.

Norway: retrieval inside the most regulated fishery

The Norwegian programme operates in the fishery least likely, on paper, to need it. Norwegian net fisheries lose mostly offshore gear at under 0.1 percent of the nets used across the capture sector, and fishers themselves recover an estimated 80–100 percent of what is lost [5]. Reporting of lost nets is mandatory — Norway's procedure is estimated to capture over 80 percent of losses — and controls on net length and soak time apply [5] [5]. Yet the survey exists, annually, at an annual cost thought to be in the order of US$260,000 (Brown et al, 2005) [5]. The justification is in the yield: across 1983–1997 the Norwegian surveys recovered thousands of Greenland halibut gillnets from a fishery that was already reporting, already regulated, already recovering most of its own losses — prompting the Marine Institute's Norman Graham to conclude that despite the mandatory reporting and the controls, "there is clearly still a need to conduct retrieval surveys" [5]. The Norwegian lesson is uncomfortable for prevention optimists: even the best-governed gillnet fishery in the review leaks enough gear, in deep enough water, to justify a standing retrieval institution. Norway's programme also anchors the review's collaborative geography, leading the Norwegian, English, and Irish projects that took retrieval into the Northeast Atlantic deepwater gillnet fishery [5].

The Baltic programme: Sweden's annual operation

Sweden's Baltic retrieval programme is the smaller sibling: annual costs estimated at US$70,000, against Norway's US$260,000 and an estimated US$185,000 for a one-off deepwater pilot retrieval in the Northeast Atlantic [5]. Its unit economics are documented at expedition level: in a 2004 expedition to retrieve lost gear along Sweden's south coast, a stern trawler built for pelagic trawling retrieved lost net at a cost of US$800 per kilometre (Tschernij and Larsson, 2003) [5]. The Baltic programme's process is the one the review chooses to tabulate in full — "Process and costs of the Baltic retrieval programme conducted by Sweden" [5] — and the shared step-cost breakdown shows where the money goes in a programme of this type [5]. Two Baltic specifics stand out. First, disposal of retrieved gear is borne by the port authorities in Sweden and Denmark — the landside end of the operation is a municipal cost, not a vessel cost [5]. Second, the hardware does not travel: retrieval gear development must be assessed region by region, because "Norwegian gear [is] not suitable to Baltic conditions" [5]. Even between neighbouring northern seas, retrieval is a local technology.

The anatomy of a retrieval budget

The step-cost table the review reproduces (for a deepwater pilot survey, on the same template as the Norwegian and Baltic tables) reads as a recipe [5]. It begins onshore and socially: determine the areas of net loss with industry — two person-days of fishermen's labour and two of scientists' — because the programme depends on good communications between industry and researchers, with information collected in advance of the retrieval operation. Then hire the platform: a normal commercial vessel — a medium-sized stern trawler with two net drums — rather than a research vessel, for 10 sea days at over €1,100 per day (12,000 Kr/day), with timing traded against the calendar (cheaper during the summer cod closure, though earlier in the year is favoured). Retrieval gear itself is cheap to buy — sweeps, hooks, and otter doors of special size, approximately €1,000 — after a development effort of two years with three people part-time (two person-months) to establish regional suitability; maintenance runs about €100 a year. Disposal passes to the port authorities. And the operation closes with evaluation: five person-days to weigh and measure the netting recovered and the fish caught in it, attempting a value-versus-cost comparison for the harvest "but [with] many uncertainties," and tracking trends through net retrieval per unit effort (NRPUE) [5]. The striking feature is the shape of the budget: the vessel hire dominates everything; the science, the gear, and the evaluation are rounding errors. Retrieval is a shipping cost with a research wrapper.

Why the comparisons refuse to settle

Set the two programmes' costs against the wider field and the range becomes vertiginous. Wiig (2005) attempted cross-programme comparison and found retrieval costs between US$65 per tonne and US$25,000 per tonne — a spread of nearly three orders of magnitude whose meaning, the review cautions, is far from clear [5]. Depth of water is a logical key determinant, but the benefit side resists the same arithmetic: what a tonne of retrieved gear was worth removing depends on how long it had been fishing, its catching efficiency, and the energy and ecosystem of the environment it sat in [5]. The comparators in the review's cost survey scatter accordingly: in Puget Sound, the Northwest Straits Initiative's 2004–2007 data put derelict net survey and removal at US$4,960 per acre of net removed and pot/trap removal at US$193 per pot; a 2003 north Hawaii expedition retrieved 120 tonnes with two

chartered boats at US$10,000 per day as the major expense; a Polish Sea Fisheries Institute retrieval in 2004 ran ten days for about US$19,000; and a 1995 Memorial University estimate priced an Atlantic-wide ghost gillnet retrieval programme at US$800,000 per year, after US$305,000 for designing and testing practical retrieval equipment [5]. Norway's US$260,000 and Sweden's US$70,000 sit inside that field as what permanence costs: not expeditions, but line items.

That is the two programmes' real significance. They convert retrieval from an event into an institution — budgeted annually, targeted by reported losses and industry consultation, evaluated by catch-per-unit-effort of nets rather than by anecdote [5] [5]. The review's own back-cover summary is blunt that the underlying information "is not comprehensive and does not allow for any global estimates" of ALDFG [5] — but in the Norwegian and Baltic cases, at least, the problem has something no other fishery in this report possesses: a denominator, a budget line, and a crew that goes out every year to close the gap between the two [5].



Exhibit. An autonomous survey vessel used in the Gulf of Mexico to locate lost fishing gear, from the NOAA Marine Debris Program. Source: S0017.

Gear marking, reporting and on-board technology to avoid or locate gear

Prevention, in the review's architecture, is the most effective class of measure — and its first instruments are informational. Every failure documented so far has an information deficit inside it: gear that cannot be traced to an owner (Section 34's abandonment problem), losses nobody reports (Section 33's missing denominators), gear that cannot be found again (Section 36's survey costs). Gear marking, loss reporting, and on-board location technology are the three fixes, and the review treats them as a suite: each works only with the others behind it [5] [5].

Marking: ownership made physical

Gear marking is gaining prominence largely through the fight against IUU fishing, and its logic is accountability: a marked pot or net has an owner, and an owner can be asked questions [5]. In pot and trap fisheries, individual traps could be adequately tagged tomorrow. The paradox arrives with the gear that matters most: "the most frequently lost or otherwise discarded items of gear are unlikely to retain identifying marks" — a tag is exactly what a torn net sheds [5]. To survive loss, identification must be intrinsic: distinct colours or markings woven within multi-strand twines, with further development needed to incorporate such identifiers into monofilament nets and lines (Kiessling, 2003); marking of FADs could likewise effectively prevent their loss, abandonment, or discarding [5]. Hence Recom-

mendation 10: “for the available technology in gear marking to be most effective, identification should be made an intrinsic feature of gear at the point of manufacture,” recorded at the most appropriate level of the supply chain — manufacturer or chandler — since manufacturers do not always sell directly to vessel owners [5]. The costs are real: labelling must be practical and must not restrict the gear’s performance; identifiers added in manufacture will raise prices to customers and add complexity for regulators, who must establish and maintain a database of gear ownership; and chip technology, if it matures, should be applied at the appropriate level within a suite of gear reporting measures [5].

The blind spot is structural. Vessels engaged in IUU fishing “would not easily be incorporated into a gear identification system,” and gear abandoned in an IUU context is highly unlikely to carry traceable marks; removing unmarked gear is a curative act by monitoring, control and surveillance authorities, not prevention [5]. For marking to be preventative rather than forensic, Recommendation 11 requires it be “supported by a comprehensive vessel and gear registration system and port inspection” — gear checked at the quay, against a registry, before it ever fishes [5]. The parallel with Section 29’s vessel registration is exact: a mark without a registry is a label; a registry without inspection is a list.

Reporting: making loss speak

Reporting is the suite’s second leg, and its weakness is human. Direct reporting by the operator of lost gear would enable the most accurate location and identification, “but such reporting is rare” [5]. The reasons are not mysterious: confidentiality about precise fishing locations, and professional pride in admitting gear loss; most reporting in practice comes from other marine users — often recreational users — who come across ALDFG, and facilitated public reporting (online forms, free phone numbers), advertised appropriately and with GPS technology enabling accurate location logging, has proved a useful information source [5]. Design responses follow: many recovery programmes promote a “no blame” approach to lost-gear reporting, as the Northwest Straits Derelict Fishing Gear Removal Project in Washington State does [5]. Where reporting is mandatory, it works: Norway’s mandatory procedure is estimated to capture in excess of 80 percent of losses [5].

The regulatory scaffold exists but covers a sliver of the fleet. Under MARPOL, ships over 400 GT and ships certified to carry 15 or more persons — “a very small fraction of the global fishing fleet” — must carry a garbage management plan and a garbage record book recording the dumping or loss of fishing gear and discharges to reception facilities, open to inspection by administrations, coast guards, fisheries MCS officers, and port state officers [5]. If smaller vessels were brought under the same conditions, the same inspection regime would follow; national administrations such as Iceland’s already issue fishing-vessel guidance on reporting gear lost at sea [5]. National capacities vary widely — Malaysia built a national inventory of net types and gear; Latvia collects loss data through its fisheries data system and fisher questionnaires; Namibia has said it needs technical and financial assistance to build such a system at all [5]. In U.S. waters, foreign fishing vessels in the EEZ operate under an express prohibition on disposing of or abandoning gear and must report accidental loss or emergency jettisoning to the Coast Guard [5].

On-board technology: findability as prevention

The third leg keeps gear from becoming lost in the first place, and finds it fast when it does. Modern GPS gives fishing vessels positional accuracy in the range of ±3 m, and improving seabed imaging lets mobile gear be towed close to the seabed and known obstacles, reducing seabed contact and thus the risk of snagging and loss; for static gear, the same technologies enable more accurate setting, location, and retrieval [5]. Acoustic instruments using combined bottom echoes extend the picture [5]. But the review is blunt about the limit case: the main determinant of successful recovery is the reason for the initial loss — where nets are trawled away, fishers report recovery at sea is virtually impossible (though Danish trawlers that catch lost nets deliver them to harbour, where tags bearing the vessel number identify them — marking closing the loop after all) [5].

Transponders are the growth technology. Already common in large-scale fisheries through satellite vessel tracking for safety and MCS, transponders fitted to gear — marker buoys, floats — improve the ability to locate gear in the water, and large vessels operating mobile gear already use transponders and sensors on the gear itself to aid deployment [5]. The constraint is cost, and it maps onto the fleet’s inequality: transponders are an added cost most likely borne where gear is larger and more expensive, with use in coastal and small-scale fleets limited by cost and technology — and often assumed unnecessary inshore, where landmarks give bearings and cheap GPS suffices [5]. The review’s recommendations push against that gradient. Recommendation 12 reframes the purpose: the “‘findabil-

ity' of fishing gear should be promoted as a preventative measure by enabling fishers to better find gear that is temporarily lost, rather than as a potentially punitive measure post-recovery" — because deliberate abandonment is difficult to prove unless marking (identify the owner) and reporting (confirm the non-compliance) combine against it [5]. Recommendation 13 directs support to affordable transponders and equipment for locating drifting gear and FADs — a use the review calls particularly appropriate — and to GPS technology and assistance for small-scale fishers marking static gear positions, noting GPS is becoming increasingly affordable and doubles as navigation equipment that can also help mobile gear users avoid agreed static-gear zones [5]. Even so, in large-scale fisheries the extension of transponder use may still require mandatory measures to ensure the right fisheries adopt them, with transponders of suitable type and position to aid rapid recovery [5].

The through-line is the review's refrain from Section 34: information measures fail in isolation. A mark needs a registry and an inspector; a report needs a no-blame channel and a recovery programme listening at the other end; a transponder needs a fisher who can afford it and a reason to switch it on. Assembled, they convert gear from anonymous debris-in-waiting into tracked property — the precondition for everything the mitigation measures of Section 39 then attempt [5] [5] [5].

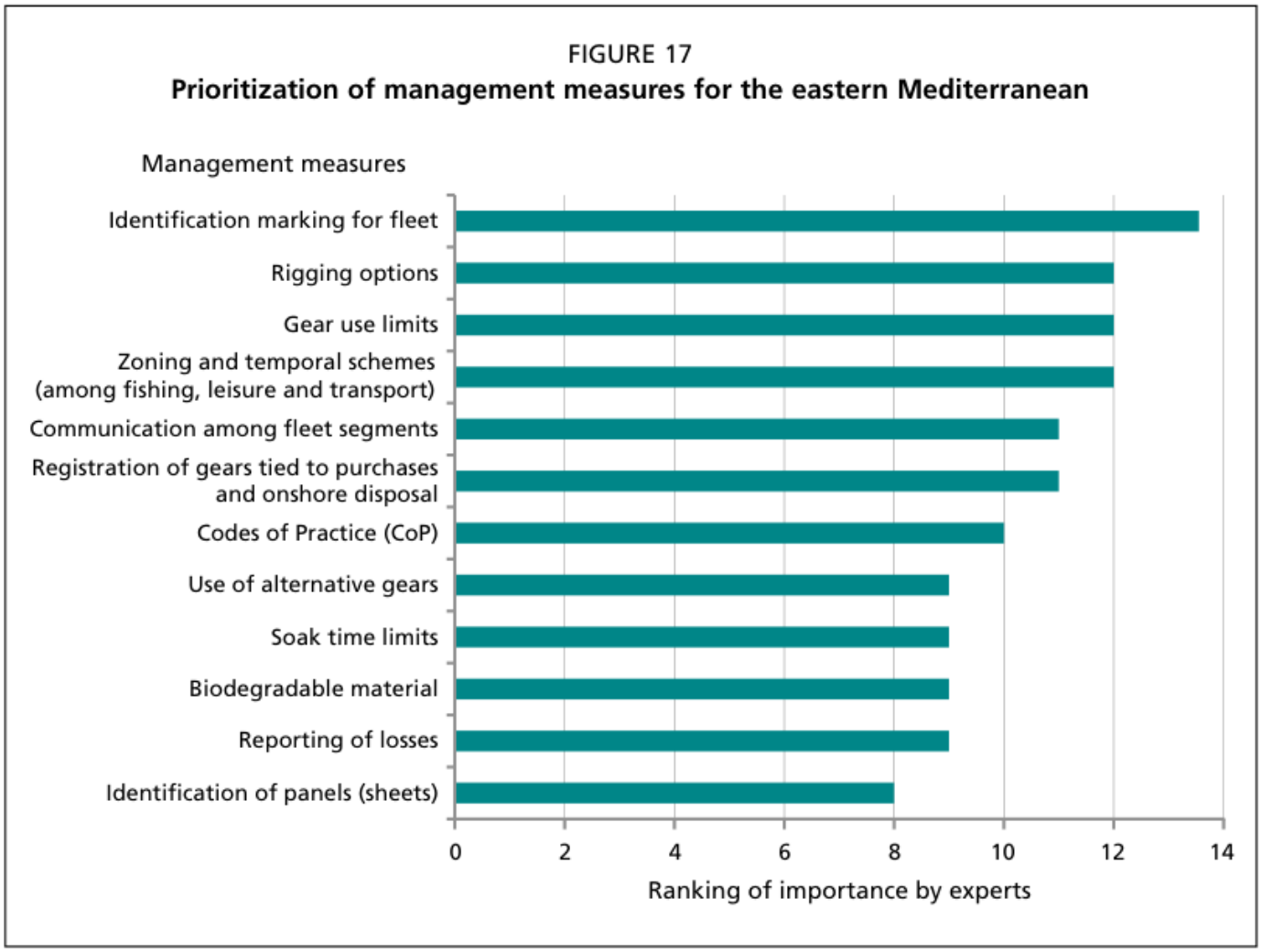


Exhibit. FIGURE 17 Prioritization of management measures for the eastern Mediterranean Management measures Identification marking for fleet Rigging options Gear use limits Zoning and temporal schemes (among fishing, leisure and transport). Source: S0017.

Reduced ghost catches through biodegradable nets and pots

Mitigation is the middle class of ALDFG measures: it accepts that gear will be lost — Sections 33 and 34 established that some loss is inevitable — and designs the gear to stop fishing once it is. The review divides the idea in two. The first branch shortens the gear's own catching life: sacrificial components that rot on schedule, opening the trap or collapsing the panel. The second, treated in the review's adjacent heading, reduces the catches of incidental species while gear is still attended. Both run into the same obstacle, stated in the executive summary: mitigation measures "are limited in their extent and application as many may increase costs through reduced effectiveness of gear

or higher gear prices," so innovative materials have developed slowly and "the return to biodegradable netting by the industry has been very limited" [5]. There is a historical irony in that phrase: fishing gear was biodegradable for millennia by default. Synthetic persistence created ghost fishing; mitigation is the attempt to engineer the old mortality back in, on a timer.

Rot cords: the trap that opens itself

The working technology is not a biodegradable net but a degradable weak link. A number of shellfish fisheries require degradable escape panels in traps: Florida's spiny lobster fishery has had such a requirement since 1982 (Matthews and Donahue, 1996) [5]. The Bering Sea fisheries management plan for king and tanner crab states the design intent verbatim: "an escape mechanism is required on all pots; this mechanism will terminate a pot's catching and holding ability in case the pot is lost" [5]. Canada's recreational fishing traps require features "to ensure that if the trap is lost, the section secured by the cord will rot, allowing captive crabs to escape and to prevent the trap from continuing to fish" (DFO, 2007), and the Pacific Region's 2008 Integrated Fisheries Management Plan for crab by traps carries biodegradable escape mechanism requirements [5]. The rot cord is an elegant inversion of Section 35's self-baiting cycle: the pot's entrance is held shut by the one component guaranteed to fail, so the trap's fishing life is capped by the cord's rotting life, not by the trap's structural life.

Then the compliance finding, which converts the technology story into a governance story. Despite the requirements, trap recovery programmes have identified that significant proportions of recovered traps lack the requisite rot cord — in Port Susan, Washington State, 40 percent of commercial traps recovered did not have rot cords (Natural Resources Consultants, Inc., 2007) [5]. The review draws the conclusion plainly: this "highlights the importance of monitoring and enforcement to support any mitigation measures that are implemented" [5]. A mitigation measure is a regulation wearing a technology's clothes; uninspected, four in ten fishers in a documented recovery sample simply did not rig it.

Biodegradable materials: bags before nets

Whole-gear biodegradation lags the weak-link approach, and the review notes the use of biodegradable materials "is less evident in net fisheries" [5]. What exists is at the periphery of the gear: efforts to develop biodegradable and oxy-degradable plastics for the fishing industry, led by the Australian and New Zealand Environment Conservation Council's national approach to biodegradable materials in bait bag manufacture (Kiessling, 2003); a biodegradable bait bag has been developed for recreational fishers in Queensland and is likely to be introduced in Western Australia, and trials have begun on biodegradable ice bags [5]. The pattern is telling — biodegradation enters through disposable accessories (bags) rather than through the load-bearing gear itself, because a net that degrades on schedule is a net whose working life is also capped, and the cost of technical gear measures falls directly on fishers through higher gear prices, reduced catch rates, and reduced handling efficiency [5]. Recall Section 36's expert verdict: in both the Baltic and the Western Channel workshops, fishers considered biodegradable nets unacceptable [5]. The constituency for a self-destructing asset is, understandably, thin.

The adjacent branch: catching fewer of the wrong animals

The review's neighbouring mitigation heading concerns incidental catch — reducing what attended and lost gear does to non-target species, cetaceans above all. Acoustic pingers have been effective in reducing cetacean bycatch since the 1990s, but their relatively high cost has hindered wider implementation, alongside concerns over reliability and whether they cause long-term noise pollution in the animals' environment [5]. The alternative direction is to make the net itself more detectable rather than noisier: Aquatec Group Ltd's Passive Porpoise Deterrent, winner of the UK prize in WWF's International Smart Gear Competition, uses resonant acoustic reflectors that increase the net's "acoustic visibility" — when a porpoise clicks, the reflectors return a stronger echo, appearing to the animal as much larger objects and alerting it to danger, in a less complicated way than pingers [5]. Trials are likewise continuing on net materials that increase sound reflectivity to reduce bycatch of cetaceans and other non-target species, with initiatives such as the Smart Gear Competition encouraging the innovation the market has been slow to supply [5].

Why the timer matters

The stakes are Section 35's, quantified once more: gillnets and traps share "a common tendency to continue fishing with a declining catch as the gear becomes less effective," over durations that vary widely with local conditions; overall ALDFG catch rates vary too greatly for a global estimate to be meaningful, but Sancho et al (2003) considered lost tangle nets to catch around 5 percent of the total commercial catch [5]. Ghost fishing, in other words, already decays — the biography of Section 35 is a decline curve. What biodegradation buys is a shorter tail: a rot cord does not prevent the loss, the entrapment, or the first weeks of catching; it forecloses the months and years. Lost trawls and longlines, by contrast, rarely ghost fish at all — their harms are smothering and entanglement instead — which is why mitigation research concentrates on the two gear families that keep working unattended [5]. The measure's limits are the report's recurring ones: it costs the fisher, it needs inspection to exist in practice, and in its whole-gear form it remains, for nets, mostly a bait-bag technology awaiting a material that is strong on the fisher's schedule and weak on the ocean's [5] [5].



Exhibit. FIGURE 11 Crab trap with rot cord Source: Fisheries and Oceans, Canada. Source: S0017.

MARPOL Annex V and international recognition of the ALDFG problem

ALDFG is transboundary by construction. Section 33's record showed Korean waters littered with gear of mixed origin, Japanese recoveries "mainly of non-Japanese origin," and Australian beaches collecting the fleets of other nations — a problem no state's programme can close alone. The FAO/UNEP review opens with exactly that premise: "the transboundary nature of the problem means that regional and international cooperation to prevent ALDFG is vital," and the recognition is institutionalised in a large number of international organisations and agreements that now focus specifically on ALDFG, above provisions of UNCLOS (Articles 1, 192, 194, 197, 207, 211 and 216–218) requiring nations to combat marine debris more generally [5].

The mandate stack: UNGA resolutions, 2004–2007

The United Nations General Assembly built the mandate in a sequence of resolutions (detailed in the review's Appendix A). Resolution A/RES/59/25 (2004) called upon States, FAO, IMO, UNEP — in particular its Regional Seas Programme — regional fisheries management organisations and other bodies that had not yet done so to take action on lost or abandoned fishing gear and related marine debris, through collection of data on gear loss, economic costs to fisheries and other sectors, and impacts on marine ecosystems [5]. Resolution A/RES/60/30 (2006, Oceans and the Law of the Sea) noted the lack of information on marine debris, encouraged further studies, urged States to integrate marine debris into national waste-management strategies, and invited the IMO to review Annex V to MARPOL [5]. A/RES/60/31 (2006) pressed for better information and collaboration, calling on States and organisations such as FAO to address lost or otherwise abandoned gear through data collection and through both preventative and curative measures [5]. A/RES/61/222 (2007) repeated the waste-strategy urging and welcomed the IMO's Annex V review; A/RES/61/105 (2007) reaffirmed the importance of ALDFG and encouraged FAO's Committee on Fisheries (COFI) to consider the issue at its 2007 meeting [5]. The sixth UN Open-ended Informal Consultative Process on Oceans and the Law of the Sea (UNICPOLOS, New York, June 2005) had already taken up marine litter and abandoned fishing gear at the Assembly's request [5].

The review's stakeholder analysis maps the division of labour this creates: the General Assembly provides the mandate, with high potential influence "if Resolutions are acted upon"; FAO is looked to for leadership on fisheries and can feed solutions back to member countries through COFI; the IMO's influence is high, especially during the Annex V review; UNEP's Regional Seas work is "the only systematic work on the marine litter problem at the regional and supra-regional level"; regional fishery bodies can legislate or encourage coherent government action — and the catching sector itself must be encouraged, voluntarily or through legislation, to buy in, since enforcement problems dog any purely legal approach [5].

Annex V: the prohibition and its seam

The operative treaty is the IMO's MARPOL Annex V — prevention of pollution by garbage from ships — which entered into force on 31 December 1988 and has been amended twice since [5]. Annex V completely prohibits certain discharges of ship-generated garbage, plastics above all, and for other garbage specifies the distances from land and manner of disposal; in designated areas, stricter requirements apply where the general rules can be shown ineffective, provided adequate reception facilities exist [5]. The disposal table is a distance gradient — inside 3 nautical miles, total restriction; plastics are illegal to dispose of at every distance [5]. For fishing gear the text is precise, and so is its limit: "the prohibition of the discharge of plastics specifically prohibits the discharge of synthetic fishing nets; however, the Annex does not apply to the accidental loss of such nets, provided that all reasonable precautions have been taken to prevent such loss" [5]. Deliberate discarding is banned; accidental loss is carved out. Most of ALDFG lives in the carve-out.

The Annex applies to all vessel types including fishing vessels of all sizes; Regulation 9 requires ships of 400 GT and over to keep records that include reporting the discharge, escape, or accidental loss of garbage — synthetic fishing material expressly included — under Regulation 6, together with the circumstances and reasons for the loss [5]. And the Annex V Guidelines reach toward Section 38's agenda, calling for fisheries managers to utilise fishing gear identification systems providing vessel name, registration number and nationality, and encouraging governments to consider developing technology for more effective gear identification [5]. The treaty root of gear marking is here.

Regulation 7: the reception-facility gap

Annex V's second operative promise is landside. Regulation 7 requires that "the Government of each Party to the Convention undertakes to ensure the provision of facilities at ports and terminals for the reception of garbage, without causing undue delay to ships, and according to the needs of the ships using them" [5]. International recognition has since focused on the gap between that undertaking and small-port reality: scale and capacity problems have prevented adequate reception facilities at many small ports and harbours, which are disproportionately fishing harbours [5]. For Pacific Island States, where 90 percent of fishing operations are foreign, a lack of port reception facilities led the South Pacific Regional Environment Programme to identify solid waste management as the number one issue facing the Pacific Island States (Kiessling, 2004) [5]. Even in the United States, the NRC (2008) found the Coast Guard's Certificate of Adequacy programme certifies that ports are capable of accepting garbage or have

providers on call — not that they actually accept it — and crews willing to pay for disposal "are not always able to secure these services, even from those ports with COAs" [5]. FAO and IMO's joint Bay of Bengal initiative first quantified and categorised the waste problem in fishery harbours, then developed guidelines for fishery harbour operation; the Chennai Declaration from FAO's 1999 expert consultation recommended charging tariffs for harbour services to fund their management — though the review notes any additional tariff on waste reception may act as a disincentive compared to burning or dumping at no immediate cost [5].

Practice shows the fee design deciding the behaviour. Where disposal costs are folded into general port charges or provided free, the incentive to discard at sea collapses: in the Baltic cod gillnet fishery, end-of-life nets go into harbour containers with disposal already inside port fees, leaving "little economic incentive for fishers to deliberately discard nets at sea"; in Greece, net fishers strip old netting from the ropes and use the municipal tip at no cost, with the same result (Brown et al, 2005) [5]. For recreational monofilament — individually trivial, collectively significant — several U.S. states encourage safe disposal through pier-side reception facilities, with education rather than enforcement as the route [5].

Incentives beyond compliance

Two programmes in the review convert reception into reward. The Republic of Korea's Waste Fishing Gear Buy-back Project, implemented since 2003 by the Ministry of Maritime Affairs and Fisheries, purchases waste fishing gear — nets, traps, lines, floats deposited in the sea and on the seabed — that fishers bring back to port, issuing large hard-wearing bags so litter collected during fishing can be deposited on the quayside instead of being collected and thrown back, as fishers previously did; prompted in part by the sinking of a passenger ferry after entanglement in discarded gear, it is reported as highly effective for recovery and disposal, though no cost-benefit analysis has been undertaken for a scheme dependent on significant public funding [5]. The North Sea's Fishing-for-Litter project, started in March 2000 by the Dutch Government's North Sea Directorate with the Dutch Fisheries Association, has fishers bring ashore litter trawled up during normal operations; by the end of its initial three years, 54 boats in four countries had collected 450 tonnes [5].

Recognition complete; implementation open

Later assessments keep building on the same scaffold. Breaking the Plastic Wave lists as enabling conditions for its ALDFG intervention: more research into the effectiveness and feasibility of measures, and increased international cooperation — harmonised reporting of gear production and losses, codes of practice, and communication protocols, "such as the newly adopted FAO guidelines in the PSMA and voluntary guidelines on the marking of fishing gear" — plus the inclusion of gear loss in fishery certification bodies' sustainability criteria [12]. Its shipping-litter analysis sits under the same Annex: deliberate dumping of plastic from vessels is illegal under international law with some exemptions (MARPOL Annex V), yet the practice is believed widespread — EU waters alone see an estimated 54,000 to 67,000 metric tons of shipping litter annually, 35 percent of maritime sources, with ALDFG the other 65 percent [12]. The review's own conclusions state the position exactly: sound international policy, legislative, and regulatory regimes have been developed and are in place (MARPOL Annex V the example), but "there are significant deficiencies in the implementation and enforcement of these regimes," and the challenge depends substantially on changing human behaviour, not only on technological fixes [5]. Recognition, in short, was the work of 2004–2007. It is finished. The remaining sections of this report — dark fleets, shadow fleets, sunken warships — are, in their different ways, all about what happens in the space between a rule and its enforcement [5] [5].

Dark Fleets: Vessels That Hide

Satellite mapping reveals extensive industrial activity at sea

The report now turns from objects the sea hides to activity people hide. Paolo et al's study in Nature (January 2024) — a Global Fishing Watch-led collaboration — begins from an asymmetry: the "blue economy" is worth 1.5–2.5 trillion dollars and growing faster than the overall global economy, more than a billion people depend on the ocean for their primary source of food, 260 million are employed by marine fisheries alone, about 80% of traded goods move by sea, and nearly 30% of the world's oil is produced offshore — yet this industrialisation is "poorly documented" [17]. On land, maps exist for almost every road and datasets are being built for every human-made structure; in the ocean, many vessels do not broadcast their location or are not detected by public monitoring systems, and offshore

infrastructure information is often held private [17]. A third of fish stocks are operated beyond biologically sustainable levels and an estimated 30–50% of critical marine habitats have been lost — under an observational regime the authors set out to replace [17].

Why the public record was blind

Existing tools each see a slice. Vessel monitoring systems (VMS) used in fishing are proprietary, limiting the ability to map and compare across regions [17]. Public mapping has centred on AIS, which broadcasts vessel coordinates for maritime safety and can reveal identities, owners, and fishing activities — but not all vessels are required to carry AIS, as regulations vary by country, vessel size, and activity; vessels engaged in illicit activities often turn off their transponders or manipulate the locations they broadcast; the largest recent cases of illegal fishing and forced labour involved fleets that mostly did not use AIS; and large blind spots emerge along coasts where satellite reception is poor, while terrestrial AIS data can be restricted by national governments [17]. The paper's term for the invisible remainder is careful: vessels not visible in publicly accessible AIS data are "not publicly tracked" — "also sometimes referred to as 'dark vessels'" [17]. Not publicly tracked is not a synonym for illegal; it is a statement about the record, and the study's discipline in keeping the two apart runs through everything that follows.

The method: radar, optics, and 53 billion positions

The team analysed 2 petabytes of satellite imagery from 2017–2021, covering more than 15% of the ocean — the coastal fraction in which more than 75% of industrial activity is concentrated [17]. Three deep convolutional neural networks did the seeing: one to identify objects (>97% accuracy) and estimate their lengths (R^2 of 0.84), one to classify offshore infrastructure into oil, wind, and other objects (>98% accuracy), and one to classify vessels as fishing or non-fishing (>90% accuracy); in total more than 67 million image tiles were classified, combining dual-polarisation synthetic-aperture radar (SAR) imagery from Sentinel-1 with optical imagery from Sentinel-2 [17]. SAR is the key instrument because it sees regardless of cloud or darkness and does not depend on a vessel's cooperation: its resolution captures most objects larger than 15 m, with detection rates above 70% for 25-m vessels and above 90% for vessels of 50 m and larger — though recall decays steeply below 25 m, and Sentinel-1's coverage embraces most coastal waters but does not sample most of the open ocean [17] [17]. Against those detections the authors matched 53 billion AIS GPS positions (from providers ORBCOMM and Spire, through Global Fishing Watch's pipeline) to determine whether each detected vessel was publicly tracked [17] [17].

The headline: three-quarters of industrial fishing is dark

During 2017–2021, on average about 63,300 vessel occurrences were detected at any given moment, roughly half of them (42–49%) fishing vessels, from 23.1 million vessel detections [17]. The central finding: about three-quarters (72–76%) of globally mapped industrial fishing did not appear in public monitoring systems, compared with one-quarter (21–30%) for other vessel activities [17]. Correcting that blindness rewrites the geography of fishing. AIS data alone wrongly suggest Europe and Asia have comparable fishing activity, with other continents at less than a fifth; the satellite map shows Asia dominating industrial fishing with 70% of all fishing vessel detections, and nearly 30% of all mapped fishing vessels concentrated in China's EEZ alone [17]. AIS suggests European Mediterranean countries log more than ten times the fishing hours of African countries in their EEZs; the mapping shows fishing vessel detections fairly balanced between the Mediterranean's northern and southern shores [17]. Activity overall is widespread but concentrated: a vessel was detected at least once in 84% of the 0.1° study cells, yet half of all vessel activity sat in less than 3% of them; 86% of fishing and 75% of non-fishing activity occurred in waters less than 200 m deep — 7% of the ocean; and by continent, Asia held about 67% of all vessel activity, followed by Europe (12%), North America (7%), Africa (7%), South America (4%), and Australia (2%) [17].

What the map shows that AIS could not

High densities of previously invisible activity appear in Indonesia, South Asia, Southeast Asia, and along the northern and western coasts of Africa [17]. The map resolves behaviour, not just presence: between Tunisia and Sicily, tracked and untracked fishing vessels aggregate along ocean banks and canyon edges — the signature of bottom trawling — and off Bangladesh, where almost no vessels are publicly tracked and no public fishing maps exist, vessels follow bathymetric contours and submarine canyons radiating from the Ganges Delta [17]. Untracked vessels were detected inside marine protected areas — including the Galápagos Marine Reserve and the Great Barrier Reef

Marine Park, two of the most iconic and well-monitored MPAs in the world, which showed on average more than 5 and more than 20 such vessels per week respectively [17]. And the study's most striking hotspot corrects the literature's geography: previous work had shown substantial illicit activity in the eastern waters of North Korea, but the global mapping shows most of the undisclosed fishing occurred in the western part of the Korean Peninsula — from 2017 to 2019 the highest density of fishing vessels in the world, about 40 vessels per 1,000 km² — a fleet whose rhythms (a May peak coinciding with China's domestic fishing moratorium, an 85% collapse when North Korea shut its borders during COVID-19) the next section examines in full [17] [17].

Non-fishing activity is far more legible: most transport and energy vessels broadcast, about a quarter missing from public systems, with Asia holding 65% of detections — though many of its non-broadcasting vessels operated where satellite AIS reception is poor, so they may have transmitted without being trackable; all other continents showed less than 20% of transport and energy vessels not publicly trackable [17]. The infrastructure census extends the picture: fixed oil and wind structures grew across 2017–2021, with China leading wind development, and in 2021 nearly 4,140,000 hours of vessel activity were associated with oil platforms and about 792,500 hours with wind turbines [17].

The study's conclusion is about capability. Fishing is "by far the ocean industry with the most activity that is not public"; with the freely available dataset and technology, hotspots of potentially illegal activity can now be shown, and industrial fishing vessels identified encroaching on artisanal grounds or other countries' EEZs, at global scale and accessible to any nation [17]. The maps can now include all vessels — not just those based on AIS tracking, which misses about three-quarters of large vessels — at much higher resolution than EEZs or statistical reporting areas [17]. For the sections that follow, the map is the new evidentiary baseline: the question is no longer whether the dark fleets can be seen, but what they are doing [17] [17].

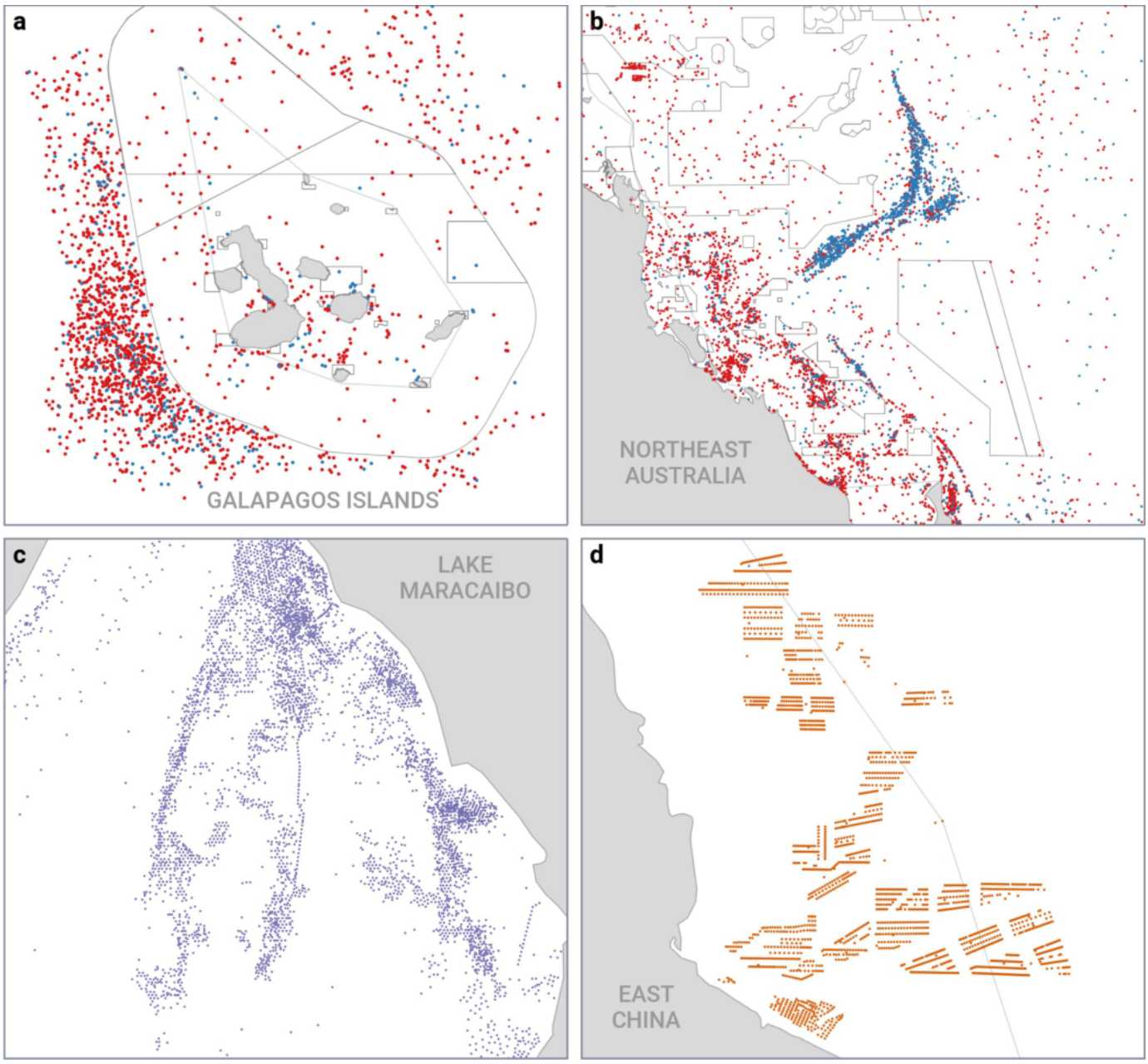


Exhibit. Fig. 7 Satellite imagery-based detection allows monitoring at local scale. Source: S0021.

Illuminating dark fishing fleets in North Korea

If Section 41 drew the global map, this section is its ground truth — the study the map's North Korea hotspot rests on, and the report's point of closest contact between "dark vessels" and literal ghost ships. Park et al (2020, Science Advances) studied the waters between North Korea, South Korea, Japan and Russia, where geopolitical tensions and disputed boundaries create "a vacuum of shared data and management" [18]. The same stock of Todarodes pacificus — Japanese flying squid, South Korea's top seafood by production value — is targeted by several fleets, including China's distant-water fleet, which has fished in North Korean waters since 2004 but whose activity and catches are only intermittently published, and not since 2016; meanwhile reported catches in South Korean and Japanese waters have dropped by 80 and 82% respectively since 2003, and the stock cannot be assessed [18]. Dark fleets — "vessels that do not appear in public monitoring systems" — epitomise the IUU problem here; many dark vessels broadcast on country-mandated VMS, but those data are tightly guarded, useless for third-party oversight or transboundary management [18].

Four satellites, one dark fleet

The study's methodological signature is fusion: four satellite technologies, each compensating for the others' blindness. The team obtained 22 days of 3-m resolution optical imagery from Planet's PlanetScope constellation covering most of the claimed North Korean EEZ in 2017 and 2018, and trained a convolutional neural network to identify

pair trawlers — the largest proportion of foreign vessels in the region, with a distinctive fishing pattern [18]. Planet's 0.72-m SkySat satellites were then tasked to verify the identifications at high resolution, and SAR imagery from three satellites verified fleet location and size — radar sees the large metallic industrial vessels but not the small wooden ones [18] [18]. The wooden fleet was caught by a fourth instrument: VIIRS nighttime imagery. Chinese "lighting vessels" — 55- to 60-m ships that lure squid with bright lights — carry up to 700 incandescent bulbs generating over 1000 lux, "equivalent to the light of some football stadiums," while North Korean lighting vessels are 10- to 20-m wooden boats with only 5 to 20 bulbs, so brightness stratifies the fleets from orbit: the brightest (>1000 nW/cm²/sr) mostly Chinese, the dimmest (<50) mostly North Korean [18] [18]. Finally, the minority of vessels that did broadcast AIS supplied identity and origin, cross-checked against South Korean Coast Guard inspections [18].

What the fusion found

At least 796 distinct pair trawlers operated in North Korean waters in 2017 and at least 588 in 2018 — all counts are minima, thinned by weather and coverage — plus a minimum of 108 Chinese-origin lighting vessels in 2017 and 130 in 2018: over 900 industrial vessels originating from China in 2017 and over 700 in 2018 [18] [19]. Only a fraction broadcast AIS, but those signals, matched to imagery for 140 vessels, demonstrate the fleet originated from Chinese ports (one tracked vessel to Shandong province) and fished in Chinese waters, corroborating the Coast Guard's inspections [18]. Estimated effort: 91,400 fishing days in 2017 (82,600 by pair trawlers, 8,800 by lighting vessels) and 67,300 in 2018 — figures that are 70% and 91% respectively of what Coast Guard crossing counts imply [18]. The estimated catch for 2017–2018 was more than 160,000 metric tons of Pacific flying squid, worth US$440 million [19].

The legal analysis is a fork with no clean exit. The fishing is "likely in violation of United Nations sanctions" — Security Council Resolutions 2371, 2375 and 2397, which from September 2017 barred the fisheries-cooperation arrangements under which such access had operated [19]. The vessels often carry no appropriate papers and may be "three-no boats" — no registration, no flag, no licence — operating outside official Chinese authority; if they lack approval from both the Chinese and North Korean governments, they are fishing illegally, since Chinese regulations require ministerial approval to fish in foreign waters and UNCLOS grants the coastal state sovereign rights over living resources in its waters [18]. But "if they are operating with approval from either or both governments, those state(s) are in violation of UN Security Council sanctions" — a scenario the Chinese government has repeatedly refuted [18]. Meanwhile the management asymmetry is stark: to prevent overfishing South Korea sets a total allowable catch for squid, limits jiggers' lighting power, bans pair trawling, and permits fewer than 40 small trawlers — while the dark fleet uses brighter lighting power, pair trawling, and far greater numbers on the same stock, whose jigger CPUEs are declining and whose larval densities have fallen drastically since 2003 [18]. A political stalemate rooted in sovereignty conflicts has prevented any joint regional management [18].

Displacement: the wooden fleet and the ghost boats

The second fleet is the tragedy's engine. In 2018 about 3,000 small North Korean squid vessels fished in the Russian EEZ — most of them unauthorised, since Russia has authorised fewer than 100 North Korean boats since 2014 and granted no permits at all in 2017 [18] [18]. Their fishing days in Russian waters rose every year from 39,000 in 2015 to 222,000 in 2018 [18]. The study reads this as displacement: competition from the larger foreign trawlers pushes the small wooden boats out of their own waters, and the squid's southward migration through the Russian EEZ in the last trimester of the year gives the North Koreans a brief window to fish before the foreign fleets deplete the stock [18] [19]. But the boats are "severely underequipped for the long-distance travel required to reach Russian fishing grounds": between 2014 and 2018, 505 North Korean boats washed ashore on Japanese coasts, incidents that frequently involve starvation and deaths, in vessels containing no crew or, in some cases, only human remains — the "ghost boats" of the Japanese press — and local media report that many fishing villages on North Korea's eastern coast have been coined "widows' villages" [18] [19].

2019: scrutiny changes nothing

Global Fishing Watch's 2019 follow-up analysis, using the same methods, found the activity continuing "at a comparable level despite growing scrutiny": close to 800 vessels originating from China (790 pair trawlers and lighting vessels combined, between 2017's and 2018's totals), an estimated 82,200 satellite-based fishing days (about 61% of the 134,000 days implied by Coast Guard observations), and an estimated catch of 76,700 metric tons of raw squid — about US$240 million, bringing the accumulated catch since the sanctions took effect in September 2017 to an

estimated US$560 million [19] [19] [19]. In 2019 that Chinese-origin catch alone represented about 80% of the total reported domestic squid catch of South Korea and Japan combined [19]. North Korean activity in Russian waters returned to 2017 levels with over 2,000 vessels, and across five years the number of ghost boats recorded on Japanese shores is highly correlated with the number of North Korean vessels fishing in Russian waters — the wash-ups are, statistically, the displaced fleet's shadow [19] [19].

The authors close on method as politics: combining satellite technologies can reveal dark fleets at scales now achievable by small, independent research groups, "filling a major gap in the management of transboundary fisheries" in an industry long dominated by "a culture of confidentiality and concealment" [18]. The next section widens the lens again — from one sea's dark fleet to the global geography of unseen vessels [19].

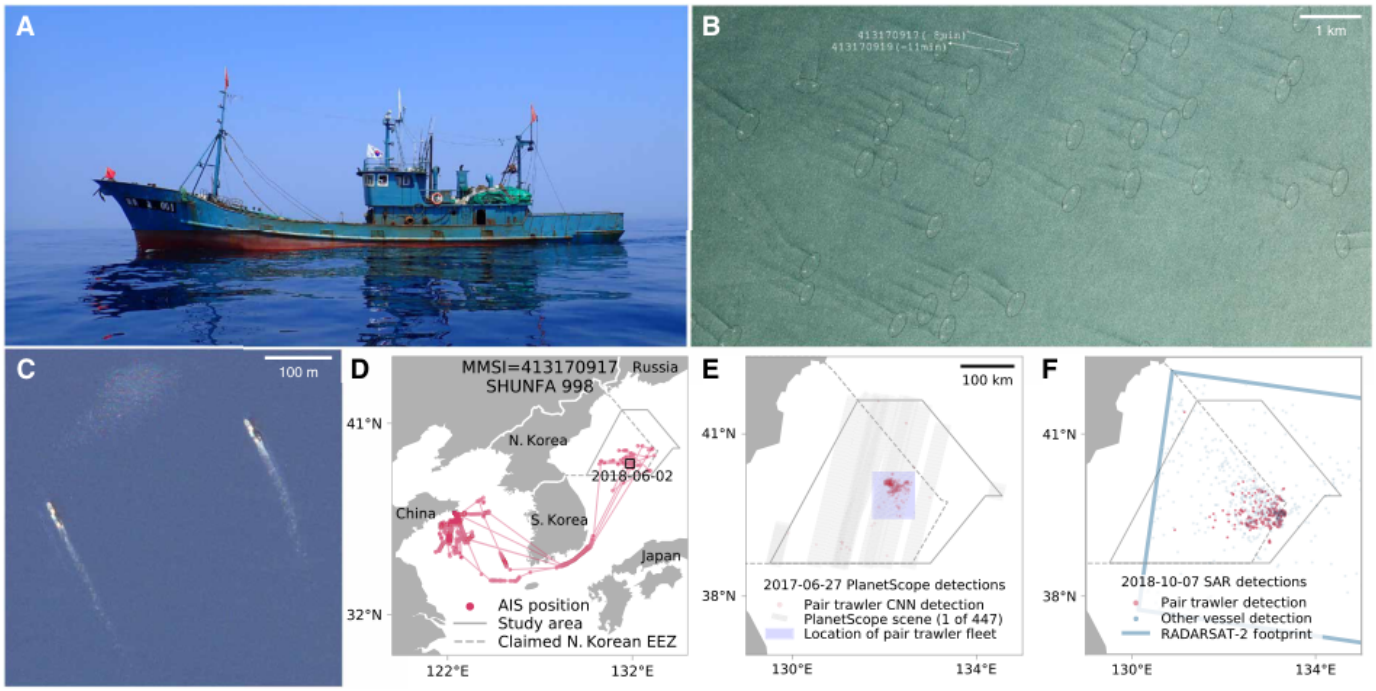


Exhibit. Fishing activity originating from Chinese ports in North Korean waters during 2017 and 2018, documented in Park et al. (2020). Source: S0018.

Hot spots of unseen fishing vessels

Sections 41 and 42 dealt with vessels that never appear. Welch et al (2022, Science Advances) study a subtler invisibility: vessels that carry AIS, broadcast normally — and then go quiet. AIS, created as a collision-avoidance tool, had revolutionised public monitoring of fishing fleets, but "the utility of AIS as a monitoring tool is impeded by vessels intentionally disabling their AIS devices," and until this study no one had quantified the scale, spatial footprint, or drivers of intentional disabling [20]. The stakes are the IUU economy itself: global losses from illegal, unreported and unregulated fishing are valued at US$10–25 billion annually, one in every five wild-caught fish is harvested illegally or unreported, and labour abuses including forced labour and trafficking are linked to the same fleets [20]. Disabling can hide unauthorised fishing inside EEZs and marine protected areas, and unauthorised transshipments — at-sea transfers of catch to refrigerated cargo vessels that, poorly monitored, can launder illegally caught seafood into the market [20]. But disabling is not per se evidence of crime: AIS is not universally mandated, vessels are not always required to keep devices on, some fleets are exempted because fishing locations are confidential, and vessels may go dark to avoid hostile interactions in piracy-prone waters [20]. The study's discipline is in its adjective — these are suspected disabling events.

Building the dataset of gaps

The authors present "the first global dataset and analysis of suspected AIS disabling in commercial fisheries": a rule-based classification model identifies which gaps in AIS transmission are likely intentional — as opposed to the reception failures that plague satellite AIS — and the analysis then maps where the obscured activity sits, by location, flag state, and gear type [20]. Only areas with sufficient satellite reception quality (more than 10 positions per day) are shown, so that silence can be read as choice rather than coverage [20]. The yield is substantial: the suspec-

ted disabling dataset "illuminates an additional 4.9 M hours (6%) of fishing vessel activity by vessels with AIS," on top of a baseline in which an estimated 50 to 80% of fishing operations more than 100 nautical miles from shore are conducted by AIS-equipped vessels and a majority of high-seas effort broadcasts [20].

The hot spot geography

The disabling hot spots cluster where rich fishing grounds meet limited management oversight — "a formula that may embolden vessels to engage in IUU activities" [20]. Three stand out, all previously flagged as areas of IUU concern. In the **Northwest Pacific** — the same waters where Park et al exposed the North Korean dark fleet — one cited study found that more than half of the vessels flagged to China operating there had suspected disabling events, and estimated that more than half of their chub mackerel catch was unreported [20]. On the **high seas adjacent to Argentina's EEZ**, transshipment is potentially subject to fewer regulations than in national waters, and it is hypothesised that non-broadcasting "dark" fleets fish without authorisation inside Argentine waters and off-load catch to high-seas refrigerated cargo vessels, avoiding port calls that might expose illegal catch [20]. Off **West Africa**, where illegal fishing costs up to US$2.3 billion a year — including authorised incursions of industrial fleets into inshore national waters — disabling that degrades monitoring is "particularly alarming" [20]. These three hot spots are largely products of distant-water fleets operating in foreign and international waters, whose flag-state VMS data local authorities are unlikely ever to see [20].

The fourth hot spot is the control case: **U.S. waters offshore of Alaska**, caused by disabling in U.S. trawlers — a fleet monitored by U.S. authorities via VMS, in "one of the most intensely managed regions in the world" [20]. There, disabling likely indicates something narrower: hiding high-quality fishing grounds from competitors, since VMS — unlike AIS — cannot legally be disabled and cannot be received by competitors [20]. Same signal, opposite meaning: the hot spot map needs the governance layer before it can be read as an illegality map.

Behaviour at the boundary

The spatial signature is consistent: disabling events were concentrated in waters adjacent to EEZ boundaries — most likely adjacent to foreign and disputed EEZs — while fishing itself occurred primarily inshore or on the high seas outside EEZ boundaries, "suggesting that some of these events may indicate vessels disabling their transponders before entering unauthorized locations to fish illegally" [20]. The disproportionate concentration near contested boundaries suggests political conflicts create "managerial blind spots, which could make IUU activities more profitable and less likely to be prosecuted" — though opaque fisheries access agreements obscure which events warrant attention [20]. Disabling was also common where transshipment activity is high, while fishing occurred where it is lower — "a desire to have these possibly unauthorized transfers obscured from public view" — a pattern particularly pronounced for squid jiggers, a gear type with heavy loitering behaviour and relatively little management oversight [20].

Three events that show what a gap can contain

The paper grounds the statistics in cases. The Oyang 77 disabled its AIS device nine times adjacent to the Argentine EEZ before being apprehended by the coast guard and charged with illegally fishing within Argentina's waters [20]. The Lu Rong Yuan Yu 668 disabled its device twice on the high seas beyond the EEZ and was likewise captured for illegally fishing inside it [20]. And one transshipment during a disabling event was made visible by accident: a gear-mounted AIS device — presumably fitted to reduce loss of expensive gear — kept broadcasting while the vessel's own AIS was off, revealing an encounter with a refrigerated cargo vessel and a transshipment the flag state confirmed; in that instance both vessels were properly authorised to transship, which "reveals that transshipment events do take place during disabling events" without itself proving an offence [20]. The legality of any given event is genuinely hard to discern — AIS rules vary by region, RFMO, flag state and vessel size, an "intricate patchwork" — but the study demonstrates strong links between suspected disabling and the two behaviours of IUU concern: potentially unauthorised border crossings and transshipments [20].

The closing inversion is the section's epitaph: goals to eliminate IUU fishing have been unattainable partly for want of data on who is involved and where and why — and "it is the absence of AIS data itself that contains a wealth of information" [20]. What predicts a vessel's decision to go dark — the drivers — is the next section's subject [20].

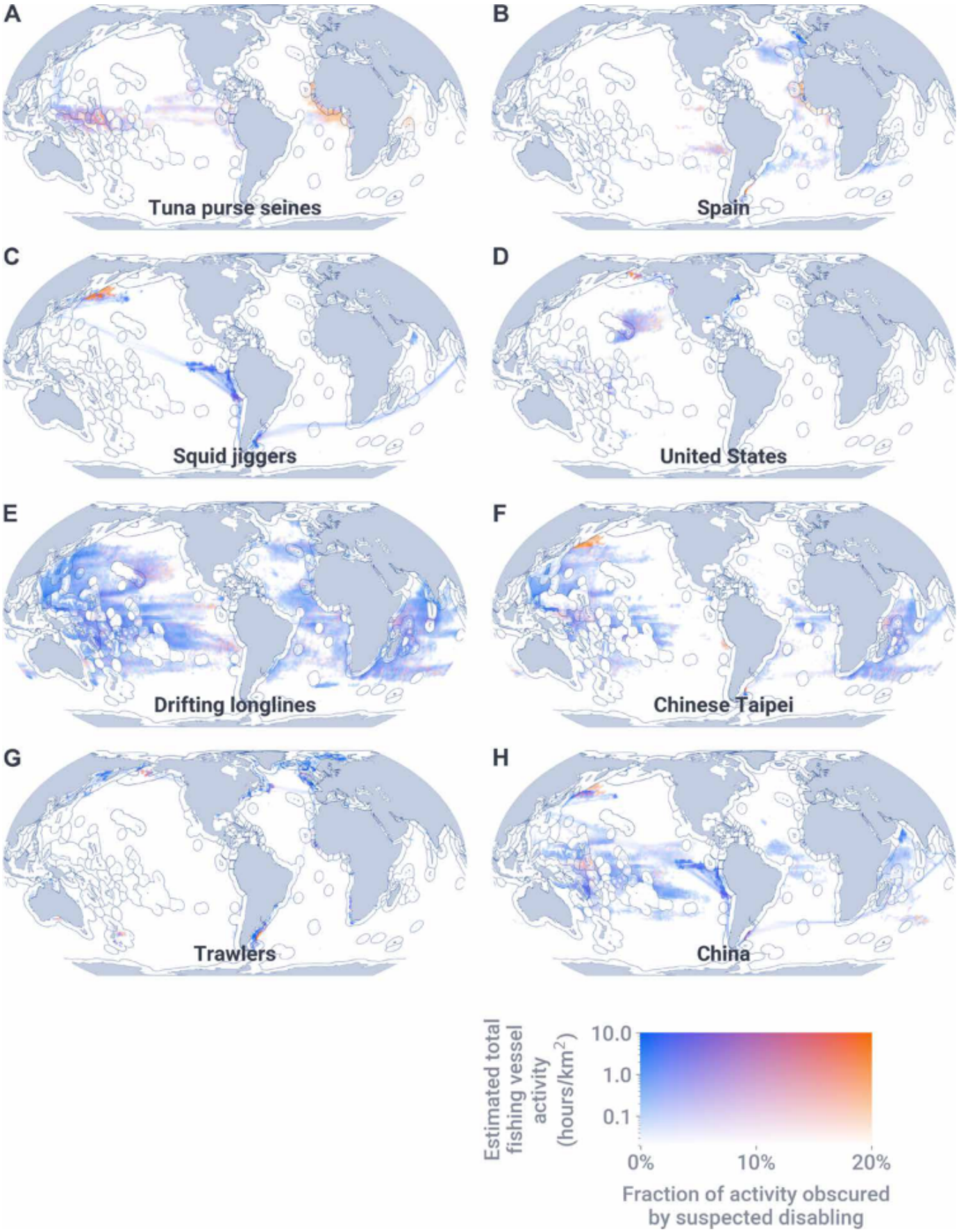


Exhibit. Estimated total fishing vessel activity and the fraction of that activity obscured by suspected AIS disabling events. Source: S0020.

A global picture of AIS disabling and its drivers

Section 43 mapped where vessels go dark; this section — the second half of Welch et al's study, whose results chapter carries this title — asks who goes dark, and why. The empirical base is the AIS record itself: more than 28 billion AIS messages from 2017 to 2019, acquired from Global Fishing Watch (satellite providers Spire and Orbcomm,

with exactEarth data for validation), processed by a convolutional neural network that identifies fishing versus non-fishing events and gear type, and aggregated into quarter-degree rasters of vessel days, fishing days, and suspected disabling events [20]. The map that results (the paper's Fig. 1) shows the highest fishing activity and the highest fractions obscured by disabling in the three regions of IUU concern — near Argentina and West Africa and in the Northwest Pacific — with intensively managed Alaska as the contrast case [20].

Who disappears: gears and flags

By gear, tuna purse seines had the highest fraction of vessel activity obscured by disabling events (up to 21%), followed by squid jiggers (up to 7%), and drifting longlines and trawlers (both up to 5%) [20]. By flag, vessels of just four nations made up 82% of the time lost to suspected disabling in waters more than 50 nautical miles from shore: Spain had the highest fraction of its activity obscured (up to 14%), followed by the United States (up to 8%), Chinese Taipei (up to 6%), and China (up to 5%) — while in absolute hours, drifting longlines among gears and China among flag states accounted for the most total time lost [20]. The authors attach a structural caveat: AIS usage is biased toward vessels flagged to upper- and middle-income countries, so these results partly reflect who uses AIS at all — "higher AIS usage leads to high opportunity to disable AIS" [20].

The driver models

To explain the geography, the study built boosted regression tree models — for the four dominant gear types and one full model — in which presences were locations with suspected disabling and absences were locations with fishing activity but no disabling, testing drivers related to fishing-ground quality, piracy, transshipment, and jurisdictional boundaries: loitering by transshipment vessels, distance to shore, distance to marine protected areas, distance to anti-shipping (piracy) activity, chlorophyll, eddy kinetic energy, sea surface temperature and its variability [20] [20]. The models discriminated strongly (32.7% mean explained deviance; 0.83 mean AUC), indicating a real capacity to separate the places vessels go dark from the places they simply fish [20].

Transshipment loitering. Loitering by refrigerated cargo vessels — a proxy for potential transshipment events — was the most important driver in the full model (49% relative importance) and the squid jigger model (37%): disabling occurrence rose with loitering activity, particularly in areas with 1,000 to 10,000 hours of loitering, whereas fishing activity was distributed in areas with 100 hours or less; and more than half of squid jiggers' disabling events occurred close enough to a loitering reefer to actually undertake a transshipment [20]. Loitering mattered much less in the longline, purse seine, and trawler models [20].

The 200-nautical-mile line. Distance to shore was the most important driver for drifting longlines (33%) and third in the full model (13%): disabling occurrence peaked at or just outside 200 nautical miles from shore — the distance that marks most EEZ boundaries — and decreased further offshore, while fishing activity was bimodally distributed inshore and beyond [20]. Peaks at 200 nm also appeared in the squid jigger and tuna purse seine models [20]. The obvious objection — that fleets simply fish near boundaries — was tested directly: more than 96% of EEZ-adjacent disabling events occurred next to foreign EEZs, and waters adjacent to the EEZs of Argentina, Russia and Peru had higher percentages of disabling than fishing activity alone would predict, with more than 47% of all disabling events adjacent to the Argentine EEZ [20]. China- and Spain-flagged vessels likewise showed EEZ-adjacent disabling shares above their fishing-activity shares [20]. And although only 11% of EEZs have contested boundaries, disabling events were twice as likely within or adjacent to those EEZs as fishing activity — "in some key locations and fleets, the high amount of suspected disabling near EEZ boundaries cannot be explained by the amount of fishing activity alone" [20].

Legitimate shadows. Not every driver points at crime. Chlorophyll — the ocean-colour proxy for productive water — was the most important driver in the trawler model (50%) and important for tuna purse seines, suggesting these gears disable to hide productive fishing grounds from competitors; trawling is far more spatially concentrated than other gears because it is restricted to continental shelves, making competitive exposure particularly costly [20] [20]. Piracy was the most important driver of tuna purse seine disabling, indicating vessels hiding their positions in dangerous waters [20]. The driver analysis thus sorts a single behaviour — switching off — into at least three motives: obscuring transfers and border crossings, protecting commercial secrets, and protecting the vessel itself.

What the picture supports

The legality of individual events remains entangled in the patchwork of AIS rules varying by region, RFMO, flag state, and vessel size; the study cannot certify which events are illegal, but it demonstrates strong links between disabling and the two behaviours of IUU concern — potentially unauthorised border crossings and transshipments [20]. Its practical claim is predictive: models with strong skill on previously unseen data could position at-sea or airborne surveillance and focus the port inspections required by the Port State Measures Agreement [20]. With that, the report's satellite arc closes: the dark can be mapped (Section 41), interrogated fleet by fleet (42), localised to hot spots (43), and now modelled well enough to anticipate (44). The next part turns from fishing vessels that hide to a fleet built to hide — the tankers of the Russian shadow fleet [20].

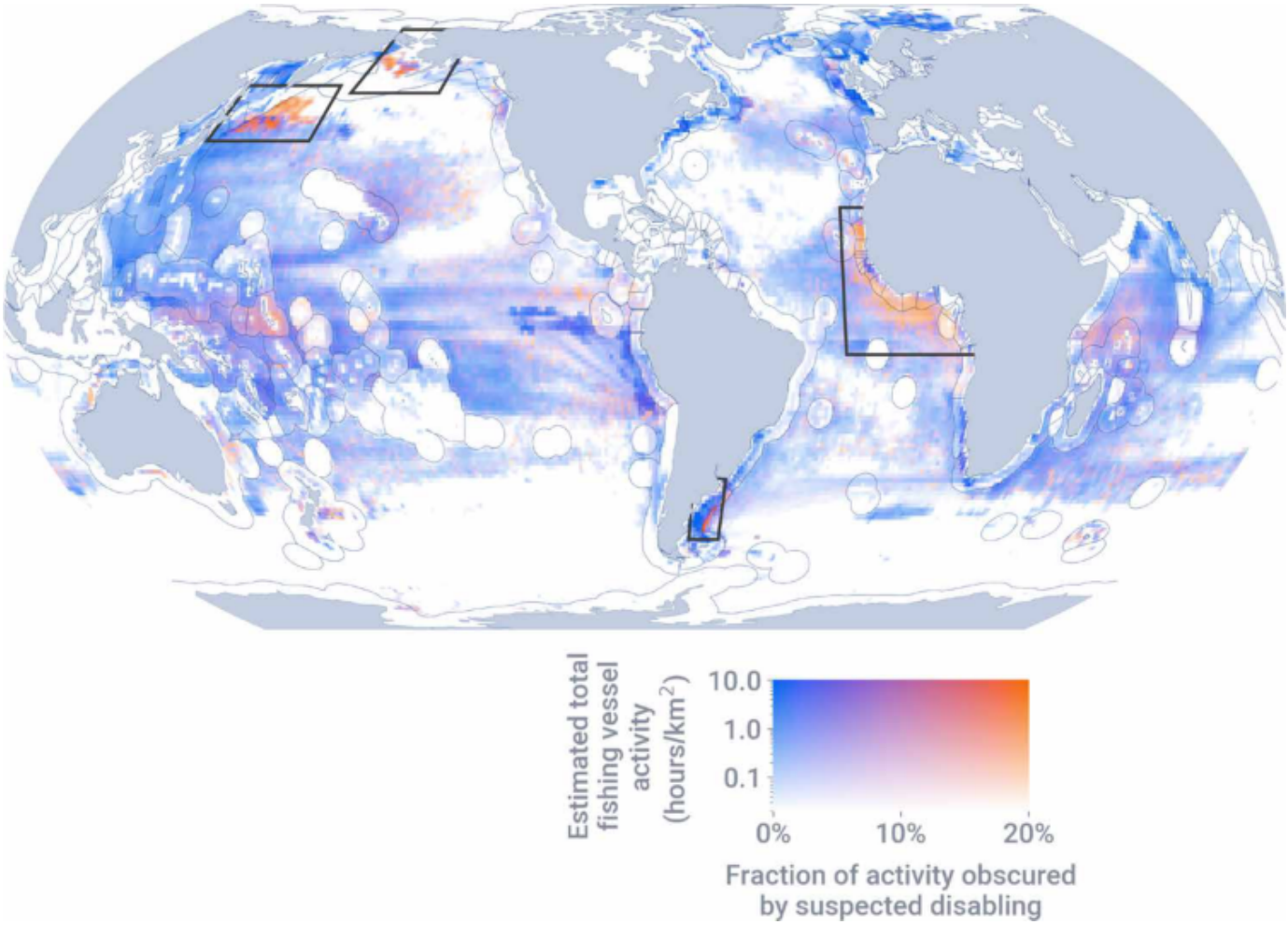


Exhibit. Fig. 1. Estimated total fishing vessel activity and the fraction of this activity obscured by suspected disabling events in areas with sufficient satellite reception quality (10 positions/day). Source: S0020.

The Russian shadow fleet: ecosystem of sellers, managers and flag states

The report's dark fleets now change cargo and motive: from fish to oil, from evading fisheries inspectors to evading sanctions. The evidence here is the Kyiv School of Economics (KSE) Institute's Russian Shadow Fleet Tracker (September 2026 edition, data through August 2026), built from Equasis, Kpler, International Group P&I club records and EU Council designations [21] [21]. Its subject is not a list of hulls but an ecosystem — the tracker organises the fleet by who sells the oil, who manages the ships, and which flags they fly — and its central structural fact is the overlap between the "shadow fleet" and the "EU-designated fleet": designated tankers' exports accounted for 53% of all Russian seaborne oil exports in August 2026 and 86% of shadow fleet volumes, so the sanctioned core is not a fringe of the shadow trade but most of it [21].

The August 2026 snapshot: substitution, not shrinkage

Russian seaborne oil exports decreased to 4.9 million barrels per day in August 2026 (−6.1% month-on-month, −12.4% year-on-year), while shadow fleet exports increased to 3.0 mb/d (+1.8% m-o-m) and EU-designated fleet

exports rose 5.2% m-o-m [21]. The fleet that did the carrying: 258 unique tankers, of which 166 (64%) were shadow tankers and 120 (47% of the total fleet) EU-designated [21]. The Baltic was the growth region — exports from Baltic ports rose to 2.5 mb/d (+6.5% m-o-m) with shadow exports up 16% m-o-m; 128 unique tankers loaded there, 65 (51%) of them shadow and 49 EU-designated [21]. As conventional capacity retreats, in other words, the opaque fleet does not merely persist; it absorbs the share.

Sellers: a concentrated, partly sanctioned front end

The selling layer is strikingly concentrated. During June–August 2026 the top 10 sellers accounted for 85% of crude oil and 56% of oil products moved by the shadow fleet (84% and 61% respectively for the EU-designated fleet) [21]. The leading crude seller for both fleets was UAE-based **Redwood Global Supply FZE LLC** — itself an EU-designated company — with 21% of shadow-fleet and 22% of designated-fleet crude over June–August, while **Rosneft** led oil products (16% and 19%) [21] [21]. Among the top 10 sellers in both fleets, Redwood and **Surgutneftegas** are subject to EU sanctions [21]. The Baltic trade is more concentrated still: the top 10 sellers took 97% of shadow-fleet crude and 75% of products (96% and 82% for the designated fleet), with **Rusexport** the top identified crude seller and Rosneft again leading products [21].

Managers: two nationalities run the fleet

Ship management shows the same pattern of concentration plus sanction entanglement. Chinese and Russian ship managers remained the largest operators of both the shadow and EU-designated fleets, jointly handling 64% of exports — and from Baltic ports, 75% of shadow and 78% of designated exports [21]. The top 10 managers moved 881 kb/d for the shadow fleet (29% of volumes) and 889 kb/d for the designated fleet (34%) [21]. Among those top-10 managers in both fleets, four are themselves EU-sanctioned: **South Fleet Ltd** and **Invest Fleet Ltd** (Russia) and **Dreamer Shipmanagement LLC-FZ** and **Albatross Shipmanagement** (UAE) [21]. In the Baltic subset, the top 10 managers handled 371 kb/d of shadow volumes (31%) and 355 kb/d of designated volumes (34%), with Dreamer Shipmanagement and South Fleet again among the sanctioned names [21].

Flags: convenience, and a "false/unknown" category

Russia and Sierra Leone were the leading flag states for both fleets, jointly accounting for 56% of shadow and 59% of designated exports; from Baltic ports, Sierra Leone and Russia together flagged 63% of shadow and 69% of designated exports, and the Equatorial Guinea flag reached a record 8% of the total [21]. Behind the leaders runs a long tail of small registries — Oman, Cameroon, São Tomé and Príncipe, Panama, Vietnam, Barbados, Guinea-Bissau, Vanuatu — and, tellingly, an explicit "False/Unknown" flag category in the tracker's own charts: part of the fleet sails under flags that are not merely convenient but fictitious or undetermined [21] [21].

Destinations and insurance: the opacity layers

China and India remained the main destinations for both fleets, jointly taking 64% of shadow and 63% of designated exports; from the Baltic, India and Singapore led the identified destinations (50% and 49%) — while 47% and 48% respectively had unknown destinations, a destination opacity that is itself part of the ecosystem's design [21]. Insurance shows the same partial visibility: in August 2026, 52 of 166 shadow tankers (31%) were identified as operating under Russian insurance, outside the International Group system that covers conventional tonnage — and 48 of those 52 (92%) were sanctioned by at least one jurisdiction among the EU, US, UK, Canada, Australia and New Zealand [21].

One aggregate hangs over the whole ecosystem, and the next section takes it up in full: between August 2024 and August 2026, 444 of 625 EU-designated oil tankers (71%) loaded at Russian ports at least once after their designation, carrying 1.2 billion barrels — 28% of Russia's seaborne exports in the period [21]. Designation, it turns out, is not removal. What post-designation loading looks like as a system is Section 46's subject [21].

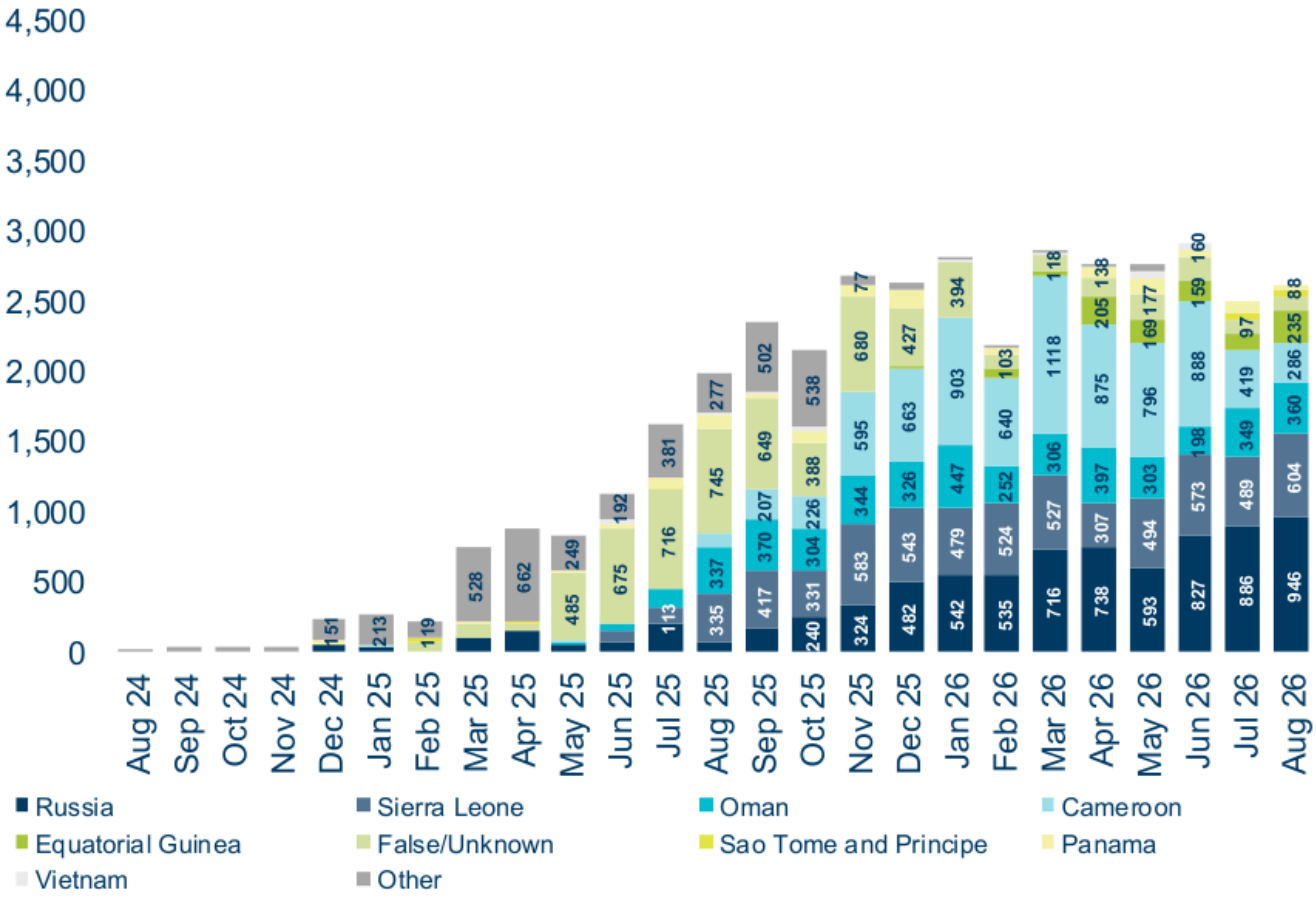


Exhibit. EU-designated shadow fleet oil volumes by flag state, in thousands of barrels per day, led by Russia, Sierra Leone, Oman, Cameroon and Panama, from August 2024 onward. Source: S0026.

EU-designated tankers and post-designation Russian oil loadings

Section 45 ended on the aggregate; this section takes it apart. First, the tracker's own definitions, which contain the finding in embryo. The KSE Institute defines the shadow fleet as tankers "without links to G7+ jurisdictions for registered ownership, management, and flagging, as well as without P&I insurance by the International Group of P&I Clubs" — and the EU-designated fleet as "tankers that conducted loadings in the respective month after being designated by the EU," including tankers also designated by other jurisdictions [21]. Read that second definition again: membership in the designated fleet is defined by post-designation activity. The category exists because the behaviour does.

The two-year ledger

Between August 2024 and August 2026, EU-designated tankers loaded 1,190.7 million barrels of Russian oil at Russian ports after their designation — the 1.2 billion barrels, 28% of Russia's seaborne exports, carried by 444 of the 625 designated tankers [21] [21]. The flow is not historical residue: in August 2026 alone, post-designation loadings totalled 81.0 million barrels, and 120 of the month's 166 shadow tankers (72% of the shadow fleet) "made voyages while under EU sanctions" [21] [21]. In the Baltic the share is higher still: 49 of 65 shadow tankers (75%) voyaged under EU sanctions in August 2026, and designated-fleet exports from Baltic ports rose 16.0% month-on-month to 41% of total Baltic exports and 85% of Baltic shadow volumes [21] [21].

Anatomy of the top 20

The tracker's table of the top 20 designated tankers by post-designation volume shows a long tail rather than a rogue's gallery: the top 20 account for roughly 259 million barrels of the 1,190.7 million total, with "Other" — the remaining hundreds of designated hulls — carrying 931.8 million [21]. The leaders are instructive. First is Nichole (IMO 9332822), flagged to São Tomé and Príncipe, managed by China-based Suntun Management Ltd and owned by Sea Swift Co Ltd, with 21.5 million barrels loaded post-designation (1.5 million in August 2026 alone) [21]. Second, Evita

(IMO 9408530), Cameroon-flagged, Chinese-managed (Aether Craft Co Ltd) and Chinese-owned (Ironclad Group Co Ltd), at 16.9 million barrels [21]. Then a Russian-flag cohort — Yuri Senkevich (16.6 mb), Kapitan Kostichev (15.3 mb) and Victor Konetsky (13.7 mb), all managed by the EU-sanctioned South Fleet Ltd, and Nikolay Zadornov (16.5 mb) and Vladimir Arsenyev (14.6 mb), managed by SCF TM Ltd — whose registered owners are Hong Kong single-purpose companies: Comitana Shipping HK, Glimmer Shipping HK, Vimena Shipping HK, Anchorstar Shipping HK [21]. A third cohort flies Oman's flag — Arrow (15.0 mb), Amulet (14.8 mb), Voyager (9.9 mb), Discovery (8.8 mb), Galaxy (8.5 mb), Respect (7.9 mb) — owned by UAE special-purpose companies (Citrine Marine SPC, White Agate Marine SPC) and rotated among UAE managers (Greenlight, Nova, Lynx, Grace, Albatross Shipmanagement) [21]. Around them: Cameroon-flagged Jun Tong and DG Hong Kong, Sierra Leone-flagged Tiral and Vellamo — the latter two with ISM managers in Azerbaijan (Blue Horizon Shipping, Nautexa Stratum) beneath Chinese ship managers [21]. Every layer of the Section 45 ecosystem — seller aside — is stacked into each hull: a convenience flag, a manager in one jurisdiction, an SPV owner in a second, a technical manager in a third.

Two further features deserve emphasis. First, designation stacking does not stop the loading: the table's sanction columns show these tankers designated not only by the EU but, in most cases, also by the UK and by several of the US, Canada, Australia and New Zealand — Nichole, Yuri Senkevich, Nikolay Zadornov and Victor Konetsky are designated by all six jurisdictions listed, and loaded anyway [21]. Second, the fleet is old: tankers older than 15 years accounted for 92% of the shadow fleet in August 2026, and 94% of the Baltic shadow fleet [21] [21]. The designated fleet that keeps Russia's exports moving is thus also the fleet least likely to be insured, maintained, or classed to conventional standards — the sanctions problem and the derelict-vessel problem of this report's earlier parts converging on the same hulls.

Where the post-designation oil goes

The Baltic cut shows the destination pattern in miniature: for both the shadow and designated fleets loading from Baltic ports in August 2026, India was the most important identified recipient (48% and 47% respectively), Singapore a distant second (2% each) — and shipments with an unknown destination accounted for 47% and 48% [21]. Half the post-designation trade, in other words, is not merely sanctioned but destination-opaque, consistent with the global picture of Section 45 [21].

The conclusion the tracker supports is uncomfortable for list-based sanctions: over two years, designation functioned less as a restraint than as a registry — a census, maintained at public expense, of a fleet that continued to load roughly three barrels in ten of Russia's seaborne exports. The gap is not in identification; the top-20 table names hulls, IMO numbers, managers and owners. The gap is between listing and interdiction [21] [21].

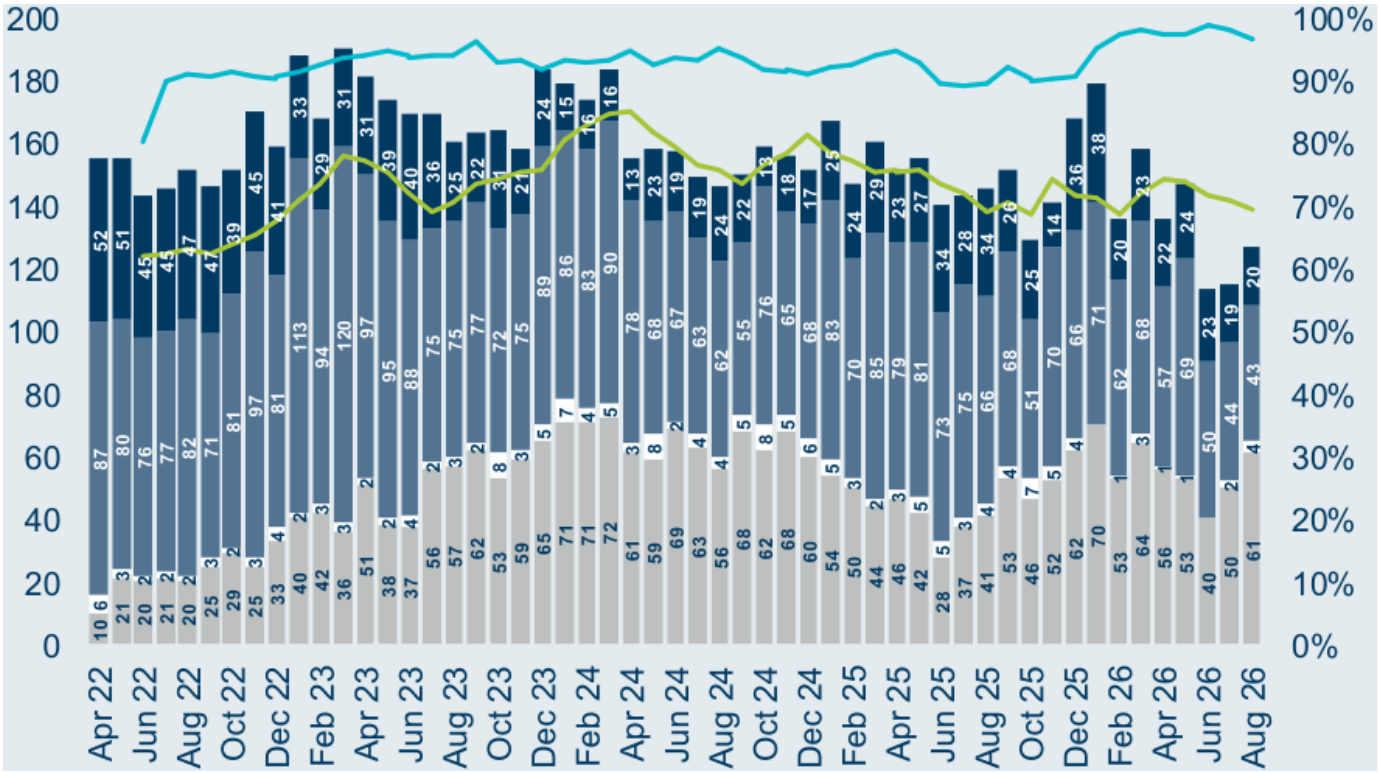


Exhibit. Shadow tankers carrying Russian oil, split between EU-designated and non-designated vessels, from April 2022 onward. Source: S0026.

Law of the Lost Ship: Salvage, Abandonment, and Sunken Warships

Immunity of sunken warships and salvage law

The report now goes below the surface: not vessels that hide or sink, but vessels long sunk — and the question of whose they still are. The anchor text is Mariano Aznar Gómez's "Legal Status of Sunken Warships Revisited," whose first part poses the doctrinal problem directly: sunken warships sit at the collision point of two bodies of law that were never designed for them — the sovereign immunity of State vessels, and the private law of salvage [22] [22].

Immunity does not sink

That warships attract sovereign immunity is historically settled — from Chief Justice Marshall in The Schooner Exchange v. McFaddon (1812) through the 1926 Brussels Convention's immunity from arrest and seizure for government vessels in public service, to UNCLOS Articles 32, 95 and 236, which "plainly recognise that warships enjoy immunity," and Article 16 of the ILC's 1991 Draft Articles on Jurisdictional Immunities, confirming immunity for warships, naval auxiliaries and other State ships in exclusively non-commercial government service [22]. The pivotal claim is about what sinking changes: "nothing in the legal texts suggests that they lose their immunity by the simple fact of their sinking" [22]. A functional counter-argument exists — a sunken warship is no longer a "ship" within the UNCLOS Article 29 definition, has lost its functional condition, and so should not retain immunity, "particularly if the warship became a wreck long time ago" — but Aznar notes it "has been not only criticised by doctrine but rejected by State practice — domestic and international — as well" [22].

Salvage law's first gate: no peril, no salvage

Salvage law has its own threshold, and a wreck fails it. In O'Keefe and Nafziger's formulation, quoted with approval: "the law of salvage relates solely to the recovery of items endangered by the sea… For underwater cultural heritage, the danger has passed; either a vessel has sunk or an object has been lost overboard. Indeed, the heritage may be in greater danger from salvage operations than from being allowed to remain where it is" [22]. The 1989 International Convention on Salvage bears this out textually: a salvage operation is an act "to assist a vessel or any other property in danger," and a vessel is a structure "capable of navigation" — an ordinary-meaning reading "plainly excludes wrecks from salvage operations" [22]. Practice is not uniform: U.S. courts recite a peril exigency (the Supreme Court's The Sabine applied salvage to property recovered "from actual peril or loss, as in cases of ship wreck, derelict or recapture") yet have allowed salvage claims for long-submerged wrecks (Platoro Ltd. v. The Unidentified Remains of a Vessel, 5th Cir. 1983), while Spanish jurisprudence has "clearly settled" a marine-peril requisite [22].

The second gate: warships are carved out of the salvage conventions

Whatever the peril rule for ordinary wrecks, warships were excluded from the salvage regime by name. Article 14 of the 1910 Brussels Convention stated flatly: "this convention does not apply to ships of war or to Government ships appropriated exclusively to a public service" [22]. A 1967 Protocol softened the exclusion — salvage services rendered by or to warships could fall within the regime — but ring-fenced the flag State: a claim against a State for services to its warship "shall be brought only before the Courts of such State" [22]. The 1989 Convention reaffirmed the exclusion in Article 4(1): it "shall not apply to warships or other non-commercial vessels owned or operated by a State and entitled, at the time of salvage operations, to sovereign immunity under generally recognised principles of international law unless that State decides otherwise" [22]. Article 29 adds that, absent the State owner's consent, nothing in the Convention grounds seizure, arrest, detention or proceedings in rem against immune State cargoes; the 1999 Arrest of Ships Convention, Article 8(2), likewise excludes warships and State non-commercial ships [22]. U.S. domestic law parallels the carve-out: the U.S. Salvage Act excludes "Government ships appropriated exclusively to a public service" and particularly "ships of war" [22]. Aznar's conclusion is that a customary rule now exists alongside the treaties: international law "excludes the application of the law of salvage to warships unless otherwise expressly decided by the flag State," because Article 4's mandatory exception covers warships "sunken or not" [22].

The reservation machinery shows how contested the edges remain. The 1989 Convention's Article 30(1)(d) reservation was read by some as implying that, absent a reservation, salvage law would reach historic wrecks; the IMO countered that, given the Convention's private-law, non-mandatory character, "the right to exclude the application of salvage law existed even without express reservation" [22]. Most specially affected States reserved anyway — Australia, Canada, China, France, Germany, Russia, the United Kingdom and others (the United States, notably, is not a party) — while the Netherlands went the other way, applying the Convention to its own State vessels under domestic-law conditions [22]. And when the IMO turned to drafting a Wreck Removal Convention, the draft again "expressly leaves aside the removal of sunken warships otherwise decided by the flag State" [22].

Title: salvage pays, finds takes — and finds needs abandonment

Underlying the salvage question is a property question. Salvage, even where it applies, "only gives the salvor a right to compensation for the services rendered, but not title to the property salved"; the law of finds, by contrast, lets a finder hold title against all but the rightful owner — which is why everything turns on abandonment [22]. For State craft the standard is demanding. In the Sea Hunt litigation — Sea Hunt Inc. had located the Spanish frigates La Galga (sunk 1750) and Juno (1802) off Virginia and taken State permits under the Abandoned Shipwreck Act — the District Court found Spain had abandoned La Galga by the 1763 Treaty of Paris; the Fourth Circuit reversed, upholding Spanish title to both wrecks and denying any salvage award [22]. The Columbus-America standard, correctly applied, requires abandonment "proved by clear and convincing evidence… such as an owner's express declaration abandoning title"; an inference of abandonment from age alone is improper once the prior owner appears and asserts ownership [22]. Critics warned the standard "would render the ASA a virtual nullity," but the Fairport line of cases expressly confined itself to privately owned vessels, taking no view on vessels "initially owned by the United States" — and U.S. salvage doctrine separately recognises an owner's right to decline assistance where only the owner's property is at stake, a thread the aircraft-salvage litigation of Sections 50–51 pulls hard [22] [23].

State practice speaks in notes verbale

The diplomatic record runs one way, with one doctrinal exception. When a Japanese company began salvaging the Admiral Nakhimov, the USSR protested on 3 October 1980 that "in accordance with international law a sunken warship is completely immune from the jurisdiction of any State other than the flag State"; Japan answered not by denying immunity but by invoking capture — rights over a captured enemy warship pass "immediately and finally to the captor State" [22]. In the USS Panay affair, Japan paid a $2,214,007.36 indemnity for sinking the U.S. gunboat in the Yangtze in 1937 and treated payment as acquiring the wrecks; the United States accepted transfer of the private vessels but rejected salvage of the gunboat outright — it found "no authority in law for acceding in any case to such a request" [22]. For HMS Birkenhead — sunk in 1852 off the Cape with 445 people and a reputed 240,000 gold sovereigns — the UK's 1989 Exchange of Notes with South Africa regulated salvage settlement while protecting the wreck as a military grave and expressly not prejudicing ownership; the Foreign Office stated the position plainly: "the Crown maintains rights and interest in ships of the Royal Navy which have sunk, wherever they may be and without time limit" [22]. The 1997 UK–Canada Memorandum on HMS Erebus and HMS Terror splits the bundle with care: "Britain, as owner of the wrecks, hereby assigns custody and control of the wrecks and their contents to the Government of Canada… Britain does not waive ownership or sovereign immunity with respect to the wrecks or their contents while they are on the seabed" [22]. And courts have held title even across regime collapse: Singapore's High Court, in the U-859 mercury case, held that "the German State has never ceased to exist despite Germany's unconditional surrender in 1945" and its property remains its property; Norwegian courts likewise held title to U-76 rested with Germany until the 1945 seizure of German assets [22].

The counter-current is political, not legal: the Council of Europe's Parliamentary Assembly, in Recommendation 1486 (2000), urged agreements "as will mitigate the sovereign immunity which states retain over vessels of war and other state-owned vessels wherever they are sunk," citing their historic value and their status as war graves, and recommended avoiding commercial recovery operations [22] [22]. The next section takes up that heritage framing directly — sunken warships as underwater cultural heritage [22].

Sunken warships as underwater cultural heritage

Section 47 asked whose a sunken warship is. This section asks what it is — and the heritage frame changes the answer's consequences. A wreck can be at once State property, a war grave, and an archaeological site; salvage law sees a recovery opportunity, heritage law sees an object whose value lies in not being recovered. Aznar Gómez's

survey shows the heritage frame is not an overlay on the immunity regime but a parallel legal order, built mostly outside the common law's salvage/finds machinery [22] [22].

Two legal cultures

Tullio Scovazzi's observation, quoted in the article, is the starting point: "the bodies of 'the law of salvage and other rules of admiralty', despite their immemorial tradition, are today typical of a few common law systems but are complete strangers to other domestic legal systems" [22]. Where the common law asks whether a wreck is abandoned (finds) or merely imperilled (salvage), most maritime States legislate the heritage directly. Even within U.S. law the common-law frame is hemmed in: admiralty courts apply a presumption against abandonment (Hener v. United States), salvage gives an award but not title where the owner retains an interest (Treasure Salvors), and finds vests title only where the owner has abandoned (Columbus-America; Martha's Vineyard Scuba Headquarters) [22].

Congress then switched the machinery off for historic wrecks. The Abandoned Shipwreck Act transfers title to abandoned shipwrecks embedded in a State's submerged lands to that State (43 U.S.C. § 2105) and provides expressly that "the law of salvage and the law of finds shall not apply" to them (§ 2106(a)) [22]. The House Report's reasoning is heritage reasoning: "admiralty principles are not well-suited to the preservation of historic and other shipwrecks to which this Act applies. Abandoned shipwrecks covered by this Act are not considered... to be in marine peril, necessitating their recovery by salvage companies" [22].

The comparative layer: vesting and permits, not finds

The domestic legislation of the "specially affected" States converges on the same move — displace salvage/finds with State ownership or permit control. Spain's heritage law (Ley 16/1985, Art. 44) expressly makes the Civil Code's finds rule inapplicable to maritime heritage finds; Portugal (Decreto-Lei 164/97) and France (Décret 61-1547; the 1989 law on biens culturels maritimes) regulate maritime cultural property directly [22]. Australia's Historic Shipwrecks Act 1976 subordinates any action directed to shipwrecks to a ministerial permit; South Africa's National Heritage Resources Act 1999 treats heritage resources as part of the national estate and requires special permits for any activity directed at them [22]. Others vest ownership outright: China's 1989 Regulations (State Council Decree No. 42, Art. 3) grant ownership of underwater cultural relics to the State; Tunisia's 1994 heritage code (Art. 73) grants the State ownership of archaeological objects found in inland waters or the territorial sea; Israel's Antiquities Law 1978 establishes State ownership of antiquities; Italy's 1999 code places cultural objects within the public domain (beni del demanio storico, artistico e archivistico); the UK's Protection of Wrecks Act 1973 protects designated wrecks and excludes them from salvage [22] [22].

Public property persists — but its character changes

The heritage frame does not displace the flag State's title; in Aznar's analysis it overlays it. Even if sunken warships were "ships no more," they remain the public property of the flag State — Anastasia Strati: with respect to sunken State vessels and warships "it must be accepted that they do retain their status as public State property so that their recovery may require the consent of the flag State" [22]. Quéneudec, analysing the Alabama case, states the rule in its strongest form: the transformation of a ship into a wreck never ipso facto extinguishes the owner's right of property — a fortiori for public vessels — and a State's property in its warship wreck "subsists as long as that State has not expressly renounced its right or has not voluntarily transferred it," the sole exception being capture by another State in the exercise of belligerent rights [22]. What changes on the seabed is the object's character: contents of a military character (documents, instruments, weapons) retain their jure imperii claim to immunity, but even where that character is "replaced by an historical, spiritual or archaeological character (old guns, human graves, charts, etc.)," the different treatment "does not necessarily impede the application of State immunity since States are still owners of these artefacts, precluding the reversion of title to the finder since a rightful owner always exists unless an express abandonment act has been proven" [22]. U.S. doctrine's vintage rule says the same in miniature: "when articles are lost at sea the title of the owner in them remains" (The Akaba, 4th Cir. 1893), and Columbus-America holds courts "hesitant to find an abandonment" of property "involuntarily taken" from its owner [22].

The counter-claim: heritage of the coastal State, heritage of humanity

Heritage language, however, cuts both ways — it can arm the coastal State against the flag State. In the political realm, Latin American and Caribbean countries asserted in the Santo Domingo Declaration (16 June 1998) that underwater cultural heritage "is the property of the State in which it is found and through this it is the heritage of the Humanity" [22]. Spain's controversy with Uruguay over the frigate San Salvador — sunk with her crew on 31 August 1812 in Maldonado Bay — is that claim in litigation posture: Uruguay, applying its own decrees (Real Decreto 14.341 of 1976; Decreto 692/986 of 1986), considers title over the wreck reverted to Uruguay; Spain, reiterating its Note verbale of 8 February 2000, "manifestly expressed that it had not abandoned the vessel," insisted that "human remains should be treated with particular respect," and proposed a general agreement to cooperate "in the preservation of common underwater cultural heritage" [22]. The classic case in the article's review, the Confederate raider CSS Alabama — sunk by USS Kearsarge on 19 June 1864, seven miles off Cherbourg — posed the same France–United States geometry a century earlier [22].

The heritage frame thus strengthens protection while relocating the dispute: from salvor versus owner to flag State versus coastal State, with "humanity" invoked by both sides. The instrument meant to settle that geometry is the 2001 UNESCO Convention on the Protection of Underwater Cultural Heritage — the subject of Scovazzi's companion analysis and of the legal-regime section that follows [22] [22].

The legal regime of UNCLOS and state practice

Sections 47 and 48 built the rule from doctrine and heritage law. This section assembles the full regime — treaty layers first, then the state practice that fills their gaps, then the scholarly fight over whether the resulting customary rule is ancient or invented. Aznar Gómez's article is again the spine, now in its second and third parts: "The New Regime and its Impact on Sunken Warships" and "The practice of States as evidence of the rule" [22] [22].

Layer one: UNCLOS protects objects, not titles

UNCLOS addresses underwater heritage in a single article, and its caution is the point. Article 303(1) provides that "States have the duty to protect objects of an archaeological and historical nature found at sea and shall cooperate for this purpose"; Article 303(2) lets the coastal State presume that removal of such objects from the contiguous-zone seabed without its approval infringes its customs/fiscal laws (via Article 33). But Article 303(3) immediately saves everything it might have settled: "Nothing in this article affects the rights of identifiable owners, the law of salvage or other rules of admiralty, or laws and practices with respect to cultural exchanges," and 303(4) is without prejudice to other agreements and rules on protection of such objects [22]. For warships specifically, UNCLOS supplies the immunity provisions (Articles 32, 95, 236) but no wreck regime — the gap commentators call UNCLOS's inadequacy, which "led to the new UNESCO Convention as an effort to define a legal regime for underwater cultural heritage, including sunken warships when suitable" [22].

Layer two: the UNESCO Convention — lex specialis with a savings clause

The 2001 UNESCO Convention (CPUCH) positions itself, in Carducci's formula, as "a sort of lex specialis for the specific purpose of [underwater cultural heritage] while the UNCLOS remains the fundamental and authoritative lex generalis for the law of the sea" — its Article 3 saves conformity with UNCLOS, and Article 6(3) preserves rights and obligations regarding sunken vessels arising from earlier bilateral, regional or multilateral agreements, "in particular, those that are in conformity with the purposes of this Convention" [22]. State vessels enter the Convention in two roles: as objects of protection (Article 1(8)) and as means of protection (Article 13 — warships and other immune State craft on non-commercial operations "shall not be obliged to report discoveries of underwater cultural heritage" under the Convention's notification articles, though States Parties must ensure compliance "as far as is reasonable and practicable" without impairing operations) [22].

Two design choices limit what the Convention can settle. The first is temporal: the Convention covers sunken warships only when they "meet the definition of underwater cultural heritage" — traces that "have been partially or totally under water, periodically or continuously, for at least 100 years" (Article 1(1)(a)) — which "leaves to one side the possible problem of recently sunken warships" [22]. The second is definitional care: "State vessels and aircraft" (Article 1(8)) means warships and other State-owned or -operated vessels or aircraft "used, at the time of sinking, only for government non commercial purposes," identified as such and meeting the heritage definition —

“an historical and functional concept” keyed to iure imperii use, under which Spain’s Carrera de Indias ships Juno and La Galga qualify as galeones de guerra, officially commanded and inscribed in the Naval Museum’s Lista Naval [22]. And on the core question — title and immunity — the Convention expressly declines to rule: Article 2(8) provides that nothing in it “shall be interpreted as modifying the rules of international law and State practice pertaining to sovereign immunities, nor any State’s rights with respect to its State vessels and aircraft.” As a drafting insider put it, Article 2(8) “simply maintains the uncertainty status quo” [22]. Where the Convention does innovate is the Area: a Coordinating State acting on heritage beyond national jurisdiction “shall act for the benefit of humanity as a whole, on behalf of all States Parties,” with “particular regard… paid to the preferential rights of States of cultural, historical or archaeological origin,” and interested States may take practicable measures, even before consultations, against immediate danger including looting (Article 12(6)) [22].

Layer three: state practice and the express-abandonment rule

With the treaties deliberately incomplete, the operative rule is customary, and its canonical statement comes from litigation: the U.S. Department of State’s Statement of Interest in Sea Hunt — “The practice of nations confirms the well-established rule of international law that title to such vessels is lost only by an express act of abandonment, gift or sale by the sovereign in accordance with relevant principles of international law and the law of the flag State governing abandonment of government property, or by international agreement or by capture or surrender during battle before sinking” [22]. Aznar decomposes the rule into “building blocks”: title remains in the flag State, governed by its domestic law; title is lost or transferred only by (a) express abandonment — unilateral or by agreement — or (b) capture or surrender under the laws of war; and special circumstances add particular status where the wreck is (a) a human grave and/or (b) a historical or cultural site [22]. Practice supplies the conventional transfers: the U.S. schooners Hamilton and Scourge, sunk in Lake Ontario in the War of 1812, were transferred to the City of Hamilton, Canada, in 1978 by the Secretary of the Navy under the Constitution’s Property Clause — U.S. courts having “consistently recognized that the federal government cannot abandon property absent an affirmative act authorized by Congress” (Royal Indemnity Co. v. United States, 1941); and the warships sunk in the atomic tests at Bikini and Kwajalein were transferred to the Marshall Islands under the Compact of Free Association [22]. The grave block is equally practical: when Virginia issued a commercial permit for Juno and La Galga, Spain’s Diplomatic Note No. 43/48 of 8 May 1998 protested the disturbance of “military graves” [22]. And the courts police the time fallacy in both common-law systems: the Fourth Circuit in Sea Hunt stressed that “[t]he mere passage of time since a shipwreck is not enough to constitute abandonment,” mostly since “technology has only recently become available for its salvage”; the Sixth Circuit on remand in Fairport made lapse of time “one factor among several”; the Ninth Circuit in Yukon Recovery counted lack of technology in judging whether inaction is abandonment; and U.S. courts have held the United States has not abandoned ships sunk over a century ago in the Civil War (Steinmetz; the USS Hatteras litigation) [22].

The dissent: implied abandonment and “instant custom”

The regime has two serious critics inside the sources. Lucius Caflisch would admit implied abandonment: immunity survives while the flag State manifests, by concrete measures, its intention to raise the wreck; failing that, jurisdiction passes to the coastal State or to the salvor’s flag State [22]. David Bederman’s attack is deeper — historiographical. The “received wisdom” (title preserved for the owning sovereign; absolute immunity from salvage) must, to be effective, “characterize itself as established customary international law” and claim ancient vintage for the express-abandonment rule; Bederman believes there has instead been “a careful and concerted effort to contrive a pattern of state practice and to pass it off as established and binding customary international law” — a phenomenon of “‘instant’ custom” that manipulates the lawmaking process in service of the heritage-protection programme [22]. Aznar answers on the same systemic ground, in the opposite direction: the genuinely hoary principle is warship immunity itself, coupled with exclusion from commercial salvage unless the flag State decides otherwise; until modern diving technology, wrecks lay beyond all but “the most crude and destructive devices,” and it is only now that salvors, “with a panoply of talented lawyers,” question the rule’s vintage — an effort, he notes, to reverse the burden of proof onto the flag State, which “international practice has not accepted” [22].

The regime that results is layered and deliberately unfinished: UNCLOS general and savings-bound; the UNESCO Convention special but self-denying on title; a customary immunity-and-title rule policed in diplomatic notes and admiralty courts; and bilateral agreements — Erebus/Terror, Birkenhead, Sussex — doing the actual settlement work, with cooperation, as Aznar puts it, “a landmark which must be achieved” for the benefit of humanity “and not solely

for the benefit of treasure hunters and auction houses" [22]. The next part tests the doctrine against a single recalcitrant case: a United States Navy aircraft on the seabed, and a salvor who raised it anyway [22].

Abandonment in law: the Navy aircraft salvage case

Doctrine becomes concrete in International Aircraft Recovery, L.L.C. v. The Unidentified, Wrecked and Abandoned Aircraft and United States of America (11th Cir., 17 July 2000) — the case that carried the express-abandonment rule from Spanish frigates to a single American aeroplane. The in rem defendant was a U.S. Navy TBD-1 torpedo bomber lying on the seabed, "submerged in a corrosive environment and slowly disintegrating" [23]. Douglas Champlin — president of Historic Aircraft Recovery, Inc., with International Aircraft Recovery (IAR) as successor interest — had located the wreck, videotaped it, and brought its canopy and radio mast to dry land; the district court permitted him to continue salvage operations [23]. The United States, which had objected to Champlin's efforts in writing since at least 1993, appealed on the ground that the plane was never abandoned and never the salvor's to take [23] [23].

Was the TBD-1 abandoned? The statutory path not taken

IAR's abandonment theory ran through paperwork: the Navy had "struck" the aircraft from its inventory, and the Surplus Property Act of 1944 had directed that surplus property "shall be disposed of." The court dismantled both limbs. Striking is an accounting act, not a disposition: the Navy's own 1944 circular on "Aircraft — Striking and Disposition Of" treats striking as removal from the active inventory, and the Ninth Circuit had already held in Kern Copters that the Army did not abandon a helicopter crashed in Guatemala by "dropping [it] from accountability records" [23]. The 1944 Act, meanwhile, "did not mention the abandonment of property; its focus instead was on the sale of surplus of World War II matériel, including salvage and scrap," with disposal to occur only as prescribed [23]. Abandonment became administratively possible only under the Surplus Property Administration's Regulation § 8319 (1945) — and then only "after an affirmative finding made by a 'responsible officer, approved by a reviewing authority,' and reduced to writing," typically with written findings on value and handling costs and a thirty-day published notice offering the property for sale or donation [23]. "The Navy neither made such findings nor compiled the written report required to abandon property pursuant to the Surplus Property Administration's regulations" — so no abandonment occurred by that route either [23]. Nor was there any indication the Surplus Property Board or Administration had ever granted case-by-case permission to abandon crashed Navy planes, as it once had for surplus submarine and torpedo netting [23].

IAR's fallback — that the Property Clause does not apply in admiralty and the common-law abandonment test should govern, citing California v. Deep Sea Research for the admiralty meaning of "abandoned" — was held inapposite: the case required no interpretation of the Abandoned Shipwreck Act (neither party argued the Act applied, the defendant being neither a shipwreck nor "embedded in the submerged lands of a State"), and Deep Sea Research "involved a privately-owned steamship with privately-insured cargo" [23] [23]. The distinction is the Sea Hunt architecture in miniature: sovereign property leaves the public patrimony only by the affirmative, authorised act the flag State's own law prescribes — the same principle the ASA's legislative history states for warships, that "the U.S. only abandons its sovereignty over, and title to, sunken U.S. warships by affirmative act" [22].

Peril conceded, consent decisive

On salvage, the court gave the salvor his premise and denied his conclusion. The TBD-1 was in marine peril — a torpedo bomber "slowly disintegrating" in a corrosive environment — and it is "settled that all manner of objects other than vessels and their cargo are subject to salvage," raising sunken craft being a recognised salvage service [23] [23]. But salvage's encouragement rationale (its purpose is "to encourage rescue") runs through presumed consent, and the presumption is rebuttable: "salvage cannot be exacted for assistance forced upon a ship" (Merritt & Chapman Derrick & Wrecking Co. v. United States, 1927) [23]. The older cases are blunt — "If the master of a burning vessel prefers to allow her to burn rather than to permit outside parties to extinguish the flames, he may do so. He has a perfect right to decline any assistance that may be offered him: he should not be assisted against his will" (New Harbor Protection Co. v. Steamer Charles P. Chouteau, 1881; accord The Indian, 5th Cir. 1908) [23]. The treatises agree: an owner may reject salvage where only ordinary property interests are involved (Norris), and a salvor who acts without the owner's express or implied consent is a "gratuitous intermeddler" (Schoenbaum) — who adds, for historic wrecks specifically, that "only in a rare case where the governmental owner gives express or implied consent to salvage, should an award be given because the government has full power to reject or prohibit the services" [23]. The historic-wreck case law aligns: a salvage award "may be denied if the salvor forces its services on a vessel despite

rejection of them by a person with authority over the vessel" (Platoro); Sea Hunt's district court barred continued salvage of the 1802 Spanish wreck without Spain's permission; and the Eleventh Circuit itself, in Klein, had denied an award for a Biscayne National Park wreck, noting "the owner of the property [the United States] may not even have desired for the property to be 'rescued'" [23]. IAR's policy argument — the plane is deteriorating, so salvage should be permitted while recovery is still possible — the court expressly did not reach: it implicated no rights "beyond that of the United States in its own property" [23].

Holding: stop now; the past is a timing question

The judgment splits past from future. "IAR has no right to continue salvage operations over the express objections of the TBD-1's owner," and the district court's order permitting continuation was **reversed** [23] [23]. But Champlin's past efforts — locating the aircraft, videotaping it, returning the canopy and radio mast — "appear to satisfy" the elements of a salvage claim (voluntary assistance, no legal duty, constructive steps toward preservation of a vessel in marine peril), so an award might lie for work done before the United States effectively rejected the salvage [23]. Everything turns on the date of rejection: the record contains federal objections "stretching back to at least 1993" — letters from the Navy's Federal Preservation Officer (June 1993), a Justice Department trial attorney (February 1995), and the Director of Naval History (November 1998) — yet also evidence that Champlin negotiated with the Navy about trading the TBD-1 for vintage aircraft in government possession, suggesting the Navy "at times acquiesced" [23] [23]. The court also noted that laws regulating public property could themselves effect a "constructive rejection" of salvage of publicly owned vessels [23]. It remanded for the district court "to consider when the United States effectively rejected the salvage efforts of IAR and its predecessors-in-interest, and to calculate a salvage award, if appropriate, for their past efforts" [23].

The case thus refines the doctrine twice over: for title, abandonment of federal property is a formal act that inventory paperwork cannot simulate; for salvage, an owner's rejection — express, or constructed from the law governing public property — terminates the salvor's licence prospectively but not retroactively. What counts as rejection, and how much official silence or inaction can look like consent, is the question Section 51 takes up [23].

Federal salvage rights and the limits of official inaction

Section 50 stated the holdings; this section works through the record that produced them — because the case's real subject is time. What does forty-seven years of official silence do to a sovereign's title? International Aircraft Recovery (11th Cir., No. 99-13117, decided 17 July 2000, Kravitch, J.) answers: nothing at all [23] [23].

The aircraft and the forty-seven years

The defendant — sued in rem as an unidentified, wrecked and abandoned aircraft lying within one marine league of 25°43′34″ N, 80°2′8″ W — was a Navy "Devastator" TBD-1 torpedo bomber, built in 1938 [23]. It flew neutrality patrol in the central Atlantic, was assigned in mid-1941 to the carrier Yorktown in the Pacific, and fought at Coral Sea — where TBD-1s sank the Japanese carrier Shoho and badly damaged Shokaku — and at Midway; Yorktown recovered this aircraft after Coral Sea, and the Navy later overhauled it and used it for training at Miami [23]. On 1 July 1943, during a torpedo-attack instruction flight, the plane developed mechanical difficulties; the pilot and crew parachuted safely, and it crashed in deep international waters approximately eight miles east of Miami Beach [23]. The Navy did not know exactly where it had crashed; it "struck" the plane from the inventory of active aircraft in September 1943 — and, in the opinion's flat summary, "[s]ince that time, the Navy has taken no steps to locate or salvage the plane" [23].

In 1990 a group searching for Spanish galleons found the TBD-1 and offered to sell its location to the National Museum of Naval Aviation, which declined for want of an acquisitions budget; the discoverers sold the location instead to Windward Aviation, a corporation controlled by Douglas Champlin, a private collector of fighter planes [23]. Champlin negotiated to salvage the plane and hand it to the museum in exchange for other aircraft, but no agreement was reached; in two brief operations — 1994 (filming the wreck, recovering part of the canopy) and 1998 (another videotape, the radio mast) — he and his companies, including IAR, invested over $130,000 [23]. A first in rem suit in 1994, filed to secure exclusive salvage rights against rival salvors, was voluntarily dismissed after the federal government objected, and Champlin turned the canopy over to the museum [23]. The second suit — the one on appeal — sought an injunction protecting IAR's "exclusive salvage rights," plus either "a full and liberal salvage award" or title to the

aircraft under the law of finds [23]. Both parties agreed the aircraft is of substantial historical value: no TBD-1 has been preserved for display or study, and the only other known example also lies submerged in deep water [23].

"Federal salvage rights" first, title second — the district court's ordering

IAR's framing on appeal was that the district court had found abandonment, reviewable only for clear error. The Eleventh Circuit read the record otherwise: the district court "did not resolve the matter." It had discussed ownership and abandonment but held them "secondary to the question of whether this Court can protect the Plaintiff's ongoing federal salvage rights as to the In Rem Defendant aircraft," retained jurisdiction, and "clearly envisioned conducting salvage award proceedings in the future" [23]. That ordering matters doctrinally: had IAR acquired title through the law of finds, "there would be no basis for salvage award proceedings because courts do not supervise the efforts of owners to preserve their own property" [23]. The two bodies of law are exclusive: under salvage, "rescuers take possession of, but not title to, the distressed vessel and its contents," and a court fashions an award; only "[a] vessel without owner" is subject to finds — "finders keepers" [23]. And admiralty starts from a presumption that "owners do not give up title to ships and cargo in marine peril, even if cargo is swept overboard or a crew has to leave its vessel on the open water" [23]. The district court's instinct that the United States had probably abandoned the plane was, the court of appeals said, an instinct it "was correct to avoid" converting into a holding on this evidence [23].

Why inaction cannot divest the sovereign

The Property Clause supplies the rule: the Constitution gives Congress the power to dispose of federal property, and "[a]s courts consistently have recognized, the federal government cannot abandon property absent an affirmative act authorized by Congress" (Royal Indemnity Co. v. United States, 313 U.S. 289, 294 (1941)) [23]. The rationale is a trust: quoting United States v. California, 332 U.S. 19, 40 (1947) — "The Government, which holds its interests... in trust for all the people, is not to be deprived of those interests by the ordinary court rules designed particularly for private disputes over individually owned pieces of property; and officers who have no authority at all to dispose of Government property cannot by their conduct cause the Government to lose its valuable property rights by their acquiescence, laches, or failure to act" [23]. Acquiescence, laches, failure to act: the opinion enumerates the three forms official inaction can take and disables each. The admiralty precedents run the same way at longer timescales — the United States "has not abandoned its interests in ships sunk over a century ago during the Civil War" (United States v. Steinmetz; Hatteras, Inc. v. The U.S.S. Hatteras) [23]. Forty-seven years of Navy silence about the TBD-1 is, against that line, not a close case.

IAR's best evidence collapsed on inspection. Its striking argument "relies heavily" on the deposition of Dr. William Dudley, the Navy's Director of Naval History, who, asked a hypothetical — how a carrier captain destroying excess planes at sea at the end of World War II would record their "final disposition" for a successor — speculated that an inventory would show them as "stricken" [23]. Dudley qualified the answer as conjectural and in need of research; it addressed inventory notation, not "how the Navy would officially divest itself of all property interests"; and later in the same deposition he clarified that the Navy could abandon title to an aircraft only by asking Congress to pass specific legislation [23]. Striking records the end of accountability; it is not, and cannot be, the affirmative act of disposal the Property Clause requires.

The asymmetry the case leaves behind

The limits of inaction are thus one-directional. For title, official inaction is legally inert — no lapse of time, no failure to search, no subordinate's acquiescence moves property out of the sovereign. For salvage awards, by contrast, official behaviour is precisely what counts: whether the United States objected, when, and whether its negotiators' willingness to discuss trading the TBD-1 for other vintage aircraft amounted to acquiescence in Champlin's work — the questions the remand (Section 50) sends back [23]. The salvor's "federal salvage rights," in other words, exist only in the owner's shadow: they can be extinguished prospectively by objection, but never enlarged by silence into ownership. IAR itself disclaimed any wish to keep the plane — it "believes that the plane belongs on display at the National Museum of Naval Aviation," the outcome the district court intimated it might reach by awarding the aircraft to the museum with compensation to IAR [23]. The law preferred the owner's veto to the artefact's rescue, and left the last unpreserved Devastator where it lay. With the wreck-law part closed, the report changes seas and subjects — from who owns the dead to who is blamed for the dying: the criminalisation of rescue in the Mediterranean [23].

The Deadliest Crossing: Forensic Oceanography and the Mediterranean

Forensic Oceanography: blaming the rescuers

By 2017, the people pulling migrants out of the central Mediterranean had become suspects. Frontex and other state actors accused the search-and-rescue NGOs of "colluding with smugglers," of "constituting a pull-factor," and of ultimately endangering the migrants they rescued. Blaming the Rescuers, the counter-analysis produced by Forensic Oceanography — the research practice of Charles Heller and Lorenzo Pezzani, whose earlier Death by Rescue had indicted the EU's policies of non-assistance — answers the accusations in the accusers' own documents: the leaked six-monthly report of the EUNAVFOR MED operation, Frontex's published risk analyses, and the statistical record of crossings and deaths. Its method is forensic in the strict sense: reconstruct what actually moved the numbers, then compare that reconstruction with the story being told [24] [24].

The story being told skips its own baseline. After the April 2015 shipwrecks, state and non-state actors had re-expanded at sea together, and through most of 2015 "complementarity prevailed over conflict": a "division of labour" in which NGO rescues let security-oriented actors concentrate on surveillance, interceptions of smugglers and destruction of vessels. The flotilla grew fast — only MOAS operated in 2014; in 2015 MSF joined with two vessels and Sea-Watch with one — and by the end of 2015 the NGOs had rescued more than 20,000 people, some 13% of all rescues. Even though 2,800 people died in the central Mediterranean between May and December 2015, the combined effort brought the mortality rate down to a level comparable to that recorded during Italy's Mare Nostrum operation. The security actors, meanwhile, claimed the credit: EUNAVFOR MED's first six-monthly report boasted that "for the first time in 3 years, we have seen a 9% reduction in the migrant flow using the central (Mediterranean) route," attributing it partly to "the deterrence effect provided in international waters by EUNAVFOR Med assets," and expected the decrease "to continue to be driven down" [24].

That prognosis was, in the report's word, deeply flawed — and Frontex's own analysis shows why. The 2015 aggregate concealed opposite movements by nationality: 153,946 detections on the central Mediterranean route, a 10% decrease on 2014, driven by the collapse in Syrian crossings (about 40,000 in 2014, fewer than 7,500 in 2015, as Syrians shifted to the eastern route), while East and West Africans "steadily increased from below 80,000 in 2014 to more than 108,000 in 2015 (+42%)." The rise that broke in 2016 was already running in 2015, before the NGO flotilla reached full strength; it was the product of "deeper regional economic and political dynamics" and of the chaos in Libya itself [24].

Against the first accusation — the pull factor — the counter-analysis is therefore chronological. The increased crossings of 2016 "were not the product of the supposed 'pull-factor' constituted by SAR NGOs, but were a continuation of a trend that had already begun independently of the presence of SAR NGOs"; the underlying trends "were reported by Frontex both internally and externally, but were not mentioned in relation to the agency's arguments against SAR NGOs." The termination of Mare Nostrum had already demonstrated the point empirically: crossings continued without dedicated rescue, because "those stuck in Libya have little choice but to attempt it, and do so with or without dedicated SAR operations." Nor was the rebuttal an NGO talking point: the commander-in-chief of EUNAVFOR MED, Admiral Credendino, the Italian Coast Guard's Rear Admiral Nicola Carlone and Italy's vice-minister for foreign affairs Mario Giro — "institutional figures with no interest in defending the NGOs" — all made the same argument, urging attention to push factors such as the appalling conditions in Libya. And the report presses the moral structure underneath: the pull-factor argument "rests on the morally questionable assumption that it would be wrong to provide migrants with a safe(r) opportunity to leave," and whoever invokes it "is also implicitly legitimising the use of the prospect of death at sea as a deterrent" [24].

The second accusation — that NGOs encouraged smugglers toward poorer boats and more dangerous tactics — inverts the documented causation. The degradation of crossing conditions dates to 2013 and is rooted in Libya's violent fragmentation; by the end of 2015 it had produced "a new and more dangerous model of militia-led smuggling." The EU's anti-smuggling operation then amplified it, "as recorded in its own internal reports": by interdicting and destroying smugglers' vessels, EUNAVFOR MED "confined them to Libyan territorial waters and contributed to the shift from wooden to cheap rubber boats." Vessel destruction was not incidental to the operation — it was its mandate, launched under Council Decision 2015/778 to "identify, capture and dispose of vessels" used by smugglers, within an EU action plan that expressly included "destroying smuggler vessels" and seizing criminal assets [25]. The timing closes the argument: the tactical shifts "were noted at the end of 2015 and in the first months of 2016, when

the presence of SAR NGOs was limited," so the NGOs cannot have caused them. What did add danger in 2016 was the Libyan Coast Guard, which, "under pressure from the EU," increasingly intercepted boats as they left the coast and "repeatedly exercised violence in the process at times leading boats to capsize." From spring 2016, it was the Italian Coast Guard that "increasingly directed" NGO vessels closer to the Libyan coastline "to avert situations of imminent distress." The report concedes the one real feedback: moving closer "may have contributed to further consolidating smugglers' new practices – such as no longer providing migrants with a satellite phone," but insists "this was the condition to rescue migrants more effectively" [24].

The third accusation — that NGOs made the crossing more dangerous despite their intentions — rests on an apparent paradox the report dissolves by disaggregating the year. 2016 was the deadliest on record, with 4,581 deaths and a mortality rate that "rose to 2.5 from 1.83 in 2015," and it also saw the highest number of SAR NGOs operating. But the mortality rate "had risen in early 2016 before NGO SAR assets returned to the central Mediterranean following their winter break, and declined in parallel to their redeployment"; it "rose again only when SAR NGOs' presence decreased at the end of the autumn." The correlation runs the opposite way to the accusation — a "striking negative correlation" between deployed NGO vessels and mortality (Pearson's R of −.314 for 2016; −.532 for 2015), and the report prints its own caveat: the 2016 figure is not statistically significant, the 2015 one only borderline. What the numbers cannot capture, a photograph does: Jugend Rettet's Iuventa over Easter weekend 2017, nominal capacity no more than 100 people, taking hundreds aboard "to make up for the absence of state-led SAR assets" [24].

Why, then, blame the rescuers? The report's conclusion is that the narrative does strategic work. It diverts attention "from their own responsibilities and failures – such as the continuing SAR gap that has made SAR NGOs essential in the first place, and the effects of the EU's anti-smuggling operations which have contributed to making the crossing more dangerous but failed to stop the smuggling business." It stands "in continuity with prior policies – such as the ending of Mare Nostrum – which have attempted to deter migrants by making the crossing more difficult, with the only effect of leading to thousands of deaths." And it clears the ground for the "real" solutions its proponents invariably offer: cooperation with regimes at the EU's periphery. Frontex director Fabrice Leggeri, asked for a solution for the central Mediterranean, cited the closing of the Atlantic route to Spain through "cooperation with African states from which migrants' boats were leaving," to which "migrants were swiftly brought back"; the European Commission's February 2017 strategic note held that "the drawbacks of SAR operations as they are currently carried out by European naval forces, coast guards and NGOs must be acknowledged with a view to stemming the numbers of irregular crossings," since "[a] purely humanitarian approach will not suffice." The counter-analysis answers that the Atlantic closure "did not stop the crossings of illegalised migrants and deaths at sea overall, but rather displaced these crossings and deaths to other routes," and that externalisation in the central Mediterranean "effectively means cooperating with (war) criminals, such as in Sudan and Eritrea," and relegating migrants to Libya, where violations are systematically documented: "targeting SAR NGOs so as to gain control over them and enabling the LCG to pull-back migrants to Libya are both sides of the same coin," and preventing departure from Libya "amounts to complicity with arbitrary detention, torture, sexual violence, forced labour and trafficking" [24] [24].

The deterrence logic underneath was stated plainly, on the record, by a senior European border official who told the journalist Zack Campbell: "In order to not create a pull factor, we are patrolling up to the SAR area of Malta. We don't cover Libya" — reasoning that if the journey seems longer and more dangerous, refugees will not risk it, "especially in winter" [24]. And the campaign against the rescuers belongs to a wider criminalisation of solidarity: in Denmark, Greece and France, people who hosted or helped migrants en route have been tried for trafficking and other "crimes of solidarity," endangering not only migrants but "the rights of EU citizens to stand in solidarity as well as their capacity to exercise civilian oversight at the EU's frontiers" [24] [24]. Frontex's own next risk analysis would quietly concede the causal geography the counter-analysis had insisted on: the central Mediterranean numbers collapsed in July 2017 "mostly due to internal developments in Libya," in a year when total detections fell to 204,719 — down 60% on 2016's 511,047 [26]. The rescuers had been blamed for a system whose levers were always elsewhere.

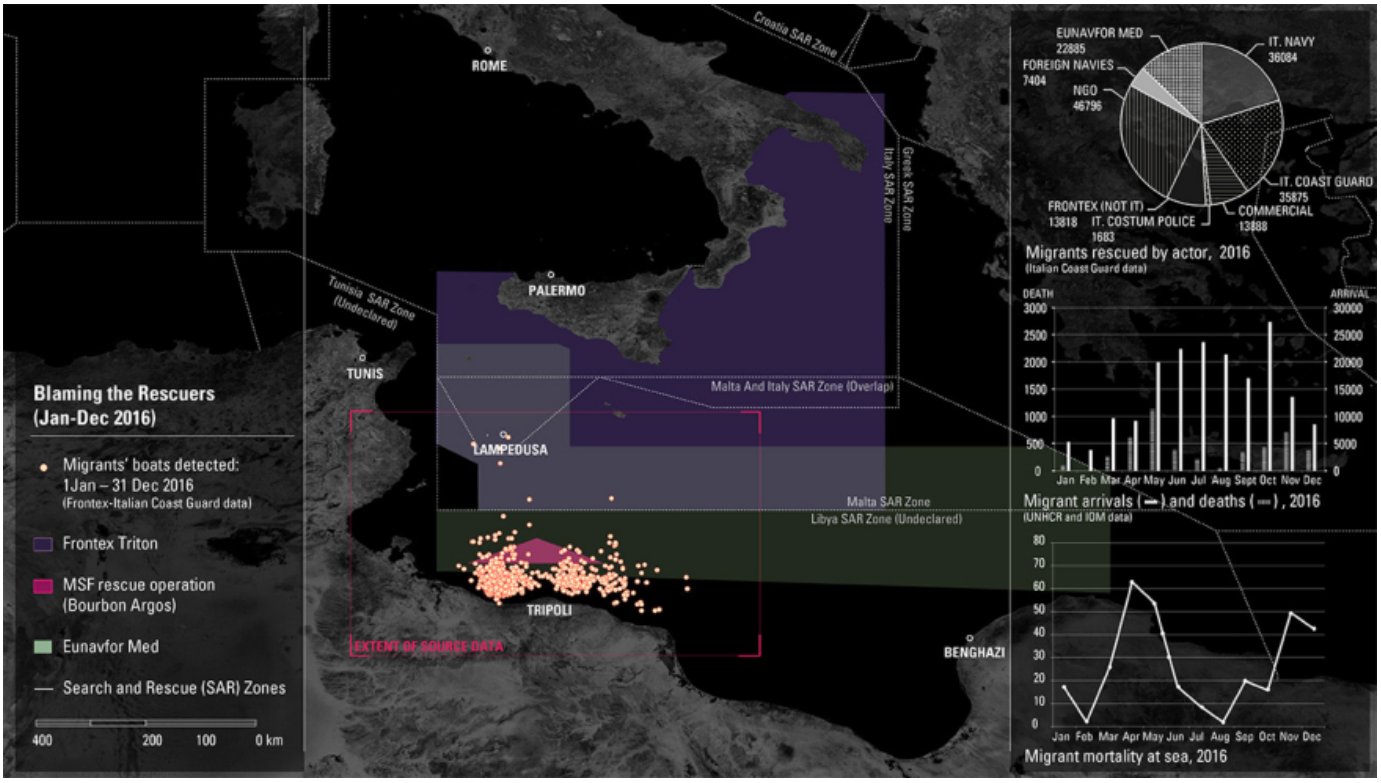


Exhibit. Map of the situation in the central Mediterranean between January and December 2016, reconstructing search-and-rescue incidents involving NGO vessels and the Libyan Coast Guard. Source: S0023.

Criminalising solidarity and re-enforcing deterrence

The subtitle of Blaming the Rescuers states its thesis as a single mechanism: criminalising solidarity is how deterrence gets re-enforced. The previous section followed the report's rebuttals of the three charges against the rescue NGOs. This one follows its dissection of the accuser — how Frontex built the narrative, what the 2016 season actually looked like while the narrative was being built, and what remained when, by Easter 2017, the rescuers were still the ones doing the rescuing.

Start with the standard Frontex sets for itself. Coordinating European border management, the agency describes its first task as "monitoring migratory flows and carrying out risk analysis," and its risk analyses as produced in the "pursuit of factual exactness, truth and exhaustive analysis" — a "reliable basis for its operational activities." Measured against that standard, the 2017 Annual Risk Analysis's treatment of rescue is a case study in selective fact. The data Frontex itself reports are striking: during 2015 and the first months of 2016, smuggling groups instructed migrants to make satellite phone calls to the Maritime Rescue Coordination Centre in Rome; more than half of all rescue operations were initiated that way, and "NGO vessels involved in less than 5% of the incidents." From June to October 2016 "the pattern was reversed": satellite phone calls "decreased sharply to 10% and NGO rescue operations rose significantly to more than 40% of all incidents" [24].

What Frontex builds on that reversal is not analysis but a narrative move, executed in two steps. First, generalisation: Libyan-based smugglers had "heavily relied on the International Convention for the Safety of Life at Sea (SOLAS), and associated SAR as well as humanitarian assistance efforts, turning it into a distinct tactical advantage," organising dangerous crossings "with the main purpose of being detected by EUNAVFOR Med/Frontex and NGO vessels." Then the conclusion, which sweeps in everyone: "Apparently, all parties involved in SAR operations in the Central Mediterranean unintentionally help criminals achieve their objectives at minimum cost, strengthen their business model by increasing the chances of success." Forensic Oceanography names the strategy precisely: the report "focuses on a single actor, SAR NGOs, isolating them from the web of interactions with other actors which together shape the dynamics of migration across the sea," establishing parallels between their activities and wider trends; it then "generalises its criticism of the 'unintended consequences' of SAR to 'all parties involved in SAR operations'" — but because the NGOs were singled out first, and because they "have become the primary SAR actor in the central Mediterranean, they emerge as the main target of the criticism." The move had a precedent: it is "how similar the argument formulated by Frontex here is to that it repeated, time and again, against Mare Nostrum in 2014," a campaign that "contributed to delegitimizing and ultimately terminating the operation" — with consequences the authors documented in Death by Rescue [24].

The season the narrative misdescribed is recoverable from the operational record. By the beginning of 2016, the effects of EUNAVFOR MED and the dynamics within Libya "had combined to make the conditions of crossing increasingly dangerous. More people were crossing, in more boats that were less sea-worthy, in more dangerous conditions, and were in distress closer to the Libyan coast." SAR events leapt from 906 in 2015 to 1,424 in 2016; smugglers sent out many boats at once, with more than 20 simultaneous cases occurring regularly over the summer and "a staggering peak of 53 SAR events in one day reached on 29 August." Average loading on rubber boats rose from 103 people in 2015 to 122 in 2016, with cases of 200; Frontex's own AFIC report notes that "the wooden planks used to reinforce the floor of the rubber dinghies are so roughly cut that, on several overcrowded boats, they broke under the migrants' weight," and suffocation deaths, once a horror of wooden vessels' holds, were now reported on rubber boats, migrants "crushed by fellow passengers." Denied easy recovery of their vessels by EUNAVFOR MED's interdictions, smugglers adopted a new tactic the operation's own report records — "a skiff towing a rubber boat without an engine, which is then left adrift" — lethal in the 26 May 2016 case, when one of two boats carrying some 500 travellers each capsized about 70 kilometres northeast of Zuwara. EUNAVFOR MED had already conceded the underlying condition in 2015: with "the limited supply and the degree of overloading, the migrant vessels are [to be considered in distress according to international conventions] from the moment they launch," and in 2016 that "the majority of migrants still die inside or very close to Libyan territorial waters" [24].

The NGOs' response to distress moving shoreward was to move shoreward themselves. The flotilla expanded from the four vessels of 2015 to a peak of twelve in 2016 — SOS Méditerranée, Sea-Eye, Jugend Rettet, Life Boat, Proactiva Open Arms, Bootvluchteling and Save the Children joining MOAS, MSF and Sea-Watch, several with smaller, lower-capacity vessels. Deployment lagged the danger: in April 2016 only SOS Méditerranée's vessel was present for most of the month, and NGOs accounted for just 4% of SAR activity in April and 12% in May. MSF and Sea-Watch initially held beyond the 24-nautical-mile limit, but "MRCC Rome however increasingly directed them to intervene within the 24nm limit, where situations of imminent distress were being signalled"; over the summer others intervened proactively still closer, "at times within, Libyan territorial waters" — inter-NGO competition may have played a part, the report allows, but "moving closer to the Libyan coast corresponded to a humanitarian need." The result, over the summer, was mortality brought down as the flotilla peaked at eleven vessels and the NGO share of rescues peaked at 35% in June — and mortality peaking again in November and December as the vessels withdrew for winter. Frontex's explanation for record crossings in October and November was that smugglers, knowing "maritime assets are patrolling close to the Libyan shore," encouraged migrants "to sail even when the sea is rough because they will be rescued soon after their departure" — an attribution the report notes was made "without providing any evidence of this causal connection," and which fails on its own terms: proximity had been constant for months, so it cannot explain a sudden increase; it omits the changing nationality mix (West and Central Africans, who continue to depart in winter); and it omits Libya itself, where Nancy Porsia recorded intense clashes for control of the port of Zawiya and confrontations in Tripoli, and where, as Mark Micallef notes, the coastal road from Tripoli toward the west coast — closed since October 2015 — was reopened for most of September, October and early November, easing the movement of migrants to the shore [24].

Easter weekend 2017 supplied the natural experiment. Two years after the April 2015 shipwrecks that killed 1,200 people, crossings hit record levels again: between 14 and 16 April, 9,262 people travelling on 55 boats were rescued, "one of the largest events of concentrated SAR operations in the Mediterranean Sea in the past few years." Of the 25 ships involved, according to MRCC data, one was operated by Frontex and one by EUNAVFOR MED; ten belonged to NGOs, six to the Italian coastguard, six were merchant vessels and one was an Italian navy ship. Frontex clarified that some coastguard assets were co-financed under Triton but admitted only six of its sixteen assets were deployed near the events; EUNAVFOR MED confirmed only one of its ships took part, its four others "either in port or busy with other tasks." The NGOs — "under constant attacks for several months" — worked, in the WatchTheMed Alarm Phone's summary, "at the limit of their capabilities": MOAS's crew alone rescued more than 1,500 people from nine boats, taking hundreds aboard the Phoenix, while Jugend Rettet's Iuventa took hundreds aboard and, "[u]nable to navigate, they were even forced to send off a MAYDAY call on Sunday" before completing the operation and returning safely to Malta [24].

What deterrence holds in place, while the argument runs, is visible in Frontex's own later accounting. The agency's Risk Analysis for 2018 records that children are the second most prevalent group among trafficking victims after women — around 25–30% of victims globally in 2012–2014 — and that 92% of all children arriving in Italy by sea in 2016 and the first two months of 2017 were believed to be unaccompanied, a population it describes as "the perfect target for unscrupulous traffickers." It likewise notes that, assuming the EU-Turkey statement holds, the decisive variable for 2018 arrivals is "the remaining large pool of migrants stranded in Libya." Enforcement, meanwhile, coun-

ted 10,213 reported facilitators in 2017, down 19% on 2016 [26] [26]. The crossings that feed those statistics were, in scale, a return to pre-crisis patterns on the central route — 170,760 in 2014 and 153,964 in 2015, after 40,000 in 2013, dwarfed in 2015 by the eastern route's 885,386 [25]. The report's closing claim follows from its evidence: the criminalisation of rescue is not an analytical error but a policy instrument — it removes the rescuers and the witnesses in one motion, re-enforces deterrence by leaving the sea emptier, and prices the policy in the mortality statistics of the months when the NGO vessels are gone [24] [24].

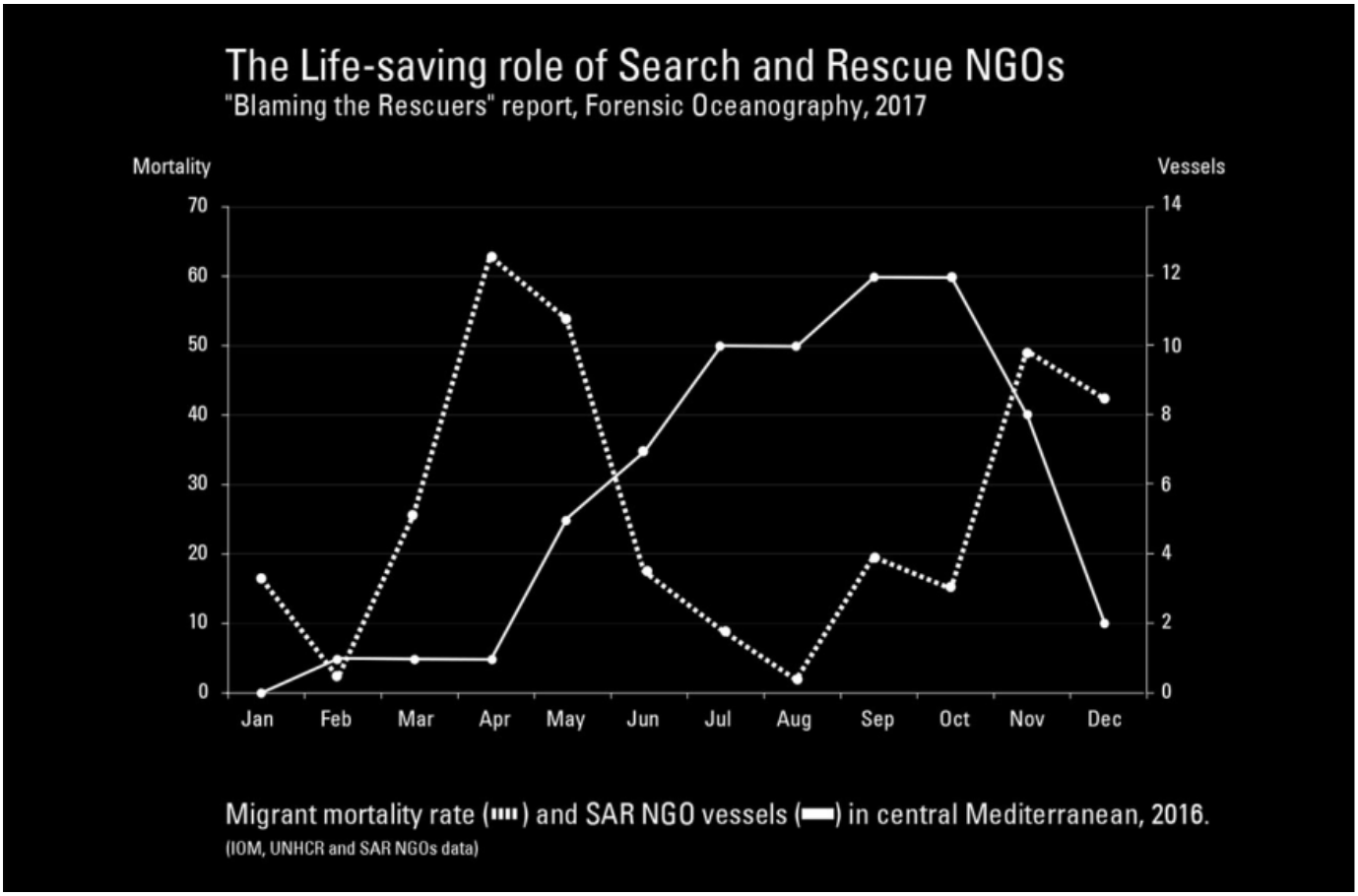


Exhibit. Mediterranean crossing deaths in 2016, the deadliest year on record, which coincided with the highest number of search-and-rescue NGOs operating in the central Mediterranean. Source: S0023.

The turmoil in Libya and the Libyan coast guard

Every account of the 2016 crossing that stays at sea level misses the cause. Blaming the Rescuers insists, in its boldest passage, that "the dynamics of Libyan smuggling are deeply shaped by the fragmented political landscape in Libya, which constitutes a causal factor in its own right." Drawing on Nancy Porsia's fieldwork in Libya and Mark Micallef's Human Conveyor Belt, the report reconstructs the Libyan side of the route: a smuggling market reorganised by militia predation and split into two tiers of danger, and a coast guard that Europe was simultaneously building up, paying for, and — in its own intelligence reporting — describing as part of the smuggling business [24].

The market's structure determined who travelled safely and who did not. Brokers from Nigeria, Eritrea and Ethiopia had set up headquarters inside Libya and could "control the entire passage from Sudan and Niger up to Europe," relying on Libyan smugglers for particular legs. Migrants from the Horn of Africa — mainly Eritreans, Ethiopians and Somalis — sealed their deals in Sudan for the whole journey; they paid higher prices, and in certain cases the smugglers were paid only once passengers arrived safely in Europe, through the "hawala" payment system. That gave Horn migrants a margin of bargaining power: they "usually refuse to leave outside of the Spring and Summer months" when bad weather is less likely. Central and West African migrants, by contrast, were usually recruited in Agadez, in Niger, and paid cash for each leg; opting for lower-budget crossings, they were "offered less high security standards, which translates in boats of lesser quality, travelling in the hold of vessels and being regularly sent off during the winter." Over 2015 the low-cost model spread to all embarkation points, but the hierarchy persisted — and the changing nationality composition of 2016 meant "a greater proportion of migrants went through the more dangerous networks," an effect on risk the report flags as real but "difficult to verify empirically," since IOM's death

records carry too many unknown nationalities [24]. Frontex's own debriefing programme corroborates the hub structure from the other side: facilitation was perceived as the most important variable by 65% of interviewed migrants on the central Mediterranean route; facilitators operated mainly in Turkey (37%) and Libya (30%); and Libyan facilitators overwhelmingly smuggled other nationalities — Moroccans, Syrians, Sudanese, Egyptians, Tunisians and Ethiopians, a 93% non-matching pattern [26].

Into this landscape, Europe set out to build a partner. Since the fall of the Qaddafi regime in 2011 — "with whom Italy had collaborated to push-back migrants intercepted at sea" — the EU and its member states had pressured whichever Libyan authority they could find to clamp down on departures. The EU's EUBAM Libya mission had been training the Libyan Coast Guard since 2014, and the LCG cooperated with the Commission, Frontex and the Seahorse project. On 20 June 2016 the European Council launched a dedicated training programme to be implemented by EUNAVFOR MED, which considered "a capable and well-resourced Libyan Coastguard who can [...] prevent irregular migration taking place from their shores" critical to its own exit strategy; training began on 26 October 2016 with 78 trainees aboard EUNAVFOR MED assets, with UNHCR and Frontex teams running specific modules. On 2 February 2017 Italy formalised the relationship in a Memorandum of Understanding with Libya's Government of National Accord, centred on fighting illegalised migration at sea and on Libya's southern border. A Tripoli court suspended the Memorandum on 22 March 2017 after a complaint by legal experts and former politicians, but "its implementation remains underway," and on 15 May 2017 Italy delivered the first four of ten patrol vessels to the LCG [24] [24].

The partner, however, existed mostly on paper. As Porsia's analysis shows, the LCG "is itself as fragmented and volatile as the Libyan political landscape": technically six sectors coordinated by a national command in Tripoli, but the Tripoli command "has no control over the units in the Eastern area, as they report to the Parliament based in Tobruq," and in the west it "has little control overall," with militias performing coast-guard functions — "making it extremely difficult to differentiate actors on the ground." Despite the fragmentation and its limited vessels, EU pressure worked on the metric Europe measured: whereas in 2015 the LCG intervened infrequently in distress situations, in 2016 its units "rescued" 18,904 people off the Libyan coast, according to IOM data, "bringing them back to Libya where migrants are subsequently detained in extremely dire conditions" [24].

The LCG's intervention, the report continues, "simultaneously conflicts with, and is embedded within, the smuggling business." While the LCG sought to demonstrate effectiveness "to tap into the opportunity for EU financial and political support, units also receive payment by smugglers and militias to let boats pass, and officials may receive payment for the release from detention centres of intercepted migrants." None of this was secret to the agencies building the partnership. Frontex noted in its internal reporting — in a biweekly update of 31 March 2016 — that "[g]athered information suggests that high ranking officers from different military branches are involved in the smuggling of irregular migrants from the west coast of Libya towards Italy," with "military officers between the ranks of Lieutenant and General involved at different stages of smuggling," and law-enforcement officers likewise identified; the same analysis appears in Frontex's AFIC 2016 report [24].

Interception on those terms made the crossing more violent. Smugglers adapted to the LCG's increasing activity by carrying heavy weapons — in April 2017 the LCG clashed with suspected smugglers and four people were killed. The Zawiya coast guard, the most active west of Tripoli with several functioning patrol vessels and rigid-hulled inflatables along the coast from Mutrud to Sabratha, "is reported to have removed the engine of boats seeking to pass without payment, leaving the boats adrift." The Times published video — initially part of Ross Kemp's documentary Libya's Migrant Hell — showing the Zawiya coast guard beating migrants with a rope as they sat packed in a rubber boat during an interception, a practice which, beyond its deplorability, "can also lead to the boat capsizing" [24]. The same maritime units around Zawiya were implicated in a sequence of incidents with the rescue NGOs. On 17 August 2016, MSF's Bourbon Argos, 24 nautical miles north of the coast, was attacked by armed men on a speedboat who "fired shots toward the Bourbon Argos from a distance of 400 to 500 metres and then boarded the vessel," leaving without harming the crew. On 9 September 2016 the crew of a Sea-Eye speedboat was arrested by the LCG after entering territorial waters near Zawiya [24]. And on 21 October 2016, the Zawiya coast guard violently interrupted a rescue Sea-Watch was conducting 14.5 nautical miles from the coast, boarding the overcrowded rubber boat and beating people, "causing panic and a rupture in the boat": over 150 people ended up in the water, of whom Sea-Watch rescued 124 and recovered four corpses — the report's photographs of the sequence record the incident as costing at least 25 lives [24].

The pattern held as the report went to press. On 10 May 2017, MRCC Rome received a distress call from a boat inside Libyan territorial waters; the Libyan authorities said they would coordinate, and Rome also tasked Sea-Watch, whose vessel arrived first, the boat by then outside territorial waters. As Sea-Watch deployed its inflatable to ap-

proach, “the LCG’s vessel almost rammed into the Sea-Watch vessel to intimidate them”; Sea-Watch retreated and watched the LCG stop the migrants’ boat “under the threat of a gun,” failing to act in accordance with established safety standards. No casualties were reported; the migrants were pulled back to Libya and detention. The report’s inference is blunt: “the destabilising intervention of the LCG is bound to increase in 2017, as the pressure and resourcing from the EU continues” [24].

The section’s verdict gathers the three evolutions — “the increasing involvement of militias in the smuggling business, the shift in composition of migrant nationalities, and the increasing interventions of the Libyan Coast Guard” — as the drivers of the downward spiral of 2015–2016, and charges that while Frontex “has analysed smuggling networks in Libya, it has kept these factors out of the analysis of the causes of the deteriorating conditions of crossing offered to migrants, blaming them instead on SAR NGOs” [24]. The interception regime’s measurable harvest came the following summer: after a first half of 2017 running at or above 2016 levels (23,461 detections in June against 22,344 a year earlier), central Mediterranean detections collapsed to 11,460 in July and 3,914 in August, for a year total of 118,962, down 34% — a fall Frontex attributes to “a variety of factors, notably internal developments in Libya,” explicitly “not a decline in the pool of migrants looking to use the route,” but a second half in which “migrants were increasingly prevented from departing from Libya” [26]. Prevention, not safety, was the product.

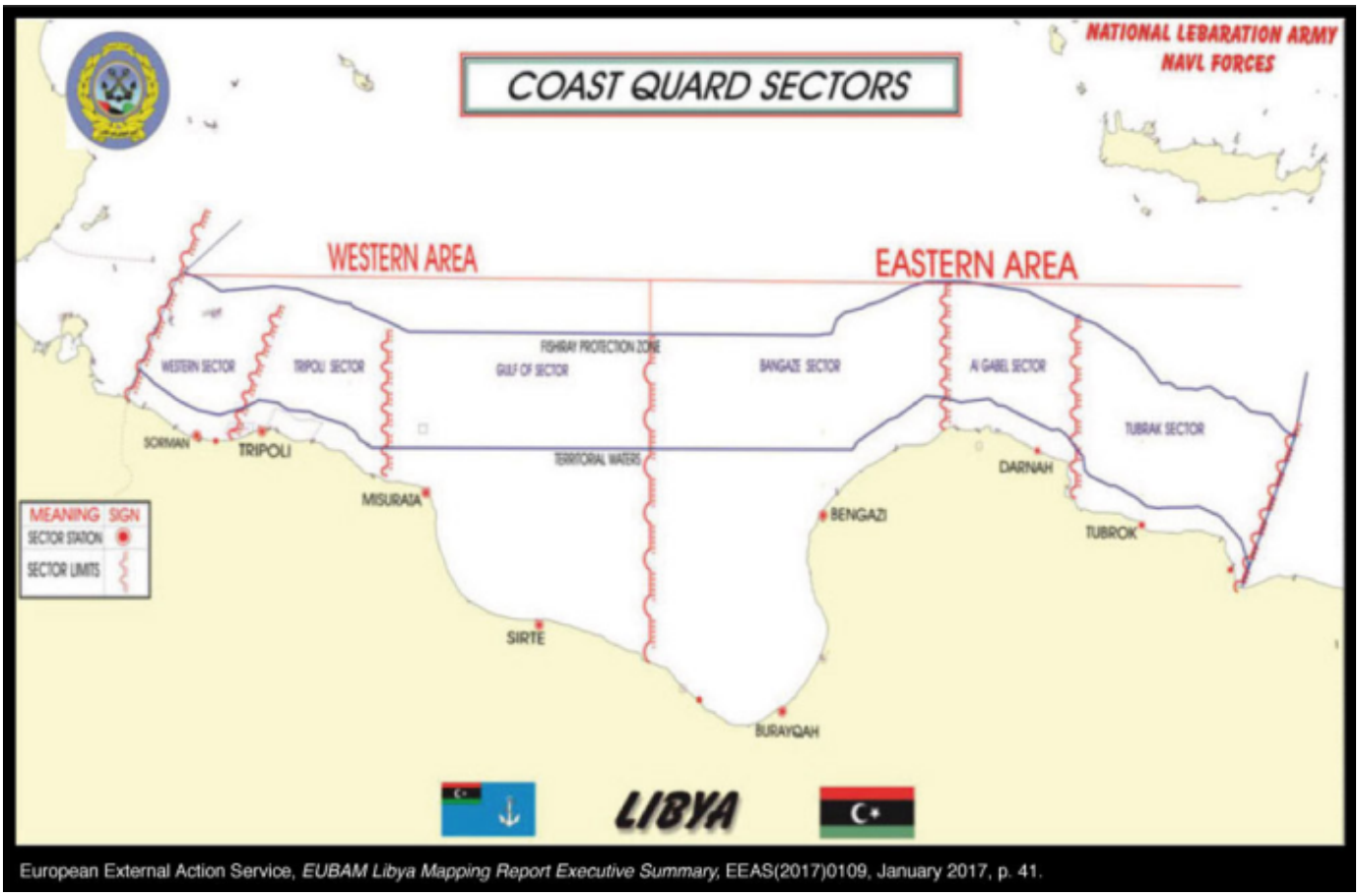


Exhibit. Map of Libyan Coast Guard sectors. Source: S0023.

Worsening smuggler tactics and the danger of the crossing

Nobody disputes the fact at the centre of this section: the crossing got more dangerous. Deaths in the central Mediterranean rose from 2,892 in 2015 to 4,581 in 2016, and the mortality rate — Forensic Oceanography’s own calculation from IOM death counts and UNHCR arrival figures — rose with them, from 1.83 to 2.5 per hundred crossings. “The increasing danger of crossing in the central Mediterranean is a fact,” the report concedes, and it “demands a serious evaluation in and of itself.” The dispute is entirely about causation: whether the rescue NGOs’ presence drove smugglers to worse tactics, or whether the tactics worsened for reasons that predate and surround the rescuers. The report’s method is built for exactly that question — interviews with state officials, SAR NGOs and migrants, newly accessed official reports, investigative journalists specialising in Libyan smuggling networks, statistical analysis and cartography, assembled by a team based within the Forensic Architecture agency at Goldsmiths, University of London, that specialises in reconstructing “the conditions that lead to deaths at sea” [24].

The tactical deterioration is itemised plainly. Over 2015 and 2016 smugglers shifted to "bad quality rubber boats instead of the more solid wooden boats"; they provided "less fuel, food and water"; departures increased "in more difficult weather conditions"; and overloading grew ever more extreme — the downward spiral the report "acknowledges" in full [24]. To that inventory the operational record adds the swarm tactic: Frontex's biweekly update of 31 March 2016 reported that "migrants continue to report simultaneous departures of rubber dinghies from the Libyan coast in what seems to be one solution to the lack of wooden boats. Using this approach smuggling networks are able to smuggle several hundred migrants at the same time aboard rubber dinghies" [24]. The European Parliament's research service describes the route's standard materiel in the same terms: crossings "usually crossed in old, often unseaworthy fishing boats or small rubber dinghies, usually with insufficient fuel and without proper navigation systems" [25].

The attribution collapses on chronology. The marked tactical shifts — the increasing use of rubber boats and the multiple simultaneous departures — "were reported by Frontex between January and March 2016. At this point in time, there was only one single NGO SAR vessel deployed (SOS Méditerranée's Aquarius), while other SAR NGO assets had not been deployed over the winter. As such, the allegation that the presence of SAR NGOs has been the driver of these tactical shifts must be ruled out." The positive attribution points the other way: "[t]he evidence provided by the evaluation of EUNAVFOR MED of its own mission between 2015 and 2016, confirmed by interviews in Libya and the trends documented by the data compiled by the Italian Coast Guard and Frontex, thus demonstrates that the shifts in smugglers' tactics registered at the end of 2015 and the beginning of 2016 cannot be attributed to NGO SAR activities since these activities were limited during these periods. Rather, EUNAVFOR MED attributes these shifts, which increased the danger of the crossing, to the effects of its own operation" [24].

The mechanism is economic, and the operation stated it itself. Destroying and interdicting smugglers' vessels changed the cost structure of the business: "(...) smugglers can no longer recover [wooden] smuggling vessels on the High seas, effectively rendering them a less economic option for the smuggling business and thereby hampering it," EUNAVFOR MED reported in January 2016 [24]. A wooden boat that cannot be recovered is a capital loss on every trip; a cheap rubber dinghy is a consumable. The report situates the episode in a general pattern documented across past operations: anti-smuggling operations "systematically lead to an evolution of smuggling tactics, usually entailing more risk for migrants, while the demand for the service of smugglers remains unchanged." What is "surprising," the authors add, is that while Frontex repeatedly noted the tactical shifts in its own biweekly reporting, it never once related them to EUNAVFOR MED, which "seems to remain entirely outside of their field of attention" — and "[k]eeping EUNAVFOR MED's effects on smugglers' tactics outside of the frame of analysis has allowed Frontex and others in turn to blame these shifts on SAR NGOs, fuelling the toxic narrative against them" [24].

A control case runs along the coast. On the western Mediterranean route from Morocco, crossings rose 46% from 2015 to 2016 "in the absence of any SAR NGO assets" — the pull-factor hypothesis predicts no such rise where no rescuers operate. And Frontex's own 2017 summary of the central route concedes the trend's continuity: "the Central Mediterranean saw the highest number of migrant arrivals ever recorded from sub-Sahara, West Africa and the Horn of Africa (181,459 migrants, increase of 18% compared with 2015). This trend, which is consistent with previous year-on-year increases, shows that the Central Mediterranean has become the main route for African migrants to the EU" [24]. The route's modern scale, in any case, long predates the flotilla: the OECD's 2015 briefing, summarised by the European Parliament's research service, records illegal maritime arrivals in Italy jumping after the fall of the Qaddafi regime from 4,406 in 2010 to 64,261 in 2011, with a record 172,114 crossings in 2014 [25].

The business models explain who absorbs the new risk. The OECD distinguishes two smuggling systems: a "full package" model that "works almost like a 'travel agent'," requiring large sums and bundling services including falsified documents and transport, and a "pay-as-you-go" model in which "the final destination is not predetermined and the speed and direction of the journey depends on the funds available to migrants at each step." It is the pay-as-you-go clientele — cash per leg, no package protections — who fill the worst boats in the worst weather. The same briefing "points to a direct link between migrant smuggling and corruption," with bribes to public officials at entry and exit points supplementing official incomes throughout the chain — the structural embeddedness that, on the Libyan coast, made the coast guard a participant rather than a referee [25]. The longer frame only sharpens the point: the IOM estimates 22,400 people died trying to reach Europe via the Mediterranean between 2000 and 2014, that in 2014 the Mediterranean accounted for 75% of all border-related fatalities worldwide, and that more than 3,770 migrants were lost at sea in 2015 alone [25].

All of this unfolded while the rescuers were being investigated as well as blamed. Beyond Frontex's subtler line — the alleged effect of proactive rescue on migration dynamics — the de-legitimisation campaign "has not only in-

volved Frontex… high-level politicians, and the media, but has also led to the opening of several exploratory inquiries by prosecutors in Italy"; its most heinous aspects, the collusion allegations, "have proven baseless or have already been effectively refuted," in parliamentary hearings and in the NGOs' own published rebuttals. "Proactive" is the operative word: Forensic Oceanography defines it as operations whose main goal is search and rescue, deploying toward the areas where migrants encounter distress — whereas "[c]urrently, all of the state-operated missions in the Central Mediterranean have security-oriented goals… and engage in SAR operations only when called upon by the competent authorities." The gap the NGOs filled was created by a decision: the termination of Mare Nostrum, which had patrolled close to the Libyan coast, whose end "left a huge gap in Search and Rescue capabilities that, partially filled by ill-equipped merchant vessels, led to a staggering increase in deaths at sea in early 2015" — and although ending Mare Nostrum was recognised, in President Juncker's words to the European Parliament, as a "serious mistake," no proactive state-led operation replaced it [24].

The stakes of attribution are therefore practical, not academic. Over Easter weekend 2017, the second anniversary of the April 2015 shipwrecks, the NGOs "took the lead in the rescue of more than 9,000 migrants," and "[w]hile states have not taken up the task of proactive SAR, without the life-saving presence of SAR NGOs, many of the migrants would have died in a tragic repetition of the events of 2015." If the criminalisation campaign succeeds in forcing the flotilla to stop or scale down, "there is a real risk that many more lives will be lost in the Mediterranean." The report's conclusion generalises the section's finding: "[a]s long as migrants are forced to resort to smugglers for lack of legal pathways, proactive Search and Rescue at sea will be a humanitarian necessity – whether it is operated by states or NGOs. Only a fundamental re-orientation of the EU's migration policies to grant legal and safe passage may bring the smuggling business… to an end" [24]. The tactics worsened; the evidence says the rescuers were the response, not the cause.

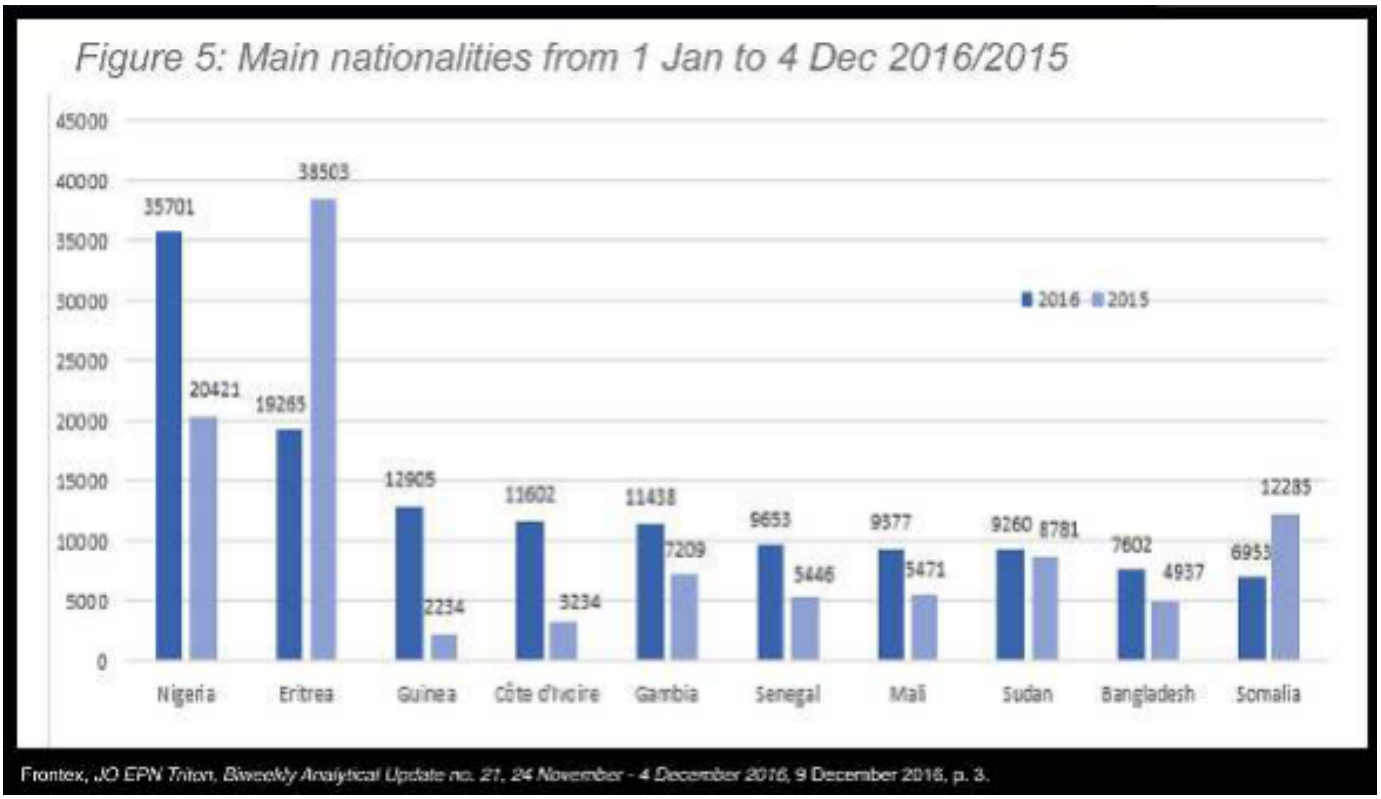


Exhibit. crossing the sea dropped by nearly half. However, migrants of di昀昀erent nationalities resort to using distinct smuggling networks, which have their respective operational modes, implying more or less danger for migrants. Source: S0023.

Central Mediterranean route: Frontex risk analysis for 2018

Frontex's Risk Analysis for 2018 is the agency's consolidated account of the year the central Mediterranean route broke — and read against the forensic counter-analysis of the preceding sections, it is a revealing document in two registers at once: what it counts, and what it no longer argues. The headline figures frame the system: 204,719 detections of illegal border-crossing across all EU external borders in 2017, down 60% on 2016 and 89% on the 1.8 million of 2015, in the agency's first full year operating as the European Border and Coast Guard Agency [26] [26]. Within

that total, the central Mediterranean remained "by far the largest" route — and its year divides into two sharply different halves.

The first half of 2017 ran at or above the elevated levels of 2016, "due to even higher migratory pressure from Libya than in the year before": June 2017 registered 23,461 detections against 22,344 in June 2016. Then the reversal. In July the numbers "dropped suddenly and markedly to less than half the level of June" — 11,460 — "followed by an even larger fall to almost a third of that level in August," 3,914, and they stayed low for the rest of the year, giving a route total of 118,962 detections, down 34% on 2016 [26]. The agency's causal language is the point. The fall is attributed to "a variety of factors, notably internal developments in Libya" — and Frontex is explicit about what did not cause it: the drop was "not a decline in the pool of migrants looking to use the route." In the second part of the year, "migrants were increasingly prevented from departing from Libya." Unrelated to the Libyan decrease, more Tunisians and Algerians embarked from their own shores in the third and fourth quarters [26]. Prevention of departure, not diminished demand, is the mechanism the agency itself records — the outcome the Italy–Libya Memorandum and the coast-guard build-up were designed to produce, stated in the neutral register of a risk analysis.

The surrounding routes display the displacement geometry. On the eastern Mediterranean, pressure in 2017 "remained roughly on a level with the months after the implementation of the EU-Turkey statement," between 1,601 and 7,136 detections a month, and the agency judges that "the statement has continued to prevent a large share of departures" — while the comparison of preventions and apprehensions suggests attempts actually rose, particularly in September. Diversion effects show at the margins: 5,540 crossings on the Greek land border with Turkey, "the highest number of detections since 2012," and a temporary Black Sea route that carried 537 migrants between August and November, "the highest number of migrants ever detected on this route since 2009," as a large pool of potential migrants remained stranded in Turkey. The Western Balkan flow continued at lower levels, concentrated at Serbia's borders, with stranded migrants making repeated attempts at different border sections. The western Mediterranean, meanwhile, "hit a new record high in 2017, more than doubling the previous record of last year," reaching new levels from June; in the latter part of the year, increased use of rubber dinghies east of the Strait of Gibraltar by sub-Saharan migrants suggested to Frontex "increased activity of people smuggling networks, as the dinghies and the required engines are unlikely to have been procured by the (mostly sub-Saharan) migrants themselves." On the western African route, 421 detections — the lowest since 2015 — are credited to the Spain–Senegal–Mauritania memorandum's joint surveillance and effective returns: the externalisation model, working as its proponents advertise, at the far edge of the system. The eastern land border reported its lowest crossings since FRAN data collection began, with Russian authorities "continuing good cooperation" and more than a third of detected migrants Vietnamese citizens [26].

The enforcement balance sheet shows the system's other bottleneck. Member States issued 279,215 return decisions in 2017, down 8.6% on 2016 — a figure the agency notes is understated, with data unavailable from Austria, France and the Netherlands — against 151,398 effective returns. The gap is structural: "the main reasons for non-return are related to practical problems in the identification of returnees and in obtaining necessary documentation from third-country authorities," and many voluntary-return decisions simply do not materialise, the persons deciding to stay illegally; of the returns that did occur, half were voluntary and half forced. The agency itself assisted in more than 14,000 returns, about a third more than the previous year. And the data infrastructure is admitted to be weak: the Commission found basic parameters — average length of detention, grounds for detention, failed returns, entry bans — available from only a limited number of Member States, with "common definitions and approaches concerning data collection frequently absent" [26].

Secondary movement, finally, persists beneath the suppressed headlines. The concerted border effort along the Western Balkan route reduced the visible flow, but "these movements continued after March 2016 and throughout 2017, although at lower levels and in a more covert way": detections of persons travelling without valid documents between Schengen countries fell only around 15% between 2016 and 2017. The asylum data expose the undercount — around 13,800 Afghan, Iraqi and Pakistani nationals were detected crossing the eastern Mediterranean illegally in 2017 and only around 8,700 the Western Balkan route, yet Member States reported some 78,000 asylum applications from those three nationalities. Detections of West African nationals staying illegally remained at 2016 levels even as their recorded border-crossings fell by a third; France detected substantially more people who had previously applied for asylum or been detected entering Italy, many transiting toward the United Kingdom; and in November 2017 Austria announced checks for migrants hiding on freight trains near the Brenner Pass, after repeated cases of people "risking their lives travelling on cargo trains" toward Germany [26].

Set beside the argument of the previous sections, the document's silences are as informative as its numbers. The pull-factor framing that structured the 2017 edition's treatment of rescue does not organise the 2018 route account; what appears instead is the causal apparatus the counter-analysis had insisted on — Libyan internal developments, departure prevention, a migrant pool that has not diminished. Forensic Oceanography had already identified the pattern in the earlier edition: Frontex itself recognised that the rising crossings of 2016 were "consistent with the increase amongst migrants of different Sub-Saharan nationalities, which had begun already over 2015 — that is, prior to NGO SAR assets becoming the biggest rescue actor in the central Mediterranean," recognised regional drivers in economic and political crises, and documented the western Mediterranean's 46% increase "in absence of any NGO SAR assets whatsoever" — yet those findings "were not mentioned in relation to arguments that focused on the role of SAR NGOs allegedly constituting a pull-factor. This occlusion contributed to the toxic narrative against SAR NGOs" [24]. The Risk Analysis for 2018 does not retract the narrative; it simply stops needing it. The route had been closed by other means — and the agency's own figures record that the people who would have used it were still there, stranded on the far shore, with the analysis's outlook naming "the remaining large pool of migrants stranded in Libya" as the decisive variable for the year ahead [26] [26].



Exhibit. Detections of illegal border-crossing at border crossing points by border section, including people hiding in vehicles, in 2017 with 2016 figures in parentheses. Source: S0022.

When the Sea Itself Is the Hazard

The Andrea storm and the Andrea wave

Just past midnight on 9 November 2007, in the middle of a North Sea storm, the down-looking lasers at the Ekofisk field recorded a single wave that stands out of its own sea state the way the Draupner wave had twelve years earlier: the Andrea wave. It is one of the very few rogue waves ever measured in the open ocean, and the paper at the centre of this closing cluster — a comparison of the measured Andrea wave with numerical simulations — begins from the storm that made it and the instrument record that caught it [27] [27].

The storm, named Andrea, was a compact, violent low. It entered the northern North Sea on 8 November 2007, covered southern Norway, and moved on the morning of 9 November towards southern Sweden. Strong westerly winds of 50–55 knots followed the low, and in the afternoon of 8 November a high wave field built up north of the Ekofisk field, with significant wave height of 10–11 metres. At Ekofisk itself the wind eased slightly in the evening, between about 18:00 and 21:00 UTC, dropping from 22 to 19 metres per second; the strongest part of the wind field passed northeast and east of Ekofisk around 06:00 UTC on 9 November and generated waves of up to 11–12 metres [27]. The Andrea wave did not arrive at the storm's climax. It was recorded "just past 00:00 UTC the 9 November 2007" — the hindcast analysis refines the moment to 00:40 — hours before the wind field peaked, in water about 75 metres deep [27] [27].

What the lasers caught qualifies, formally, as the most extreme category the literature recognises. Wave scientists describe a record by its significant wave height Hs, its spectral peak period Tp, the maximum crest height Cmax, the maximum zero-downcrossing wave height Hmax associated with that crest, and two ratios: the crest factor CF (the crest criterion) and the height factor HF (the height criterion). Simplified definitions call a wave rogue within a 20-minute record if CF exceeds 1.3 (or 1.2) and HF exceeds 2; a wave fulfilling both criteria is, following Krogstad and colleagues, a "double rogue wave." Both the Andrea wave and its famous predecessor qualify. The predecessor is the New Year wave — the Draupner wave — measured by Statoil at the Draupner platform on 1 January 1995, and the paper tabulates the two side by side: comparable in characteristics, both recorded in the North Sea at platforms standing in water depths of about 75 metres. The Andrea wave's crest ratio, Cmax/Hs = 1.63, is the signature number: a crest standing more than one and a half times the significant wave height of the sea around it [27] [27].

A single wave, however well measured, would be a curiosity if the field were rich in data. It is not, and the introduction is frank about why studies like this one must lean on simulation. Rogue waves are rarely recorded at all; standard field practice samples the sea surface for 17–30 minutes every third hour, so any individual record carries "sampling variability" — uncertainty from the limited number of observations — and the met-ocean conditions between measurements have to be assumed stationary for want of information. Continuous measurement stations exist but "are still spare." The statistics that diagnose rogue-wave physics are precisely the ones sampling hurts most: the higher-order moments, skewness and kurtosis, "are more unstable than the estimates of significant wave height," and obtaining reliable estimates of them can require 250–350 repetitions of a 17-minute record. "Therefore numerical wave models remain an important supporting tool in an analysis of field data" — in this study, the spectral model WAM, which describes a sea state as a two-dimensional spectrum but "does not give any information about the instantaneous position of the sea surface," coupled to a phase-resolving nonlinear model solving the Euler equations by the Higher Order Spectral Method, which reconstructs the surface itself and, by repeating simulations, can average down the sampling uncertainty that a single 20-minute laser record cannot escape [27].

The storm's own evidence already narrows the mystery of what made the wave. The recognised generation mechanisms are five: linear Fourier superposition (frequency or angular focusing), wave–current interactions, crossing seas, quasi-resonance nonlinear interactions — modulational instability — and shallow-water effects. Linear focusing is rare and, in a study addressed to nonlinear waves, set aside. No strong current was reported in the Ekofisk area, an intermediate-depth zone, which removes wave–current interaction and, with it, shallow-water effects. Crossing seas require two wave trains of similar energy and frequency travelling at particular angles — the kurtosis that measures extreme-wave probability peaks for crossing angles between 40° and 60° — and the Andrea storm fails that condition decisively: "wind sea dominates clearly the total sea during the growth, peak and decay of the storm," with swell significant wave height "only slightly above 1 m" against a wind sea reaching mostly 10 metres at the peak, the two comparable only at the storm's start and end. The directional spectrum through the night of the recording, from 18:00 UTC on 8 November to 06:00 UTC on 9 November, shows a single wave system. What remains is modulational instability: the hindcast "seems to point out that the Andrea wave might have occurred in the sea state more prone to extreme waves as a result of modulational instability (i.e. the sea state with relative high steep-

ness and not particularly broad spectrum)" — the storm's steepness reached 0.14 at its maximum, and, suggestively, "the same high steepness is observed before the significant wave height reaches its maximum." One caveat is built into the evidence itself: with hindcast sampling every six hours, the steepness at 00:40, the moment of the wave, cannot be directly observed [27].

So the Andrea wave stands as both a measurement and a problem. A double rogue wave, crest ratio 1.63, rose out of a wind-sea-dominated storm in 75 metres of water at 00:40 UTC, hours before the storm's own significant wave height peaked — in a sea state whose steepness was already high and whose spectrum, the simulations would test, stayed narrow enough for nonlinear focusing to do its work. The paper's conclusions add the unsettling corollary that Ekofisk was not uniquely unlucky: across the storm's history, sea states capable of producing rogue waves arose several times and in several places — during the storm's development and decay, notably, rather than at its maximum height [27]. Whether the numbers can reproduce the wave itself is the question the simulations take up next.

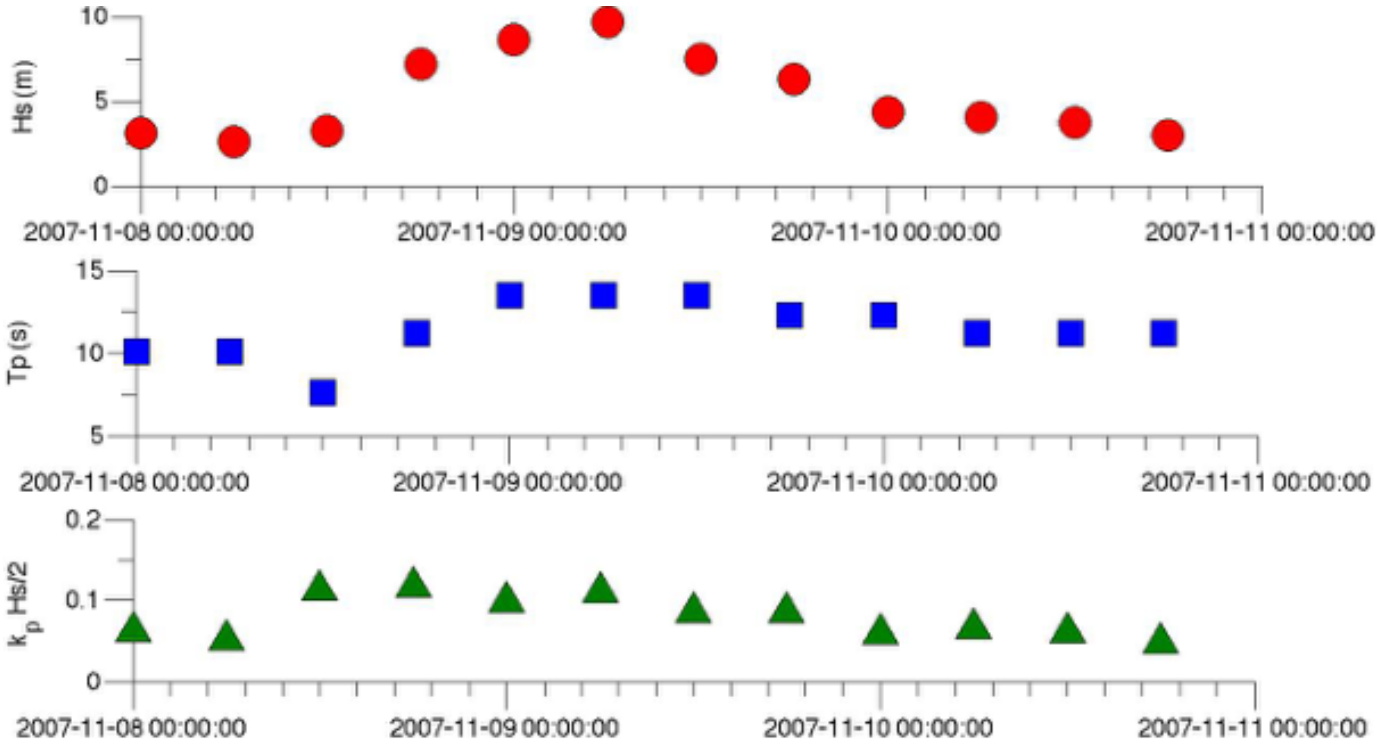


Exhibit. History of significant wave height, spectral wave period and sea state steepness for the total sea during the Andrea storm. Source: S0025.

Comparing the measured rogue wave with numerical simulations

The comparison at the heart of this study is asymmetric by construction. On one side stands a measurement: twenty minutes of laser data from Ekofisk, containing one crest with a ratio Cmax/Hs of 1.63. On the other side stands an ensemble: hundreds of simulated sea surfaces, generated from hindcast spectra at a location some 50 kilometres from Ekofisk, none of which is the Andrea wave itself. The question the simulations answer is therefore not "can the model rebuild that wave?" but the more useful one — do the sea states the storm actually produced, evolved under the nonlinear physics the models encode, generate crest statistics that reach toward what the lasers recorded? [27]

The machinery runs in two stages. The spectral side supplies the storm's evolving directional wave spectrum across its history — the spectra shown for the period from 00:00 UTC on 8 November to 18:00 UTC on 10 November 2007 "have been an input to the present simulations." Each spectrum is then realised as an initial surface for the phase-resolving model, which solves the Euler equations by the Higher Order Spectral Method. The numerical details are chosen to protect exactly the quantities under study: 256 modes were applied in the downscaling "to maintain unchanged the peak period," because with too few modes "the new wavenumbers may not necessarily coincide with the original modes," shifting the peak; the domain's grid spacings were selected "to ensure 10 wavelengths in the physical domain"; and each initial surface "has been let to evolve... for 70 peak periods." Because a single realisation would reproduce the field data's sampling problem inside the computer, "[i]n order to have statistically signific-

ant results, 150 repetitions of the wave surface with different sets of random amplitudes and phases... for the initial surface have been simulated" [27].

The headline comparison is close, and honestly short. Tracking the crest ratio CF = Cmax/Hs, the skewness and the kurtosis through the storm days of 8–11 November, the simulations at the study location produce a maximum crest ratio of 1.4, "corresponding to the kurtosis value of 3.35." Set against the measurement: "[t]he ratio Cmax/Hs = 1.4 is comparable to the one characterising the Andrea wave Cmax/Hs = 1.63 recorded at the Ekofisk field (see Table 1), however, not as high." The gap deserves a plain reading. The ensemble demonstrates that the storm's sea states, under nonlinear evolution, generate crest ratios deep into the rogue range together with the elevated kurtosis that marks a sea prone to extremes — the regime is reproduced. The single largest observed crest, a one-in-a-storm event sampled by chance at a platform 50 kilometres away, sits above the simulated maximum; the models confirm the conditions for the Andrea wave without counterfeiting the wave itself [27].

The timing evidence cuts the same way, once the geometry is respected. Within the 70-peak-period evolution windows, the simulated sea's kurtosis at the study location reaches its maximum early — "already after 30 Tp" in the simulations initialised at 18:00 UTC on 8 November, the evening the storm was building over Ekofisk. The authors note the two reasons this cannot be read as a mismatch with a wave recorded at 00:40 on 9 November: a 70 Tp window "is not sufficient to include the time when the Andrea wave was recorded at the Ekofisk," and, more fundamentally, "[t]he location used in the analysis is not the same as the Ekofisk one therefore the kurtosis and wave crest could reach their maximum prior to the time of the occurrence of the Andrea wave at Ekofisk." Rogue-capable sea states, meanwhile, are not a single moment in the ensemble: "[i]t is interesting to note that the sea states being able to trigger rogue waves occur several times during the Andrea storm history at the location considered." Judged by steepness alone, such states are unremarkable in the North Sea — the Hs–Tp scatter diagrams "indicate that sea states with similar steepness do not occur very seldom" — but the authors warn that scatter diagrams "provide limit description of sea states as they do not give any information about the frequency-directional wave spectrum," which is where the rogue-relevant structure lives [27].

The paper's own verdict is measured. The coupled approach delivers, from "the actual hindcast/forecast spectrum, whether this is bimodal or unimodal," the quantities field campaigns struggle to stabilise — "sea surface elevation, the maximum wave crest, skewness and kurtosis when rogue waves are present" — and "[t]he results obtain agree satisfactory with the observations recorded at the Ekofisk field." The method "is of general character and can be applied to investigations of any other storm," serving as "a good supporting tool for an analysis of field data," and the authors propose it "can be considered to be used for forecasting purposes," computationally intense though both models are. The limitations are stated with equal plainness. HOSM "cannot be applied for prediction of rogue waves for very steep sea states because it does not include wave breaking." The six-hour sampling of the spectral input "may present a limitation when studying extreme and rogue waves. Important information between sampling intervals maybe missed." And "[u]ncertainties associated with the WAM and HOSM model, which will affect the presented results, have not been investigated in this study" — further work is called for to evaluate their impact [27].

That is the real content of the comparison: calibration in both directions. The measurement disciplines the models — any credible simulation of this storm must produce a rogue-prone sea and crest ratios approaching 1.63, and the shortfall to 1.4 marks where breaking physics, sampling gaps and unassessed model uncertainty still live. The models discipline the measurement — the lone laser record becomes a population statement: across the storm's development and decay, at the study location as at Ekofisk, the North Sea that week repeatedly entered sea states in which a wave like Andrea was not an anomaly but an expectation [27] [27].

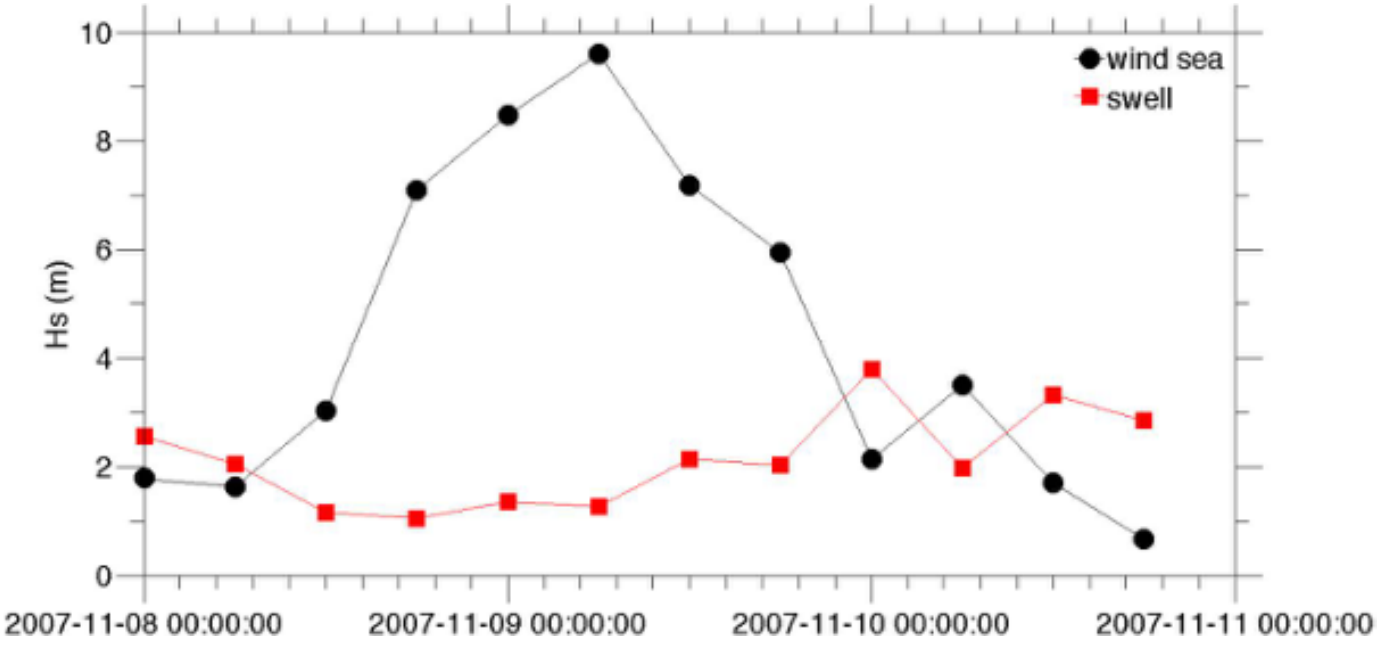


Exhibit. Fig. 3. History of significant wave height for wind sea and swell during the Andrea storm. Source: S0025.

Hindcast data and storm-driven dispersal

Every object this report has followed — the derelicts, the toys, the tsunami's armada of houses and docks — poses the same predictive problem: the sea scatters in a day what observers then spend years finding, one beach at a time. The working answer, across the whole corpus, is hindcasting: reconstruct the winds and currents that actually blew, release the debris into the model, and compute where the dispersal must have gone. The Canadian science review of the Tōhoku debris, updated in March 2013, shows the method maturing in public — and the global mass-budget modelling shows the same machinery answering the largest question in the field, where all the plastic goes.

The update exists because the first advice had been overtaken. The original assessment's simulations had modelled objects with essentially zero windage, driven by climatological statistics — the average Pacific. The new simulations, the review notes, "examine the influence of windage on the transport of debris" and, decisively, "rather than climatological statistics, the new model runs make use of the actual wind forcing over the Pacific from March 2011 through August 2012." That substitution — the weather that happened for the weather that usually happens — is the hindcast move, and it changed the forecasts' resolution: arrival could now be predicted by debris class and season, not just in aggregate [2].

The physics the hindcast encodes starts with the conveyor. Debris deposited along the northern Honshu coast entered the region where the Oyashio and Kuroshio currents merge into the Kuroshio Extension, "a swift eastward-flowing current" that carried it toward the central North Pacific, where the flow "becomes very broad and sluggish" as the North Pacific Current, the West Wind Drift. Approaching North America, that current bifurcates — drifter studies place the split at about 50°N, with year-to-year variability — into the northeastward Alaska Current, feeding Canadian waters, and the southeastward California Current, feeding the US west coast and Hawaii. Currents alone do not beach anything: "[c]urrents can only bring debris close to the shore. Debris is then deposited on the shoreline by the action of retreating tides, onshore winds, and waves." And the delivery is seasonal — summer's northerly winds drive upwelling and offshore transport that "help to protect the coast," while winter's southerlies create "downwelling conditions and onshore transport of surface waters which favours deposition of debris along the coast" [2].

Superimposed on the conveyor is diffusion, and the review quantifies it. Eddies — "ocean analogues to the atmospheric systems that produce familiar day-to-day weather patterns" — spread an initially compact cluster with a lateral eddy diffusivity that Zhurbas and Oh measured at 5,000–10,000 m²/s in the Kuroshio Extension. At the low end of that range, the dispersion scale reaches some 400 kilometres after six months: "a circle with a radius of 400 km encompasses an area of about 500,000 km², which is comparable to the area of the province of Alberta… or the state of California." Within half a year, then, the tsunami's debris field was "a very diffuse cloud that extends over hundreds of thousands of square kilometres" — though not a uniform one: "patchy clumps and elongated filaments of debris are likely to be observed" [2].

What sorts the cloud is windage — how much of an object stands above the waterline to catch the wind. An object's velocity decomposes into the water's velocity plus a wind-driven slip, and the simple drag-balance model gives the slip's scale: an object with equal area above and below the waterline drifts relative to the water "at a rate of 2.5% of the 10-metre wind speed," in the wind's direction, which need not match the current's. Docks, containers, drums, buoys and vessels ride high and cross fast; the review judges such objects "a relatively small fraction of all the debris," but an outsized one, "since, in many instances, large, potentially hazardous, objects will have significant windage" [2] [2]. The forecasts split accordingly: high-windage debris "started to arrive along the BC coast during the winter of 2011-12," exactly when the seasonal winds favour deposition; the low-to-moderate-windage bulk moved more slowly, "expected to approach the coast of North America during the second half of 2012, through the first half of 2013," arriving for years from Alaska to northern California, mostly as small pieces — while most low-windage debris would never beach at all, remaining at sea to collect in the Subtropical Convergence Zone. The Strait of Georgia was written out of the forecast entirely, barred by seaward surface flow at the Juan de Fuca mouth and the island barrier behind it [2].

The global models run the identical experiment with the whole planet's coastline as the source. In the mass-budget study, Lagrangian particles — 2,510,918 of them in the stranding analysis — are released from coastal point sources in proportion to each country's mismanaged waste, timed randomly through the year, and advected by "sea surface currents, stokes drift and variable influences of wind" through two decades of modelled circulation. Stranding is made operational: a particle counts as shore-bound if it spends two consecutive days within one model cell (1/16°, a few kilometres) of land — at least one full tidal cycle. The result is the dispersal system's master asymmetry: with no wind influence, 96% of particles have spent two consecutive days near a shoreline within a year; with a windage coefficient of 2% of the 10-metre wind, 98% have. Wind drives debris ashore; what accumulates in the gyres is the low-windage residue — "in good agreement with observations reporting mainly low-windage debris accumulating in oceanic gyres." The budget's converged parameters put the stranding probability at 97%, offshore transport at 33% and coastal release at 1% [10].

Those numbers are what let the hindcast arbitrate the field's standing paradox. A pure emission–degradation model had predicted that 99.8% of all plastic mass entering the ocean since 1950 has degraded into micro- and nanoplastics and settled, and that under zero emissions "almost all plastic would be removed from the ocean surface layer within three years." The dispersal evidence refutes the premise twice over: debris "requires on average a minimum of several years to reach such remote oceanic areas," so a sub-three-year surface lifetime could never stock the gyres at all — and the gyres' own contents testify against it, with "a significant number of decades-old objects, dating as far back as the 1970s" identified by production dates on collected macroplastics, in a medium where solar UV degradation "is severely retarded when floating in seawater." The mass-budget authors' resolution is coastal dynamics: "a significant time interval, on the order of several years to decades, between terrestrial emissions and representative accumulation in offshore waters," from which it follows that "the current generation of secondary microplastics in the global ocean is mostly a result of the degradation of objects produced in the 1990s and earlier" [10] [10].

The tsunami itself becomes the tracer that validates the machinery. Of the estimated 4.5 million tonnes washed to sea in 2011, some 70% may have sunk rapidly, leaving about 1.4 million tonnes to travel at the surface; the GPGP survey's collected objects carry the origin evidence — Japan the main production country for 34% of labelled items, Japanese the most common inscription language at 30% — and the dispersal model, assuming the tsunami debris shared the global sources' low-windage fraction, attributes "around 10–20% of the debris input into the GPGP post 2011" to the Tōhoku event [8]. A single storm-driven injection, hindcast forward, lands as a measurable share of an ocean basin's accumulation.

That is the thread on which the report's disparate fleets all hang. Whether the question is a rogue wave's sea state at 00:40 on a November night or a dock's landfall two winters after a tsunami, the evidence arrives as a record too sparse to speak alone — a laser's twenty minutes, a satellite's month of tracking, a sighting log — and it is hindcast data, the archived weather of the real ocean, that turns the record into a reconstruction: diffuse clouds with filaments, winter-weighted arrivals, windage-sorted fates, and error bars honestly the size of provinces [2] [2].

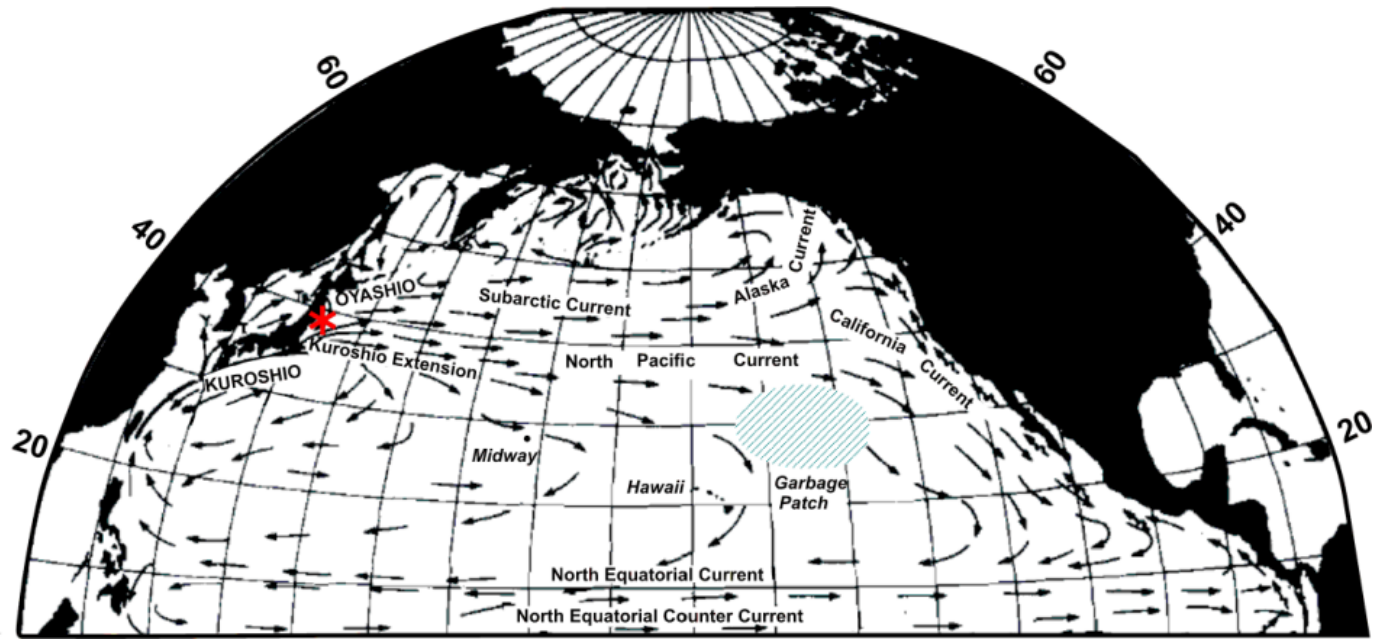


Exhibit. Figure 2. Schematic diagram showing major currents of the North Pacific. Source: S0012.

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